

A Bit of Algebra

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1

Axiomatic Set Theory

1.1 Why Axiomatic Set Theory ?

In every mathematical theory, there are corresponding primitive concepts (the reason being that every definition introduces new concepts, in other words, defining concepts is an endless process unless we manually stop it, so we must have primitive concepts in order to begin our study): for example, “point” in geometry; “number” in number theory, and so on. Mathematicians describe the behavior and properties of these primitive concepts through axioms.

In what is commonly called *naïve set theory*, the fundamental primitive concept is that of a *set*. Intuitively, a set is understood as any collection of all objects that satisfy some given property. Based on this intuitive understanding, one might adopt the following informal axiom:

AXIOM 1 (Naive Comprehension). If φ is a property, then there exists a set

$$Y = \{x : \varphi(x)\}$$

consisting of all elements that possess the property φ .

There is a concept “property” here that has not yet been explained; its standard definition will be discussed in detail in the next section. This axiom for “naïve” set theory, however, is false ! It follows from the following theorem of Russell known as *Russell’s Paradox*.

RUSSSELL’S PARADOX. Consider the set S whose elements are all those (and only those) sets that are not members of themselves: $S = \{X : X \notin X\}$. Does S belong to S ? If S belongs to S , then S is not a member of itself, and so $S \notin S$. On the other hand, if $S \notin S$, then S belongs to S . In either case, we have a contradiction. Thus, we conclude that

$$\{X; X \notin X\} \text{ is not a set.}$$

This paradox and other similar to it, convinced mathematicians that we cannot rely on our intuition when dealing with abstract notions such as “set”. To avoid this trouble, “naïve” set theory has been replaced with axiomatic set theory.

Finding a single “universal” system of axioms for set theory which matches our intuition and avoids paradoxes is difficult. But after nearly a century of discussion, a set of ten axioms known as **ZFC** came to be seen as the most natural foundation. It is worth noting that **ZFC** is far from “perfect”. It could be expected that a “perfect” set of axioms should be

1. complete: for any statement φ expressed in the language of set theory, either φ or its negation is a consequence of axioms;
2. consistent: the system should not lead to a contradiction.

Unfortunately, we cannot prove either of these properties for the **ZFC** axioms. More precisely, we do “believe” that the **ZFC** axioms are consistent. However, if this belief is correct, we can not prove it using the **ZFC** axioms alone. Does it mean that we should search for a better system of set theory axioms that would be without such a deficiency ? Unfortunately, there is no use in searching for it, since no “reasonable” set of axioms of set theory can prove its own consistency. This follows from the following celebrated theorem of Gödel:

1.1.1 THEOREM. *Gödel’s Second Incompleteness Theorem* Let T be a set of axioms expressed in a formal language $\mathcal{L}$ and assume that T has the following “reasonable” properties.

1. T is consistent.
2. There is an effective algorithm that decides for an arbitrary sentence of the language $\mathcal{L}$ whether it is in T or not.
3. T is complicated enough to encode simple arithmetic of the natural numbers.

Then there is a sentence φ of the language $\mathcal{L}$ that encodes the statement “ T is consistent.” However, φ is not a consequence of the axioms T .

In other words, it is impossible to prove internal consistency within a system that is sufficiently powerful to describe its own rules of operation and is itself consistent; one can only prove it using some external system.

1.1.2 THEOREM. *Gödel’s First Incompleteness Theorem* Let T be a set of axioms expressed in a formal language $\mathcal{L}$ and assume that T has the following “reasonable” properties

1. T is consistent.
2. There is an effective algorithm that decides for an arbitrary sentence of the language $\mathcal{L}$ whether it is in T or not.
3. T is complicated enough to encode simple arithmetic of the natural numbers.

Then there is a sentence φ of the language $\mathcal{L}$ such that neither φ nor its negation $\neg\varphi$ can be deduced from the axioms T .

1.2 Language of Sets

In the previous section, many abstract or vague concepts were mentioned, such as “property.” In order to standardize these vague concepts, we need to specify a language for set theory to describe various propositions about sets, and so on. When describing the set theory, we use the “variable symbols”, “logical symbols” (including the five connectives and two quantifiers), and “non-logic symbols” provided by *first order logic*. In detail, the language of set theory includes: variables used to denote sets; the logical symbols needed to form formulas (see definition 1.2.1); and the non-logic symbol “ $\in$ ” that represents the membership relation between sets and the non-logic symbol “ $=$ ” represents equality. (By the way, it is stipulated that membership is also a primitive concept in set theory, that is, set theory has two primitive concepts: set and membership.):

1. $\neg$: It is used to denote “negation” and is called the negation symbol.
2. $\wedge$: It is used to denote “and” and is called the conjunction symbol.
3. $\vee$: It is used to denote “or” and is called the disjunction symbol.
4. $\rightarrow$: It is used to denote “implies” and is called the implication symbol.
5. $\leftrightarrow$: It is used to denote “equivalence” and is called the equivalence symbol
6. $\forall$: It is used to denote “for all” and is called the universal quantifier.
7. $\exists$: It is used to denote “there exists” and is called the existential quantifier.

1.2.1 DEFINITION. 1. The **atomic formulas**: $(x \in y)$, $(x = y)$;

2. The **formulas** are recursively defined by the following five rules:

- (a) The atomic formulas are formulas;
- (b) If φ is a formula, then $\neg\varphi$ is a formula;
- (c) If φ, ψ are formulas, then $(\varphi \wedge \psi)$, $(\varphi \vee \psi)$, $(\varphi \rightarrow \psi)$, $(\varphi \leftrightarrow \psi)$ are formulas;

- (d) If φ is a formula and x is a variable, then $(\forall x\varphi)$ and $(\exists x\varphi)$ are formulas;
- (e) Any formula is obtained only by one of the above four rules.

Only variables, the relations $=$ and $\in$, and the logic symbols already mentioned are allowed in formal formulas. However, we will often use some other symbols. But it will be tacitly understood that each such formula can be written in a form that only involves $=, \in$ as nonlogical symbols. There are there commonly used “illegal” symbols:

1. $a \subset b : \forall x(x \in a \rightarrow x \in b)$;
2. $x \neq y : \neg(x = y)$;
3. $x \notin y : \neg(x \in y)$.

1.2.2 DEFINITION. In the formulas $\forall x\varphi$ and $\exists x\psi$, the occurrence of x is called a **bound variable**. If a variable in a formula is not a bound variable, it is called a **free variable**. For example, in the formula $\forall y\neg(x = y)$, y is a bound variable, and x is a free variable. A formula containing no free variables is called a **sentence**. For example, $\forall x\forall y(x = y)$.

1.2.1 REMARK. Note that the difference between sentences and general formulas: a sentence has a definite truth value, whereas the truth value of a general formula is often indeterminate.

1.2.2 REMARK. Concerning formulas with free variables, we adopt the notational convention that all free variables of a formula $\varphi(u_1, \dots, u_n)$ are among $u_1, \dots, u_n$ (possibly some u_i are not free, or even do not occur in φ).

AXIOM Logic. (Equality). In more general logical axioms, there are three axioms of equality. However, we are now only discussing axiomatic set theory. For this, there are only two axioms of equality, as shown below (the reason is that the third axiom involves function symbols, which are not symbols in the language of set theory).

1. $\forall x(x = x)$;
2. $\forall p_1 \dots \forall p_n \forall x \forall y ((x = y) \rightarrow (\varphi(x, p_1, \dots, p_n) \rightarrow \varphi(y, p_1, \dots, p_n)))$.

In the second axiom, $\varphi(y, p_1, \dots, p_n)$ is obtained by replacing the free variable x in φ with y . For example: $(x = y) \rightarrow \forall z(z \in x \rightarrow z \in y)$. In fact, such a substitution need not replace all occurrences; it can replace one or more. However, to avoid introducing too many definitions, we shall simply replace all of them. The reader only needs to understand the idea; for details, any textbook on mathematical logic may be consulted.

Before we formally begin explaining the **ZFC** axioms, we introduce a concept that does not belong to the **ZFC** axiomatic system. In the **ZFC** system, there is only one type of object we study, namely sets. However, other axiomatic system (for example, NBG) introduce another concept, namely **classes**.

1.2.3 DEFINITION. If $\varphi(x, p_1, \dots, p_n)$ is a formula, for each fixed choice of these parameters $p_1, \dots, p_n$, we may define

$$C = \{x : \varphi(x, p_1, \dots, p_n)\}$$

and it is called a **class**. Members of the class C are all those sets x that satisfy $\varphi(x, p_1, \dots, p_n)$, that is, $(x \in C) \leftrightarrow \varphi(x, p_1, \dots, p_n)$. (This is entirely defined in the language of set theory and is “legal”.)

We use the notion of a class merely as a name before determining whether it is a set, and we will not elaborate further on it. We shall avoid the issue of what sort of object a class is.

1.3 Five Basic Axioms

Axioms of a theory are always sentences, that is, they have no free variables. In expositions, an axiom may be presented with free variables, but it is understood that the actual axiom is the **universal closure**, obtained by universally quantifying each free variable.

AXIOM ZF0. (Set Existence). $\exists x(x = x)$.

Regarding this axiom, it actually does not need to be listed, because it can be derived from the general logical axioms. It is included here only to ensure that set theory is not a trivial theory.

AXIOM ZF1. (Extensionality). If X and Y have the same elements, then $X = Y$. In other words, a set is determined by its elements:

$$\forall X \forall Y (\forall u (u \in X \leftrightarrow u \in Y) \rightarrow X = Y) \quad (1-1)$$

1.3.1 REMARK. Since “=” is the equality predicate, $X = Y$ implies $\forall u (u \in X \leftrightarrow u \in Y)$, i.e., $(X = Y) \rightarrow \forall u (u \in X \leftrightarrow u \in Y)$. Thus, we have $(X = Y) \leftrightarrow \forall u (u \in X \leftrightarrow u \in Y)$. In other words, we can take $(X = Y) \leftrightarrow \forall u (u \in X \leftrightarrow u \in Y)$ as the first axiom, and the predicate calculus of the logical language ensures that we only need to provide another implication.

AXIOM ZF2. (Separation Schema). Let $\varphi(u, p)$ be a formula with free variables u, p . For any given sets X and p , the class $Y = \{u : u \in X \wedge \varphi(u, p)\}$ is a set:

$$\forall X \forall p \exists Y \forall u (u \in Y \leftrightarrow u \in X \wedge \varphi(u, p)) \quad (1-2)$$

1.3.2 REMARK. For each formula $\varphi(u, p)$, the formula (1-2) is an axiom of Separation. It is easy to verify that the set Y in (1-2) is unique by Extensionality.

1.3.3 REMARK. Readers may observe that this axiom seems similar to the first axiom in the previous section, except that here a set is given in advance, and then the property to be satisfied, i.e., the formula. Intuitively, this is a kind of “restriction”; without these restrictions, the previously mentioned Russell’s Paradox would arise. In other words, to define a set, one cannot rely only on a property; one must already have a set in advance, and then use the property within that set to form a new set. In this way, Russell’s paradox can be avoided.

1.3.4 REMARK. According to the Axiom of Set Existence, sets do exist, but so far we have not yet seen a concrete set. Now, if we take a fixed set X , we can use this axiom to define a set (uniquely):

$$Y = \{u \in X : u \neq u\},$$

this set is called the **empty set**, denoted by $\emptyset$. The empty set has no elements (otherwise, there exists u and $u \neq u$, which contradicts $u = u$), so by the Axiom of Extensionality, it does not depend on the chosen set X used in its definition.

1.3.5 REMARK. Using Axiom 2, we can define the **difference** of two sets: for given sets X and Y , the class $Z = \{u \in X : u \notin Y\}$ is a set automatically and is denoted as $X - Y$.

It is perhaps curious that, given three sets a, b, c , we can now define the ordered triple (a, b, c) as a set, whereas we cannot yet justify that there is a set with precisely these

elements. Less surprising is that although we can construct an ordered pair as a set, we cannot yet construct interesting sets of ordered pairs, like the **Cartesian products** $X \times Y$. The batch of axioms in the next section will rectify this situation and enable us to do much more.

Using the first two axiom, we can obtain the empty set. Are there any others ? The first two axioms cannot provide any further information: Axiom 2 only allows us to construct new sets within a known set. Intuitively, this new set will not be “larger” than the original set. In short, whether there exists any sets other than the empty set. The first two axioms do not give us the answer. Below, we introduce another axiom:

AXIOM ZF3. (Pairing). For any given sets a and b , the class $C = \{x : x = a \vee x = b\}$ is a set and is denoted as $\{a, b\}$, that is:

$$\forall a \forall b \exists C \forall x (x \in C \leftrightarrow (x = a \vee x = b)) \quad (1-3)$$

1.3.6 REMARK. By Axiom of Extensionality such set c is unique, thus, we can define $\{a, b\} = c$ such that $\forall x (x \in c \leftrightarrow x = a \vee x = b)$. Use the axiom we can define the **singleton** $\{a\}$ is the set $\{a, a\}$; we can also see that in the set a, b , the positions of a and b are symmetric, i.e., $a, b = b, a$. This reflects the concept — **unordered**. So how can we distinguish the positions of a and b ? We may define the concept of **ordered pairs**:

$$(a, b) = \{\{a\}, \{a, b\}\}.$$

It is not important how we define the ordered basis, but it must express the distinction between the positions of a and b , so that unless $a = b$, we cannot have $(a, b) = (b, a)$. This is illustrated by the following proposition:

1.3.1 PROPOSITION. $(a, b) = (c, d) \leftrightarrow (a = c \wedge b = d)$

Proof. Our proof will make existence use if axiom of extensionality and the axiom of equality. Suppose that $(a, b) = (c, d)$, i.e., $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$. By extensionality every element of the set on the left side of this equation is an element of the right side, and vice versa:

First take the element $\{a\}$ of the left side. This must equal one of the elements $\{c\}$ and $\{c, d\}$ of the right side. If $\{a\} = \{c\}$, by **ZF1**, we have $a = c$. Now consider the element $\{a, b\}$ of the left side. This must equal one of $\{c\}$ and $\{c, d\}$: if $a = b$, then by **ZF1** we have $\{a, b\} = \{a\}$, so that by **ZF1**

$$\{\{a\}, \{a, b\}\} = \{\{a\}, \{a\}\} = \{\{a\}\}.$$

Since the $\{c, d\}$ must be an element of the left side, this forces, again by **ZF1**, $\{c, d\} = \{a\}$, which in turn forces that $c = a$ and $d = a$, by **ZF1**. This gives $a = b = c = d$, so that $a = c, b = d$ as desired. If $a \neq b$, then we cannot have $\{a, b\} = \{c\}$, since **ZF1** would force that $a = c$ and $b = c$, giving $a = b$, a contradiction. Thus we must have $\{a, b\} = \{c, d\}$. Since $a = c$, this gives $\{a, b\} = \{a, d\}$. By **ZF1**, the element b of $\{a, b\}$ equals one of the elements of $\{a, d\}$. Since $b \neq a$, we can only have $b = d$. Thus, $a = c$ and $b = d$. Hence if $\{a\} = \{c\}$ we must have $a = c \wedge b = d$.

Now, if $\{a\} = \{c, d\}$, by **ZF1**, $c = a$ and $d = a$, so that $c = d$. Thus, by **ZF1**, $\{c, d\} = \{c\}$, so that

$$\{\{c\}, \{c, d\}\} = \{\{c\}, \{c\}\} = \{\{c\}\}.$$

Then $\{\{a\}, \{a, b\}\} = \{\{c\}\}$. The elements $\{a\}$ and $\{a, b\}$ must both be elements of $\{\{c\}\}$. Thus, by **ZF1**, $\{a\} = \{a, b\} = \{c\}$. By **ZF1** again, $a = b = c$. Since $a = d$, this gives that $a = c$ and $b = d$. $\square$

1.3.1 EXAMPLE. Using the axiom 3, we can construct these sets from the empty set $\emptyset$: $\{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \{\{\emptyset\}\}, \{\emptyset, \{\{\emptyset\}\}\}$. According to the axiom of Extensionality, these sets are distinct from each other.

1.3.7 REMARK. Using the ordered pairs we may further define the ordered triples, quadruples, etc., as follows (Here the symbols $1, 2, \dots, n+1$ all refer to the set of natural numbers or the smallest inductive set, which will be understood later. For now, we may use them without objection, and we simply regard this as the repeated application of **ZF3**.):

$$\begin{aligned}(a, b, c) &= ((a, b), c) \\ (a, b, c, d) &= ((a, b, c), d) \\ &\vdots \\ (a_1, \dots, a_n, a_{n+1}) &= ((a_1, \dots, a_n), a_{n+1})\end{aligned}$$

It is easy to verify that two order n -tuples $(a_1, \dots, a_n)$ and $(b_1, \dots, b_n)$ are equal if and only if $a_1 = b_1, \dots, a_n = b_n$.

1.3.8 REMARK. Note that a more general version of axiom 2 can be proved using the ordered n -tuples: Let $\psi(u, p_1, \dots, p_n)$ be a formula. Let $\varphi(u, p)$ be the formula defined as follows

$$\exists p_1 \dots \exists p_n (p = (p_1, \dots, p_n) \wedge \psi(u, p_1, \dots, p_n))$$

and then, given the sets X and $p_1, \dots, p_n$, let $Y = \{u \in X : \varphi(u, (p_1, \dots, p_n))\}$. Then we may obtain the general version of (1-2):

$$\forall X \forall p_1 \dots \forall p_n \exists Y \forall u (u \in Y \leftrightarrow u \in X \wedge \psi(u, p_1, \dots, p_n)).$$

The new sets obtained using the **ZFC.3** may be complex, but they always have only two members (Here, “member” is used to distinguish it from an element within a set, indicating that it itself is a set. In other words, axiom 3 deals with binary sets composed of sets). Next, we introduce a new axiom which provide a new method of construction: combining all the elements of a set to form a new set, called the union.

AXIOM ZF4. (Union). For any given set X , the class $Y = \{u : \exists(z \in X \wedge u \in z)\}$ is a set and is denoted as $\bigcup X$, and is called the **union** of the set X :

$$\forall X \exists Y \forall u (u \in Y \leftrightarrow \exists z (z \in X \wedge u \in z)) \quad (1-4)$$

The form of the union axiom does not initially resemble the binary union $X \cup Y$ commonly seen in naive set theory. The reason we do not adopt the binary union symbol is that it is too restrictive. Specifically, we may encounter a situation where we need to take the union of “infinitely” (We will not give rigorous definitions of the concepts of “finite” and “infinite” at this point, but it is harmless to use them in this way. The purpose here is merely to deepen our understanding of the axiom.) many sets, $A_0, \dots, A_n, \dots$. If we only had the binary union symbol, we would have to use an infinitely long formula as follows:

$$\dots A_0 \cup A_1 \cup A_2 \cup A_3 \cup \dots$$

This does not conform to the definition of a formula in standard set theory, where formulas must be of finite length. However, if there exists a set, denoted by X , whose elements are exactly these sets, then the union axiom allows us to express this union with a finite formula:

$$\bigcup X$$

1.3.9 REMARK. Nevertheless, the binary union symbol is indeed more convenient. In safe cases, we may use this symbol: define $X \cup Y = \bigcup \{X, Y\}$, $X \cup Y \cup Z = (X \cup Y) \cup Z$, etc. Then, $\bigcup \{\{a, b\}, c\} = \{a, b\} \cup \{c\} = \{u : u = a \vee u = b \vee u = c\}$ and this set is simply denoted as $\{a, b, c\}$. Similarly,

$$\begin{aligned}\{a, b, c, d\} &= \bigcup \{\{a, b, c\}, d\} = \{a, b, c\} \cup \{d\} \\ &= \{u : u = a \vee u = b \vee u = c \vee u = d\}\end{aligned}$$

1.3.10 REMARK. Using axiom 4 and axiom 2 we can also define the *intersection*: for any given set X , the class $Y = \{u \in \bigcup X : \forall z(z \in X \cap u \in z)\}$ is a set automatically and is denoted as $\bigcap X$. The role of axiom 2 here is to indicate that the intersection is defined as a subset of the union. Intuitively, the intersection is the set consisting of all the common elements of the members of X . Similarly, we have the following similar notational conventions:

$$X \cap Y = \bigcap \{X, Y\}, \quad X \cap Y \cap Z = (X \cap Y) \cap Z$$

AXIOM ZF5. (Power Set). For any set X , the class $Y = \{u : u \subset X\}$ is a set and is denoted as $P(X)$, and is called the *power set* of X :

$$\forall X \exists Y \forall u (u \in Y \leftrightarrow u \subset X) \quad (1-5)$$

1.3.11 REMARK. Intuitively, the power set of X is the set of all subsets of X . Since the empty set is a subset of any set, the power set of any set includes the empty set $\emptyset$.

1.4 Relations and Structures

Having presented the above five basic axioms, we have defined concepts such as singleton sets, ordered pairs, the empty set, intersections, set differences, and so on. Beyond these, there are many more set-theoretic structures that can be defined collectively by these five axioms:

Relations

1.4.1 PROPOSITION. $(x \in X \wedge y \in Y) \rightarrow (x, y) \in P(P(X \cup Y))$.

Proof.

$$\begin{aligned} (x \in X \wedge y \in Y) &\rightarrow \{x\}, \{x, y\} \subset X \cup Y \\ &\rightarrow \{x\}, \{x, y\} \in P(X \cup Y) \\ &\rightarrow \{\{x\}, \{x, y\}\} \subset P(X \cup Y) \\ &\rightarrow (x, y) \in P(P(X \cup Y)). \end{aligned}$$

□

This axiom is not meaningless; in fact, this proposition ensures that the following definition is legitimate, and this is guaranteed by axiom 2, because the power set is a set.

1.4.1 DEFINITION. Let X and Y be two sets, the set $\{(x, y) : x \in X \wedge y \in Y\}$ is called the *product* of X and Y and is denoted as $X \times Y$.

From now on, for the sake of brevity, when describing sets we will not explicitly emphasize the role of axiom 2. However, the reader should be aware of which known set the set in question is a subset of.

1.4.1 REMARK. Further, we may define $X \times Y \times Z = (X \times Y) \times Z$ and in general $X_1 \times \cdots \times X_{n+1} = (X_1 \times \cdots \times X_n) \times X_{n+1}$, that is,

$$X_1 \times \cdots \times X_n = \{(x_1, \dots, x_n) : x_1 \in X_1 \wedge \cdots \wedge x_n \in X_n\}.$$

1.4.2 DEFINITION. Let X and Y be two sets. A *relation* from X to Y is a subset R of $X \times Y$. A relation from X to X is called a *binary relation* on X . Generally, we may define the *n-ary relation* R as the subset of X^n .

Since $\emptyset \subset X \times Y$, $\emptyset$ is a relation from X to Y , and is the “smallest relation” from X to Y . Naturally, the “maximal relation” from X to Y is $X \times Y$ itself. Sometimes, we write $(x, y) \in R$ as xRy or $R(x, y)$. And we define the **domain** of R and the **range** of R the sets respectively:

$$\begin{aligned}\text{dom}(R) &= \{u : \exists v \in Y \wedge (u, v) \in R\} \\ \text{ran}(R) &= \{v : \exists u \in X \wedge (u, v) \in R\}\end{aligned}$$

1.4.3 DEFINITION.

1. Assume that $R \subset X \times Y$, the **inverse** of R denoted as R^{-1} is the set

$$R^{-1} = \{(y, x) \in P(P(X \cup Y)) : (x, y) \in R\}.$$

One can easily prove that $\text{dom}(R^{-1}) = \text{ran}(R)$ and $\text{ran}(R^{-1}) = \text{dom}(R)$.

2. Assume that $R \subset X \times Y$, the **restriction** of R to a set C is the relation

$$R|_C = \{(x, y) \in R : x \in C\}$$

One can easily prove that $\text{dom}(R|_C) = \text{dom}(R) \cap C$

3. Assume that $R \subset X \times Y$, the **image** of a set C by f is the set

$$R(C) = \{y \in Y : \exists x \in X \wedge (x, y) \in R\}$$

4. Assume that $R \subset X \times Y$, the **inverse image** of a set C by f is the set

$$R_{-1}(C) = \{x \in X : \exists y \in C \wedge (x, y) \in R\}$$

5. Assume that $R \subset X \times Y, S \subset Y \times Z$, the **composition** of R and S is the set

$$S \circ R = \{(x, z) : \exists y((x, y) \in R \wedge (y, z) \in S)\}$$

One can easily prove that $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.

1.4.4 DEFINITION. Let X and Y be two sets, and let $f \subset X \times Y$ be a relation from X to Y . f is a **function from X to Y** if the following sentence is true

$$\forall x(x \in X \rightarrow \exists! y(y \in Y \wedge (x, y) \in f)).$$

The unique y such that $(x, y) \in f$ is the **value** of f at x ; we use the standard notation $y = f(x)$ or its variation $f : x \mapsto y$, etc. for $(x, y) \in f$. In this case, the image of a set $C \subset X$ by f is the set $f(C) = \{f(x) : x \in C\}$ and the inverse image of a set $C \subset Y$ by f is the set $f_{-1}(C) = \{x \in X : f(x) \in C\}$.

1.4.2 REMARK. The set of all functions from X to Y is denoted as Y^X . Y^X is a set: indeed, $Y^X \subset P(X \times Y)$.

1.4.3 REMARK. A function f from X to Y is a special kind of relation. Intuitively, a function satisfies the following three conditions:

1. $f \subset X \times Y$;
2. $\text{dom}(f) = X$;
3. $(x, y_1) \in f \wedge (x, y_2) \in f \rightarrow y_1 = y_2$.

The first point is noting special to discuss. Note that the second conditions implies that a function must satisfy $\text{dom}(f) = X$ but not necessarily $\text{ran}(f) = Y$; if the latter holds, we call such a function is **onto** (or f is **surjective**). The third condition states that a function must assign a unique value to each element, but these values need not be distinct; if different elements always have different values, we call such a function **one-to-one** (or f is **injective**).

1.4.4 REMARK. Let $f \subset X \times Y$ be a function. It is clear that the inverse $f^{-1} = \{(y, x) : (x, y) \in f\}$ may not be a function from Y to X . However, if f is one-to-one, then one can easily verify that f^{-1} is a function from $\text{ran}(f)$ to X .

1.4.5 REMARK. By the definition of a function f from X to Y , the relation f is special relation from X to Y . If $Y \subset Z$, then the function f also can be identified with a function from X to Z . But they are not the “same” function: if $f : X \rightarrow Y$ is surjective, then $f : X \rightarrow Z$ may not be surjective. But this does not cause major problems; with a bit of additional definition, for example, defining it using ordered pairs, such ambiguity can be completely avoided.

1.4.6 REMARK. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two functions. We may obtain the composition of f and g , namely $g \circ f$. By the definition of the composition of relations,

$$\begin{aligned} g \circ f &= \{(x, z) : \exists y(y = f(x) \wedge z = g(y))\} \\ &= \{(x, z) : z = g(f(x))\} \end{aligned}$$

that is, $g \circ f(x) = z = g(f(x))$ for all $x \in X$. One may easily verify that the composition of two injective (surjective) functions is injective (surjective).

Structures

1.4.5 DEFINITION. A **structure** is a non-empty set X together with a set of relations on X (the arities of these relations are all fixed), a set of operations on X (namely, the functions $f : X^n \rightarrow X$) and a set of distinguished constants (elements of X). It is denoted as

$$\langle X, R, \dots, f, \dots, c, \dots \rangle$$

Sometimes this structure is simply written as X and we say that X is a structure for the language $\mathcal{L} = \{R, \dots, f, \dots, c, \dots\}$ or simply that X is an $\mathcal{L}$ -structure.

The concept of structure described above has one characteristic: a structure is always associated with some language. In many cases, there are multiple structures for the same language $\mathcal{L}$. If necessary, they are distinguished notationally, for example as

$$\langle A, R_A, \dots, f_A, \dots, c_A, \dots \rangle, \quad \langle B, R_B, \dots, f_B, \dots, c_B, \dots \rangle$$

where R_A and R_B respectively denote the n -ary relations on A and on B corresponding to the same relation symbol R (where n is the arity of R), f_A and f_B respectively denote the operations on A and on B corresponding to the same function symbol f (with arity the same as that of f), and c_A and c_B respectively denote the distinguished elements of A and B (corresponding to the same constant symbol c).

Now we introduce the concepts of isomorphism. When we talk about this concept, we are considering a function between two structures that share the same language and satisfy certain conditions. Two structures of the same language are said to be isomorphic under the following conditions. First, there exists a bijection between the two structures. Second, this correspondence has the following properties: preserving the operations, relations and constants. Formally,

1.4.6 DEFINITION. Suppose A and B are two structures of the same language $\mathcal{L} = \{R, \dots, f, \dots, c, \dots\}$. We say that they are **isomorphic** denoted $A \cong B$, if there exists a bijection $F : A \rightarrow B$ satisfying:

1. For every relation symbol R (suppose it is n -ary) of $\mathcal{L}$ and for any n elements $x_1, \dots, x_n \in A$, $R_A(x_1, \dots, x_n) \leftrightarrow R_B(F(x_1), \dots, F(x_n))$.
2. For every function symbol f (suppose it is n -ary) of $\mathcal{L}$ and for any n elements $x_1, \dots, x_n \in A$, $F(f_A(x_1, \dots, x_n)) = f_B(F(x_1), \dots, F(x_n))$.

3. For every constant symbol c of $\mathcal{L}$, $F(c_A) = c_B$.

In this case, the bijection F is called an **isomorphic mapping** from A to B , or simply an isomorphism from A to B ; we say that A and B are isomorphic and write $F : A \cong B$.

Perhaps this seems a bit abstract, but if the reader has a little knowledge of abstract algebra, the following example may be helpful.

1.4.1 EXAMPLE. The language of group theory is $\mathcal{L} = \{*, e\}$, where $*$ is a binary operation symbol and e is a constant symbol (called the identify element of the group). Group theory has only three sentences:

- $G_1. \forall x, y, z((x * y) * z) = x * (y * z);$
- $G_2. \forall x((x * e = x) \wedge (e * x = x));$
- $G_3. \forall x \exists y((x * y = e) \wedge (y * x = e)).$

1.5 Natural Numbers

Axiom of Infinity

It should be reminded again that the previous axioms we have seen still cannot directly provide any sets other than the empty set and its derivatives, such as $\{\emptyset\}$, $\{\emptyset, \{\emptyset\}\}$, and so on (in fact, the entire **ZFC** axiom system cannot directly provide them either). Nevertheless, it is very powerful; for example, it can independently support **Peano arithmetic**. Peano arithmetic will be introduced in this section. Before that, we introduce another axiom.

AXIOM ZFC.6. (Infinity). There exists an *infinite* set.

To give a precise formulation of axiom 6, we have to define first the notion of “finite”. The most natural definition of finiteness uses the notion of a “natural number”, which is as yet undefined. So, let’s forget that description for the moment.

1.5.1 DEFINITION. Let x be a set. We call $x^+ = x \cup \{x\}$ is the **successor** of x . In other words, the successor of x is obtained by adding x itself as a new element to the set x .

1.5.2 DEFINITION. A set S is an **inductive set** if

$$\emptyset \in S \wedge (\forall x(x \in S \rightarrow (x \cup \{x\} \in S))).$$

From this definition, it can be seen that if an inductive set exists, then the process of “successor” can continue indefinitely. Specifically, we can start this process with the emptyset:

$$\begin{aligned} \emptyset^+ &= \emptyset \cup \{\emptyset\} = \{\emptyset\}; \text{ then } \{\emptyset\} \in S; \\ (\emptyset^+)^+ &= \emptyset^+ \cup \{\emptyset^+\} = \{\emptyset, \{\emptyset\}\}; \text{ then } \{\emptyset, \{\emptyset\}\} \in S; \\ &\dots \end{aligned}$$

The question is: does an inductive set necessarily exist? From the perspective of the first five axioms, we cannot be sure. To assert its existence, we need a new axiom.

AXIOM ZFC.6. (Infinity). There exists a inductive set.

Axiom 6 is not like the previous five axioms, which construct a class and then assert that this class is a set. Axiom 6, like Axiom 0, directly stipulates a set. Using the axiom 6, there exists an inductive set, say S . Note that $\emptyset \in S \wedge (\forall x(x \in S \rightarrow (x \cup \{x\} \in S)))$ is itself a formula, by axiom 2 we may construct a new set (By axiom 1, ω is unique):

$$\omega = \{x : x \in S \wedge (\forall E(E \text{ is an inductive set} \rightarrow x \in E))\}$$

In other words, ω is the intersection of all inductive sets, which is a subset of each of them, must be a subset of S in particular.

1.5.1 THEOREM. ω is the smallest inductive set.

Proof. Clearly, $\omega \subset E$ for any inductive set E . It is clear that $\emptyset \in \omega$. If $x \in \omega$, then $x \in S$ and for any inductive set E , $x \in E$, then $x^+ \in S$ and $x^+ \in E$. It follows that $x^+ \in \omega$. These say that ω is an inductive set. $\square$

1.5.3 DEFINITION. A set T is a **transitive set** if

$$(t \in T \wedge x \in t) \rightarrow (x \in T).$$

That is, every element of T is a subset of T . Fortunately, transitive sets obviously exist: $\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \{\{\emptyset\}\}, \dots$ are transitive sets. Non-transitive sets obviously also exist: $\{\{\emptyset\}\}$ is not a transitive set (since $\emptyset \notin \{\{\emptyset\}\}$). Here are several equivalent definitions of a transitive set.

1.5.2 PROPOSITION. The following statements are equivalent for a set T :

1. T is a transitive set;
2. $\bigcup T \subset T$;
3. $t \in T \rightarrow t \subset T$;
4. $T \subset P(T)$.

Proof. (1) $\rightarrow$ (2): suppose that $t \in \bigcup T$, then $\exists x \in T, t \in x$. Since T is transitive, $t \in T$, which follows (2). (2) $\rightarrow$ (3): suppose that $t \in T$, then $t \subset \bigcup T$ which implies $t \subset T$. (3) $\rightarrow$ (4): Clearly. (4) $\rightarrow$ (1): suppose that $(x \in T) \wedge (t \in x)$, then $x \subset T$. Since $t \in x$, $t \in T$. $\square$

1.5.3 PROPOSITION. If x is a transitive set, then x^+ is a transitive set.

Proof. Assume that $t \in x^+ = x \cup \{x\}$ and $y \in t$. It suffices to show that $y \in x^+$: if $t \in x$, then $y \in x$ since x is transitive, then $y \in x^+$; if $t = x$, then $y \in t$ means $y \in x$, it follows that $y \in x^+$. $\square$

1.5.1 REMARK. We have already seen that the emptyset $\emptyset$ is a transitive set, so $\emptyset^+, (\emptyset^+)^+, ((\emptyset^+)^+)^+, \dots$ are all transitive sets. Moreover, by definition, these transitive sets are all elements of the smallest inductive set ω . The question is: besides containing these transitive sets, does ω also include any non-transitive sets?

1.5.4 THEOREM. Every element of ω is a transitive set.

Proof. Define a set $\alpha = \{x \in \omega : x \text{ is a transitive set}\}$. Now, we show that α is a inductive set: first, $\emptyset \in \alpha$ since $\emptyset \in \omega$ and $\emptyset$ is a transitive set. For any $x \in \alpha$, x^+ is transitive by the proposition 1.4.2; $x^+ \in \omega$ since ω is inductive. Therefore, $x^+ \in \alpha$. Hence α is an inductive set and $\alpha \subset \omega$. However, ω is the smallest inductive set, $\alpha = \omega$, it follows that every element in ω is transitive. $\square$

1.5.5 THEOREM. ω is itself a transitive set.

Proof. Define a set $\beta = \{x \in \omega : x \subset \omega\}$. Clearly $\emptyset \subset \omega$, so $\emptyset \in \beta$. Suppose $n \in \beta$, i.e., $n \in \omega$ and $n \subset \omega$. Then $n^+ = n \cup \{n\} \in \omega$. Since $\{n\} \subset \omega$ and $n \subset \omega$, $n^+ \subset \omega$. Hence $n^+ \in \beta$. Therefore, $\beta = \omega$. This shows that $\bigcup \omega \subset \omega$, thus, ω is a transitive set. $\square$

1.5.6 THEOREM. $\forall x(x \in \omega \wedge x \notin x)$.

Proof. Define a set $\beta = \{x \in \omega : x \notin x\}$. We show that β is an inductive set: clearly, $\emptyset \in \beta$ since $\emptyset \in \omega$ and $\emptyset \notin \emptyset$. If $x \in \beta$, then $x \in \omega$ and $x \notin x$. Since ω is inductive, $x^+ \in \omega$. Assume that $x^+ \in x^+ = x \cup \{x\}$, there are two cases: if $x^+ = x$, then $x \in x$, which contradicts $x \notin x$. If $x^+ \in x$, then $x \in x^+ \in x$ implies $x \in x$, also a contradiction. Therefore, β is an inductive set and $\beta \subset \omega$. So, $\beta = \omega$. $\square$

1.5.7 THEOREM. *The set ω satisfies:*

1. $\emptyset \in \omega$;
2. $x \in \omega \rightarrow x^+ \in \omega$;
3. $x \in \omega \rightarrow x^+ \neq \emptyset$;
4. $x = y \leftrightarrow x^+ = y^+$;
5. $((a \subset \omega) \wedge ((\emptyset \in a) \wedge (x \in a \rightarrow x^+ \in a))) \rightarrow a = \omega$.

Proof. (1). and (2). are from the definition of ω . (3). is clear (since $\emptyset$ contains no elements). (4). Clearly $(x = y) \rightarrow (x^+ = y^+)$. Assume that $x^+ = y^+$, that is $(x \cup \{x\} = y \cup \{y\})$. So $\bigcup x^+ = \bigcup y^+$. Since x, y are transitive sets, it follows that $\bigcup x^+ = x$ and $\bigcup y^+ = y$. So $x = y$. (5). Clearly a is an inductive set, which implies $a = \omega$. $\square$

If the reader is familiar with the Peano arithmetic axioms, these five properties should feel very familiar. In fact, these are exactly all the properties that Peano arithmetic must satisfy. Now let us take another look at Peano arithmetic.

AXIOM (G. Peano). The Peano axioms define the “natural numbers”, often denoted as $\mathbf{N}$. They were introduced in 1889 by Giuseppe Peano.

1. $0 \in \mathbf{N}$;
2. $n \in \mathbf{N} \rightarrow S(n) \in \mathbf{N}$;
3. $n \in \mathbf{N} \rightarrow S(n) \neq 0$;
4. $n = m \leftrightarrow S(n) = S(m)$;
5. $((K \subset \mathbf{N}) \wedge ((0 \in K) \wedge (n \in K \rightarrow S(n) \in K))) \rightarrow K = \mathbf{N}$.

These axioms formalize the idea of “counting”. Intuitively, to count a collection of things, we start with not having anything counted i.e., 0, which is property (1). Property (2) is a way of saying that after every number n , there is a next number $S(n)$ (which we think of as $n + 1$, but addition is not directly defined by the axioms). Property (3) tells us that there is natural number “before” zero, and property (4) tells us that each number has exactly one successor. Property (5) tells us that $\mathbf{N}$ corresponds exactly to things that we can count, given an arbitrary long amount of time. We have to start counting (i.e., 0, $S(0) = 1, S(1) = 2, \dots$) and at each step, we are always able to count one more thing. It tells us that every natural number can “reached” by enough applications of the “successor”.

1.5.2 REMARK. In other words, using the first six axioms, we can construct a set ω that satisfies all the requirements of the Peano arithmetic axioms for the set of natural numbers. Therefore, we naturally call ω *the natural number set*, and its elements are

called natural numbers. Specifically, we proceed as follows:

$$\begin{aligned} 0 &= \emptyset; \\ 1 &= 0^+ = 0 \cup \{0\} = \{0\}; \\ 2 &= 1^+ = 1 \cup \{1\} = \{0, 1\}; \\ 3 &= 2^+ = 2 \cup \{2\} = \{0, 1, 2\}; \\ &\dots \end{aligned}$$

1.5.4 DEFINITION. A set X is **finite** if there is a one-to-one function from X onto some $n \in \omega$; A set X is **infinite** if it is not finite.

For our definition of finite and infinite set to correspond to our firm intuitions, we would hope to be able to show that ω is infinite and that there is no bijection between different natural numbers m and n . Proving these two facts requires the pigeon-hole principle, which we will not prove here.

1.5.8 THEOREM. ω is infinite.

Proof. It suffices to show that there is no bijection $f : n \rightarrow \omega$ for every $n \in \omega$. Suppose for contradiction, there is such a bijection $f : n \rightarrow \omega$. Then its inverse function $f^{-1} : \omega \rightarrow n$ is also a bijection. Consider the restriction of this map to the subset n of ω :

$$f^{-1}|_n : n \rightarrow n, \quad i \mapsto f^{-1}(i).$$

As f^{-1} is one-to-one, so is $f^{-1}|_n$. But then, by the pigeon-hole principle, $f^{-1}|_n$ is also onto. Thus wherever f^{-1} maps elements of the complement $\omega - n$ to images in n , f^{-1} can no longer be one-to-one, which is a contradiction, so that ω is indeed infinite. $\square$

Induction and Recursion on ω

1.5.9 THEOREM (Induction on). *Let $\varphi(x)$ be a formula. If it satisfies $\forall n \in \omega (\varphi(0) \wedge (\varphi(n) \rightarrow \varphi(n^+)))$, then $\forall n \in \omega \varphi(n)$.*

Proof. This is called the (weak) induction principle for the ω . This is a direct corollary of the last of the five theorems above concerning ω . Let $\beta = \{x \in \omega : \varphi(x)\}$, then β is an inductive set, and so $\beta = \omega$. $\square$

1.5.10 THEOREM (Strong Induction). *Let $\varphi(x)$ be a formula. If it satisfies $\forall n \in \omega (\forall m \in n \varphi(m) \rightarrow \varphi(n))$, then $\forall n \in \omega \varphi(n)$.*

Proof. Define a formula $\psi(n) \leftrightarrow (\forall m \in n \varphi(m))$ and we prove $\forall n \psi(n)$ by induction. First $\psi(0)$ holds since there are no $m \in 0$. Suppose $\psi(n)$ holds, then $\varphi(m)$ holds for all $m \in n$ and so $\varphi(n)$ holds by the hypothesis of the theorem. But then $\varphi(m)$ holds for all $m \in n^+$, i.e., $\psi(n^+)$ holds. $\square$

Let X be a set, $h : X \rightarrow X$ a function and $a \in X$. Intuitively, starting from a and iterating step by step via h , we can generate: $a_0 = a, a_1 = h(a_0), a_2 = h(a_1), \dots, a_{n+1} = h(a_n), \dots$. In other words, there exists a function $f : \omega \rightarrow X$ such that $(f(0) = a) \wedge (\forall n (f(n^+) = h(f(n))))$. However, such an iteration would require infinitely many steps to complete. So, under the existing axioms, can we guarantee that the envisaged function indeed exists? We will prove below that it legitimately exists. The main method is to construct the various segments of h and finally obtain the required function.

1.5.11 THEOREM (Recursion). *Let X be a set, $a \in X$ and $g : X \rightarrow X$ a function. Then there exists a unique function $f : \omega \rightarrow X$ such that*

$$(f(0) = a) \wedge (\forall n (f(n^+) = g(f(n))))$$

Proof. Let $n \in \omega$, a function $h : n \rightarrow X$ will be called n -nice if $n = 0$, or if $0 \in n$ then $h(0) = a$ and $h(m^+) = g(h(m))$ for all m with $m^+ \in n$. A function $h : \omega \rightarrow X$ will be called very nice if, for each n , the restriction of h to n is n -nice. The conclusion of the theorem may be restated as follows:

There is a unique very nice function $f : \omega \rightarrow X$.

Claim 1. For each n there exists an n -nice function.

Proof. By induction, for $n = 0$, there is nothing to prove. If $n = 0$, we set $h(0) = a$ to define a 1-nice function: define h_1 in which $\text{dom}(h) = 1 = \{0\}$ and $\text{ran}(h) = a$, then for any m with $m^+ \in 1$, there is only one case, namely $m^+ = 0$, but this is impossible, which means the proposition is true and h_1 is 1-nice. Suppose that the assertion is true for n with $0 \in n$, then there is a n -nice function h_n . Now we define a function $h_{n^+} = h_n \cup \{\langle n, g(h_n(k)) \rangle\}$ in which $k^+ = n$. Then $\text{dom}(h_{n^+}) = n \cup \{n\} = n^+$ and $h_{n^+}(0) = a$, now we show that $h_{n^+}(m^+) = g(h_{n^+}(m))$ for all $m^+ \in n^+$: if $m^+ \in n$, then $h_n(m^+) = g(h_n(m))$, clearly that $h_{n^+}(m^+) = g(h_{n^+}(m))$. If $m^+ = n$, then $m = k$ since $k^+ = n$. Clearly $h_{n^+}(m^+) = h_{n^+}(n) = g(h_n(k)) = g(h_n(k)) = g(h_{n^+}(m))$. Therefore, h_{n^+} is n^+ -nice. $\square$

Claim 2. Suppose h_1, h_2 are n_1 -nice, n_2 -nice respectively, and $m \in n_1 \cup n_2$. Then we have $h_1(m) = h_2(m)$.

Proof. By induction on m . For $m = 0$ the statement holds because $0 \in n_1 \cap n_2$ implies $h_1(0) = x_0 = h_2(0)$. Suppose the assertion is true for m , and suppose that $m^+ \in n_1 \cap n_2$. By induction hypothesis:

$$h_1(m^+) = g(h_1(m)) = g(h_2(m)) = h_2(m^+)$$

$\square$

Claim 3. For each n there is a unique n -nice function.

Claim 4. a very nice function, if it exists, is unique.

Proof. Suppose that $f_1, f_2 : \omega \rightarrow X$ are very nice. For any n , the restriction of f_1, f_2 to domain n are n -nice, and so agree. So $f_1 = f_2$. $\square$

Claim 5. There exists a very nice function.

Proof. An n -nice function is an element of $P(\omega \times X)$, and among all elements of $P(\omega \times X)$ we can “comprehend” the subset N of functions which are n -nice for some n , i.e., $N = \{h : \exists n \in \omega, h \text{ is a } n\text{-nice function}\}$. As N is a set of functions, its union $f = \bigcup N$ is a set of ordered pairs (a subset of $\omega \times X$). Now we show that f is a function from ω to X : suppose that $n \in \omega$. Then there exists an $(n+1)$ -nice function h , so there is some pair $\langle n, x \rangle \in \bigcup N$, where $x = h(n) \in X$. If $\langle n, x' \rangle \in \bigcup N$ then there is a k -nice function h' with $n \in k$ and $h'(n) = x'$. By Claim 2 we must have $x' = h'(n) = h(n) = x$. So f is a function $\omega \rightarrow X$. Finally, we must show that f is very nice: let $n \in \omega$ and let $h : n^{++} \rightarrow X$ be n^{++} -nice, then $h \in N, h \subset f = \bigcup N$, so $f(m) = h(m)$ whenever $\langle m, h(m) \rangle \in h$. We have $h(0) = x_0$ which implies $f(0) = x_0$; and $f(n^+) = h(n^+) = g(h(n)) = g(f(n))$. Clearly f has the required properties. So f is indeed very nice. $\square$

$\square$

1.5.12 COROLLARY. Suppose that A, X are sets, $h : A \rightarrow X$ is a function and $g : A \times X \rightarrow X$ is a function. Then there exists a unique function $f : A \times \omega \rightarrow X$ such that

1. $f(a, 0) = h(a)$ for all $a \in A$;
2. $f(a, n^+) = g(a, f(a, n))$ for all $a \in A, n \in \omega$.

Proof. For each $a \in A$, apply the theorem with $x_0 = h(a)$ and $g_a : X \rightarrow X$ given by $g_a(x) = g(a, x)$ to get a unique function $f_a : \omega \rightarrow X$ with $f_a(0) = h(a)$ and $f_a(n^+) = g_a(f_a(n))$ for all $n \in \omega$. We now define $f : A \times \omega \rightarrow X$ by $f(a, n) = f_a(n)$. $\square$

Arithmetic on ω

1.5.5 DEFINITION. There exists a unique function $+$: $\omega \times \omega \rightarrow \omega$ (we call **addition**.) with the following properties (by the corollary 1: take $h(n) = n$ and $g(n, m) = m^+$):

1. $n + 0 = n$ for all $n \in \omega$;
2. $n + m^+ = (n + m)^+$ for all $n, m \in \omega$.

1.5.6 DEFINITION. There exists a unique function $\cdot$: $\omega \times \omega \rightarrow \omega$ (we call **multiplication**) with the following properties (take $h(n) = 0$ and $g(n, m) = m + n$, not $n + m$):

1. $n \cdot 0 = 0$ for all $n \in \omega$;
2. $n \cdot m^+ = n \cdot m + n$ for all $n, m \in \omega$.

1.5.7 DEFINITION. There exists a unique function $\hat{}$: $\omega \times \omega \rightarrow \omega$ (we call **exponentiation**) with the following properties (take $h(n) = 0$ and $g(n, m) = m + n$, not $n + m$):

1. $n \hat{} 0 = 0$ for all $n \in \omega$;
2. $n \hat{} m^+ = n \hat{} m \cdot n$ for all $n, m \in \omega$.

1.5.1 EXAMPLE. Using the definition above to compute $0 \cdot 3$. Multiplication is the more complicated operation, as it defined in terms of the more basic operation of addition. When we make a use of the definition of multiplication that results in an expression involving $+$, it makes sense to then simplify the latter as much as possible using the definition of addition, before going back to the more complicated multiplication operation:

$$\begin{aligned}
 0 \cdot 3 &= 0 \cdot 2^+ = (0 \cdot 2) + 0 \quad (\text{definition of } n \cdot m^+) \\
 &= 0 \cdot 2 \quad (\text{definition of } n + 0) \\
 &= 0 \cdot 1^+ = (0 \cdot 1) + 0 \quad (\text{definition of } n \cdot m^+) \\
 &= 0 \cdot 1 \quad (\text{definition of } n + 0) \\
 &= 0 \cdot 0^+ = (0 \cdot 0) + 0 \quad (\text{definition of } n \cdot m^+) \\
 &= 0 \cdot 0 \quad (\text{definition of } n + 0) \\
 &= 0 \quad (\text{definition of } n \cdot 0).
 \end{aligned}$$

We expect that the addition, multiplication, and exponentiation we define will satisfy the properties we use in everyday life, such as commutativity, associativity, distributivity, and so on. Since for small m, n, k we can "brute force" compute the corresponding results, it is natural to think of using induction to prove the general case:

1.5.13 PROPOSITION. *The following hold for all $n, m, k \in \omega$:*

1. $(n + m) + k = n + (m + k)$;
2. $n + m = m + n$;
3. $n \cdot (m + k) = n \cdot m + n \cdot k$;
4. $(n \cdot m) \cdot k = n \cdot (m \cdot k)$;
5. $n \cdot m = m \cdot n$;

Proof. The proof is not particularly meaningful; most of the methods can be proved using induction. If interested, the reader may try it—it wouldn't hurt. $\square$

1.6 Axiom of Replacement, Axiom of Regularity

Before starting the next chapter, we give the remaining two axioms apart from the Axiom of Choice. One of them is clearly more technical in nature, supporting us in carrying out transfinite recursion within ZF, and so on; the other is mainly for ruling out strange sets and for giving a well-founded structure to the universe of sets.

1.6.1 DEFINITION. Let $\varphi(x, y, p_1, \dots, p_n)$ be a formula with free variables among $x, y, p_1, \dots, p_n$ such that $\forall p_1 \dots \forall p_n \exists! y \varphi(x, y, p_1, \dots, p_n)$. Then for each fixed choice of these parameters $p_1, \dots, p_n$, we may define a class

$$F = \{(x, y) : \varphi(x, y, p_1, \dots, p_n)\},$$

and we call this class the **class function**. We shall apply all the notation for functions to the concept of class functions as well. This is perfectly unobjectionable; in fact, the only difference between a class function and a function lies in the “domain”. If we restrict the “domain” of a class function to a subset of a known set, then we obtain a function.

AXIOM ZFC.7. (Replacement). If F is a class function, then for every set X , $F(X)$ is a set. That is, for each formula $\varphi(x, y, p)$:

$$\begin{aligned} \forall p \forall x, y, z (\varphi(x, y, p) \wedge \varphi(x, z, p) \rightarrow y = z) \\ \rightarrow \forall X \exists Y \forall y (y \in Y \rightarrow (\exists x \in X \wedge \varphi(x, y, p))) \end{aligned} \quad (1-7)$$

We shall look at a major application of the axiom of replacement later in this section, where it plays a crucial role in defining functions within **ZF** by recursion. Meanwhile there are a couple of useful technical dodges in its use. For example, the statement of the axiom, writing $\varphi(x, y, p)$ just involving three free variables, make it look as though it can obly be used to define functions of one variables. But for instance a function of two variables x_1, x_2 could be represented by a formula $\varphi(x, y, p)$ where x is the ordered pair (x_1, x_2) . For simplicity we would usually write the formula as $\varphi(x_1, x_2, y, p)$ rather than $\varphi((x_1, x_2), y)$.

Now, we would like to introduce one further, even more powerful, version of the recursion principle. In fact, the Recursion Principle would essentially have a flaw. Specifically, Zermelo discovered that the first six axioms cannot prove that the class

$$\{\mathbf{N}, \mathcal{P}(\mathbf{N}), \mathcal{P}(\mathcal{P}(\mathbf{N})), \dots\}$$

is a set, even though intuitively it should be, because its “size” is akin to that of $\mathbf{N}$. According to our earlier treatment of classes, a class of this size ought to be a set. Zermelo therefore proposed the Axiom of Replacement. We can use the recursion principle to argue this (find the class function). If the function $h : \mathbf{N} \times Y \rightarrow Y$ in the recursion principle is defined by $h(n, y) = \mathcal{P}(y)$, then the recursion principle define a function $f : \mathbf{N} \rightarrow \mathbf{V}$ by

$$\begin{aligned} f(0) &= \mathbf{N} \\ f(n^+) &= \mathcal{P}(f(n)) \end{aligned}$$

then the “range” of f , $\{\mathbf{N}, \mathcal{P}(\mathbf{N}), \mathcal{P}(\mathcal{P}(\mathbf{N})), \dots\}$ is a set by the axiom of replacement. In this case, Y would have to include as a subset precisely the class $\{\mathbf{N}, \mathcal{P}(\mathbf{N}), \dots\}$. This means that this class is already a subset of Y from the very beginning, hence it is itself a set; there is no need to recursively find a function to show that it is a set. However, without finding a function and then using the Axiom of Replacement, one cannot prove that it is indeed a set. We can modify the recursion principle as follows to fix this flaw.

1.6.1 THEOREM (Recursion, class form). Let $F : \mathbf{N} \times \mathbf{V}$ be a class function, and let x_0 be a given set. Then there is a unique function $f : \mathbf{N} \rightarrow \mathbf{V}$ such that

$$\begin{aligned} f(0) &= x_0, \text{ and} \\ f(n^+) &= F(n, f(n)), \text{ for every } n \in \mathbf{N}. \end{aligned}$$

Proof. The proof is exactly the same as the proof of non-class form. The only place that one should prove is that $g_{n+} = g_n \cup \{(n^+, h(n, g_n(n)))\}$ is actually a set. Since the induction hypothesis at this stage of the proof was that g_n is a function, this means that there is a unique z , namely, $h(n, g_n(n)) = z$. Then by the ordered pair construction we have that $(n^+, h(n, g_n(n)))$ is a set, so that the singleton $\{(n^+, h(n, g_n(n)))\}$ is a set, giving that the union $g_n \cup \{(n^+, h(n, g_n(n)))\}$ is a set. $\square$

Next, we introduce the eighth axiom, the axiom of Regularity (or Foundation). It is introduced to exclude certain “pathological sets”, such as cyclic membership relations, infinite descending chains, and self-membership:

AXIOM ZFC.8. (Regularity). Every nonempty set contains an $\in$ -minimal element. That is, an element $y \in x$ which does not contain any element of x :

$$\forall x \exists y ((y \in x) \wedge (x \cap y) = \emptyset).$$

1.6.2 PROPOSITION. Let x be a set. Then $x \notin x$.

Proof. Assume that $x \in x$. Put $X = \{x\}$. Then X violates this axiom: for x is the only element of X but $x \in x$ and $x \in X$ means that $x \cap X = \{x\} \neq \emptyset$. $\square$

1.6.3 PROPOSITION. Suppose that $\langle x_n \rangle$ is a sequence (i.e., the function with domain $\mathbf{N}$). Then this sequence cannot form what is called an infinite descending $\in$ -chain, i.e., we cannot have

$$\cdots \in x_{n+1} \in x_n \in \cdots \in x_2 \in x_1 \in x_0.$$

Proof. Suppose that the sequence does form an infinite descending $\in$ -chain. Taking x to be the set $\{x_n : n \in \mathbf{N}\}$. By the axiom there is some $y \in x$ such that $x \cap y = \emptyset$. Since $y \in x$ we must have $y = x_n$ for some n . But, then x_{n+1} is an element both of x and of y , so that $x_{n+1} \in x \cap y$, contradicting that $x \cap y = \emptyset$. $\square$

1.6.1 REMARK. With this axiom, the concept of ordinals becomes simpler. According to the previous definition of ordinals, a set should satisfy the following five properties to be called an ordinal:

1. $\forall x \in \alpha (x \notin \alpha)$;
2. $\forall x, y, z \in \alpha (x \in y \wedge y \in z \rightarrow x \in z)$;
3. $\forall x, y \in \alpha (x \in y \vee x = y \vee y < x)$;
4. $\forall a \subset \alpha (a \neq \emptyset \rightarrow \exists x \in a \forall y \in a (x \leq y))$;
5. $\forall x \in \alpha (y \in x \rightarrow y \in \alpha)$.

With the Axiom of Regularity, condition 4 is automatically satisfied. And with the above proposition, condition 1 is also automatically satisfied.

2

Ordinality and Cardinality

2.1 Order Sets

Well-Ordered Sets

2.1.1 DEFINITION. Let X be a set and R a binary relation of $X \times X$. R is a strictly parital ordering on X if it has the following properties:

1. $\forall x \in X \neg(xRx)$;
2. $\forall x, y, z \in X (xRy \wedge yRz \rightarrow xRz)$.

The set X equipped with a strictly parital ordering R forms a structure $\langle X, R \rangle$ and it is called the R -strictly paritally ordered set. If $Y \subset X$, it is clear that $\langle Y, R \cap (Y \times Y) \rangle$ is also a strictly paritally ordered set.

2.1.1 REMARK. Suppose R is a strictly parital ordering on a set X , and $x, y \in X$. It is easy to see that the following three cases all contradicts the irreflexivity of the strictly parital ordering:

1. $xRy \wedge x = y \rightarrow yRy$ (by the axiom of equality)
2. $x = y \wedge yRx \rightarrow yRy$ (by the axiom of equality)
3. $xRy \wedge yRx \rightarrow xRx$ (by transitivity)

Thus, among the three cases, $xRy, x = y, yRx$ at most one can hold; however it is possible that none of them holds, that is, the law of trichotomy does not necessarily hold for a strictly partial ordering.

2.1.2 REMARK. Sometimes a strictly parital ordering R on X is written as $<$. Then the conditions for the strictly partially ordered set $\langle X, < \rangle$ can be written as:

1. $\forall x \in X \neg(x < x)$;
2. $\forall x, y, z \in X ((x < y \wedge y < z) \rightarrow x < z)$.

If we now use $x \leq y$ to mean $x < y \vee x = y$, then one may check that the new binary relation $\leq$ on X satisfies the conditions

1. $\forall x \in X (x \leq x)$;
2. $\forall x, y \in X ((x \leq y \wedge y \leq x) \rightarrow x = y)$;
3. $\forall x, y, z \in X ((x \leq y \wedge y \leq z) \rightarrow x \leq z)$.

Usually $\leq$ is called the *parital ordering*, and we call the structure $\langle X, \leq \rangle$ the paritally ordered set. Conversely, if $\leq$ satisfies (1), (2) and (3), then the relation $<$ defined as $x \leq y \wedge x \neq y$ is a relation satisfying the irreflexivity and transitivity, that is, $<$ is a strictly parital ordering on X . In conclusion, strictly partial ordering and partial ordering are uniquely determined by each other. Since it is natural to discuss the case of equality, we will not strictly distinguish between the two concepts in the following; we will uniformly refer to them as partial orderings. The reader only needs to consider equality or not consider equality as appropriate. (**Whatever is most convenient.**)

2.1.1 EXAMPLE. Let $\langle X, < \rangle$ be a partially ordered set, define the binary relation $>$ on $X \times X$ by $x > y$ if and only if $y < x$, then one may check that $\langle X, > \rangle$ is also a paritally ordered set. So where we deal with a set with a parital ordering, we always have a choice of two ways of regarding the order. Simiarly, we can also define the inverse ordering $\geq$ of $\leq$. Although we use the symbol $<$ when we comparing elements in an ordered structure $\langle X, < \rangle$, sometimes, we also use its inverse order $>$ to facilitate our presentation.

Partially ordered sets are the most fundamental structure of ordered sets. All subsequent new ordered sets presuppose partially ordered sets. Before that, we first introduce a few additional concepts used in ordered sets.

2.1.2 DEFINITION. Let $\langle X, < \rangle$ be a partially ordered set, Y is a nonempty subset of X and $x \in X$. Note the following few terms,

1. x is a *maximal element* of Y if $x \in Y$ and $\forall y \in Y (x \not< y)$;
2. x is a *minimal element* of Y if $x \in Y$ and $\forall y \in Y (y \not< x)$;
3. x is the *greatest element* of Y if $x \in Y$ and $\forall y \in Y (y \leq x)$;
4. x is the *least element* of Y if $x \in Y$ and $\forall y \in Y (x \leq y)$;
5. x is an *upper bound* if $\forall y \in Y (y \leq x)$, the least upper bound is called the *supremum*;
6. x is a *lower bound* if $\forall y \in Y (x \leq y)$, the greatest lower bound is called the *infimum*;

2.1.3 REMARK. A least (greatest) element is always a minimal (maximal) element, but the converse is not true. Note also that, if a least element or a greatest element exists, it is unique; maximal or minimal elements may not be unique. For example, the partially ordered set $\langle \{\emptyset, \{x\}, \{y\}\}, \subset \rangle$ has no greatest element, but it has two maximal elements $\{x\}$ and $\{y\}$.

2.1.3 DEFINITION. Let $\langle X, <_X \rangle$ and $\langle Y, <_Y \rangle$ be two partially ordered sets. Then the function $f : X \rightarrow Y$ is said to be an *order-preserving* function (figuratively speaking, we also call it a strictly increasing function) if

$$\forall x_1, x_2 \in X (x_1 <_X x_2 \rightarrow f(x_1) <_Y f(x_2)).$$

Then by the definition of “isomorphism”, $\langle X, <_X \rangle$ and $\langle Y, <_Y \rangle$ are *order-isomorphic*, written $\langle X, <_X \rangle \cong \langle Y, <_Y \rangle$, if there is a bijection $f : X \rightarrow Y$ such that $\forall x_1, x_2 \in X (x_1 <_X x_2 \leftrightarrow f(x_1) <_Y f(x_2))$. The map f is said to be an *order-isomorphism*.

2.1.4 REMARK. One may check that “order-isomorphic” is an equivalence relation on partially ordered sets, i.e., $\cong$ is reflexive, symmetric and transitive. But, in general, the collection of partially ordered sets order-isomorphic to a given partially ordered set is a proper class, not a set. This is very intuitive, for a fixed partially ordered set $\langle X, < \rangle$, we can define the partially ordered set $\langle X \times \{Y\}, < \rangle$ where $<$ is defined by $(x_1, \{Y\}) < (x_2, \{Y\})$ if

$$x_1 < x_2$$

This generates a “great many” partially ordered sets. Consider the class of all ordered sets isomorphic to $\langle X, < \rangle$: $C_X = \{\langle Z, < \rangle : Z \cong X\}$. If this class is a set, then the class $\{\langle X \times \{Y\}, < \rangle \in C_X : Y \in \mathbf{V}\}$ would indeed be a set. So the class of the second coordinates of the product it contains, namely the universe $\mathbf{V}$. But $\mathbf{V}$ is not a set, so neither is the class corresponding to “isomorphic to $\langle X, < \rangle$ ”.

2.1.4 DEFINITION. Let $\langle X, < \rangle$ be a partially ordered set. A subset S of X is an *initial segment* of X if for every $x, y \in X$, $x < y \wedge y \in S \rightarrow x \in S$. An initial segment S of X is proper if $S \neq X$.

2.1.5 REMARK. It is clear that if $f : X \rightarrow Y$ is an order-isomorphism, then $S \subset X$ is an initial segment of X if and only if $f[S]$ is an initial segment of Y . Note also that for every element x_0 of a partially ordered set $\langle X, < \rangle$, the set

$$\text{seg}_X x_0 = \{x \in X : x < x_0\}$$

is a proper initial segment of X . However, not every proper initial segment of a partially ordered set is of the form $\text{seg}_X x_0$. Later, we will also introduce well-ordered sets, where the situation will be different. In other words, every proper initial segment of a well-ordered set is of the form $\text{seg}_X x_0$ for some $x_0 \in X$.

2.1.5 DEFINITION. A parital ordering $<$ on a set X is called a *linear oredering* on X if $<$ satisfies the law of trichotomy: $\forall x, y \in X (x < y \vee x = y \vee y < x)$, that is:

1. $\forall x \in X \neg(x < x)$;
2. $\forall x, y, z \in X (x < y \wedge y < z \rightarrow x < z)$;
3. $\forall x, y \in X (x < y \vee x = y \vee y < x)$.

A linear ordering is also called a total ordering. And the structure $\langle X, < \rangle$ is called a linearly ordered set. If $\langle X, < \rangle$ is a linearly ordered set and $Y \subset X$, one may check that $\langle Y, <_Y \rangle$ in which $<_Y = < \cap (Y \times Y)$ is also a linearly ordered set. As for why it is called a linear (total) ordering, the reader can draw a Hasse diagram and will find that every two points can be directly connected by a line. In other words, every two elements can be compared (ordered with respect to each other).

2.1.6 DEFINITION. A linear ordering $<$ on a set X is called a *well-oredering* if it satisfies the well-foundedness, that is, every non-empty subset of X has an $<$ -minimal element, i.e.,

1. $\forall x \in X (x \not< x)$;
2. $\forall x, y, z \in X (x < y \wedge y < z \rightarrow x < z)$;
3. $\forall x, y \in X (x < y \vee x = y \vee y < x)$;
4. $\forall Y \subset X (Y \neq \emptyset \rightarrow \exists y \in Y \forall x \in Y (x \not< y))$

The structure $\langle X, < \rangle$ is called a *well-ordered set*. Becasue of trichotomy, the minimal element is in fact the least element, so every non-empty subset of a well-ordered set has a least element. In a similar way, if $\langle X, < \rangle$ is a well-ordered set and $Y \subset X$, then $\langle Y, < \cap Y \times Y \rangle$ also a well-ordered set.

2.1.6 REMARK. When we introduced paritally ordered sets, we had the concept of the inverse ordering. At that time, we noted that the inverse paritally ordered set is also a paritally ordered set. The reader may check that this conclusion also holds for learnly ordered sets. However, this is not true for well-ordered sets. Intuitively, the least element of every subset in a well-ordered set becomes the greastest element in the inverse ordered set. Nevertheless, for fintie well-ordered sets, the conclusion still holds.

2.1.2 EXAMPLE. $\langle \omega, \in \rangle$ is a well-ordered set. For each property, we can use induction to prove it. As an exercise, I will provide a detailed proof for each one.

Proof. First, we show that $\in$ is irreflexive. Define a subset A of ω by $A = \{n \in \omega : n \notin n\}$. Since $0 = \emptyset$ and $\emptyset$ contains on elements, then we have, in particular, that 0 is not an element of 0 , i.e., $0 \notin 0$. Thus, $0 \in A$. Now, suppose that $n \in A$ and show that $n^+ \in A$, i.e., $n^+ \notin n^+$. Suppose, for contradiction, $n^+ \in n^+$, then $n^+ \in n \cup \{n\}$. This gives two possibilities: in the first case, that $n^+ \in n$, we have, as $n \in n \cup \{n\} = n^+$, $n \in n^+ \in n$. But $\in$ is transitive on ω , so that $n \in n$, contradicting that $n \in A$. In the other case, that $n^+ = n$, we have that $n \in n \cup \{n\} = n^+ = n$. i.e., $n \in n$, again contradicting that $n \in A$. In both cases we have a contradiction, so that $n^+ \notin n^+$. Thus, A is inductive, and hence $A = \omega$.

Next, we define a subset B of ω by $B = \{p \in \omega : \text{if } m \in n \in p \text{ then } m \in p\}$ in which m, n are fixed. Since $0 = \emptyset$ and $\emptyset$ contains no elements, it cannot be true that $m \in n \in 0$, so that the statement is vacuously true. Thus $0 \in B$. Now suppose that $p \in B$ and show that $p^+ \in B$. Suppose that $m \in n \in p^+$. Since $n \in p^+ = p \cup \{p\}$, we have two possibilities. In the first case, that $n \in p$, we have $m \in n \in p$, so that, as $p \in B$, we have $m \in p \in p^+$. In the case that $n = p$, then $m \in n \rightarrow m \in p$, so that, as in the first case, $m \in p^+$. Therefore, $p^+ \in B$. Thus B is inductive, it follows that $B = \omega$.

Third, we show that for all $m, n \in \omega$, $m \in n$ or $m = n$ or $n \in m$. We fix $m \in \omega$. Define a subset C of ω by $C = \{n \in \omega : m \in n \vee m = n \vee n \in m\}$. For $n = 0$ we cannot

have $m \in 0 = \emptyset$. If $m = 0$, we are done, and if $m \neq 0$, then $0 \in m$. Thus the result holds for $n = 0$. Now, suppose that $n \in C$, so that $m \in n \vee m = n \vee n \in m$, and show that $n^+ \in C$. In the case that $m \in n$, then $m \in n \in n^+$. And in the case that $m = n$, then $m = n \in n^+$, we have $m \in n^+$. For the third case, that $n \in m$; this means that $m \neq 0$, so $m = k^+$ for some $k \in \omega$. This gives that $n \in m = k \cup \{k\}$. If $n \in k$, then $n^+ \in k^+$, i.e., $n^+ \in m$; and if $n = k$, then $n^+ = k^+$, i.e., $n^+ = m$. In all cases, we have $m \in m^+ \vee m = n^+ \vee n^+ \in m$. Thus, C is inductive, it follows that $C = \omega$.

Finally, we show that ω is well-founded. Let M be a nonempty subset of ω . It is clear that $\langle M, \in \rangle$ is linear ordered. We show that M has a least element. Suppose, for contradiction, M does not have a least element. Define a subset D of ω by $D = \{n \in \omega : \forall m \leq n (m \notin M)\}$. We shall show that $D = \omega$, so that $M = \emptyset$, contradicting that $M \neq \emptyset$. 0 cannot be in M , as otherwise 0 would be the least element of M . Thus, $0 \in D$. Now, suppose that $n \in D$ and show that $n^+ \in D$. Since $n \in D$, we have $m \notin M$ for all $m \leq n$. For n^+ to be in D , all we need to show is that $n^+ \notin M$. If it were the case that $n^+ \in M$, then, as $m \notin M$ for all $m \leq n^+$, n^+ would be the least element of M ; this would contradict that M has no least element, so we conclude that $n^+ \notin M$. Hence $n^+ \in D$. Then by induction, $D = \omega$. $\square$

2.1.1 THEOREM. *Let $\langle X, < \rangle$ be a well-ordered set. Then every proper initial segment of S of X is of the form $\text{seg}_X x_0$ for some $x_0 \in X$.*

Proof. Since S is a proper subset of X , the set $X - S$ is nonempty. Let $x_0 = \min\{X - S\}$. We show that $S = \text{seg}_X x_0$. If $x \in S$, then $x < x_0$ (otherwise $x_0 \leq x$, which would imply $x_0 \in S$, contradicting $x_0 \in X - S$). Hence $x \in \text{seg}_X x_0$; on the other hand, if $x \in \text{seg}_X x_0$, then $x < x_0$. Thus $x \notin X - S$ (otherwise it would contradict the minimality of x_0), so $x \in S$. Therefore $S = \text{seg}_X x_0$. $\square$

Transfinite Induction on Well-Ordered Sets

2.1.2 THEOREM (Transfinite Induction). *Let $\langle X, < \rangle$ be a well-ordered set, and let $\varphi(x)$ be a formula. If $\forall x \in X (\forall y < x \varphi(y) \rightarrow \varphi(x))$, then $\forall x \in X \varphi(x)$.*

Proof. Define a set $A = \{x \in X : \neg \varphi(x)\}$. We shall prove that $A = \emptyset$. Suppose, for contradiction that $A \neq \emptyset$, i.e., there exists $x \in X$ such that $\neg \varphi(x)$. Then A has a least element. Let the least element of A be x_0 . Since $x_0 \in A$, $\neg \varphi(x_0)$. But x_0 is the least, every y smaller than x_0 satisfies $\varphi(y)$; that is, $\forall y < x_0 \varphi(y)$. Then by the hypothesis, $\varphi(x_0)$ must also hold, a contradiction. Therefore, $A = \emptyset$. $\square$

2.1.3 COROLLARY. *Transfinite induction on well-ordered sets also has the following form, which is very common. Let $\langle X, < \rangle$ be a well-ordered set, and let Y be a subset of X . If for every $x \in X$, the set Y satisfies the condition $\text{seg}_X x \subset Y \rightarrow x \in Y$, then $Y = X$.*

2.1.4 THEOREM. *If $f : X \rightarrow X$ be an order-preserving function defined on a well-ordered set $\langle X, < \rangle$, then $x \leq f(x)$ for all $x \in X$.*

Proof. Let $Y = \{x \in X : x \leq f(x)\}$. We will show that $X = Y$. Let $x \in X$ be such that $\text{seg}_X x \subset Y$. We shall show that $x \in Y$. So take $y \in \text{seg}_X x$. Then $y \leq f(y)$. Moreover, $f(y) < f(x)$, since $y < x$. Hence $y < f(x)$ for every $y \in \text{seg}_X x$. In particular, $f(x) \in X - \text{seg}_X x$. Since x is the first element of $X - \text{seg}_X x$ we conclude that $x \leq f(x)$. Thus, $x \in Y$. $\square$

2.1.5 COROLLARY. *If well-ordered sets $\langle A, < \rangle$ and $\langle B, < \rangle$ are isomorphic, then the isomorphism $f : A \rightarrow B$ is unique.*

Proof. Let $g : A \rightarrow B$ be another order-isomorphism. We will show that $f = g$. So let $x \in A$. The function $g^{-1} \circ f : A \rightarrow A$ is also an isomorphism. By the theorem,

we conclude that $x \leq g^{-1}(f(x))$, and applying function g to both sides, $g(x) \leq f(x)$. Similarly, $f(x) \leq g(x)$. Thus, $f(x) = g(x)$. Since x is arbitrary, $f = g$. $\square$

2.1.6 COROLLARY. *No well-ordered set is isomorphic to any of its proper initial segments.*

Proof. If a well-ordered set X were isomorphic to an initial segment $\text{seg}_X x$ for $x \in X$, and $f : X \rightarrow \text{seg}_X x$ were an isomorphism establishing this fact, then we would have $f(x) \in \text{seg}_X x$, that is, $f(x) < x$, contradicting the theorem. $\square$

2.1.7 THEOREM. *If $\langle A, < \rangle$ and $\langle B, < \rangle$ are well-ordered sets then either*

1. $A \cong B$;
2. $A \cong \text{seg}_B b$ for some $b \in B$;
3. $B \cong \text{seg}_A a$ for some $a \in A$.

Proof. Let $Z = \{x \in A : \exists y_x \in B (\text{seg}_A x \cong \text{seg}_B y_x)\}$. Notice that if $x \in Z$, then y_x from the definition of the set Z is unique. Then we may define a function $f : Z \rightarrow B$ by $f(x) = y_x$, that is,

$$f = \{(x, y) \in A \times B : \text{seg}_A x \cong \text{seg}_B y\}.$$

First we shall show that Z is an initial segment of A . To see it, let $x < z$ and $z \in Z$. We have to show that $x \in Z$. Since $z \in Z$, there is an isomorphism h between $\text{seg}_A z$ and $\text{seg}_B f(z)$. The restriction of h to $\text{seg}_A x$ gives an isomorphism, whose range must be a proper initial segment S of $\text{seg}_B f(z)$. Since every proper initial segment of a well-ordered set is of the form seg , $S = \text{seg}_{\text{seg}_B f(z)} y$ for some $y \in \text{seg}_B f(z)$. Thus, $x \in Z$ and we conclude that Z is an initial segment of A . This argument shows also that f is strictly increasing, since $x < z$ and $z \in Z$ imply that $\text{seg}_A x$ is isomorphic to $\text{seg}_{\text{seg}_B f(z)} y$ for some $y \in \text{seg}_B f(z)$, that is $f(x) = y < f(z)$.

Similarly as for Z we can also prove that $f[Z]$ is an initial segment of B . So $f : Z \rightarrow f[Z]$ is an order-isomorphism.

If $Z = A$, then f establishes an isomorphism between A and either B or a proper initial segment of B . Similarly, if $f[Z] = B$, then f^{-1} shows that B is isomorphic to either A or a proper initial segment of A . Thus, it is enough to prove that either $Z = A$ or $f[Z] = B$. Suppose, for contradiction that $Z \neq A$ and $f[Z] \neq B$. Then Z is a proper initial segment of A and $f[Z]$ is a proper initial segment of B . Hence $Z = \text{seg}_A x$ for some $x \in A$ and $f[Z] = \text{seg}_B y$ for some $y \in B$. But, then f is an isomorphism between $Z = \text{seg}_A x$ and $\text{seg}_B y$, contradicting the fact that $x \notin Z$. $\square$

This theorem is a very strong result. It says that any two well-ordered sets are essentially the same or one of them is just the other with extra elements tacked on at the end. It suggests that one can organize well-ordered sets in terms of their “length” as orderings.

Arithmetic on Ordered Sets

First, we shall define the “addition” of two ordered sets X and Y . One natural idea is to take the elements of X in ascending $<_X$ order and then add on, as greater elements, all the elements of Y in ascending $<_Y$ order.

$$\cdots < x_1 < x_2 < \cdots < y_1 < y_2 < \cdots$$

So, we would define the “sum” of X and Y to be $X \cup Y$ with this order. However, if X and Y have elements in common, what should we do? The solution is to create *copies* of X and Y which are disjoint and then the sum as follows:

2.1.7 DEFINITION. Let $\langle X, <_X \rangle$ and $\langle Y, <_Y \rangle$ be linearly ordered sets. We define the *order sum* of X and Y , written as $X + Y$ to be the ordered set $(X \times \{0\}) \cup (Y \times \{1\})$ with order $<$ (one may check that $\langle X + Y, < \rangle$ is a linearly ordered set) defined by

$$\begin{aligned} (x_1, 0) &< (x_2, 0) \text{ if } x_1, x_2 \in X \text{ and } x_1 <_X x_2 \\ (x_1, 0) &< (y_1, 1) \text{ if } x_1 \in X \text{ and } y_1 \in Y \\ (y_1, 1) &< (y_2, 1) \text{ if } y_1, y_2 \in Y \text{ and } y_1 <_Y y_2 \end{aligned}$$

Now, we shall define the “multiplication” of ordered sets. If we view natural numbers n and m as ordered sets, we would aim to define their ordered set product so that it is at least order-isomorphic to the natural number $m \cdot n$. Since $m \cdot n$ is isomorphic to $m \times n$, this would suggest defining the order product to be some sort of order on the Cartesian product $m \times n$. And this is what we shall do in general. One sort of order on $X \times Y$ copies the way one orders words in an English dictionary.

2.1.8 THEOREM. Let m, n be natural numbers with their usual orderings $\in$. Then their order-sum $m + n$ is order-isomorphic to their arithmetic sum $m + n$. To make a distinction, we shall tentatively denote the latter sum as $m \oplus n$.

Proof. □

Since $\oplus$ is commutative and associative on ω , this theorem implies that the order sum $+$ on finite well-ordered sets is also commutative and associative. Specifically,

$$m + n \cong m \oplus n = n \oplus m \cong n + m.$$

What happens when we add infinite sets? Consider the ordered sets $\omega + 2$ and $2 + \omega$. We shall show that $\omega + 2 \not\cong 2 + \omega$. However, order addition does have other familiar properties, for example, associativity:

2.1.9 THEOREM. If X, Y and Z are linearly ordered sets, then $X + (Y + Z) \cong (X + Y) + Z$. In general, $X + Y \not\cong Y + X$. However, if X, Y are finite, then $X + Y \cong Y + X$.

2.1.8 DEFINITION. Let $\langle X, <_X \rangle$ and $\langle Y, <_Y \rangle$ be linearly ordered sets. Then the *lexicographic ordering* on $X \times Y$ is defined by $(x_1, y_1) < (x_2, y_2)$ if and only if

$$\begin{aligned} &x_1 <_X x_2 \text{ or} \\ &x_1 = x_2 \text{ and } y_1 <_Y y_2. \end{aligned}$$

2.1.9 DEFINITION. Let $\langle X, <_X \rangle$ and $\langle Y, <_Y \rangle$ be linearly ordered sets. Then the *product* of these ordered sets is the set $X \times Y$ with the *anti-lexicographic ordering* defined by $(x_1, y_1) < (x_2, y_2)$ (one may check that $\langle X \times Y, < \rangle$ is a linearly ordered set.) if and only if

$$\begin{aligned} &y_1 <_Y y_2 \text{ or} \\ &y_1 = y_2 \text{ and } x_1 <_X x_2. \end{aligned}$$

$$\begin{array}{ccccccc} & & & Y & & & \\ & & & \underbrace{\hspace{10em}} & & & \\ \cdots & <_Y & \bullet & <_Y & \bullet & <_Y & \bullet & <_Y & \cdots \\ & & y & & y' & & y'' & & \\ & & \hline X & & X & & X & & \cdots \end{array}$$

2.2 Ordinality

If we want to single out the order type of a given partially ordered set, a natural approach is to find the common characteristic of all ordered sets that are isomorphic to this partially

ordered set. A very straightforward idea for this is: take the “equivalence class” of this well-ordered set (the class of all sets isomorphic to it), and then select a representative as its order type. The problem is: this equivalence class is not a set (as we have seen in the previous section). Moreover, when we discuss all order types, we find that the class of these representatives is also not a set. Except for knowing that it is a class, we cannot obtain any more useful properties.

However, for well-ordered sets, the situation is very good. This comes from the last theorem of the previous section: any two well-ordered sets can be compared. In more detail, suppose we first choose representatives A and B from two isomorphism classes. By the theorem from the previous section, A and B can be compared via the relation “is order-isomorphic to some initial segment of”. In a very special case, if we manage to choose all these representatives to lie within the same class (which has all the properties of a well-order but need not be a set), and this is possible, then the representatives can be compared directly by the relation “is an initial segment of”. In this case, $A \subset B$; that is, of the two representatives, one is a subset of the other.

In this section, we shall look at a construction developed by von Neumann which gives a particular set out of each order-type of well-ordered sets in a systematic way, without any need for axioms beyond those in ZF . We shall then reconstruct Cantor’s arithmetic on his ordinals using the von Neumann ordinals.

Ordinal Numbers

2.2.1 DEFINITION. A set α is an **ordinal number** (an ordinal) if it is transitive and well-ordered by $\in$. (Here, we define $\alpha < \beta \leftrightarrow \alpha \in \beta$). That is

1. $\forall x \in \alpha (x \notin x)$;
2. $\forall x, y, z \in \alpha (x \in y \wedge y \in z \rightarrow x \in z)$;
3. $\forall x, y \in \alpha (x \in y \vee x = y \vee y < x)$;
4. $\forall a \subset \alpha (a \neq \emptyset \rightarrow \exists x \in a \forall y \in a (x \leq y))$;
5. $\forall x \in \alpha (y \in x \rightarrow y \in \alpha)$.

The class of all ordinals is denoted by **ON** (since **ON** is a set of sets, we can not currently determine whether it is a set) Ordinal numbers defined in this way are usually called von Neumann ordinal numbers.

2.2.1 REMARK. $0 = \emptyset$ is an ordinal number. We have seen that every element of ω is transitive and $\langle n, \in \rangle$ is a well-ordered set, which means every natural number is an ordinal. Furthermore, we have seen that ω is transitive and $\langle \omega, \in \rangle$ is a well-ordered set, which means ω is itself an ordinal.

2.2.1 PROPOSITION. Let α, β be two ordinals.

1. $\alpha \neq 0 \rightarrow 0 \in \alpha$;
2. If $x \in \alpha$, then x is an ordinal;
3. If $x \in \alpha$, then $\text{seg}_\alpha x = x$;
4. If $\alpha \cong \beta$, then $\alpha = \beta$;
5. $\alpha \in \beta \vee \alpha = \beta \vee \beta \in \alpha$.

Proof. (1). Let α be an $\in$ -well-ordered set, and let α_0 be its $\in$ -minimum. We claim that $\alpha_0 = 0$: suppose $\alpha_0 \neq 0$, i.e., $\alpha_0 \neq \emptyset$. Then there exists $t \in \alpha_0$. Since α is transitive, $t \in \alpha$. This contradicts the $\in$ -minimality of α_0 in α .

(2). Since α is transitive, $x \in \alpha \rightarrow x \subset \alpha$. Hence x (as a subset of the $\in$ -well-ordered set α) is itself $\in$ -well-ordered. We now show that x is transitive. Take $z \in y$ and $y \in x$. Then $y \in \alpha$ (since α is transitive and $x \in \alpha$), and then $z \in \alpha$ (again by transitivity of α). By the $\in$ -transitivity property of ordinals, $z \in x$. Thus, x is transitive and well-ordered by $\in$, so $x \in \mathbf{ON}$.

(3). By definition $\text{seg}_\alpha x = \{t \in \alpha : t \in x\}$. If $t \in \text{seg}_\alpha x$, then $t \in x$. Conversely, if $t \in x$, then since $x \in \alpha$ and α is transitive, $t \in \alpha$; and $t \in x$ by assumption, so $t \in \text{seg}_\alpha x$.

(4). Suppose $f : \alpha \cong \beta$. First we show that $\forall x \in \alpha (f(x) = x)$. This is a universal statement. Since α is an ϵ -well-ordered set, we use the induction on x : for $x = 0$, since 0 is the least element, by the ordered-preserving property of f , $f(0)$ is the least element in β , hence $f(0) = 0$. Assume that for all $y \in x$, $f(y) = y$ and we prove that $f(x) = x$. First, let $y \in x$. By the order-preserving property of f , $f(y) \in f(x)$. By the induction hypothesis, $y \in f(x)$. Next, let $y \in f(x)$. In β , $f(x) = \text{seg}_\beta f(x)$. Hence $y \in \beta$. Since f is surjective, let $u \in \alpha$ be the preimage of y : $y = f(u)$. Then $f(u) \in f(x)$. By order-preserving property of f , $u \in x$. Finally, by the induction hypothesis, $u = f(u) = y$, so $y \in x$. Since $\forall x \in \alpha (f(x) = x)$, we have $\alpha \subset \beta$. Similarly, $\beta \subset \alpha$. Hence $\alpha = \beta$.

(5). Since the ordinals are ϵ -well-ordered sets, by the fundamental theorem on well-ordered sets, exactly one of the following three cases holds:

$$\alpha \cong \text{seg}_\beta y \vee \alpha \cong \beta \vee \beta \cong \text{seg}_\alpha x.$$

By proposition (3), this becomes $\alpha \cong y \vee \alpha \cong \beta \vee \beta \cong x$. By proposition (4), $\alpha \in \beta \vee \alpha \vee \beta \in \alpha$. The three cases are mutually exclusive; otherwise, they would immediately contradict ϵ -irreflexivity. $\square$

2.2.2 THEOREM. *The class $\mathbf{ON}$ is $\in$ -well-ordered in the following sense:*

1. $\forall \alpha \in \mathbf{ON} (\alpha \notin \alpha)$;
2. $\forall \alpha, \beta, \gamma \in \mathbf{ON} (\alpha \in \beta \wedge \beta \in \gamma \rightarrow \alpha \in \gamma)$;
3. $\forall \alpha, \beta \in \mathbf{ON} (\alpha \in \beta \vee \alpha = \beta \vee \beta \in \alpha)$;
4. *For any non-empty set X of ordinals contains a least element.*

Proof. For (4). If X is a non-empty set of ordinals, then it contains at least one ordinal, say α . Consider the set $Y = \{\beta \in X : \beta \in \alpha\}$. If Y is empty, then α is the least element of X . And if $Y \neq \emptyset$ then, since Y is a non-empty subset of the well-ordered set α , Y has a least element γ , which must also be the least element of X . $\square$

2.2.2 REMARK. Assume $\mathbf{ON}$ is a set, then $\langle \mathbf{ON}, \in \rangle$ is well-ordered by this theorem. Moreover, $\mathbf{ON}$ is transitive by proposition 1.6.5. That means $\mathbf{ON}$ is an ordinal, i.e., $\mathbf{ON} \in \mathbf{ON}$. However, $\mathbf{ON}$ is well-ordered, $\mathbf{ON} \notin \mathbf{ON}$, which is a contradiction. This is known as the **Burail-Forti pradox** (1897). In axiomatic set theory, we do not regard $\mathbf{ON}$ as a set. In other words, $\mathbf{ON}$ is a class and indeed a proper class.

2.2.3 THEOREM.

1. *If α is an ordinal, so is α^+ ; furthermore, α^+ is the least ordinal greater than α .*
2. *If X is a set of ordinals, then $\bigcup X$ is an ordinal and the least upper bound of X .*

Proof. To prove (2), suppose that X is a set of ordinals. To show that $\bigcup X$ is an ordinal, we have, as ever, to show that it is $\in$ -well-ordered and that each of its members is a subset. Each member of $\bigcup X$ is in some ordinal α in the set X , so is itself an ordinal. Thus, $\bigcup X$ is a set of ordinals and is then, by theorem, $\in$ -well-ordered. Now, suppose that $\beta \in \bigcup X$, we shall show that $\beta \subset \bigcup X$. Since $\beta \in \bigcup X$, we have $\beta \in \alpha$ for some $\alpha \in X$. Since $\alpha \in X$, then $\alpha \subset \bigcup X$; and since α is an ordinal, we have then $\beta \subset \alpha$. Thus, $\beta \subset \bigcup X$ are required.

Now, we show that $\bigcup X$ is an upper bound of X . For this, suppose that $\alpha \in X$. Then $\alpha \subset \bigcup X$, so $\alpha \leq \bigcup X$. We show that $\bigcup X$ is the least such upper bound, suppose that

the ordinal γ is an upper bound for X . We shall show that $\bigcup X \subset \gamma$, so that we have $\bigcup X \leq \gamma$. Take any $\beta \in \bigcup X$. Then $\beta \in \alpha$ for some $\alpha \in X$. Since γ is an upper bound for X , we have $\alpha \leq \gamma$, then $\alpha \subset \gamma$. Thus, $\beta \in \alpha \subset \gamma$, so that $\beta \in \gamma$. $\square$

2.2.3 REMARK. We have previously proved that ω is an ordinal. Here, the above proposition more generally shows that ω is indeed an ordinal. Starting from ω and repeatedly applying the successor operation yields a sequence of new ordinals that have not seen before:

$$\omega^+, \omega^{++}, \omega^{+++}, \dots$$

We cannot help but ask: is the class $\{\omega^{(n)} : n \in \omega\}$ a set? If it is a set, then naturally, $\bigcup\{\omega^{(n)} : n \in \omega\}$ is an ordinal. Intuitively, it looks very much like a set, namely ω . We know that the elements of ω include $0, 1, 2, \dots$. In other words, the two have the same “size”. Based on past experience, such a class whose size is “controllable” (meaning it can be “bounded” by some other known set) can usually be defined as a set. However, the first six axioms alone cannot prove this. Therefore, we need another axiom for this purpose. But before that, we still need to discuss more about ordinals.

2.2.2 DEFINITION. Ordinals of the form α^+ are called the **successor ordinals** of α . An ordinal α which is neither 0 nor a successor ordinal is called a **limit ordinal**.

2.2.4 PROPOSITION. α is a limit ordinal if and only if

$$\alpha \neq 0 \wedge (\beta \in \alpha \rightarrow \beta^+ \in \alpha)$$

Proof. Suppose that α is a limit ordinal, then $\alpha \neq 0$. Suppose that $\beta \in \alpha$. Then $\beta^+ = \beta \cup \{\beta\} \subset \alpha$, but then α is not a successor ordinal, so $\beta^+ \neq \alpha$, so we must have $\beta^+ \in \alpha$. Conversely, if $\alpha \neq 0$ and α is a successor ordinal, say $\alpha = \beta^+$, then $\beta \in \alpha$, but then $\beta^+ \in \alpha$, contradicting $\alpha = \beta^+$. $\square$

2.2.5 PROPOSITION. An ordinal λ is a limit ordinal if and only if it is non-empty set and $\lambda = \bigcup \lambda$.

Proof. First we show that (λ) is a limit ordinal implies $\bigcup \lambda = \lambda$. Suppose that $\beta \in \bigcup \lambda$, then $\beta \in \alpha$ for some $\alpha \in \lambda$. Since λ is an ordinal we then have $\beta \in \lambda$. Thus, $\bigcup \lambda \subset \lambda$. Suppose that $\alpha \in \lambda$, then since λ is a limit ordinal we have $\alpha^+ \in \lambda$. But $\alpha \in \alpha^+$, so that $\alpha \in \bigcup \lambda$. Thus, $\lambda \subset \bigcup \lambda$.

Conversely, if $\bigcup \lambda = \lambda$, we show that λ is a limit ordinal. Suppose, for contradiction that λ is not a limit ordinal, i.e., $\lambda = \beta^+$. Since $\bigcup \beta^+ = \beta$, $\lambda = \bigcup \lambda = \bigcup \beta^+ = \beta \in \beta^+ = \lambda$, contradicting the fact that λ is an ordinal. $\square$

Transfinite Induction and Recursion on Ordinals

The essential reason why the induction principle holds for both ω and well-ordered sets is that they are both $\in$ -well-ordered. In other words, every non-empty subset has a least element, which allows us to select the elements of the entire set from smallest to largest. Moreover, we have already proved that the ordinal universe **ON** is also $\in$ -well-ordered; however, it is a proper class. Nevertheless, we can still attempt to establish a corresponding induction principle on it.

2.2.6 THEOREM (Transfinite Induction). *Let φ be a formula. If $\forall \alpha (\forall \beta < \alpha \varphi(\beta) \rightarrow \varphi(\alpha))$, then $\forall \alpha \varphi(\alpha)$.*

Proof. If not, $\neg \varphi(\alpha)$ for some ordinal α . Taking such a α . But then $\varphi(\beta)$ for all $\beta < \alpha$, so that $\varphi(\alpha)$ by the hypothesis, which is a contradiction. $\square$

2.2.7 THEOREM (Order Type Theorem). *Every well-ordered set is isomorphic to one and only one ordinal.*

Proof. Suppose that $\langle W, < \rangle$ is a well-ordered set. Uniqueness is simple, since isomorphism between ordinals is identical with equality. If there exist α, β such that $W \cong \alpha$, and $W \cong \beta$, then $\alpha \cong \beta$ and hence $\alpha = \beta$. For $w \in W$, let $\text{seg}_W[w] = \text{seg}_W(w) \cup \{w\}$ and define

$$Z = \{w \in W : \text{seg}_W[w] \text{ is isomorphic to some ordinal } \alpha_w\}$$

Notice that the ordinal α_w and the isomorphism $f_w : \text{seg}_W[w] \rightarrow \alpha_w$ are unique. Now, if $v, w \in Z$ and $v \leq w$, then $f_w[\text{seg}[v]]$ is an ordinal α , as an initial segment of α_w . In particular, $f_w|_{\text{seg}_w[v]} : \text{seg}_W[v] \rightarrow \alpha$ is an order-isomorphism and, by the preceding uniqueness remark, $f_w|_{\text{seg}_W[v]} = f_v$. So we have shown that

$$f_v \subset f_w, \quad \forall v, w \in Z, v \leq w.$$

We will prove by transfinite induction, that $Z = W$. So assume that $\text{seg}_W(w) \subset Z$. Then, $f = \bigcup_{v \in \text{seg}_W(w)} f_v$ is a strictly increasing function from $\bigcup_{v \in \text{seg}_W(w)} \text{seg}_W[v]$ onto $\bigcup_{v \in \text{seg}_W(w)} \alpha_v$. But $\bigcup_{v \in \text{seg}_W(w)} \text{seg}_W[v] = \text{seg}_W(w)$ and $\alpha = \bigcup_{v \in \text{seg}_W(w)} \alpha_v$ is an ordinal. So $f : \text{seg}_W(w) \rightarrow \alpha$ is an order-isomorphism. Extend f to a function $F : \text{seg}_W[w] \rightarrow \alpha^+$ by putting $F(w) = \alpha$. Then the existence of F proves that $w \in Z$. We have shown that $Z = W$.

Now, the function $f = \bigcup_{w \in W} f_w$ is an isomorphism between $W = \bigcup_{w \in W} \text{seg}_W[w]$ and an ordinal number $\bigcup_{w \in W} \alpha_w$. $\square$

2.2.3 DEFINITION. For a well-ordered set W let $\text{Otp}(W)$ stand for the **order type** of W , that is, the unique ordinal number that is order isomorphic to W .

2.2.8 COROLLARY. If α and β are ordinal number and $f : \alpha \rightarrow \beta$ is an order-isomorphism. Then $\alpha \leq \beta$. Moreover, $x \leq f(x)$ for every $x < \alpha$.

Proof. If $\beta < \alpha$, then $\beta \subset \alpha$ and f is a order-preserving from α into β . But then $f(\beta) \in \beta$, that is, $f(\beta) < \beta$, contradicting the lemma. So, $\alpha \leq \beta$. The additional part follows from the lemma. $\square$

2.2.9 COROLLARY. If W is a well-ordered set and $B \subset W$ then $\text{Otp}(B) \leq \text{Otp}(W)$.

Proof. Let $f : \text{Otp}(B) \rightarrow B$ and $g : W \rightarrow \text{Otp}(W)$ be order-isomorphisms. Then $g \circ f : \text{Otp}(B) \rightarrow \text{Otp}(W)$ is order-isomorphism and by the corollary above, $\text{Otp}(B) \leq \text{Otp}(W)$. $\square$

Recall the Recursion on ω ; its core idea is that the value of n^+ is determined by n . However, in the ordinal universe **ON**, limit ordinals are not the successor of any ordinal. How should we establish the Recursion principle on the ordinal universe **ON**? A natural thought is: the value at an ordinal should be determined by all ordinals β satisfies $\beta < \alpha$:

2.2.10 THEOREM (Transfinite Recursion). Let $h : \mathbf{ON} \times \mathbf{V} \rightarrow \mathbf{V}$ be a classes function, and let δ be an ordinal. Then there is a unique function $f : \delta \rightarrow \mathbf{V}$ such that

$$\forall \beta \in \delta (f(\beta) = h(\beta, f|_\beta))$$

Proof. We shall see that a function g is a γ -function where γ is an ordinal if its domain is γ and satisfies the rule $g(\beta) = h(\beta, g|_\beta)$ for all $\beta \in \gamma$. So g is just like the required function f except that its domain is γ rather than δ .

1. Claim: suppose g is a γ -function and g' is a γ' -function, then $g|_{\gamma \cap \gamma'} = g'|_{\gamma \cap \gamma'}$.
2. Claim: for all $\gamma \in \delta^+$, if γ -function exists, is unique.
3. Claim: for all $\gamma \in \delta^+$, there is a γ -function g_γ . For 0 and successor ordinals, induction can still be used to prove the statement, which is similar to induction on natural numbers. However, ordinals also have the special case of limit ordinals, which requires special explanation.

Proof. Suppose that g_γ exists for all $\gamma \in \lambda$ where λ is a limit ordinal. By the claim 1, if $\gamma \in \gamma' \in \lambda$, then $g_\gamma = g_{\gamma' \upharpoonright \gamma}$, so that $g_\gamma \subset g_{\gamma'}$. Thus we may have a $\subset$ -chain for $\gamma \in \lambda$:

$$\cdots \subset g_\gamma \subset g_{\gamma^+} \subset g_{\gamma^{++}} \subset \cdots$$

Taking the union of this chain would give good prospects of giving a function with domain λ with the extra property needed to be a λ -function. First of all, we must check that we are dealing with sets within **ZF**: The class $\{g_\gamma : \gamma \in \lambda\}$ is a set by the axiom of replacement, We can thus define a set F within **ZF** by

$$F = \bigcup \{g_\gamma : \gamma \in \lambda\}.$$

Now, we show that F is a function. Suppose that (β, x) and (β, x') are both in F . Then there must be $\gamma, \gamma' \in \lambda$ such that $(\beta, x) \in g_\gamma$ and $(\beta, x') \in g_{\gamma'}$. Without loss of generality $\gamma \in \gamma'$, so that $g_\gamma \subset g_{\gamma'}$. This means that both ordered pair belong to $g_{\gamma'}$, which is a function, forcing that $x = x'$. Thus, F is a function.

Next, we show that the domain of F is λ . Since F is a union of functions, its domain is just the union of their domains, i.e.,

$$\text{dom}(F) = \bigcup \{\text{dom}(g_\gamma : \gamma \in \lambda)\} = \bigcup \{\gamma : \gamma \in \lambda\} = \lambda$$

Thus, F has domain λ . Finally, we show that F is the λ -function, all that remains is to show that $F(\beta) = h(\beta, F|_\beta)$ for all $\beta \in \lambda$. For any $\beta \in \lambda$, we also have $\beta^+ \in \lambda$ because λ is limit. By the definition of F , the value of $F(\beta)$ is the same as that of $g_{\beta^+}(\beta)$; and the restriction $F|_\beta$ is the same as $g_{\beta^+}|_\beta$. Thus, $F(\beta) = g_{\beta^+}(\beta) = h(\beta, g_{\beta^+}|_\beta) = h(\beta, F|_\beta)$, which holds because g_{β^+} is a β^+ -function. $\square$

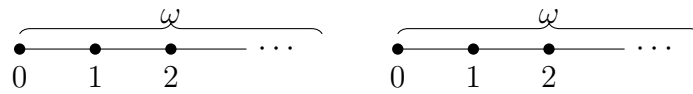
$\square$

2.2.11 THEOREM (Transfinite Recursion, class form). *Let $h : \mathbf{ON} \times \mathbf{V} \rightarrow \mathbf{V}$ be a class function, Then there is a unique class function $f : \mathbf{ON} \rightarrow \mathbf{V}$ such that*

$$\forall \beta \in \mathbf{ON} (f(\beta) = h(\beta, f|_\beta)).$$

Arithmetic on Ordinal Numbers

Now that we have some ways of constructing ordinals, we might be in a better position to show that each isomorphism class of well-ordered sets contains an ordinal. We have seen the ordinals include $0, 1, \dots, n, \dots, \omega, \omega^+, \omega^{++}, \dots, \dots, \omega^{++\dots+}, \dots$, but whether we have an ordinal order-isomorphic to the order sum $\omega + \omega$. Consider its shape



It should seem reasonable to say that $\omega + \omega$ is a “smallest” well-ordered set which contains initial segments order-isomorphic to each of the order sum $\omega, \omega + 1, \omega + 2, \dots, \omega + n, \dots$ for all $n \in \omega$. Now, we can show that

$$\forall n \in \omega (\omega + n \cong \omega^{++\dots+})$$

Then surely $\omega + \omega$ will be order-isomorphic to the least upper bound of all these $\omega^{++\dots+}$, which by the theorem 2.2.3 would be the ordinal

$$\bigcup \{\omega, \omega^+, \omega^{++}, \dots\}.$$

2.2.4 DEFINITION. For all ordinals α :

1. $\alpha \oplus 0 = \alpha$;
2. $\alpha \oplus \gamma^+ = (\alpha \oplus \gamma)^+$;
3. $\alpha \oplus \lambda = \bigcup \{\alpha \oplus \gamma : \gamma \in \lambda\}$ for every limit ordinal λ .

Now, we shall use the transfinite restriction principle to justify such a definition of ordinal sum is legitimate. Fix α , we aim to construct a function $f : \mathbf{ON} \rightarrow \mathbf{V}$ satisfying

$$f(\beta) = \alpha \oplus \beta$$

by transfinite induction. To do this, we define a recursion rule h such that for any ordinal β , $f(\beta) = h(\beta, f|_\beta)$, where $f|_\beta$ denotes the restriction of f to β . The rule h is defined according to the type of β :

1. If $\beta = 0$, $h(0, y) = \alpha$ for any set y ;
2. If $\beta = \gamma^+$, $h(\beta, y) = (y(\gamma))^+$, provided y is a function with $\beta \subset \text{dom}(y)$;
3. If β is a limit ordinal, $h(\beta, y) = \bigcup \text{ran}(y|_\beta)$, provided y is a function with $\beta \subset \text{dom}(f)$; otherwise $h(\beta, y) = \emptyset$.

By the transfinite recursion principle, there exists a unique function $f : \mathbf{OV} \rightarrow \mathbf{V}$ satisfying the recursion $f(\beta) = h(\beta, f|_\beta)$ for all β . By transfinite induction, one can show that $f(\beta)$ is always an ordinal. Thus, we successfully defined $\alpha \oplus \beta$ for all ordinals β . For a limit ordinal λ , we want

$$f(\lambda) = \alpha \oplus \lambda = \bigcup \{\alpha \oplus \gamma : \gamma \in \lambda\} = \bigcup \{f(\gamma) : \gamma \in \lambda\} = \bigcup \text{ran}(f|_\lambda).$$

Thus, we need $h(\lambda, f|_\lambda)$ to equal $\bigcup \text{ran}(f|_\lambda)$, which we can achieve by defining, for a limit ordinal

$$h(\beta, y) = \begin{cases} \bigcup \text{ran}(y|_\beta), & \text{if } y \text{ is a function with } \beta \subset \text{dom}(y); \\ \emptyset, & \text{otherwise} \end{cases}$$

2.2.12 PROPOSITION. $\alpha \oplus 1 = \alpha^+$ for all ordinal α .

Proof. $\alpha \oplus 1 = \alpha \oplus (0^+) = (\alpha \oplus 0)^+ = \alpha^+$. □

Now, we can stop using the notation α^+ , using $\alpha+1$ instead. Of course, when appropriate, the notation of successor ordinals is also used.

2.2.13 THEOREM. Let α and β be ordinals, then the order sum $\alpha + \beta$ is order-isomorphic to the ordinal sum $\alpha \oplus \beta$.

Proof. We shall prove this for a fixed α by transfinite induction on all β . Define a subset A of all ordinals $\beta \in \delta$ for which the result holds:

$$A = \{\beta \in \delta : \alpha + \beta \cong \alpha \oplus \beta\}.$$

For $\beta = 0$, both $\alpha + \beta$ and $\alpha \oplus \beta$ equal α . So the result holds trivially for $\beta = 0$. Next, suppose that the result holds for all ordinals up to, and including, γ , and show that it holds for γ^+ . On the one hand

$$\begin{aligned} \alpha + \gamma^+ &= (\alpha \times \{0\}) \cup (\gamma^+ \times \{1\}) \\ &= (\alpha \times \{0\}) \cup ((\gamma \cup \{1\}) \times \{1\}) \\ &= ((\alpha \times \{0\}) \cup (\gamma \times \{1\})) \cup (\{1\} \times \{1\}) \\ &= (\alpha + \gamma) \cup (\{1\} \times \{1\}). \end{aligned}$$

which is the ordered set $\alpha + \gamma$ with the extra element $(\gamma, 1)$ at the top (the ordering $<$ is the anti-lexicographic ordering). On the other hand

$$\alpha \oplus \gamma^+ = (\alpha \oplus \gamma)^+ = (\alpha \oplus \gamma) \cup \{\alpha \oplus \gamma\},$$

which consists of the ordinal $\alpha \oplus \gamma$, i.e., all the members of this ordinal, with the extra element $\alpha \oplus \gamma$ at the top. By the hypothesis, there is an order-isomorphism from $\alpha + \gamma$ to $\alpha \oplus \gamma$. It should be clear that this order-isomorphism can be extended, by mapping $(\gamma, 1)$ to the element $\alpha \oplus \gamma$, to give an order-isomorphism from $\alpha + \gamma^+$ to $\alpha \oplus \gamma^+$.

Now, suppose that $\alpha + \gamma \cong \alpha \oplus \gamma$ for all $\gamma \in \lambda$, where λ is a limit ordinal. Since λ is a limit ordinal, $\lambda = \bigcup \lambda = \bigcup \{\gamma : \gamma \in \lambda\}$, so that

$$\begin{aligned} \alpha + \lambda &= (\alpha \times \{0\}) \cup (\lambda \times \{1\}) \\ &= (\alpha \times \{0\}) \cup (\bigcup \{\gamma : \gamma \in \lambda\} \times \{1\}) \\ &= (\alpha \times \{0\}) \cup \bigcup \{\gamma \times \{1\} : \gamma \in \lambda\} \\ &= \bigcup \{(\alpha \times \{0\}) \cup (\gamma \times \{1\}) : \gamma \in \lambda\} \\ &= \bigcup \{\alpha + \gamma : \gamma \in \lambda\}. \end{aligned}$$

For each $\gamma \in \lambda$, let f_γ be the ordered-isomorphism from $\alpha + \gamma$ to $\alpha \oplus \gamma$. Now, put $f = \bigcup \{f_\gamma : \gamma \in \lambda\}$. If $\gamma, \gamma' \in \lambda$ with $\gamma < \gamma'$, then γ is an initial segment of γ' , so that $f_\gamma \subset f_{\gamma'}$. This means that f is a function. The domain of f is

$$\bigcup \{\text{dom}(f_\gamma) : \gamma \in \lambda\} = \bigcup \{\alpha + \gamma : \gamma \in \lambda\} = \alpha + \lambda.$$

And its image set is

$$\bigcup \{\alpha \oplus \gamma : \gamma \in \lambda\} = \alpha \oplus \lambda.$$

One may check that f is an order-isomorphism, so that $\alpha + \lambda \cong \alpha \oplus \lambda$. Thus, by transitive induction $\alpha + \beta \cong \alpha \oplus \beta$ for all $\beta \in \delta$. $\square$

2.2.5 DEFINITION. For all ordinals α :

1. $\alpha \otimes 0 = 0$;
2. $\alpha \otimes (\gamma \oplus 1) = (\alpha \otimes \gamma) \oplus \alpha$;
3. $\alpha \otimes \lambda = \bigcup \{\alpha \otimes \gamma : \gamma \in \lambda\}$ for every limit ordinal λ .

2.2.14 THEOREM. Let α and β be ordinals, then the order product $\alpha \times \beta$ is order-isomorphic to the ordinal product $\alpha \otimes \beta$. (If we define the ordering of $\alpha \times \beta$ by the lexicographic ordering, rather than antis, the result would be $\alpha \times \beta \cong \beta \otimes \alpha$.)

In view of the facts stated in the two theorems, we shall no longer distinguish between the addition and multiplication of ordered sets and those of ordinals in the following discussion. We use the symbols $+$, $\cdot$ uniformly, and readers are expected to tell them apart by context.

We may similarly define ordinal exponentiation in analogy with the exponentiation of natural numbers. This function is well-defined, i.e., uniquely determined by transfinite recursion, and we omit the detailed proof.

2.2.6 DEFINITION. For all ordinals α :

1. $\alpha^0 = 1$;
2. $\alpha^{\gamma+1} = (\alpha^\gamma) \cdot \alpha$;
3. $\alpha^\lambda = \bigcup \{\alpha^\gamma : \gamma \in \lambda\}$ for every limit ordinal λ .

Next, we introduce several basic properties of ordinal operations. First, ordinal addition and multiplication are both non-commutative. We illustrate this with the following examples.

2.2.1 EXAMPLE. $1 + \omega = \omega$, while $\omega + 1 \neq \omega$. $2 \cdot \omega = \omega$, while $\omega \cdot 2 \neq \omega$.

Proof. $1 + \omega = \bigcup\{1 + n : n \in \omega\}$ by the definition of ordinal addition. Since the finite ordinals are simply the natural numbers, we know that $1 + n = n + 1$ for all $n \in \omega$, so that $1 + \omega = \bigcup\{n + 1 : n \in \omega\}$. By theorem $\bigcup\{n + 1 : n \in \omega\}$ is the ordinal which is the least upper bound of the set $\{n + 1 : n \in \omega\}$. So we shall show that ω is this least upper bound.

First, it is clear that $n + 1 \leq \omega$ for each $n \in \omega$, which means ω is an upper bound. For a smaller ordinal $m < \omega$, we have

$$m < m + 1 \in \{n + 1 : n \in \omega\},$$

so that m is not an upper bound for the set. Thus ω is indeed the least upper bound of the set, as desired. However, $\omega + 1 = \omega + 0^+ = (\omega + 0)^+ = \omega^+ = \omega \cup \{\omega\} \neq \omega$.

The equality $2 \cdot \omega = \omega$ can be proved in the same way. However, $\omega \cdot 2 = \omega \cdot 1 + \omega = \omega + \omega = \bigcup\{\omega + n : n \in \omega\} \neq \omega$. $\square$

2.2.15 PROPOSITION. *Let α, β and γ be ordinals.*

1. $(\alpha + \beta) + \gamma = \alpha + (\alpha + \beta)$;
2. $(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma)$;
3. $\alpha \cdot (\beta + \gamma) = (\alpha \cdot \beta) + (\alpha \cdot \gamma)$.

2.2.16 THEOREM. *Let α, β and γ be ordinals, then*

1. *If $\beta < \gamma$, then $\alpha + \beta < \alpha + \gamma$;*
2. *If $\alpha > 0$ and $\beta < \gamma$, then $\alpha \cdot \beta < \alpha \cdot \gamma$;*
3. *If $\alpha > 1$ and $\beta < \gamma$, then $\alpha^\beta < \alpha^\gamma$.*

Proof. We shall use transfinite induction on γ , for fixed α and β . The result is vacuously true for all γ with $\gamma \leq \beta$. The first case is $\gamma = \beta^+$. Then

$$\alpha + \beta^+ = (\alpha + \beta)^+ > \alpha + \beta.$$

so that the result holds for β^+ . Suppose that the result holds up to, and including, γ , where $\beta < \gamma$. Then

$$\alpha + \gamma^+ = (\alpha + \gamma)^+ > \alpha + \gamma + \alpha + \beta$$

so that the result holds for γ^+ . Finally, suppose that the result holds for all $\gamma \in \lambda$, where λ is a limit ordinal with $\beta < \lambda$. Then

$$\alpha + \beta \in \alpha + \beta^+ \subset \bigcup\{\alpha + \gamma : \gamma \in \lambda\} = \alpha + \lambda$$

so that the result holds for λ . Both multiplication and exponentiation can be proved by transfinite induction, and we omit the proofs here. $\square$

Based on this theorem and our prior study, we see that ordinals can be listed in ascending order as follows (all ordinals are constructed starting from the empty set 0):

$$\begin{aligned} 0 &< 1 < 2 < \cdots < \omega < \\ \omega + 1 &< \cdots < \omega + n < \cdots < \omega + \omega = \omega \cdot 2 < \\ \omega \cdot 2 + 1 &< \cdots < \omega \cdot 2 + \omega = \omega \cdot 3 < \cdots < \omega \cdot \omega = \omega^2 < \\ \omega^2 + \omega &< \cdots < \omega^2 + \omega^2 = \omega^2 \cdot 2 < \cdots < \omega^2 \cdot \omega = \omega^3 < \cdots < \omega^\omega < \cdots < \omega^{\omega^\omega} < \cdots \end{aligned}$$

2.2.7 DEFINITION. The *cumulative hierarchy of sets*, $\mathbf{V}_\alpha$ for all ordinals α :

$$\begin{aligned} \mathbf{V}_0 &= \emptyset \\ \mathbf{V}_{\gamma^+} &= \mathcal{P}(\mathbf{V}_\gamma) \\ \mathbf{V}_\lambda &= \bigcup\{\mathbf{V}_\gamma : \gamma \in \lambda\} \text{ (for a limit ordinal } \lambda) \end{aligned}$$

$$\begin{aligned}
 \mathbf{V}_0 &= 0 \\
 \mathbf{V}_1 &= \mathcal{P}(\mathbf{V}_0) = \{0\} = 1 \\
 \mathbf{V}_2 &= \mathcal{P}(\mathbf{V}_1) = \{0, 1\} = 2 \\
 \mathbf{V}_3 &= \mathcal{P}(\mathbf{V}_2) = \{0, 1, \{1\}, 2\} \\
 &\dots\dots
 \end{aligned}$$

The transfinite recursion principle ensures that there is a class function $f : \mathbf{ON} \rightarrow \mathbf{V}$ such that $f(\alpha) = \mathbf{V}_\alpha$ for all ordinals α , with each $\mathbf{V}_\alpha$ being a set. Introducing the temporary notation $\mathbf{V}'$ for the class of all sets which are in at least one of the $\mathbf{V}_\alpha$, i.e.,

$$\mathbf{V}' = \bigcup \{\mathbf{V}_\alpha : \alpha \in \mathbf{ON}\}.$$

Note that for any set x in $\mathbf{V}'$, so that $x \in \mathbf{V}_\alpha$ for some ordinal α , there must be a least ordinal for which this is true (by the well-foundedness of $\mathbf{ON}$). What can we say about the least ordinal β such that $\beta \in \mathbf{V}_\beta$? First of all, $\beta \neq 0$ because $\mathbf{V}_0 = \emptyset$.

2.2.17 PROPOSITION. *Suppose that β is the least ordinal such that $x \in \mathbf{V}_\beta$, then β cannot be a limit ordinal.*

Proof. Suppose for contradiction that β is a limit ordinal, then by definition of $\mathbf{V}_\beta$, it would equal to $\bigcup \{\mathbf{V}_\gamma : \gamma \in \beta\}$. Thus, if $x \in \mathbf{V}_\beta$, x would be in $\mathbf{V}_\gamma$ for some $\gamma \in \beta$, contradicting the fact that β is the least ordinal such that $x \in \mathbf{V}_\beta$. $\square$

Thus, the least ordinal β such that $x \in \mathbf{V}_\beta$ must be a successor, i.e., $\beta = \alpha^+$ for some ordinal α . This allows us to associate a unique ordinal with the set x as in the following definition.

2.2.8 DEFINITION. For any set x in $\mathbf{V}'$, the *rank* of x , denoted as $\rho(x)$, is the least ordinal α such that $x \in \mathbf{V}_{\alpha^+}$.

2.2.18 LEMMA. $X \subset \mathbf{V}' \rightarrow X \in \mathbf{V}'$.

Proof. Since X is a set and $\rho(x)$ is defined for all $x \in X$, the axiom of replacement gives that $\{\rho(x) : x \in X\}$ is a set of ordinals. Thus, $\bigcup \{\rho(x) : x \in X\}$ is an ordinal, say β . Suppose that λ is a limit ordinal greater than β . Note that since λ is a limit and $\beta \in \lambda$, we also have $\beta^+ < \lambda$. Now, for any $x \in X$, we have

$$x \in \mathbf{V}_{\rho(x)^+} \subset \bigcup \{\mathbf{V}_\gamma : \gamma \in \lambda\} = \mathbf{V}_\lambda,$$

because $\rho(x) \leq \beta \rightarrow \rho(x)^+ < \beta^+ < \lambda$. So that $X \subset \mathbf{V}_\lambda$. But then the definition of $\mathbf{V}_{\lambda^+} = \mathcal{P}(\mathbf{V}_\lambda)$, we have $X \in \mathbf{V}_{\lambda^+}$. So that, $X \in \mathbf{V}'$. $\square$

2.2.19 THEOREM. $\mathbf{V} = \mathbf{V}'$.

Proof. We shall show that every set is an element of $\mathbf{V}_\alpha$ for some ordinal α . Let Y be any set. Note that if every element of Y is in $\mathbf{V}'$, then the lemma gives that Y is in $\mathbf{V}'$, as desired. Now, suppose that not every element of Y is in $\mathbf{V}'$. We shall show that this leads to a contradiction. First we form the set

$$T(Y) = \bigcup \{\cup^n Y : n \in \omega\} = Y \cup (\cup Y) \cup (\cup \cup Y) \cup \dots$$

$T(Y)$ is a transitive set: if $v \in x \in T(Y)$, then $x \in \cup^n Y$ and it follows that $y \in \cup^{n+1} Y$, so that $y \in T(Y)$. Let Y' be the complement $T(Y) - \mathbf{V}'$ (taking the complement of a class in a set doesn't legal, but all is well, we can express Y' as a set using the axiom of separation $Y' = \{y \in T(Y) : \neg \exists \alpha ((\alpha \text{ is an ordinal}) \wedge y \in \mathbf{V}_\alpha)\}$.)

Since $Y \subset T(Y)$ and there are elements of Y not in $\mathbf{V}'$, we have $Y' \neq \emptyset$. By the axiom of regularity Y' contains an $\in$ -minimal element y_0 , meaning that $y_0 \in Y'$ (so that also $y_0 \in T(Y)$) and $y_0 \cap Y' = \emptyset$.

Since $y_0 \in Y' \subset T(Y)$ and $T(Y)$ is $\in$ -transitive, we have that if $x \in y_0$ then $x \in T(Y)$, so that $y_0 \subset T(Y)$. Since $y_0 \cap Y' = \emptyset$ we thus have

$$y_0 \subset T(Y) - Y' = T(Y) \cap \mathbf{V}',$$

so that every element of y_0 is in $\mathbf{V}'$. Using the lemma above this gives that $y_0 \in \mathbf{V}'$, contradicting that $y_0 \in Y'$. This contradiction means that every element of a set Y was in $\mathbf{V}'$. $\square$

2.3 Axiom of Choice

AXIOM ZFC.9. Axiom of Choice. For every family (set) $\mathcal{F}$ of nonempty disjoint sets there exists a choice function. A choice function for the family is a function f on $\mathcal{F}$ such that $f(X) \in X$ for every $X \in \mathcal{F}$.

This axiom was proposed by Zermelo in 1908. Its formulation is so intuitive that Cantor, in his work, used it as an obvious fact. Nevertheless, this axiom is the most discussed and most controversial among mathematicians. It has also been proven that the Axiom of Choice is independent of the ZF axioms. Consequently, readers will find that in some works, authors accept the Axiom of Choice on the basis of ZF, while in others, they do not. Of course, most of the mathematical theories we learn in daily life—at least at the undergraduate level, whether in algebra, analysis, or geometry—mostly accept the Axiom of Choice. If readers have some understanding of functional analysis, they may already be familiar with the Axiom of Choice

2.3.1 REMARK. This is a non-constructive axiom; it merely asserts the existence of a choice function, without providing a specific definition. More precisely, the Axiom of Choice is primarily applicable in cases where a choice function cannot be explicitly constructed—we simply assert that one exists. If we consider a family of singleton sets, the choice function is unique, and the Axiom of Choice is completely unnecessary. If the family consists of non-singleton sets but a choice function can be explicitly constructed, then the Axiom of Choice is also unnecessary. For example, given infinitely many pairs of shoes, selecting one shoe from each pair does not require the Axiom of Choice, because we can directly construct a function—for instance, always choose the left shoe from each pair. However, if we have infinitely many pairs of socks and want to select one sock from each pair simultaneously, we cannot construct a choice function to make a global selection all at once. Attempting to do so would require infinitely many choices (choose one from the first pair, then extend the function to the second pair, then the third, ...), which is not permissible. Hence, the Axiom of Choice is needed.

2.3.1 THEOREM (WO: Well-Ordering Theorem). *Every set X can be well-ordered.*

Proof. Let C be a choice function on the family $P(X) - \{\emptyset\}$. Choose an arbitrary $p \notin X$, for example, $p = X$, and define $F : P(X \cup \{p\}) \rightarrow X \cup \{p\}$ by

$$F(Z) = \begin{cases} C(Z), & \text{for } Z \in P(X) - \{\emptyset\} \\ p, & \text{otherwise} \end{cases}$$

Let $\mathcal{G}$ be the family of all ordinal numbers ξ such that there is a one-to-one function from ξ to X . Notice that $\mathcal{G}$ is a set by the replacement axiom:

$$\mathcal{G} = \{\text{Otp}(Y) : Y \subset X, < \in P(X \times X), \langle Y, < \rangle \text{ is well-ordered}\}$$

Let α be an ordinal number that does not belong to $\mathcal{G}$. For example, take $\alpha = (\bigcup \mathcal{G})^+$. By the recursion theorem, there exists a function $f : \alpha \rightarrow X \cup \{p\}$ such that

$$\forall \xi < \alpha (f(\xi) = F(X - \{f(\zeta) : \zeta < \xi\}))$$

The meaning of this recursion is to select, step by step, the elements of X that have not yet been chosen. Once all elements have been selected, the first time the empty set is encountered, it outputs p , and all subsequent steps will output p as well. To achieve this goal (at least to reach the situation where the recursion encounters the empty set), α must be sufficiently large. Choosing α not belonging to $\mathcal{G}$ directly guarantees this requirement. Notice that for every $\beta \leq \alpha$, if $f[\beta] \subset X$, then $f|_\beta$ is one-to-one. This is the case, since $f(\xi) \in X - \{f(\zeta) : \zeta < \xi\}$ for $\xi < \beta$. In particular, $f[\alpha] \not\subset X$ since there is no one-to-one function from α to X . So the set $S = \{\beta \leq \alpha : f(\beta) = p\}$ is nonempty. Let $\beta = \min S$. Then $f|_\beta$ is one-to-one and it establishes a well ordering of X of type β by

$$x \leq y \leftrightarrow f^{-1}(x) \leq f^{-1}(y).$$

□

2.3.2 THEOREM. WO $\leftrightarrow$ AC.

Proof. Let $\mathcal{F}$ be an arbitrary family of nonempty sets. We need to construct a choice function for $\mathcal{F}$. Define $X = \bigcup \mathcal{F}$. X is a set by the axiom of union. By **WO**, there exists a well-ordering $<$ on X , that is, $\langle X, < \rangle$ is a well-ordered set. For each set $F \in \mathcal{F}$, since $F \neq \emptyset$ and $F \subset X$, the well ordering $<$ on X restricts to F , and F has a least element under $<$, denote this least element by $\min F$. Now, define the function $f : \mathcal{F} \rightarrow X$ by $f(F) = \min F$ for every $F \in \mathcal{F}$, this function f is a choice function for $\mathcal{F}$. □

For a paritally ordered set $\langle P, < \rangle$ we say that a set $S \subset P$ is a chain in P if S is linearly ordered by $<$. Let us recall that an element $b \in P$ is said to be an upper bound of a set $S \subset P$ if $s \leq b$ for every $s \in S$. Simiarly, $b \in P$ is a lower bound of $S \subset P$ if $b \leq s$ for every $s \in S$; $m \in P$ is a maximal element of $\langle P, < \rangle$ if there is no $p \in P$ such that $m < p$.

2.3.3 THEOREM (Zorn's Lemma). *If $\langle P, < \rangle$ is a partially ordered set such that every chain in P has an upper bound, then P has a maximal element m .*

Proof. The proof is similar to theorem before. Let C be a choice function on the family $\mathcal{P}(P) - \{\emptyset\}$. Choose an arbitrary $x \notin P$, for example, $x = P$, and define $F : \mathcal{P}(P \cup \{x\}) \rightarrow P \cup \{x\}$ by

$$F(Z) = \begin{cases} C(Z), & \text{for } Z \in \mathcal{P}(P) - \{\emptyset\} \\ x, & \text{otherwise} \end{cases}$$

Let $\mathcal{G}$ be the family of all ordinal number ξ such that there is a order-preserving $p : \xi \rightarrow P$. Let α be an ordinal number that does not belong to $\mathcal{G}$. For example, take $\alpha = (\bigcup \mathcal{G})^+$. Define a function h on the family of all function p from ξ to $P \cup \{x\}$, $\xi < \alpha$, by putting

$$h(p) = F(B - \{p(\zeta) : \zeta < \xi\}),$$

where $B = \{b \in P : b \text{ is an upper bound of } \{p(\zeta) : \zeta < \xi\}\}$. By the recursion theorem, there is a function f such that

$$\forall \xi < \alpha \quad f(\xi) = h(f|_\xi)$$

The meaning of this recursion is to step by step select, among all upper bounds of the currently obtained chain, those upper bounds that have not yet been selected, thereby forming a longer chain. Once all such upper bounds have been selected, the first time the empty set is encountered, it outputs x , and all subsequent steps will output x as well. To achieve this goal (at least to reach the situation where the recursion encounters the empty set), α must be sufficiently large. Choosing α not belonging to $\mathcal{G}$ guarantees this requirement: because then there exists no order-preserving function from α to P ; that is, the length of α is greater than the length of every possible chain in P . Notice that for every $\beta \leq \alpha$, if $f[\beta] \subset P$, then $f|_\beta$ is order-preserving. This is the case, since then, $f(\xi) \in B - \{p(\zeta) : \zeta < \xi\}$ for $\xi < \beta$. In particular, $f[\alpha] \not\subset P$ since there is no order-preserving from α to P . So the set $S = \{\beta \leq \alpha : f(\beta) = x\}$ is nonempty. Let $\beta = \min S$.

Then $f|_\beta$ is order-preserving, so $B = \{b \in P : b \text{ is an upper bound of } \{p(\xi) : \xi < \beta\}\}$ is nonempty, $\{p(\xi) : \xi < \beta\}$ being a chain. Let $m \in B$. It is enough to argue that m is a maximal element of P . But if there is a $y \in P$ with $y > m$, then $y > m > p(\xi)$ for every $\xi < \beta$. In particular, $y \in B - \{p(\xi) : \xi < \beta\}$, so $B - \{p(\xi) : \xi < \beta\} \neq \emptyset$. Thus, $f(\beta) \in P$, contradicting the choice of β . $\square$

2.3.4 COROLLARY (Hausdorff Maximal Principle). *If $\mathcal{F}$ is a nonempty family of subsets of a set A such that for every chain $S \subset \mathcal{F}$ its union $\bigcup S$ belongs to $\mathcal{F}$, then $\mathcal{F}$ has an $\subset$ -maximal element.*

Proof. Let $S \subset \mathcal{F}$ be a chain, i.e., for any $X, Y \in S$, either $X \subset Y$ or $Y \subset X$. By the hypothesis, $\bigcup S \in \mathcal{F}$. Since for every $X \in S$, we have $X \subset \bigcup S$. Hence $\bigcup S$ is upper bound of S in $(\mathcal{F}, \subset)$. Now use the Zorn's lemma, $\mathcal{F}$ contains a maximal element M with respect to $\subset$. $\square$

2.3.5 THEOREM. $\mathbf{HMP} \rightarrow \mathbf{AC}$. *It follows that $\mathbf{AC} \leftrightarrow \mathbf{WO} \leftrightarrow \mathbf{HMP}$.*

Proof. Let $\mathcal{G}$ be a nonempty family of pairwise disjoint nonempty sets. Let $\mathcal{F}$ be the family of all $T \subset \bigcup \mathcal{G}$ such that $T \cap G$ has at most one element for every $G \in \mathcal{G}$. Clearly, $\mathcal{F}$ is nonempty because $\emptyset \in \mathcal{F}$; and $(\mathcal{F}, \subset)$ is partially ordered. Now, we show that $\mathcal{F}$ satisfies the hypothesis of **HMP**: take any chain $S \subset \mathcal{F}$, let $U = \bigcup S$. We show that $U \in \mathcal{F}$. Suppose, for contradiction, that there exists some $G \in \mathcal{G}$ with $|U \cap G| \geq 2$. Then there exist distinct $x, y \in U \cap G$. Hence $x \in C_1, y \in C_2$ for some $C_1, C_2 \in S$. Since S is a chain, either $C_1 \subset C_2$ or $C_2 \subset C_1$. Without loss of generality, $C_1 \subset C_2$. Then both x and y belong to C_1 . But $C_2 \in \mathcal{F}$, so $|C_2 \cap G| \leq 1$, a contradiction. Therefore $|U \cap G| \leq 1$ for every $G \in \mathcal{G}$, i.e., $\bigcup S \in \mathcal{F}$. By the **HMP**, $\mathcal{F}$ has a $\subset$ -maximal element M , that is there is no $N \in \mathcal{F}$ such that $M \subsetneq N$.

We now show that M is a selector for $\mathcal{G}$, that is $|M \cap G| = 1$ for every $G \in \mathcal{G}$: assume that there exist some $G_0 \in \mathcal{G}$ with $G_0 \cap M = \emptyset$. Since G_0 is nonempty, pick an element $g \in G_0$. Consider $M' = M \cup \{g\}$, then $|M' \cap G_0| = 1$. For any other $G \neq G_0$, $|M' \cap G| = |M \cap G| \leq 1$ since $g \notin G$. Hence $M' \in \mathcal{F}$. But $M \subset M'$, contradicting the maximality of M . Therefore $|M \cap G| = 1$ for every $G \in \mathcal{G}$. For each $G \in \mathcal{G}$, let $f(G)$ be the unique element in $M \cap G$. Then f is the choice function for $\mathcal{G}$. $\square$

A subset S of a vector space V over a field K is linearly independent if for every finite number of distinct elements $v_1, \dots, v_n$ of S and every $k_1, \dots, k_n \in K$, the condition $k_1 v_1 + \dots + k_n v_n = 0$ implies $k_1 = \dots = k_n = 0$. A subset S of a vector space V over a field K spans V if every $v \in V$ can be represented as $v = k_1 v_1 + \dots + k_n v_n$ for some $v_1, \dots, v_n \in S$ and $k_1, \dots, k_n \in K$. A basis of V is a linearly independent subset of V that spans V .

2.3.6 THEOREM. *If S_0 is a linearly independent subset of a vector space V over K then there exists a basis $\mathcal{B}$ of V that contains S_0 . In particular, every vector space has a basis.*

Proof. $\square$

2.4 Cardinality

Cardinal Numbers

Let X and Y be two sets. If there is a bijection between them, we write $X \approx Y$. It is easy to see that for every X, Y, Z : $X \approx X$; if $X \approx Y$, then $Y \approx X$; and if $X \approx Y$ and $Y \approx Z$, then $X \approx Z$. Thus, the “relation” of having the same cardinality is an “equivalence relation”: note that the “domain” of this “relation” is the class of sets. However, the

restriction of $\approx$ to any set X is an equivalence relation on X . Thus, we will use the term relation for $\approx$ in this sense.

By **WO** and the order type theorem, for every set A there exists an ordinal number α such that $A \approx \alpha$. The smallest (Although a well-ordered set is isomorphic to only one ordinal number, the Well-Ordering Theorem does not state that a set has only one well-ordering.) ordinal with this property is called the **cardinality** of A and is denoted by $|A|$, that is,

$$|A| = \min\{\alpha : \alpha \text{ is an ordinal and } A \approx \alpha\}.$$

In particular, $A \approx |A|$ for every A . And it is clear that $A \approx B$ if and only if $|A| = |B|$. Thus, we say that “there exists a bijection between A and B , i.e., $A \approx B$ ” means *A and B have the same cardinality, i.e., $|A| = |B|$* . Since every set can use an ordinal to represent its cardinality (informally, the size of the set), we can give these special ordinal numbers a special name — cardinal numbers.

2.4.1 DEFINITION. An ordinal number κ is said to be a *cardinal number* (or just a cardinal) if $\kappa = |A|$ for some set A .

Not all ordinals are cardinals (as will be seen in later examples). The proposition below provides a basic criterion for distinguishing ordinals from cardinals. Since $\alpha \approx \alpha$ and $|\alpha|$ is the smallest ordinal β such that $\alpha \approx \beta$, $|\alpha| \leq \alpha$ for every ordinal number α . Also, this implies that $|\alpha| \leq |\beta|$ for all ordinals $\alpha \leq \beta$.

2.4.1 PROPOSITION. For an ordinal number α , the following are equivalent:

1. α is a cardinal number;
2. $|\alpha| = \alpha$
3. $\forall \beta < \alpha (\beta < |\alpha|)$;
4. $\forall \beta < \alpha (|\beta| < |\alpha|)$;
5. $\forall \beta < \alpha (|\beta| \neq |\alpha|)$.

Proof. (1) $\rightarrow$ (2): if α is a cardinal, then there exists a set A such that $|A| = \alpha$. This implies that $A \approx \alpha$, that is, $|A| = |\alpha|$. So $|\alpha| = |A| = \alpha$. (2) $\rightarrow$ (3) is obvious. (3) $\rightarrow$ (4): since $|\beta| \leq \beta$ for every ordinal β . (4) $\rightarrow$ (5) is obvious. (5) $\rightarrow$ (6): (5) implies that $\beta \approx \alpha$ for every $\beta < \alpha$. Thus, α is the smallest ordinal such that $\alpha \cong \alpha$, which means $\alpha = |\alpha|$. $\square$

2.4.1 REMARK. Property (5) of this proposition explains why we often say that the cardinal numbers are the initial ordinals, that is, ordinals are the smallest ordinal of a given cardinality.

2.4.2 THEOREM. Let A and B be arbitrary sets. The following are equivalent:

1. $|A| \leq |B|$;
2. there exists a one-to-one function $g : A \rightarrow B$;
3. there exists a function $f : B \rightarrow A$ from B onto A if $A \neq \emptyset$.

Proof. If $A = \emptyset$, the equivalence of (1) and (2) is obvious. So we assume that $A \neq \emptyset$. (1) $\rightarrow$ (3): let $h : B \rightarrow |B|$ and $g : |A| \rightarrow A$ be bijections. Pick $a \in A$. Notice that $|A| \subset |B|$ and $h^{-1}(|A|) \subset B$. Define a function $f : B \rightarrow A$ by

$$f(b) = \begin{cases} g(h(b)), & b \in h^{-1}(|A|), \\ a, & \text{otherwise} \end{cases}$$

Then $A \supset f[B] \supset f[h^{-1}(|A|)] = (g \circ h)[h^{-1}(|A|)] = g[|A|] = A$. So f is onto A . For (3) $\rightarrow$ (2), let $f : B \rightarrow A$ be onto. Let G be a choice function for the family $\{f^{-1}(a) : a \in A\}$. Then $g(a) = G(f^{-1}(a))$ is a one-to-one function from A into B .

For (2) $\rightarrow$ (1): let $g : A \rightarrow B$ be a one-to-one function and $h : B \rightarrow |B|$ be a bijection. Then $(h \circ g) : A \rightarrow |B|$ is one-to-one function from A onto $(h \circ g)[A] \subset |B|$. Hence $|A| = |(h \circ g)[A]|$ and $\text{Otp}((h \circ g)[A]) \leq \text{Otp}(|B|) = |B|$. So

$$|A| = |(h \circ g)[A]| = |\text{Otp}((h \circ g)[A])| \leq \text{Otp}((h \circ g)[A]) \leq |B|.$$

□

2.4.3 COROLLARY. *If $A \subset B$, then $|A| \leq |B|$.*

2.4.4 COROLLARY. *If $A \subset B \subset C$ and $|A| = |C|$, then $|A| = |B| = |C|$.*

2.4.5 THEOREM. *ω is a cardinal; and $\forall n \in \omega$, n is a cardinal.*

Proof. First we show that every $n \in \omega$ is a cardinal number. So, let $n \in \omega$. By the proposition above, it suffices to show that $|k| < |n|$ for every $k < n$. Thus, it suffices to show that the inequality $|n| \leq |k|$ is false, that is, by theorem above, that

there is no one-to-one function $f : n \rightarrow k$.

We prove this by induction on $n \in \omega$. Notice that for $n = 0$, there is nothing to prove. So assume that it is true for some $n < \omega$ and show that this implies for $n + 1$. Suppose, for contradiction, for some $k < n + 1$ there is a one-to-one function $f : n + 1 \rightarrow k$. If $k < n$, then $f|_n : n \rightarrow k$ is one-to-one, contradicting our hypothesis. So $k = n$ and $f : n + 1 \rightarrow n$ is one-to-one and $f[n + 1] \not\subseteq (n - 1)$, that is, $n - 1 \in f[n + 1]$. ($n - 1$ exists, since the existence of $f : n + 1 \rightarrow n$ implies that $n \neq 0$.) Let $i < n + 1$ be such that $f(i) = n - 1$. Define $g : n \rightarrow n - 1$ by putting $g(j) = f(j)$ for $j < n, j \neq i$ and $g(i) = f(n)$. (if $i = n$ this last condition is redundant and can simply be ignored.) Notice that the values of g indeed belong to $n - 1$ and that g is one-to-one, which is a contradiction.

To see that ω is a cardinal number notice that $n \subset n + 1 \subset \omega$ for every $n < \omega$. Hence, by corollary, $|n| \leq |n + 1| \leq |\omega|$. But by what we have already proved, $|n| = n < n + 1 = |n + 1|$. So $|n| \leq |n + 1| \leq |\omega|$, that is, $|n| < |\omega|$ for every $n < \omega$. Therefore, ω is a cardinal number. □

2.4.6 THEOREM (Cantor's Theorem). *$|X| < |P(X)|$ for every set X .*

Proof. Fix a set X . First, notice that $|X| \leq |P(X)|$ since the function $f : X \rightarrow P(X)$ defined by $f(x) = \{x\}$ for $x \in X$ is one-to-one. Now, we show that $|X| \neq |P(X)|$. So let $f : X \rightarrow P(X)$. We show that f is not a bijection. To see it, put

$$Y = \{x \in X : x \notin f(x)\}.$$

Then $Y \in P(X)$ is not in the range of f , since if there were a $z \in X$ such that $f(z) = Y$ then we would have $z \in Y \leftrightarrow z \in \{x \in X : x \notin f(x)\} \leftrightarrow z \notin f(z) \leftrightarrow z \notin Y$. □

2.4.2 REMARK. Cantor's theorem tell us that for any cardinal number κ there is a cardinal number λ larger than κ , namely, $\lambda = |P(\kappa)|$. We will denote this cardinal by 2^κ .

According to our original notation, 2^κ is the family of all functions from κ into the set $2 = \{0, 1\}$. However, this does not cause any ambiguity. Specifically, $|2^X| = |P(X)|$:

Consider a bijection $\chi : P(X) \rightarrow 2^X$ defined by $\chi(A) = \chi_A$, where χ_A is the characteristic function of A , that is, $\chi_A(x) = 1$ for $x \in A$ and $\chi_A(x) = 0$ for $x \in X - A$. So in this case, $2^\kappa = |P(\kappa)| = |2^\kappa|$. The reader needs to distinguish the meaning of this notation.

2.4.3 REMARK. Notice that for every cardinal number κ there is the smallest cardinal number λ greater than κ , namely, a cardinal equal to the minimum of the set

$$K = \{\mu \in 2^{2^\kappa} : \mu \text{ is a cardinal and } \kappa < \mu\}.$$

The minimum exists, since K is a nonempty ($2^\kappa \in K$) subset of a well-ordered set 2^{2^κ} . This unique cardinal number is denoted by κ^+ and is called the *cardinal successor* of κ . Starting from the cardinal ω , we then obtain the following cardinals:

$$\omega_0 (= \omega), \omega_1 (= \omega_0^+), \omega_2 (= \omega_1^+), \dots$$

2.4.7 PROPOSITION. *If $\mathcal{F}$ is a family of cardinals, then $\alpha = \bigcup \mathcal{F}$ is also a cardinal.*

Proof. We have seen that $\alpha = \bigcup \mathcal{F}$ is an ordinal. Moreover, if $\beta < \alpha$, then $\beta \in \alpha$ and there exists a $\kappa \in \mathcal{F}$ such that $\beta \in \kappa$. So $\beta < \kappa = |\kappa| \leq |\alpha|$. Thus, α is a cardinal. $\square$

2.4.4 REMARK. By the replacement axiom, $\omega_\omega = \bigcup \{\omega_n : n \in \omega\}$ is a set, and by the proposition, it is also a cardinal. The previous remark constructs new cardinals using natural numbers. More generally, we can perform such an operation not only for natural numbers but also for any ordinal.

1. $\omega_0 = \omega$;
2. $\omega_{\xi+1} = \omega_\xi^+$ for every $\xi < \alpha$;
3. $\omega_\lambda = \bigcup_{\xi < \lambda} \omega_\xi$ for every limit $\lambda \leq \alpha$.

2.4.8 THEOREM. *Every cardinal larger than ω is necessarily of the form ω_α .*

Proof. Let κ be any cardinal larger than ω . Define a set $B = \{\beta : \beta \leq \kappa \leq \omega_\beta\}$. One may check that $\kappa \leq \omega_\kappa$, hence $\kappa \in B$; thus $B \neq \emptyset$. Let α be the smallest ordinal in B . We will prove that $\kappa = \omega_\alpha$.

Since $\alpha \in B$, we have $\kappa \leq \omega_\alpha$. It suffices to show that $\kappa \not< \omega_\alpha$. Assume for contradiction that $\kappa < \omega_\alpha$. We derive a contradiction by induction on α :

If $\alpha = 0$, then $\kappa \leq \omega_0$, which contradicts the hypothesis that $\kappa > \omega$. If $\alpha = \beta + 1$, then $\beta < \alpha \leq \kappa < \omega_{\beta+1}$, it follows that $\kappa \leq \omega_\beta$ (since $\omega_{\beta+1}$ is the smallest cardinal larger than ω_β .) That means $\beta \in B$, which contradicts the minimality of α . If α is a limit ordinal, by the definition of ω_α , $\kappa < \omega_\alpha$ implies there exists some $\gamma < \alpha$ such that $\kappa < \omega_\gamma$. Then $\gamma \in B$ again contradicting the minimality of α . $\square$

2.4.2 DEFINITION. According to the above theorem, we can give an equivalent definition of cardinal numbers. A set X is called a cardinal number if $X \in \omega$ or $X = \omega_\alpha$ for some ordinal number α . And we define $\aleph_\alpha$ to be the cardinal number of ω_α for all α .

2.4.3 DEFINITION. The sets with cardinality less than ω , that is, equal to some $n < \omega$, are called *finite sets*. A set A is *countable* if $|A| = \omega = \aleph_0$. A set is *infinite*, that is, $\aleph_0 \leq |A|$; if it is not finite. A set is *uncountable* if it is infinite and not countable, that is, $|A| = \aleph_\alpha$ for some ordinal $\alpha \neq 0$.

2.4.5 REMARK. Denote the universe of all cardinal numbers by **CN**. If **CN** were a set, then by the proposition, $\bigcup \mathbf{CN}$ would also be a cardinal, and it would be the largest cardinal. But we know, that a largest cardinal does not exist by the Cantor's theorem. This is the *Cantor's paradox*. The source of the paradox is the assumption that **CN** is a set. We do not see **CN** a set in the axiomatic set theory, it is a proper class.

2.4.9 THEOREM (CC: Cardinal Comparability). **AC** $\rightarrow$ **CC**, namely, if X, Y are sets, then $|X| \leq |Y|$ or $|Y| \leq |X|$.

Proof. By the linearity of the usual ordering on **ON**, either $|X| \leq |Y|$ or $|Y| \leq |X|$. $\square$

2.4.10 LEMMA (Hartogs' Theorem). *Let X be a set. Then there is an ordinal α such that $|\alpha| \not\leq |X|$, i.e., it is not the case that $|\alpha| \leq |X|$.*

Proof. $\square$

2.4.11 THEOREM. **CC** $\rightarrow$ **WO**. So that, $\mathbf{AC} \leftrightarrow \mathbf{WO} \leftrightarrow \mathbf{CC}$.

Proof. Assume **CC**. Let X be any set. Then by Hartogs' theorem there is an ordinal α such that $|\alpha| \not\leq |X|$. By **CC**, we have that either $|\alpha| \leq |X|$ or $|X| \leq |\alpha|$, so we must have $|X| \leq |\alpha|$. This means that there is a one-to-one function f from X into α . Define $<$ on X by setting $x < y$ if and only if $f(x) < f(y)$. This is a well-ordering on X . $\square$

Arithmetic on Cardinal Numbers

2.4.12 LEMMA. *If $|A| = |B|$ and $|C| = |D|$, then*

1. $|A \times C| = |B \times D|$;
2. $|A \cup C| = |B \cup D|$ if $A \cap C = B \cap D = \emptyset$;
3. $|A^C| = |B^D|$.

Proof. Let $f : A \rightarrow B$ and $g : C \rightarrow D$ be bijections. Define the function $F_1 : A \times C \rightarrow B \times D$ by $F_1(a, c) = (f(a), g(c))$, it is clear that F_1 is a bijection. Define the function $F_2 : A \cup C \rightarrow B \cup D$ by $F_2 = f \cup g$ and $F_3 : A^C \rightarrow B^D$ by $F_3(h)(d) = f(h(g^{-1}(d)))$ for every $d \in D$. Also, they are bijections. $\square$

2.4.4 DEFINITION. Let κ, λ be two cardinal numbers and we define the cardinal sum, cardinal product and cardinal exponentiation as follows

$$\begin{aligned}\kappa \oplus \lambda &= |(\kappa \times \{0\}) \cup (\lambda \times \{1\})| \\ \kappa \otimes \lambda &= |\kappa \times \lambda| \\ \kappa^\lambda &= |\kappa^\lambda|\end{aligned}$$

2.4.6 REMARK. Perhaps the reader can accept the definitions of cardinal multiplication and exponentiation, but many have some doubts about cardinal addition. However, a brief thinking will reveal why it is defined in that way: because this approach forces the two sets to be made disjoint (thereby ensuring that for any two sets A, B with $|A| = \kappa$ and $|B| = \lambda$, they are necessarily made disjoint, so the definition is well-defined and independent of the choice of A and B .) This is a very classic method in set theory and is quite common in point-set topology.

2.4.13 PROPOSITION. *Let κ, λ, μ be cardinals, then*

1. $\kappa \oplus \lambda = \lambda \oplus \kappa$;
2. $\kappa \oplus (\lambda \oplus \mu) = (\kappa \oplus \lambda) \oplus \mu$;
3. $\kappa \otimes \lambda = \lambda \otimes \kappa$;
4. $\kappa \otimes (\lambda \otimes \mu) = (\kappa \otimes \lambda) \otimes \mu$;
5. $\kappa \otimes (\lambda \oplus \mu) = \kappa \otimes \lambda \oplus \kappa \otimes \mu$.

2.4.14 PROPOSITION. *If m and n are natural numbers, then addition and multiplication in the ordinal sense and in the cardinal number sense are the same.*

Proof. We denote ordinal addition by $+$, and cardinal addition by $\oplus$. With $m \in \omega$ fixed we prove that $m + n = m \oplus n$ by induction on n . The case $n = 0$ is clear. Now we suppose that $m + n = m \oplus n$. Then $m + (n + 1) = (m + n) + 1 = (m + n) \cup \{m + n\}$. On the other hand,

$$\begin{aligned}m \oplus (n + 1) &= |\{(i, 0) : i \in m\} \cup \{(i, 1) : i \in n^+\}| \\ &= |\{(i, 0) : i \in m\} \cup \{(i, 1) : i \in n \cup \{n\}\}| \\ &= |\{(i, 0) : i \in m\} \cup \{(i, 1) : i \in n\} \cup \{(n, 1)\}| \\ &= |\{(i, 0) : i \in m\} \cup \{(i, 1) : i \in n\}| \oplus 1 \\ &= (m \oplus n) \oplus 1 = (m + n) \oplus 1 \\ &= (m + n) + 1 = m + (n + 1).\end{aligned}$$

For multiplication, we denote ordinal multiplication by $\cdot$, and cardinal multiplication by $\otimes$. With $m \in \omega$ fixed we prove that $m \cdot n = m \otimes n$ by induction on n . The case $n = 0$ is clear. Now we suppose that $m \cdot n = m \otimes n$. Then

$$\begin{aligned} m \cdot (n + 1) &= m \cdot n + m = m \otimes n + m \\ &= m \otimes n + m \otimes 1 \\ &= m \otimes (n \oplus 1) = m \otimes (n + 1). \end{aligned}$$

□

2.4.15 LEMMA. *Every infinite cardinal (i.e., ω_α) is a limit ordinal.*

Proof. First, notice that $|\omega \cup \{x\}| = |\omega|$. If $x \in \omega$, then this is obvious. Otherwise, this is justified by a function $f : \omega \cup \{x\} \rightarrow \omega$ defined by $f(x) = 0$ and $f(n) = n + 1$ for $n < \omega$. Hence, by the lemma, for every infinite ordinal α , $|\alpha + 1| = |\alpha \cup \{\alpha\}| = |(\alpha - \omega) \cup (\omega \cup \{\alpha\})| = |(\alpha - \omega) \cup \omega| = |\alpha|$. Therefore, $\alpha + 1$ is not a cardinal number for any infinite ordinal α . □

2.4.16 THEOREM (Hessenberg). *If κ is an infinite cardinal, then $\kappa \otimes \kappa = \kappa$.*

Proof. Let κ be an infinite cardinal. We show that by transfinite induction on $\omega \leq \alpha \leq \kappa$ that $|\alpha| \otimes |\alpha| = |\alpha|$. So let $\omega \leq \lambda \leq \kappa$ such that $|\alpha| \otimes |\alpha| = |\alpha|$ for every $\omega \leq \alpha < \lambda$. We show that $|\lambda| \otimes |\lambda| = |\lambda|$:

if λ is not a cardinal, that is, there is an $\alpha < \lambda$ such that $|\alpha| = |\lambda|$ then $|\lambda| \otimes |\lambda| = |\alpha| \otimes |\alpha| = |\alpha| = |\lambda|$. Hence we may assume that λ is a cardinal number.

Now notice that for every $\alpha < \lambda$, $|\alpha \times \alpha| < \lambda$: If α is finite, clearly $|\alpha \times \alpha| = \alpha \otimes \alpha = \alpha \cdot \alpha < \omega < \lambda$; if α is an infinite ordinal, then $|\alpha \times \alpha| = |\alpha| \otimes |\alpha| = |\alpha|$ by our hypothesis. Define a well ordering $\prec$ (and $\alpha \preceq \beta = \alpha \prec \beta \vee \alpha = \beta$) on $\lambda \times \lambda$ by putting $(\alpha, \beta) \prec (\gamma, \delta)$ if and only if

$$\begin{aligned} &(\max\{\alpha, \beta\} < \max\{\gamma, \delta\}) \vee \\ &(\max\{\alpha, \beta\} = \max\{\gamma, \delta\} \wedge \alpha < \gamma) \vee \\ &(\max\{\alpha, \beta\} = \max\{\gamma, \delta\} \wedge \alpha = \gamma \wedge \beta < \delta) \end{aligned}$$

It is easy to check that $\prec$ is a well-ordering. Now, let $(\gamma, \delta) \in \lambda \times \lambda$ and let $\epsilon = \max\{\gamma + 1, \delta + 1\}$. Since λ is a limit ordinal by the lemma, $\gamma < \lambda$ implies $\gamma + 1 < \lambda$, which follows $\epsilon < \lambda$. Moreover, the initial segment $\text{seg}_{\lambda \times \lambda}(\gamma, \delta)$ is a subset of $\epsilon \times \epsilon$, since $(\alpha, \beta) \prec (\gamma, \delta)$ implies $\max\{\alpha, \beta\} \leq \max\{\gamma, \delta\} < \epsilon$. Hence

$$|\text{Otp}(\text{seg}_{\lambda \times \lambda}(\gamma, \delta))| = |\text{seg}_{\lambda \times \lambda}(\gamma, \delta)| \leq |\epsilon \times \epsilon| = |\epsilon| < \lambda.$$

and so $\text{Otp}(\text{seg}_{\lambda \times \lambda}(\gamma, \delta)) < \lambda$, otherwise, we would have $\lambda = |\lambda| \leq |\text{Otp}(\text{seg}_{\lambda \times \lambda}(\gamma, \delta))|$. Since every ordinal number $\beta \in \text{Otp}(\lambda \times \lambda, \prec)$ is of the form $\text{Otp}(\text{seg}_{\lambda \times \lambda}(\gamma, \delta))$ we can conclude that $\text{Otp}(\lambda \times \lambda, \prec) \leq \lambda$. So

$$|\lambda \times \lambda| = |\text{Otp}(\lambda \times \lambda, \prec)| \leq \text{Otp}(\lambda \times \lambda, \prec) \leq \lambda = |\lambda|.$$

Thus, $|\lambda \times \lambda| = |\lambda|$. □

2.4.17 COROLLARY. *If λ, κ are infinite cardinals then $\kappa \oplus \lambda = \kappa \otimes \lambda = \max\{\kappa, \lambda\}$.*

Proof. We assume that $\lambda \leq \kappa$, then

$$\kappa \oplus \lambda = |(\kappa \times \{0\}) \cup (\lambda \times \{1\})| \leq |\kappa \times \{0, 1\}| \leq |\kappa \times \lambda| = \kappa \otimes \lambda.$$

and

$$\kappa \otimes \lambda = |\kappa \times \lambda| \leq |\kappa \times \kappa| = \kappa.$$

Since evidently $\kappa \leq \kappa \oplus \lambda$ we obtain $\kappa \leq \kappa \oplus \lambda \leq \kappa \otimes \lambda \leq \kappa$. □

2.4.18 COROLLARY. *A countable union of countable sets is countable.*

Proof. Let $\mathcal{F}$ be a countable family of countable sets, $\mathcal{F} = \{F_n : n \in \omega\}$. Then for every $n < \omega$ there is a bijection $f_n : \omega \rightarrow F_n$. Define function $f : \omega \times \omega \rightarrow \bigcup \mathcal{F}$ by putting $f(n, m) = f_n(m)$. Notice that f is onto. So, $\omega = |F_n| \leq |\bigcup \mathcal{F}| \leq |\omega \times \omega| = \omega$. $\square$

2.4.19 COROLLARY. *If κ is an infinite cardinal and $|X_\alpha| \leq \kappa$ for all $\alpha < \kappa$, then $|\bigcup_{\alpha < \kappa} X_\alpha| \leq \kappa$.*

Proof. For every $\alpha < \kappa$, there exists a surjection $g_\alpha : \kappa \rightarrow X_\alpha$ since $|X_\alpha| \leq \kappa$. Now we define a function $f : \kappa \times \kappa \rightarrow \bigcup_{\alpha < \kappa} X_\alpha$ by $f(\alpha, \beta) = g_\alpha(\beta)$. It is clear that f is onto. So, $|\bigcup_{\alpha < \kappa} X_\alpha| \leq |\kappa \times \kappa| = \kappa$. $\square$

2.4.20 COROLLARY. $|\mathbf{Q}| = |\mathbf{Z}| = |\mathbf{N}| = \omega$.

Proof. It is clear that $\omega = |\mathbf{N}| \leq |\mathbf{Z}| \leq |\mathbf{Q}|$. To see that $|\mathbf{Q}| \leq \omega$, it suffices to show that $|\mathbf{Q}| \leq |2 \times \omega \times \omega|$, since $|2 \times \omega \times \omega| = 2 \otimes \omega \otimes \omega = \omega$. So let $f : 2 \times \omega \times \omega \rightarrow \mathbf{Q}$ be defined by $f(i, m, n) = (-1)^i \frac{m}{n+1}$. It is easy to see that f is onto $\mathbf{Q}$. $\square$

2.4.21 THEOREM. $|\mathbf{R}| = |P(\mathbf{N})| = 2^\omega$.

Proof. The first inequality $|2^\omega| \leq |\mathbf{R}|$ is established by a one-to-one function $f : 2^\omega \rightarrow \mathbf{R}$ defined by (where $\forall n \in \omega (a_n = 0 \vee a_n = 1)$)

$$f(a_0, a_1, \dots) = \sum_{n < \omega} \frac{2a_n}{3^{n+1}}.$$

To see that $|\mathbf{R}| \leq |2^\omega|$ define a one-to-one function $g : \mathbf{R} \rightarrow P(\mathbf{Q})$ by $g(r) = \{q \in \mathbf{Q} : q < r\}$, i.e., $g(r)$ is the Dedekind cut associated with r . So $|\mathbf{R}| \leq |P(\mathbf{Q})| = |2^{\mathbf{Q}}| = 2^{|\mathbf{Q}|} = 2^\omega$. Therefore, $|\mathbf{R}| = 2^\omega$. $\square$

2.4.22 THEOREM. *If λ and κ are cardinal numbers such that $\omega \leq \lambda$ and $2 \leq \kappa \leq \lambda$, then $\kappa^\lambda = 2^\lambda$. In particular, $\lambda^\lambda = 2^\lambda$ for every infinite cardinal number λ .*

Proof. Notice that $2^\lambda \subset \kappa^\lambda \subset \lambda^\lambda \subset P(\lambda \times \lambda)$. Hence $2^\lambda \leq \kappa^\lambda \leq \lambda^\lambda \leq |P(\lambda \times \lambda)| = |2^{\lambda \times \lambda}| = 2^{|\lambda \times \lambda|} = 2^\lambda$. $\square$

The exponentiation of cardinal is very “mysterious”. Although we know that every cardinal greater than ω is of the form ω_α , within the **ZFC** axiom system we cannot determine which $\aleph_\gamma$ the expression $\aleph_\alpha^{\aleph_\beta}$ actually equals. We cannot even determine which $\aleph_\alpha$ equals $2^{\aleph_0}$. Apart from a very few facts (such as the few theorems mentioned above), there are almost no effective conclusions.

2.4.7 REMARK (CH: Continuum Hypothesis). $2^{\aleph_0} = \aleph_1$. We often denote $2^{\aleph_0}$ by $\mathfrak{c}$.

As for why this assumption can be made, Gödel proved that CH is consistent with ZFC—that is, ZFC cannot refute CH. Cohen also proved that $\neg$ CH is consistent with ZFC—that is, ZFC cannot prove CH either.

2.4.8 REMARK (GCH: Generalized Continuum Hypothesis). $2^{\aleph_\alpha} = \aleph_{\alpha+1}$.

Cofinality

As mentioned in the previous section, cardinal exponentiation is quite mysterious. We introduce a new concept, cofinality, to study cardinal exponentiation. We say a subset A of an ordinal number α is *unbounded* in α if A has no strict upper bound in α , that is, when there is no $\gamma \in \alpha$ with $\xi < \gamma$ for all $\xi \in A$, namely,

$$\forall \gamma \in \alpha \exists \xi \in A (\gamma \leq \xi).$$

Intuitively, every element of α can be surpassed by some element in A . For any original number α , let $\text{cf}(\alpha)$ be the smallest ordinal number β such that there exists a function $f : \beta \rightarrow \alpha$ with the property that $f[\beta]$ is unbounded in α , that is

$$\forall \gamma \in \alpha \exists \xi \in \beta (\gamma \leq f(\xi)).$$

Such a function f is called a **cofinal map** in α . The number $\text{cf}(\alpha)$ is called the **cofinality** of α , that is, $\text{cf}(\alpha) = \min\{\beta : \text{there is a cofinal map } f : \beta \rightarrow \alpha\}$. Intuitively, cofinality is a way of measuring how hard an ordinal is to approach from below.

According to the intuitive understanding, the cofinality of an ordinal is the minimum number of “steps” needed, starting from 0, to approach its top infinitely. Thus, it is easy to see why the cofinality of a successor ordinal $\beta + 1$ is 1: we only need to take one step, to β , and we have reached the maximum. For limit ordinals, however, the situation is much more mysterious. Consider ω : as a set, ω has no greatest element. No matter how large the n we take in each step, there remain infinitely many elements still untouched. Intuitively, we need to take infinitely many steps so that no matter how large an element in ω , we can eventually reach and surpass it. Indeed, it can be proved that $\text{cf}(\omega) = \omega$.

2.4.1 EXAMPLE. If α is a successor ordinal, i.e., $\alpha = \beta + 1$, then $\text{cf}(\alpha) = 1$ since $f : 1 \rightarrow \beta + 1$, $f(0) = \alpha$ is the cofinal map in α . It is also easy to see that if $\text{cf}(\alpha) = 1$, then $\forall \gamma \in \alpha (\gamma \leq f(0))$, i.e., α has a greatest element. Thus, $\text{cf}(\alpha) = 1$ if and only if α is a successor ordinal. Moreover, we show that $\text{cf}(\alpha) < \omega$ if and only if $\text{cf}(\alpha) = 1$: let $\text{cf}(\alpha) = n < \omega$, then there exist a cofinal map $f : n \rightarrow \alpha$, $\forall \xi < \alpha \exists m < n (\xi \leq f(m)) < \max\{f(i) : i < n\}$, which means α has the greatest element, so α is a successor ordinal. In conclusion, $\text{cf}(\alpha) = 1$ if and only if $\text{cf}(\alpha) < \omega$ if and only if α is a successor ordinal. Hence, the cofinality of every successor ordinal is 1, and cofinality is only interesting for limit ordinals.

2.4.23 LEMMA. *For every ordinal number α there is an strictly increasing (order-preserving) cofinal map $f : \text{cf}(\alpha) \rightarrow \alpha$.*

Proof. For a successor ordinal, the cofinal map $f : \text{cf}(\alpha) = 1 \rightarrow \alpha$ defined by $f(0) = \alpha - 1$ is a order-preserving. If α is a limit ordinal and $g : \text{cf}(\alpha) \rightarrow \alpha$ is a cofinal map. Define by transfinite induction on $\eta < \text{cf}(\alpha)$

$$f(\eta) = \max\{g(\eta), \bigcup_{\xi < \eta} (f(\xi) + 1)\}$$

Clearly $f(\xi) < f(\xi) + 1 \leq f(\eta)$ for every $\xi < \eta < \text{cf}(\alpha)$. Thus, f is order-preserving. Now, we show that f is a cofinal map. The fact that

$$\forall \eta < \text{cf}(\alpha) (f(\eta) \in \alpha)$$

is proved by induction on $\eta < \text{cf}(\alpha)$. If for some $\eta < \text{cf}(\alpha)$ this condition holds for all $\xi < \eta$ then $f(\eta) = \max\{g(\eta), \bigcup_{\xi < \eta} (f(\xi) + 1)\} \leq \alpha$. But $f(\eta)$ can not be equal to α since otherwise the restriction of f to η would be a cofinal map in α and this would contradicting the minimality of $\text{cf}(\alpha)$. Thus, $\eta < \text{cf}(\alpha)$. Finally, f is cofinal in α since for every $\xi \in \alpha$, there is an $\eta < \text{cf}(\alpha)$ with $\xi \leq g(\eta) \leq f(\eta)$. $\square$

2.4.24 COROLLARY. $\text{cf}(\text{cf}(\alpha)) = \text{cf}(\alpha)$ for every ordinal α .

Proof. Let $f : \text{cf}(\text{cf}(\alpha)) \rightarrow \text{cf}(\alpha)$ and $g : \text{cf}(\alpha) \rightarrow \alpha$ be order-preserving cofinal maps. Then $g \circ f : \text{cf}(\text{cf}(\alpha)) \rightarrow \alpha$ is also cofinal, since for every $\xi < \alpha$, there exist $\eta \in \text{cf}(\alpha)$ with $\xi \leq g(\eta)$ and $\zeta \in \text{cf}(\text{cf}(\alpha))$ with $\eta \leq f(\zeta)$, so

$$\xi \leq g(\eta) \leq g(f(\zeta)) = (g \circ f)(\zeta).$$

Thus, by the definition of $\text{cf}(\alpha)$, $\text{cf}(\alpha) \leq \text{cf}(\text{cf}(\alpha))$. And the inequality $\text{cf}(\text{cf}(\alpha)) \leq \text{cf}(\alpha)$ holds because $\text{cf}(\alpha) \leq \alpha$ for every ordinal α . So $\text{cf}(\text{cf}(\alpha)) = \text{cf}(\alpha)$. $\square$

2.4.25 THEOREM. *If $\text{cf}(\alpha) = \alpha$, then α is a cardinal. In particular, $\text{cf}(\alpha)$ is a cardinal.*

Proof. Put $\beta = |\text{cf}(\alpha)|$ and choose a bijection $f : \beta \rightarrow \text{cf}(\alpha)$ and a cofinal function $g : \text{cf}(\alpha) \rightarrow \alpha$. Then $(g \circ f) : \beta \rightarrow \alpha$ is cofinal in α , since $(g \circ f)[\beta] = g[f[\beta]] = g[\text{cf}(\alpha)]$. Hence, by the definition of $\text{cf}(\alpha)$, $\text{cf}(\alpha) \leq \beta = |\text{cf}(\alpha)|$. $\square$

2.4.26 LEMMA (König). *If $\kappa \geq \omega \wedge \text{cf}(\kappa) \leq \lambda$, then $\kappa^\lambda > \kappa$. In particular, $\kappa^{\text{cf}(\kappa)} > \kappa$.*

Proof. It suffices to show that any function $g : \kappa \rightarrow {}^\lambda \kappa$ is not onto, hence $\kappa \neq \kappa^\lambda$. Since $\lambda \geq \text{cf}(\kappa)$, there exists a cofinal map $f : \lambda \rightarrow \kappa$. Fix an arbitrary $\alpha \in \lambda$. For any $\beta \in f(\alpha) \in \kappa$, we have

$$\beta \in \kappa, \quad g(\beta) \in {}^\lambda \kappa, \quad (g(\beta))(\alpha) \in \kappa.$$

For a fixed $\alpha \in \lambda$, define $b = \{g(\beta)(\alpha) : \beta \in f(\alpha)\}$. Then $b \subset \kappa$ and there is an obvious function from $f(\alpha)$ onto b . Hence

$$|b| \leq |f(\alpha)| < \kappa.$$

Now, construct $h : \lambda \rightarrow \kappa$ such that $h \notin g[\kappa]$, proving g is not onto. From the above, put $h(\alpha) = \min\{\kappa - b\}$, $\alpha \in \lambda$ ($|b| < \kappa$ implies $\{\kappa - b\}$ is nonempty), we show that $h \notin g[\kappa]$. Suppose, for contradiction, $h \in g[\kappa]$, i.e., there exists $\beta_0 \in \kappa$ such that $h = g(\beta_0) \in {}^\lambda \kappa$. By the definition of h :

$$(g(\beta_0))(\alpha) = h(\alpha) \notin b.$$

It follows that $\beta_0 \geq f(\alpha)$ (otherwise, $g(\beta_0)(\alpha) \in b$). By the arbitrariness of α , $\beta_0 + 1$ is the upper bound of $f[\lambda]$, which is a contradiction. $\square$

2.4.27 THEOREM. *If $\kappa \geq \omega$, then $\text{cf}(2^\kappa) > \kappa$. In particular, $\text{cf}(\mathfrak{c}) > \omega$.*

Proof. Suppose, for contradiction, $\text{cf}(2^\kappa) \leq \kappa$, then $(2^\kappa)^\kappa > 2^\kappa$ by the lemma. Notice that $(2^\kappa)^\kappa = 2^{\kappa \otimes \kappa} = 2^\kappa$ which is a contradiction. $\square$

2.4.28 LEMMA. *If κ is a limit cardinal and $\lambda \geq \text{cf}(\kappa)$, then*

$$\kappa^\lambda = \left(\bigcup_{\mu < \kappa, \mu \text{ is a cardinal}} \mu^\lambda \right)^{\text{cf}(\kappa)}.$$

Proof. By Theorem 11.48, let $\gamma : \text{cf}(\kappa) \rightarrow \kappa$ be strictly increasing with $\text{rng}(\gamma)$ unbounded in κ , and with $0 < \gamma_0$. We define $F : {}^\lambda \kappa \rightarrow \prod_{\alpha < \text{cf}(\kappa)} {}^\lambda \gamma_\alpha$ as follows. If $f \in {}^\lambda \kappa$, $\alpha < \text{cf}(\kappa)$, and $\beta < \lambda$, then

$$((F(f))_\alpha)_\beta = \begin{cases} f(\beta) & \text{if } f(\beta) < \gamma_\alpha, \\ 0 & \text{otherwise.} \end{cases}$$

Now F is one-to-one. Indeed, suppose $f, g \in {}^\lambda \kappa$ and $f \neq g$. Pick $\beta < \lambda$ such that $f(\beta) \neq g(\beta)$. Choose $\alpha < \text{cf}(\kappa)$ with $f(\beta), g(\beta) < \gamma_\alpha$ (possible because $\text{rng}(\gamma)$ is unbounded in κ). Then $((F(f))_\alpha)_\beta = f(\beta) \neq g(\beta) = ((F(g))_\alpha)_\beta$, hence $F(f) \neq F(g)$. Therefore F is injective. Consequently,

$$\kappa^\lambda = |{}^\lambda \kappa| \leq \left| \prod_{\alpha < \text{cf}(\kappa)} {}^\lambda \gamma_\alpha \right| \leq \left| \prod_{\alpha < \text{cf}(\kappa)} \left(\bigcup_{\mu < \kappa \text{ cardinal}} {}^\lambda \mu \right) \right| = \left(\bigcup_{\mu < \kappa \text{ cardinal}} \mu^\lambda \right)^{\text{cf}(\kappa)}.$$

On the other hand, note that for each $\mu < \kappa$ we have $\mu^\lambda \leq \kappa^\lambda$, so $\bigcup_{\mu < \kappa \text{ cardinal}} \mu^\lambda \subseteq \kappa^\lambda$. Taking cardinalities,

$$\left(\bigcup_{\mu < \kappa \text{ cardinal}} \mu^\lambda \right)^{\text{cf}(\kappa)} \leq (\kappa^\lambda)^{\text{cf}(\kappa)} = \kappa^{\lambda \cdot \text{cf}(\kappa)}.$$

Since $\lambda \geq \text{cf}(\kappa)$, we have $\lambda \cdot \text{cf}(\kappa) = \lambda$ (because λ is an infinite cardinal and $\text{cf}(\kappa) \leq \lambda$; for cardinals, $\lambda \cdot \lambda = \lambda$, and here $\lambda \cdot \text{cf}(\kappa) \leq \lambda \cdot \lambda = \lambda$, while clearly $\lambda \leq \lambda \cdot \text{cf}(\kappa)$). Thus $\kappa^{\lambda \cdot \text{cf}(\kappa)} = \kappa^\lambda$. Therefore

$$\left(\bigcup_{\mu < \kappa \text{ cardinal}} \mu^\lambda \right)^{\text{cf}(\kappa)} \leq \kappa^\lambda.$$

□

2.4.29 THEOREM (Hausdorff). *If κ and λ are infinite cardinals, then $(\kappa^+)^{\lambda} = \kappa^{\lambda} \cdot \kappa^+$.*

Proof. If $\kappa^+ \leq \lambda$, then both sides are equal to 2^λ (since for infinite λ , $(\kappa^+)^{\lambda} = 2^\lambda$ and $\kappa^\lambda = 2^\lambda$, so the right-hand side is $2^\lambda \cdot \kappa^+ = 2^\lambda$). Now suppose $\lambda < \kappa^+$. Then

$$(\kappa^+)^{\lambda} = |\lambda^{\kappa^+}| = \left| \bigcup_{\alpha < \kappa^+} \lambda^\alpha \right|$$

because every function from λ to κ^+ has its range contained in some $\alpha < \kappa^+$ (since κ^+ is the least ordinal of cardinality κ^+ , and $\lambda < \kappa^+$ implies the range has cardinality $\leq \lambda < \kappa^+$, hence is bounded in κ^+). Therefore

$$(\kappa^+)^{\lambda} \leq \sum_{\alpha < \kappa^+} |\alpha|^{\lambda}.$$

For each $\alpha < \kappa^+$, $|\alpha| \leq \kappa$, so $|\alpha|^{\lambda} \leq \kappa^{\lambda}$. Hence

$$\sum_{\alpha < \kappa^+} |\alpha|^{\lambda} \leq \sum_{\alpha < \kappa^+} \kappa^{\lambda} = \kappa^{\lambda} \cdot \kappa^+.$$

Thus $(\kappa^+)^{\lambda} \leq \kappa^{\lambda} \cdot \kappa^+$. Conversely, $\kappa^{\lambda} \leq (\kappa^+)^{\lambda}$ and $\kappa^+ \leq (\kappa^+)^{\lambda}$ (since $\lambda \geq \omega$ implies $2^\lambda \geq \lambda$ and $(\kappa^+)^{\lambda} \geq 2^\lambda \geq \lambda \geq \omega$, and clearly $\kappa^+ \leq (\kappa^+)^{\lambda}$ for $\lambda \geq 1$). Therefore $\kappa^{\lambda} \cdot \kappa^+ \leq (\kappa^+)^{\lambda}$. Combining the two inequalities yields $(\kappa^+)^{\lambda} = \kappa^{\lambda} \cdot \kappa^+$. □

2.4.30 THEOREM. *Let κ and λ be cardinals with $2 \leq \kappa$ and $\lambda \geq \omega$. Then*

1. *If $\kappa \leq \lambda$, then $\kappa^{\lambda} = 2^\lambda$;*
2. *If $\kappa \geq \omega$ and there is a $\mu < \kappa$ such that $\mu^{\lambda} \geq \kappa$, then $\kappa^{\lambda} = \mu^{\lambda}$;*
3. *Assume that κ is infinite and $\mu^{\lambda} < \kappa$ for all $\mu < \kappa$. Then $\lambda < \kappa$ and*
 - (a) *if $\text{cf}(\kappa) > \lambda$, then $\kappa^{\lambda} = \kappa$;*
 - (b) *if $\text{cf}(\kappa) \leq \lambda$, then $\kappa^{\lambda} = \kappa^{\text{cf}(\kappa)}$.*

Proof. We have already seen (1). Under the hypothesis of (2), $\kappa^{\lambda} \leq (\mu^{\lambda})^{\lambda} = \mu^{\lambda+\lambda} = \mu^{\lambda} \leq \kappa^{\lambda}$. Now assume the hypothesis of (3). In particular $2^\lambda < \kappa$, so of course $\lambda < \kappa$. Next, assume the hypothesis of (3.a): $\text{cf}(\kappa) > \lambda$. Then

$$\kappa^{\lambda} = |\lambda^{\kappa}| = \left| \bigcup_{\alpha < \kappa} \lambda^\alpha \right| \leq \sum_{\alpha < \kappa} |\alpha|^{\lambda} \leq \kappa.$$

Finally, assume that hypothesis of (3.b): $\text{cf}(\kappa) \leq \lambda$. Since $\lambda < \kappa$, it follows that κ is singular, so in particular it is a limit cardinal, then by the lemma

$$\kappa^{\lambda} = \left(\bigcup_{\mu < \kappa \text{ cardinal}} \mu^{\lambda} \right)^{\text{cf}(\kappa)} \leq \kappa^{\text{cf}(\kappa)} \leq \kappa^{\lambda}.$$

□

2.4.31 COROLLARY. In **GCH**. Let κ and λ be cardinals with $2 \leq \kappa, \lambda \geq \omega$. Then

1. If $\kappa \leq \lambda$, then $\kappa^\lambda = \lambda^+$;
2. If $\text{cf}(\kappa) \leq \lambda < \kappa$, then $\kappa^\lambda = \kappa^+$;
3. If $\lambda < \text{cf}(\kappa)$, then $\kappa^\lambda = \kappa$.

2.4.5 DEFINITION. Let κ be an infinite cardinal $\aleph_\alpha$.

1. κ is *regular* if $\text{cf}(\kappa) = \kappa$;
2. κ is *singular* if $\text{cf}(\kappa) < \kappa$.

2.4.32 THEOREM. κ^+ is regular for every infinite cardinal number κ .

Proof. Since $\text{cf}(\kappa^+) \leq \kappa^+$, it suffices to show that $\text{cf}(\kappa^+) \neq \alpha$ for every $\alpha < \kappa^+$. Let $\alpha < \kappa^+$ and $f : \alpha \rightarrow \kappa^+$; we show that $f[\alpha]$ can not be unbounded in κ^+ . Then $|\alpha| \leq \alpha \leq \kappa$ and $|f(\xi)| \leq \kappa$ for every $\xi < \alpha$. Hence $|\bigcup_{\xi < \alpha} [f(\xi) + 1]| \leq \kappa$ by the corollary. Thus, $\bigcup_{\xi < \alpha} [f(\xi) + 1] \neq \kappa^+$, that is, $\text{cf}(\kappa^+) \neq \alpha$ for every $\alpha < \kappa^+$. So $\text{cf}(\kappa^+) = \kappa^+$. $\square$

This theorem almost tells the full story when a cardinal is regular. Examples of singular cardinals are $\aleph_{\omega+\omega}$ and $\aleph_{\omega_1}$. But it is conceivable that there are regular cardinals not covered by this theorem.

2.4.6 DEFINITION. Let κ be a regular cardinal.

1. κ is called a *weakly inaccessible cardinal* if it is a limit cardinal.
2. κ is called a *strongly inaccessible cardinal* if $\omega < \kappa \wedge \forall \lambda < \kappa (2^\lambda < \kappa)$.

2.4.33 THEOREM. It is clear that every strongly inaccessible cardinal is weakly inaccessible (suppose, for contradiction that κ^+ is a successor ordinal. Then $\kappa^+ \leq 2^\kappa$, which is a contradiction.) In **GCH**. A regular cardinal number κ is weakly inaccessible if and only if strongly inaccessible.

3

Category Theory I

3.1 Grothendieck Universe and Categories

Grothendieck Universes

Before formally discussing category theory, we need to address a minor detail. Although it is not crucial for actual learning, filling in some logical gaps can always provide a sense of reassurance. Let us first give the conventional definition of a category as it appears in current textbooks. Once again, we reiterate that our study of basic algebra in these notes is still based on the **ZFC** axiom system.

3.1.1 DEFINITION. A *category* $\mathbf{C}$ consists of:

1. a set $\text{Ob}(\mathbf{C})$, whose elements are called the *objects* of $\mathbf{C}$;
2. a set $\text{Mor}(\mathbf{C})$, whose elements are called the *morphisms* of $\mathbf{C}$, together with a pair of functions $\text{Mor}(\mathbf{C}) \xrightarrow{s} \text{Ob}(\mathbf{C}), \text{Mor}(\mathbf{C}) \xrightarrow{t} \text{Ob}(\mathbf{C})$ where s and t give the *source* and *target* of a morphism, respectively. For $X, Y \in \text{Ob}(\mathbf{C})$, we denote $\text{Hom}_{\mathbf{C}}(X, Y) = \{s^{-1}(X) \cap t^{-1}(Y)\}$ called the *Hom-set*, its elements are called morphisms from X to Y ;
3. for each object X , a designated element $\text{id}_X \in \text{Hom}_{\mathbf{C}}(X, X)$ called the *identity morphism* from X to itself;
4. for any objects $X, Y, Z \in \text{Ob}(\mathbf{C})$, a *composition* function

$$\begin{aligned} \circ : \text{Hom}_{\mathbf{C}}(Y, Z) \times \text{Hom}_{\mathbf{C}}(X, Y) &\rightarrow \text{Hom}_{\mathbf{C}}(X, Z) \\ (f, g) &\mapsto f \circ g \end{aligned}$$

(often written simply as fg when no confusion arises). This composition must satisfy these two conditions:

- (a) (Associativity): for any morphisms $h, g, f \in \text{Mor}(\mathbf{C})$ such that the composites $f \circ (g \circ h)$ and $(f \circ g) \circ h$ are defined $f \circ (g \circ h) = (f \circ g) \circ h$. Hence both sides can be written as $f \circ g \circ h$ or fgh ;
- (b) (Identity Law): for any morphism $f \in \text{Hom}_{\mathbf{C}}(X, Y)$, $f \circ \text{id}_X = f = \text{id}_Y \circ f$.

When we discuss categories, we inevitably speak of “the category of all $\dots$ ”. For example, the category of all sets, the category of all groups, and even the category of all categories. Specifically, when we discuss categories, we are faced with two classes: $\text{Ob}(\mathbf{C})$ and $\text{Mor}(\mathbf{C})$. When we speak of the category of all sets, $\text{Ob}(\mathbf{C})$ is the class of all sets, which is illegal within the **ZFC** axiomatic system — to be precise, it is not logically well-formed. As discussed in the previous section, we must be very careful with such phrasing — at least within the **ZFC** axiomatic system, we cannot formulate it that way. Grothendieck proposed a method to allow us to discuss categories “safely” within the **ZFC** framework, namely the Grothendieck universe.

3.1.2 DEFINITION. A *Grothendieck universe* is a nonempty set $\mathbf{U}$ with properties:

1. $x \in u \in \mathbf{U} \rightarrow x \in \mathbf{U}$;
2. $u, v \in \mathbf{U} \rightarrow \{u, v\} \in \mathbf{U}$;
3. $u \in \mathbf{U} \rightarrow \mathcal{P}(u) \in \mathbf{U}$;
4. If $I \in \mathbf{U}$ and $\{x_i : i \in I\}$ is a set of elements of $\mathbf{U}$, then $\bigcup \{x_i : i \in I\} \in \mathbf{U}$;

There are two simple examples of Grothendieck universes: the empty set and the set of all hereditarily finite sets $\mathbf{V}_\omega$. We call the $\mathbf{V}_0$ and $\mathbf{V}_\omega$ the trivial universes. Other examples are more difficult to construct.

3.1.1 PROPOSITION. *The following properties are clear:*

1. If $x \in \mathbf{U}$, then $\{x\} \in \mathbf{U}$;
2. If $x \subset y$, where $y \in \mathbf{U}$, then $x \in \mathbf{U}$, in particular $\emptyset \in \mathbf{U}$;
3. If $x, y \in \mathbf{U}$, then $(x, y) = \{x, \{x, y\}\} \in \mathbf{U}$;
4. If $x, y \in \mathbf{U}$, then $x \cup y$ and $x \times y$ are elements of $\mathbf{U}$;
5. If $I \in \mathbf{U}$ and $\{x_i\}_{i \in I}$ is a family of elements of $\mathbf{U}$, then the set $\prod_{i \in I} x_i \in \mathbf{U}$;
6. If $x \in \mathbf{U}$, then $x \cup \{x\} \in \mathbf{U}$ and therefore $\mathbf{N} \subset \mathbf{U}$;
7. If $x \in \mathbf{U}$, then $\bigcup x \in \mathbf{U}$.

Clearly, if $\mathbf{U}$ is an universe, then we can start from the elements of this universe and perform almost all set-theoretic operations while ensuring that we never leave this universe. More precisely, the **ZFC** axioms strictly limit the ways in which new sets can be produced from known sets, but there is one exception: the Axiom of Replacement, which can construct a set outside this universe. To address this issue, Grothendieck propose the following axiom:

AXIOM (UA). Every set is an element of some universe.

He suggested to use **ZFC** together with requirement that every set X be contained in some Grothendieck universe guaranteeing at least an infinite sequence of Grothendieck universes: $\mathbf{U}_0 \in \mathbf{U}_1 \in \dots \in \mathbf{U}_n \in \mathbf{U}_{n+1} \in \dots$. As Grothendieck universes are transitive such a sequence is also cumulative in the sense that

$$\mathbf{U}_0 \subset \mathbf{U}_1 \subset \dots \subset \mathbf{U}_n \subset \mathbf{U}_{n+1} \subset \dots$$

3.1.3 DEFINITION. An **U-category** is a category whose objects and morphisms form **U**-classes, and whose morphisms sets are **U**-classes, i.e., $\text{Ob}(\mathbf{C})$ and $\text{Mor}(\mathbf{C})$ are **U**-classes, and $\text{Hom}_{\mathbf{C}}(X, Y)$ is a **U**-set for any $X, Y \in \text{Ob}(\mathbf{C})$.

Clearly, the category of all **U**-sets is a **U**-category: since $\mathbf{U} \subset \mathbf{U}$, the object class is a **U**-class. Every morphism $f : X \rightarrow Y$ with $X, Y \in \mathbf{U}$ is a subset of $X \times Y$; since $X \times Y \in \mathbf{U}$ by universe closure and any subset of an element of $\mathbf{U}$ is again an element of $\mathbf{U}$ (proposition 1.(2)), each morphism lies in $\mathbf{U}$, so the class of all morphisms is also a subset of $\mathbf{U}$ and therefore a **U**-class. Finally, since $X \times Y \in \mathbf{U}$ and $\mathcal{P}(X \times Y) \in Y$, by proposition 1.(2) we obtain that the set $\text{Hom}_{\mathbf{C}}(X, Y)$ of all functions from X to Y is indeed an element of $\mathbf{U}$, hence a **U**-set. We denote this category by $\text{Set}_{\mathbf{U}}$.

On the other hand, if we want to discuss a larger category, such as the category of all **U**-classes, then since the cardinality of $\mathcal{P}(U)$ is strictly greater than that of $\mathbf{U}$, its object set is not a **U**-class. What should we do then? By making reasonable use of the Universe Axiom (UA), we can take a universe $\mathbf{U}'$ that is larger than $\mathbf{U}$. It is easy to verify that the original **U**-classes become $\mathbf{U}'$ -sets in this new universe, and naturally all **U**-classes form a $\mathbf{U}'$ -set, hence they form a $\mathbf{U}'$ -class. This resolves the issue. We will not discuss the remaining details for now, such as the definition of subcategories, etc. Similarly, if we wish to construct even larger categories, we can always find a larger universe in this manner.

3.1.2 THEOREM. *The following two axioms are equivalent:*

1. **UA**: for every set x , there is an universe $\mathbf{U}$ such that $x \in \mathbf{U}$.
2. **BC**: for every cardinal κ , there is a strongly inaccessible cardinal $\lambda > \kappa$.

Proof. The proof of this theorem is not of great significance in these notes. Interested readers may consult Wikipedia or other relevant sources. $\square$

In fact, any non-trivial Grothendieck universe is of the form $\mathbf{V}_\kappa$ where κ is the strongly inaccessible cardinal. Since the existence of strongly inaccessible cardinals cannot be proved from the axioms of **ZFC**, the existence of universes other than the empty set and $\mathbf{V}_\omega$ cannot be proved from **ZFC** either.

The properties of strongly inaccessible cardinals tell us that starting from elements of $\mathbf{V}_\kappa$, no operation within the **ZFC** system can go beyond V_κ . If the reader has some acquaintance with mathematical logic, a more precise statement can be made: for a strongly inaccessible cardinal κ , the universe $\mathbf{U} := \mathbf{V}_\kappa$ together with the membership relation $\in$ constitutes a model of **ZFC**. This is tantamount to running a copy of set theory virtually inside set theory. One may view it as follows: if the elements of $\mathbf{V}_\kappa$ are regarded as “sets”, then from the perspective of the model $(V_\kappa, \in)$, the “classes” are precisely the elements of $\mathbf{V}_{\kappa+1} = \mathcal{P}(\mathbf{V}_\kappa)$.

3.1.4 DEFINITION. Let $\mathbf{U}$ be a Grothendieck universe, an *U-set* is an element of $\mathbf{U}$. An *U-class* is a subset of $\mathbf{U}$. From now on, when we need a universe in our discussion, we will typically fix a universe $\mathbf{U}$ and use the following translated terminology, which we call the *conglomerate convention for $\mathbf{U}$* (**CCU**):

1. a *U-set* in the sense of **ZFC** is called a “set” in **CCU**;
2. a *U-class* in the sense of **ZFC** is called a “class” in **CCU**;
3. any set (not necessary elements or subsets of $\mathbf{U}$) in the sense of **ZFC** is called a “conglomerate” in **CCU**.

The concept “conglomerate” has been created to deal with “collections of classes”. As a result, in this terminology, we have the class $\mathbf{U}$ of all sets, and the conglomerate $\mathcal{P}(\mathbf{U})$ of all classes, but there is no “collection” of all conglomerates.

From now on, when we discuss categories, we will default to fixing an inaccessible cardinal, thereby giving a non-trivial universe $\mathbf{U}$. Categories will be understood as $\mathbf{U}$ categories by default. Under **CCU**, we can speak of “the category of all sets” and “the category of all classes”. When a larger category is needed, it suffices to consider a larger inaccessible cardinal.

In daily life and common mathematical theories, a single non-trivial universe is sufficient for carrying out rich investigations. However, this requirement is still quite demanding (since it assumes the existence of an inaccessible cardinal). In fact, the universe we more commonly encounter is the “ordinary mathematics” universe $\mathbf{V}_{\omega+\omega}$, which is a model of Zermelo set theory (**ZF** without the axiom of replacement). The reason is that $\mathbf{V}_{\omega+1}$ is sufficient for constructing the integers, $\mathbf{V}_{\omega+2}$ is sufficient for constructing the real numbers.

Categories

The above treatment of the details concerning categories is referred to as the set-theoretic foundation of category theory. At this point, we officially begin our theoretical study of category theory. Here are some common symbol simplifications.

$$\begin{aligned} gf &\text{ to mean } g \circ f \\ A \in \mathbf{C} &\text{ to mean } A \in \text{Ob}(\mathbf{C}) \\ f : A \rightarrow B \text{ or } A \xrightarrow{f} B &\text{ to mean } f \in \text{Hom}_{\mathbf{C}}(A, B) \end{aligned}$$

3.1.1 REMARK. We sometimes figuratively call morphisms arrows. The composition of morphisms corresponds to head-to-tail concatenation of arrows. Diagrams with arrows provide a convenient language for discussing categories. Among these, the most commonly used is the notion of a *commutative diagram*, where “commutative” means that different

compositions of arrows lead to the same result. For example, the commutativity of the diagram

$$\begin{array}{ccc} A & \xrightarrow{u} & B \\ x \downarrow & & \downarrow v \\ C & \xrightarrow{y} & D \end{array}$$

is equivalent respectively to $vu = yx$. The names of morphisms are often omitted from diagrams when they are self-evident or unimportant.

3.1.1 EXAMPLE. We could define a category **Pos** whose objects are all partially ordered sets $\langle P, \leq \rangle$. A morphism $\langle P, \leq \rangle \rightarrow \langle Q, \leq \rangle$ is an increasing function $f : P \rightarrow Q$, that is, for all $x, y \in P$, $x < y \rightarrow f(x) < f(y)$. One may check that **Pos** is indeed a category.

3.1.2 EXAMPLE. (See *Algebra: Chapter 0*, Paolo Aluffi, P20, Examples 3.3) Suppose that S is a set and $\sim$ is a relation on S satisfying the reflexive and transitive properties. Then we can encode this data into a category **C**:

- $\text{Ob}(\mathbf{C}) = \{x : x \in S\}$;
- $\text{Mor}(\mathbf{C}) = \bigcup_{a,b \in S} \text{Hom}_{\mathbf{C}}(a, b)$, where $\text{Hom}_{\mathbf{C}}(a, b) = \{(a, b) : a \sim b\}$.

Note that there are very “few” morphisms: at most one for any pair of objects. How to define the “composition” of two arrows ?

First, do we have “identities” ? If $a \in S$, we need to find an element $1_a \in \text{Hom}_{\mathbf{C}}(a, a)$. This is precisely why we are assuming that $\sim$ is reflexive: this tells us that $a \sim a$ for all $a \in S$, that is, $\text{Hom}_{\mathbf{C}}(a, a)$ consists of the single element (a, a) . We have no choice; we must let

$$1_a = (a, a) \in \text{Hom}_{\mathbf{C}}(a, a).$$

As for composition, let $a, b, c \in S$ and $f \in \text{Hom}_{\mathbf{C}}(a, b), g \in \text{Hom}_{\mathbf{C}}(b, c)$; we have to define a corresponding morphism $gf \in \text{Hom}_{\mathbf{C}}(a, c)$. Now, $f \in \text{Hom}_{\mathbf{C}}(a, b)$ tells us that $\text{Hom}_{\mathbf{C}}(a, b)$ is nonempty, which means $a \sim b$ and $f = (a, b)$. Similarly, $b \sim c$ and $g = (b, c)$. Now

$$a \sim b \wedge b \sim c \rightarrow a \sim c$$

since $\sim$ is transitive. This tells us that $\text{Hom}_{\mathbf{C}}(a, c)$ consists of the single element (a, c) . Thus we again have no choice: we must let

$$gf = (a, c) \in \text{Hom}_{\mathbf{C}}(a, c)$$

Is this composition associative ? If $f \in \text{Hom}_{\mathbf{C}}(a, b), g \in \text{Hom}_{\mathbf{C}}(b, c)$ and $h \in \text{Hom}_{\mathbf{C}}(c, d)$, then necessarily $f = (a, b), g = (b, c)$ and $h = (c, d)$, it follows that $gf = (a, c)$ and $hg = (b, d)$, and hence

$$h(gf) = (a, d) = (hg)f.$$

One last question, are these identities satisfy the identity law ? If $f \in \text{Hom}_{\mathbf{C}}(a, b)$, then $f1_a \in \text{Hom}_{\mathbf{C}}(a, b)$ and $1_b f \in \text{Hom}_{\mathbf{C}}(a, b)$, we have no choice that $f1_a = f$ and $1_b f = f$.

The most trivial instance of this construction is the category obtained from a set taken with the equivalence relation “=”, that is, the only morphisms are the identity morphisms: $\text{Hom}_{\mathbf{C}}(a, b) = \{(a, a) = 1_a : a \in S\}$ and $\text{Hom}_{\mathbf{C}}(a, b) = \emptyset$ if $a \neq b$. These categories are called the *discrete categories*.

3.1.3 EXAMPLE. Every category **C** has an *opposite category* $\mathbf{C}^{op}$, defined by “reversing the arrows”. Formally, $\text{Ob}(\mathbf{C}^{op}) = \text{Ob}(\mathbf{C})$ and $\text{Hom}_{\mathbf{C}^{op}}(X, Y) = \text{Hom}_{\mathbf{C}}(Y, X)$ for all objects X, Y . Identities in $\mathbf{C}^{op}$ are the same as in **C**; composition in $\mathbf{C}^{op}$ is the same as in **C** but with the arguments reversed: let $f \in \text{Hom}_{\mathbf{C}^{op}}(X, Y), g \in \text{Hom}_{\mathbf{C}^{op}}(Y, Z)$, define the composition of morphisms by $g \circ_{\mathbf{C}^{op}} f := f \circ_{\mathbf{C}} g$. One may check that it satisfies the associativity and identity law.

3.1.4 EXAMPLE. To give readers a sense of the extremely abstract nature of category theory, we now introduce a novel category. Let $\mathbf{C}$ be a category, and let $A \in \mathbf{C}$. We try to define a new category $\mathbf{C}_A$ whose objects are some morphisms in $\mathbf{C}$:

- $\text{Ob}(\mathbf{C}_A) = \{f \in \text{Hom}_{\mathbf{C}}(Z, A) : Z \in \mathbf{C}\}$
- $\text{Mor}(\mathbf{C}_A) = \bigcup_{f_1, f_2} \text{Hom}_{\mathbf{C}_A}(f_1, f_2)$, how to define the Hom-sets ? For objects $f : Z_1 \rightarrow A$ and $f_2 : Z_2 \rightarrow A$, the Hom-set $\text{Hom}_{\mathbf{C}_A}(f, g)$ consists of all the commutative diagrams of the form

$$\begin{array}{ccc} Z_1 & \xrightarrow{\sigma} & Z_2 \\ & \searrow f_1 & \swarrow f_2 \\ & A & \end{array}$$

For now, we won't rush to explain why it is defined this way. First, let us define a suitable composition so that it becomes a category. Let $f \in \mathbf{C}_A$, the identity $1_f \in \text{Hom}_{\mathbf{C}_A}(f, f)$ is the commutative diagram

$$\begin{array}{ccc} Z & \xrightarrow{1_Z} & Z \\ & \searrow f & \swarrow f \\ & A & \end{array}$$

Let $f_1 \in \text{Hom}(Z_1, A)$, $f_2 \in \text{Hom}(Z_2, A)$ and $f_3 \in \text{Hom}(Z_3, A)$. The two morphisms $f_1 \rightarrow f_2$ and $f_2 \rightarrow f_3$ in $\mathbf{C}_A$ correspond to putting the two commutative diagrams side-by-side:

$$\begin{array}{ccccc} Z_1 & \xrightarrow{\sigma} & Z_2 & \xrightarrow{\tau} & Z_3 \\ & \searrow f_1 & \downarrow f_2 & \swarrow f_3 & \\ & A & & & \end{array}$$

and then we define the composition of $f_1 \rightarrow f_2$ and $f_2 \rightarrow f_3$ by removing the central arrow, and we obtain the commutative diagram

$$\begin{array}{ccc} Z_2 & \xrightarrow{1_Z} & Z_3 \\ & \searrow f_2 & \swarrow f_3 \\ & A & \end{array} \circ \begin{array}{ccc} Z_1 & \xrightarrow{1_Z} & Z_2 \\ & \searrow f_1 & \swarrow f_2 \\ & A & \end{array} = \begin{array}{ccc} Z_1 & \xrightarrow{\tau\sigma} & Z_3 \\ & \searrow f_1 & \swarrow f_3 \\ & A & \end{array}$$

One may check that $\mathbf{C}_A$ is indeed a category. Categories constructed in this fashion are called the **slice categories** in the literature. Define a function $\Phi : \{\sigma \in \text{Hom}_{\mathbf{C}}(Z_1, Z_2) : f_1 \in \text{Hom}_{\mathbf{C}}(Z_1, A), f_2 \in \text{Hom}_{\mathbf{C}}(Z_2, A), f_2 = f_1\sigma\} \rightarrow \text{Hom}_{\mathbf{C}_A}(f_1, f_2)$ by $\Phi(\sigma) :=$

$$\begin{array}{ccc} Z_1 & \xrightarrow{\sigma} & Z_2 \\ & \searrow f_1 & \swarrow f_2 \\ & A & \end{array}$$

It is clear that Φ is a bijection. In other words, each morphism in $\mathbf{C}_A$ uniquely determines a morphism in $\mathbf{C}$, and vice versa. For convenience, we simply denote the elements of $\text{Hom}_{\mathbf{C}_A}(f_1, f_2)$ as those morphisms in $\mathbf{C}$ that satisfy the commutative condition, i.e., $\text{Hom}_{\mathbf{C}_A}(f_1, f_2) = \{\sigma \in \text{Hom}_{\mathbf{C}}(Z_1, Z_2) : f_1 = f_2\sigma\}$.

3.1.5 EXAMPLE. We can similarly construct new categories in an analogous way. Let $\mathbf{C}$ be a category, and let $A \in \mathbf{C}$. We define a category $\mathbf{C}^A$:

- $\text{Ob}(\mathbf{C}^A) = \{f \in \text{Hom}_{\mathbf{C}}(A, Z) : Z \in \mathbf{C}\}$;
- $\text{Mor}(\mathbf{C}^A) = \bigcup_{f_1, f_2} \text{Hom}_{\mathbf{C}^A}(f_1, f_2)$. The Hom set $\text{Hom}_{\mathbf{C}^A}(f_1, f_2)$ consists of all the commutative diagrams of the form

$$\begin{array}{ccc} & A & \\ f_1 \swarrow & & \searrow f_2 \\ Z_1 & \xrightarrow{\sigma} & Z_2 \end{array}$$

3.1.6 EXAMPLE. As an instance of a category in example 4. Let $\mathbf{C} = \mathbf{Set}$ and $A = \{*\}$. Call the resulting category $\mathbf{Set}^*$. An object in $\mathbf{Set}^*$ is a morphism $f : \{*\} \rightarrow S$ where S is any set in $\mathbf{Set}$. Clearly, f is determined by the element $s = f(*) \in S$. Thus, we may denote objects of $\mathbf{Set}^*$ as pairs (S, s) . A morphism between two such objects $(S, s) \rightarrow (T, t)$ corresponds then to a function $\sigma : S \rightarrow T$ such that $\sigma(s) = t$.

3.1.7 EXAMPLE. Similarly, we can further introduce analogous categories. Let $\mathbf{C}$ be a category, and given two objects $A, B \in \mathbf{C}$. We can define a new category $\mathbf{C}_{A,B}$:

- $\text{Ob}(\mathbf{C}_{A,B}) = \left\{ (f, g) := \begin{array}{ccc} & A & \\ f \swarrow & & \searrow g \\ Z & & B \end{array} : f \in \text{Hom}_{\mathbf{C}}(Z, A), g \in \text{Hom}_{\mathbf{C}}(Z, B) \right\}$
- $\text{Mor}(\mathbf{C}_{A,B}) = \bigcup_{(f_1, g_1), (f_2, g_2)} \text{Hom}_{\mathbf{C}_{A,B}}((f_1, g_1), (f_2, g_2))$. How to define the Hom-sets? For objects (f_1, g_1) and (f_2, g_2) , the Hom-set $\text{Hom}_{\mathbf{C}_{A,B}}((f_1, g_1), (f_2, g_2))$ consists of all the commutative diagrams of the form

$$\begin{array}{ccc} & A & \\ f_1 \swarrow & & \searrow f_2 \\ Z_1 & \xrightarrow{\sigma} & Z_2 \\ g_1 \swarrow & & \searrow g_2 \\ & B & \end{array} \quad \text{or} \quad \begin{array}{ccc} & A & \\ f_1 \swarrow & & \searrow f_2 \\ Z_1 & \xrightarrow{\sigma} & Z_2 \\ g_1 \swarrow & & \searrow g_2 \\ & B & \end{array}$$

Since most of the descriptions in category theory are about morphisms, we define here some terminology concerning morphisms, whose definitions are obtained directly by generalizing the notions of bijection, surjection, and injection for functions between sets.

Unlike in set theory, where injectivity, surjectivity, and bijectivity are defined using elements, a general category may have objects that are not sets and morphisms that are not functions. Therefore, we must define the categorical analogues — monomorphisms, epimorphisms, and isomorphisms — using only the composition operation. Incidentally, the set-theoretic notions can be rephrased in terms of composition as well, confirming the consistency of the categorical definitions.

3.1.5 DEFINITION. Let $\mathbf{C}$ be a category. A morphism $f \in \text{Hom}_{\mathbf{C}}(A, B)$ is an *isomorphism* if it has a two-sided inverse under composition, that is, there exists $g \in \text{Hom}_{\mathbf{C}}(B, A)$ such that

$$gf = 1_A, \quad fg = 1_B.$$

3.1.3 PROPOSITION. (See *Algebra: Chapter 0*, Paolo Aluffi, P28, Proposition 4.2) *The inverse of an isomorphism is unique.*

Proof. Let $g_1, g_2 : B \rightarrow A$ act as the inverses of a given isomorphism $f : A \rightarrow B$. We need to verify that $g_1 = g_2$ in this case: $g_1 = g_1 1_B = g_1(fg_2) = (g_1f)g_2 = 1_A g_2 = g_2$, as desired. $\square$

Note that the argument really proves that if f is a morphism with a left-inverse g_1 and a right-inverse g_2 , then necessarily f is an isomorphism; $g_1 = g_2$; and this morphism is the unique inverse of f . Since the inverse of f is uniquely determined by f , there is no ambiguity in denoting it by f^{-1} .

3.1.4 PROPOSITION.

1. Each identity 1_A is an isomorphism and is its own inverse.
2. If f is an isomorphism, then f^{-1} is an isomorphism and further $(f^{-1})^{-1} = f$.
3. If $f \in \text{Hom}_{\mathbf{C}}(A, B)$, $g \in \text{Hom}_{\mathbf{C}}(B, C)$ are isomorphisms, then the composition gf is an isomorphism and $(gf)^{-1} = f^{-1}g^{-1}$.

Two objects A, B of a category are *isomorphic* if there is an isomorphism $f : A \rightarrow B$. An immediate corollary of this proposition is that “isomorphism” is an equivalence relation. If two objects A, B are isomorphic, one write $A \cong B$.

3.1.8 EXAMPLE. Identities may be the only isomorphisms in a category: for example, this is the case in the category $\mathbf{C}$ obtained from the relation $\leq$ on $\mathbf{Z}$. Indeed, for $a, b \in \mathbf{C}$, there is a morphism $f : a \rightarrow b$ and a morphism $g : b \rightarrow a$ only if $a \leq b$ and $b \leq a$, that is, if $a = b$. So an isomorphism in $\mathbf{C}$ necessarily acts from an object a to itself; but in $\mathbf{C}$, there is only one such morphism, that is, 1_a .

3.1.9 EXAMPLE. On the other hand, there are categories in which every morphism is an isomorphism; such categories are called *groupoids*. See Example 3.1.2. If this relation also satisfies symmetry, i.e., it is an equivalence relation, then the category defined in the same way is a groupoid.

Below we define monomorphisms and epimorphisms, which correspond to injective and surjective maps in **Set**. The following definitions may not be what the reader expects, or they may raise some questions as to why they are defined this way:

3.1.6 DEFINITION. Let $\mathbf{C}$ be a category.

1. A morphism $f \in \text{Hom}(A, B)$ is a *monomorphism* if for all objects Z and all morphisms $g, h \in \text{Hom}(Z, A)$, $(fg = fh \iff g = h)$.
2. A morphism $f \in \text{Hom}(A, B)$ is a *epimorphism* if for all objects Z and all morphisms $g, h \in \text{Hom}(B, Z)$, $(gf = hf \iff g = h)$.

3.1.2 REMARK. Let $\mathbf{C}$ be a category and let $f : A \rightarrow B$ be an isomorphism. Then there exists $g : B \rightarrow A$ such that $gf = 1, fg = 1$. Clearly, f is a monomorphism: if $fh_1 = fh_2$, then $gf h_1 = gf h_2$ which implies $h_1 = h_2$. Similarly, f is also epimorphism. Does the converse also hold, as in the category of sets **Set**, where a bijection is equivalent to an injective and surjective map? This is clearly false. A simple example is the category of posets **Pos**, which we mentioned earlier. Let $f : a \rightarrow b$ be the unique morphism from a to b . It is a monomorphism because there is at most one morphism from any c to a , so the cancellation law holds trivially; similarly, it is an epimorphism. But f is clearly not an isomorphism. Later there will be another typical example: the category of rings **Ring** also has this property.

3.2 Universal Properties

The main purpose of category theory is to unify different branches of mathematics, which often share similar structures. For example, in set theory, the “product” generally refers to the Cartesian product; in point-set topology, the “product” refers to the product spaces; and in group theory, the “product” refers to the direct product of groups and so on. While the construction details of these “products” differ, they can all be defined by the universal property that we will study in this section. This is precisely the role of universal properties: they avoid discussing the specific details of what a product is in each context and instead focus on its “function” within the corresponding category.

3.2.1 DEFINITION. Let $\mathbf{C}$ be a category.

- We say that an object I of $\mathbf{C}$ is *initial* in $\mathbf{C}$ if for every object A of $\mathbf{C}$, there exists exactly one morphism $I \rightarrow A$ in $\mathbf{C}$, that is, $\text{Hom}_{\mathbf{C}}(I, A)$ is a singleton for every object A of $\mathbf{C}$.
- We say that an object F of $\mathbf{C}$ is *final* in $\mathbf{C}$ if for every object A of $\mathbf{C}$, there exists exactly one morphism $A \rightarrow F$ in $\mathbf{C}$, that is, $\text{Hom}_{\mathbf{C}}(A, F)$ is a singleton for every object A of $\mathbf{C}$.

3.2.1 EXAMPLE. A category need not have initial or final objects. Consider the special case of the category in example 1, where S is $\mathbf{Z}$ and the relation $\sim$ is $\leq$. This category has no initial or final object. Indeed, an initial object in this category would be an integer i such that $i \leq a$ for all $a \in \mathbf{Z}$, there is no such integer. Similarly, a final object would be an integer larger than every integer, and there is no such integer.

3.2.2 EXAMPLE. In $\mathbf{Set}$, the empty set $\emptyset$ is initial (the unique function from $\emptyset$ to $S \in \mathbf{Set}$ is the empty relation $f = \emptyset$). And $\emptyset$ is the unique initial object in $\mathbf{Set}$: taking any nonempty set A , then there is no function $A \rightarrow \emptyset$. Thus, A cannot be an initial object of $\mathbf{Set}$.

Moreover, $\mathbf{Set}$ also has final objects: for every set A , there is a unique function from A to a singleton $\{p\}$, that is the constant function. Thus, every singleton is final in $\mathbf{Set}$, and the final objects are not unique in this category.

3.2.1 THEOREM. Let $\mathbf{C}$ be a category.

1. If I_1, I_2 are both initial objects in $\mathbf{C}$, then $I_1 \cong I_2$.
2. If F_1, F_2 are both final objects in $\mathbf{C}$, then $F_1 \cong F_2$.

Proof. Since F_1, F_2 are final in $\mathbf{C}$, there exists a unique morphism $f : F_1 \rightarrow F_2$. Also there exists a unique morphism $g : F_2 \rightarrow F_1$. Consider $gf : F_1 \rightarrow F_1$, and this is clearly an identity 1_{F_1} by definition of $\text{Hom}(F_1, F_1)$. Similarly, $fg = 1_{F_2}$. The proof for initial objects is entirely analogous. $\square$

3.2.3 EXAMPLE. We emphasize again Example 2.2. Let 1 denote a singleton. (It does not matter what this element is called.) Then 1 has the property: for all sets $S \in \mathbf{Set}$, there exists a unique function from S to 1 (the constant function).

Phrases of the form “there exists a unique such-and-such satisfying some condition” are common in category theory. Properties such as this are called “universal” because they state how the object being described (in this case, the set 1) relates to the entire category which it lives (in this case, the set of all sets).

3.2.2 DEFINITION. Let $\mathbf{C}$ be a category, we say that an object $A \in \mathbf{C}$ is *universal with respect to some property* if A is terminal in $\mathbf{C}$.

For example 2.3, a singleton 1 is universal with respect to the property: for all $Z \in \mathbf{Set}$, there exists a unique function $Z \rightarrow 1$. And we call this property of “mapping from sets” informally. Similarly, $\emptyset$ is universal with respect to the property: for all $Z \in \mathbf{Set}$, there exists a unique function $\emptyset \rightarrow Z$, and we call this property of “mapping to sets”.

3.2.2 PROPOSITION. Consider the category $\mathbf{Set}$, let $A, B \in \mathbf{C}$. Then the product $A \times B$ satisfies the property: for all $Z \in \mathbf{Set}$ and two functions $f_A : Z \rightarrow A$ and $f_B : Z \rightarrow B$, there exists a unique function $\sigma : Z \rightarrow A \times B$ such that the diagram commutes (where π_A and π_B are the natural projections defined by $\pi_A(a, b) = a, \pi_B(a, b) = (b)$)

$$\begin{array}{ccc}
 & & A \\
 & \nearrow f_A & \nearrow \pi_A \\
 Z & \xrightarrow{\sigma} & A \times B \\
 & \searrow f_B & \searrow \pi_B \\
 & & B
 \end{array}$$

Proof. To prove it, we define a function $\sigma : Z \rightarrow A \times B$ by $\forall z \in Z \sigma(z) = (f_A(z), f_B(z))$. This function makes the diagram commute: $\pi_A \sigma(z) = \pi_A(f_A(z), f_B(z)) = f_A(z)$, showing that $\pi_A \sigma = f_A$ and similarly $\pi_B \sigma = f_B$. Further, the definition is forced by the commutativity of the diagram; so σ is unique. $\square$

In fact, we may consider the proposition in the slice category $\mathbf{Set}_{A,B}$. Then the original proposition is restated as: for all $(Z, f_A, f_B) \in \mathbf{Set}_{A,B}$ there exists a unique commutative diagram in $\mathbf{Hom}((Z, f_A, f_B), (A \times B, \pi_A, \pi_B))$. And this proof tells us that in the category $\mathbf{Set}_{A,B}$, the object $(A \times B, \pi_A, \pi_B)$ is universal with respect to the property: for all $(Z, f_A, f_B) \in \mathbf{Set}_{A,B}$, there exists a unique function σ such that the diagram commutes

$$\begin{array}{ccc}
 & & \begin{array}{c} \xrightarrow{f_A} A \\ \nearrow \pi_A \end{array} \\
 Z & \xrightarrow{\sigma} & A \times B \\
 & & \begin{array}{c} \searrow \pi_B \\ \xrightarrow{f_B} B \end{array}
 \end{array}$$

Similarly, what is described here is actually that $(A \times B, \pi_A, \pi_B)$ is universal with respect to the property. Strictly speaking, if we want to define the product of two sets via a universal property, it is actually defined as $(A \times B, \pi_A, \pi_B)$, but defining it directly as $A \times B$ is unambiguous. In summary, we can use the universal property to define the product of a family of objects in a general category:

3.2.3 DEFINITION. Let $\mathbf{C}$ be a category and A and B objects of $\mathbf{C}$. A *product* of A and B is an object P of $\mathbf{C}$ together with two morphisms $\pi_A : P \rightarrow A$ and $\pi_B : P \rightarrow B$ satisfying the universal property: for all objects $Z \in \mathbf{C}$ and morphisms $f_A : Z \rightarrow A, f_B : Z \rightarrow B$, there exists a unique morphism $\sigma : Z \rightarrow P$ such that the diagram commutes:

$$\begin{array}{ccc}
 & & \begin{array}{c} \xrightarrow{f_A} A \\ \nearrow \pi_A \end{array} \\
 Z & \xrightarrow{\sigma} & P \\
 & & \begin{array}{c} \searrow \pi_B \\ \xrightarrow{f_B} B \end{array}
 \end{array}$$

This is the first concept we formally define using a universal property. It should be noted that concepts defined by universal properties are not unique. In many cases, different objects can satisfy the same universal property (for example, in $\mathbf{Set}$, both $(A \times B, \pi_A, \pi_B)$ and $(B \times A, \pi'_A, \pi'_B)$ satisfy the above universal property; they are both products of A and B). However, precisely because of this, they will simultaneously serve as final or initial objects in the corresponding category, and then by Theorem 3.2.1 they are isomorphic. In other words, they are “essentially” the same.

The following example illustrates well that products are not always of the form of a Cartesian product in different categories (in fact, some categories cannot even define a Cartesian product, such as those that do not include ordered pairs).

3.2.4 EXAMPLE. Consider the example 3.2.1. Does this category have products? Objects of this category are simply the integers $a, b \in \mathbf{Z}$. Call $a \times b$ the product of a and b . For all $z \in \mathbf{Z}$ and $f_a = (z, a), f_b = (z, b)$, that is $z \leq a$ and $z \leq b$, we have $z \leq \min\{a, b\}$, so there exists a morphism $\sigma = (z, \min\{a, b\})$ in $\mathbf{Hom}_{\mathbf{C}}(z, \min\{a, b\})$. Since σ is the only possible element in $\mathbf{Hom}_{\mathbf{C}}(z, \min\{a, b\})$ and $\pi_a \sigma = (z, a) = f_a$ and $\pi_b \sigma = (z, b) = f_b$, $\min\{a, b\}$ is the product of a and b .

3.2.4 DEFINITION. Let $\mathbf{C}$ be a category and A and B objects of $\mathbf{C}$. A *coproduct* of A and B is an object S of $\mathbf{C}$ together with two morphisms $\pi_A : A \rightarrow S$ and $\pi_B : B \rightarrow S$ satisfying the universal property: for all objects $Z \in \mathbf{C}$ and morphisms $f_A : A \rightarrow Z, f_B :$

$B \rightarrow Z$, there exists a unique morphism $\sigma : S \rightarrow Z$ such that the diagram commutes:

$$\begin{array}{ccccc} A & & f_A & & \\ & \searrow \iota_A & & \searrow & \\ & S & \xrightarrow{\sigma} & Z \\ & \nearrow \iota_B & & \nearrow & \\ B & & f_B & & \end{array}$$

3.2.3 PROPOSITION. *The disjoint union is a coproduct in $\mathbf{Set}$.*

Proof. Let A, B be any two sets in $\mathbf{Set}$, and let $A' = \{0\} \times A, B' = \{1\} \times B$. The disjoint union $A \coprod B$ of A and B is the set $A' \cup B' = (\{0\} \times A) \cup (\{1\} \times B)$. And the functions ι_A, ι_B are defined by $\iota_A(a) = (0, a), \iota_B(b) = (1, b)$. Now, let $f_A : A \rightarrow Z, f_B : B \rightarrow Z$ be arbitrary morphisms. If the diagram commutes:

$$\begin{array}{ccccc} A & & f_A & & \\ & \searrow \iota_A & & \searrow & \\ & A \coprod B & \xrightarrow{\sigma} & Z \\ & \nearrow \iota_B & & \nearrow & \\ B & & f_B & & \end{array}$$

then the function $\sigma : A \coprod B \rightarrow Z$ would satisfy $\sigma(c) = f_A(a)$ if $c = (0, a) \in A'$ and $\sigma(c) = f_B(b)$ if $c = (1, b) \in \{1\} \times B$. This definition is forced upon the commutativity, proving that σ is unique. $\square$

This proposition also illustrates once again that, in the same category, there may be different objects satisfying the same universal property. Here, as long as the two defined sets A' and B' are disjoint, they both satisfy the above universal property, and therefore they are isomorphic.

3.2.4 PROPOSITION. *Consider the category $\mathbf{Set}$, let $\sim$ be an equivalence relation defined on a set A . Then the quotient $A/\sim$ satisfies the property: for all $Z \in \mathbf{Set}$ and the function $\varphi : A \rightarrow Z$ satisfying $(a_1 \sim a_2) \rightarrow \varphi(a_1) = \varphi(a_2)$, there exists a unique function $\bar{\varphi} : A/\sim \rightarrow Z$ such that the diagram commutes (where $\pi : A \rightarrow A/\sim$ is the canonical projection defined by $\pi(a) = [a]_\sim$, so that $(a_1 \sim a_2) \rightarrow \pi(a_1) = \pi(a_2)$):*

$$\begin{array}{ccc} A/\sim & \xrightarrow{\bar{\varphi}} & Z \\ & \swarrow \pi \quad \searrow \varphi & \\ & A & \end{array}$$

Proof. To prove it, let $[a]_\sim$ be an arbitrary element of $A/\sim$. If the diagram is commutative, then necessarily

$$\bar{\varphi}([a]_\sim) = \varphi(a),$$

this tells us that $\bar{\varphi}$ is indeed unique if it exists. Hence we have to check that $\bar{\varphi}$ is well-defined, that is, if $[a_1]_\sim = [a_2]_\sim$, then $\varphi(a_1) = \varphi(a_2)$; and indeed

$$[a_1]_\sim = [a_2]_\sim \rightarrow a_1 \sim a_2 \rightarrow \varphi(a_1) = \varphi(a_2).$$

$\square$

In fact, we may consider the proposition in a category $\mathbf{C}$, in which $\text{Ob}(\mathbf{C}) = \{f \in \text{Hom}_{\mathbf{C}}(A, Z) : Z \in \mathbf{C} \wedge (a_1 \sim a_2 \rightarrow f(a_1) = f(a_2))\}$, and $\text{Mor}(\mathbf{C}) = \bigcup_{f_1, f_2} \text{Hom}_{\mathbf{C}}(f_1, f_2)$, the Hom-set $\text{Hom}_{\mathbf{C}}(f_1, f_2)$ consists of all the commutative diagrams of the form

$$\begin{array}{ccc} Z_1 & \xrightarrow{\sigma} & Z_2 \\ & \swarrow f_1 \quad \searrow f_2 & \\ & A & \end{array}$$

Then the original proposition is restated as: for all $\varphi \in \mathbf{C}$, there exists a unique commutative diagram in $\text{Hom}_{\mathbf{C}}(\pi, \varphi)$. From this proof, we have the conclusion that, in the category $\mathbf{C}$ defined above, the object $(A/\sim, \pi)$ (ordered pairs are used to denote objects here to emphasize the quotient set) is universal with respect to the property: for all $(Z, f) \in \mathbf{C}$, there exists a unique function $\bar{\varphi}$ such that the diagram commutes:

$$\begin{array}{ccc} A/\sim & \xrightarrow{\exists! \bar{\varphi}} & Z \\ & \swarrow \pi \quad \searrow \varphi & \\ & A & \end{array}$$

In fact, what is described here is actually that $(A/\sim, \pi)$ is universal with respect to the property. Strictly speaking, if we want to define the quotient set of a set via a universal property, it is actually defined as $(A/\sim, \pi)$, but usually we directly call it $A/\sim$ without ambiguity.

One could attempt to define quotients in a general category using universal properties, but that would require heavy abstraction. Since we prefer to think of quotients via equivalence relations, we shall focus on categories that are closely related to sets — namely, **concrete categories**.

3.2.5 DEFINITION. A *concrete category* is a category $\mathbf{C}$ together with a function σ that assigns to each object A of $\mathbf{C}$ a set $\sigma(A)$ called the *underlying set of A* in such a way that

1. every morphism $A \rightarrow B$ of $\mathbf{C}$ is a function on the underlying sets $\sigma(A) \rightarrow \sigma(B)$;
2. the identity morphism 1_A of $\mathbf{C}$ is the identity function on the underlying set $\sigma(A)$;
3. composition of morphisms in $\mathbf{C}$ agrees with the composition of functions on the underlying sets.

3.2.5 EXAMPLE. Consider the category $\mathbf{Pos}$ of all the partially ordered sets. Define the function $\sigma : \text{Ob}(\mathbf{Pos}) \rightarrow \text{Ob}(\mathbf{Set})$ by $\sigma(\langle P, \leq_P \rangle) = P$. One may check that the category $\mathbf{Pos}$ together with the function σ is indeed a concrete category.

3.2.6 DEFINITION. Let $\mathbf{C}$ be a concrete category and A an object of $\mathbf{C}$. We can talk about $P \subset \mathcal{P}(\sigma(A))$, a set of nonempty disjoint subsets of $\sigma(A)$. A *quotient* of A by P is the object $\bar{A}$ in $\mathbf{C}$ endowed with a morphism $\pi : A \rightarrow \bar{A}$ which satisfies $\pi(a) = \pi(b)$ for all $S \in P$ and $a, b \in S$ satisfying the universal property: for all $Z \in \mathbf{C}$ and morphisms $\varphi : A \rightarrow Z$ such that for all $S \in P$, $\varphi(a) = \varphi(b)$ for all $a, b \in S$, there exists a unique morphism $\bar{\varphi} : \bar{A} \rightarrow Z$ such that the diagram commutes:

$$\begin{array}{ccc} \bar{A} & \xrightarrow{\exists! \bar{\varphi}} & Z \\ & \swarrow \pi \quad \searrow \varphi & \\ & A & \end{array}$$

Concrete categories are useful since one has available not only the properties of a category, but also certain properties of sets, subsets, etc. Since in virtually every concrete category we are interested in, the function σ assigns to an object its underlying set in the usual sense (as in the examples above), we shall denote both the object and its underlying set by the same symbol and omit any explicit reference to σ .

4

Group I: Three Basic Constructions in Group Theory

In this chapter, we introduce the category **Grp** of groups and group homomorphisms. We then discuss three fundamental constructions of groups: direct products, free products, and quotient groups, which may be informally viewed as “parallel combination,” “free amalgamation,” and “compression/collapsing,” respectively. Prior to the discussion of free products, we briefly introduce the notion of free groups, whose underlying idea is shared with that of free products. The chapter concludes with an introduction to group presentations, as an application of quotient groups and free groups.

4.1 Mon and Grp

Monoids and Groups

We have known that an n -ary operation on a set X is a function from the Cartesian product $X^n = X \times X \times \cdots \times X$ of n copies into X . For binary operations $f : X \times X \rightarrow X$, two notations are in wide use. In the *additive notation*, $f(x, y)$ is denoted by $x + y$; then f is an *addition*. In the *multiplicative notation*, $f(x, y)$ is denoted by xy or $x \cdot y$; then f is a *multiplication*. In this chapter, we mostly use the multiplicative notation.

4.1.1 DEFINITION. Let S be a set with a binary operation, written multiplicatively, i.e., the structure $\langle S, \cdot \rangle$. An *identity element* of S is an element e of S such that $ex = x = xe$ for all $x \in S$.

It is easy to prove that there is at most one identity element; that is, if an identity element exists, it is unique. In the multiplicative notation, we usually denote the identity element, if it exists, by 1.

Given the structure $\langle S, \cdot \rangle$, let $n \geq 1$ and let $x_1, \dots, x_n \in S$. Now, we define the *product* of $x_1, \dots, x_n \in S$. If $n = 1$, then $x \in S$ is a product of x_1 if and only if $x = x_1$. If $n \geq 2$, then $x \in S$ is a product of $x_1, \dots, x_n$ if and only if, for some $1 \leq k < n$, x is a product $x = yz$ of a product y of $x_1, \dots, x_k$ and a product z of $x_{k+1}, \dots, x_n$. If S has an identity element 1, we may define 1 is the product of $x_1, \dots, x_n$.

In this definition, if $n = 2$, then $k = 1$, so that $x \in S$ is a product of x_1, x_2 if and only if $x = x_1x_2$. If $n = 3$, then $k = 1$ or $k = 2$, and $x \in S$ is a product of x_1, x_2, x_3 if and only if $x = yz$, where either $y = x_1$ and $z = x_2x_3$, or $y = x_1x_2$ and $z = x_3$; that is, either $x = x_1(x_2x_3)$ or $x = (x_1x_2)x_3$. This shows that the product of x_1, x_2, x_3 can have two different values. As n gets larger, the situation becomes even worse. And there is a special case that we are very Interested in.

4.1.2 DEFINITION. A binary operation on a set S (wirtten multiplicatively) is associative when $(xy)z = x(yz)$ for all $x, y, z \in S$.

Then, associativity states that the products with 3 terms do not depend on the placement of parentheses. This may extend to all products (one may check this), that is, under

the associative multiplication, all the products of n given elements $x_1, \dots, x_n$ (in that order) are equal. Then the product of $x_1, \dots, x_n$ (in that order) is denoted by $x_1 x_2 \cdots x_n$.

It should be noted that, regarding the definition of the product, we do not have commutativity like we do for addition of natural numbers and this does not even hold for the case $n = 2$. And we are also interested in this special case.

4.1.3 DEFINITION. A binary operation on a set S (written multiplicatively) is *commutative* when $xy = yx$ for all $x, y \in S$.

4.1.4 DEFINITION. A *semigroup* is a structure $\langle S, \cdot \rangle$ where S is a set and $\cdot$ is an associative operation on S . A semigroup with an identity element is a *monoid*, i.e., the structure $\langle S, \cdot, 1 \rangle$. A semigroup or monoid is commutative if $\cdot$ is commutative.

4.1.1 REMARK. Let $\mathbf{C}$ be a category and $A \in \mathbf{C}$ an object. The set of all morphisms from A to itself is denoted $\text{End}_{\mathbf{C}}(A) = \text{Hom}_{\mathbf{C}}(A, A)$. One may check that $\langle \text{End}_{\mathbf{C}}(A), \circ \rangle$ is a monoid in which $\circ$ is the composition of $\mathbf{C}$:

- the composition of two morphisms $f, g \in \text{End}_{\mathbf{C}}(A)$ is also contained in $\text{End}_{\mathbf{C}}(A)$;
- the composition is associative by the definition of category;
- necessarily, $1_A \in \text{End}_{\mathbf{C}}(A)$ and $f1_A = f = 1_A f$ for all $f \in \text{End}_{\mathbf{C}}(A)$.

In other words, every object A of every category $\mathbf{C}$ determines a monoid $\text{End}_{\mathbf{C}}(A)$, the monoid of morphisms from A to A . In particular, a category with a single object is a monoid. Now, given a monoid M , we may construct a category $\mathbf{C}_M$ whose object set is a singleton $\{*\}$ and with $\text{Hom}(*, *) = M$, one may check that $\mathbf{C}_M$ is a category.

When no confusion arises, we usually denote $\langle S, \cdot \rangle$ or $\langle S, \cdot, 1 \rangle$ simply by its underlying set in the usual sense, namely the set S . Then $\mathbf{N}, \mathbf{Z}, \mathbf{Q}, \mathbf{R}$ and $\mathbf{C}$ are commutative monoids under multiplication and commutative monoids under addition. Since the multiplication in the sense of matrices is associative but not commutative, $M_n(\mathbf{R})$ is a non-commutative monoid under the multiplication.

4.1.1 PROPOSITION. In a semigroup S , we define the n th power a^n of $a \in S$ the product of $x_1, \dots, x_n$ in which $x_1 = \dots = x_n = a$. If S is a monoid, we would consider the 0th power a^0 of a . Then the following properties hold for all $a, b \in S$ and $m, n \geq 1$:

1. $a^m a^n = a^{m+n}$;
2. $(a^m)^n = a^{mn}$;
3. if S is a monoid, then $\forall n \in \mathbf{N}, a^0 = 1 = 1^n$;
4. if S is commutative, then $(ab)^n = a^n b^n$

In some cases, we will also make extensive use of additive notation. Here, we first lay down some conventions. In a semigroup whose operation is denoted additively, we denote the identity element (if it exists) 0; the product of $x_1, x_2, \dots, x_n$ becomes their sum $x_1 + \dots + x_n$; the n th power of $a \in S$ becomes the *integer multiple* na (the sum $x_1 + \dots + x_n$ in which $x_1 = \dots = x_n = a$). Under additive notation, Proposition 1 becomes the following form:

4.1.2 PROPOSITION. In a semigroup $\langle S, + \rangle$, the following properties hold for all $a \in S$ and all the integers $m, n \geq 1$ ($m, n \geq 0$ if S is a monoid)

1. $ma + na = (m + n)a$;
2. $m(na) = (mn)a$;
3. if S is a monoid, then $\forall n \in \mathbf{N}, 0a = 0 = n0$;
4. if S is commutative, then $n(a + b) = na + nb$.

For products, there is another case we are interested in: 1 is the product of two elements. Consider a monoid M . Suppose $1 = x_1y$ and $1 = x_2y$; we say y is *left invertible*. Then we can prove $x_1 = x_2$. This only requires proving that a left invertible element y satisfies the left cancellation law: assume $ya = yb$, then $x_1(ya) = x_1(yb)$, i.e., $(x_1y)a = (x_1y)b$, so $a = b$. In particular, $x_1 = x_2$, which is called the *left inverse* of y . Similarly, we can prove that if $1 = xy_1 = xy_2$, we call x *right invertible*; then a right invertible element satisfies the right cancellation law, so $y_1 = y_2$, which is called the *right inverse* of x .

The reader can also prove that when x has both a left inverse and a right inverse, they are equal, and are denoted x^{-1} ; in this case x is said to be *invertible*. The set of all invertible elements of a monoid M is denoted by $M^\times$, one may check that $M^\times$ is a group and is called the *group of units of M* . There certainly exists a monoid in which every element is invertible, such a monoid is precisely a *group*.

4.1.5 DEFINITION. A *group* $\langle G, \cdot, 1 \rangle$ is a monoid in which every element is invertible. Formally, a group is a set G together with a binary operation on G such that

- G_1 . $\forall x, y, z((xy)z) = x(yz)$;
- G_2 . $\forall x((x1 = x) \wedge (1x = x))$;
- G_3 . $\forall x \exists y((xy = 1) \wedge (yx = 1))$.

4.1.2 REMARK. An *automorphism* of an object of a category $\mathbf{C}$ is an isomorphism from A to itself. The set of all the automorphisms of A is denoted $\text{Aut}_{\mathbf{C}}(A)$. One may check that $\langle \text{Aut}_{\mathbf{C}}(A), \circ, 1_A \rangle$ is a group in which $\circ$ is the composition of $\mathbf{C}$ and 1_A is the identity morphism:

- the composition of two isomorphisms $f, g \in \text{Aut}_{\mathbf{C}}(A)$ is also an isomorphism;
- the composition is associative by the definition of category;
- necessarily, $1_A \in \text{Aut}_{\mathbf{C}}(A)$ and $f1_A = f = 1_Af$ for all $f \in \text{Aut}_{\mathbf{C}}(A)$;
- every automorphism $f \in \text{Aut}_{\mathbf{C}}(A)$ has an inverse by the definition of isomorphism.

In other words, every object A of every category $\mathbf{C}$ determines a group $\text{Aut}_{\mathbf{C}}(A)$, the group of automorphisms of A . In particular, a groupoid with a single object is a group. Now, given a group G , we may construct a category $\mathbf{C}_G$ whose object set is a singleton $\{*\}$ and with $\text{Hom}(*, *) = G$. Defining the composition $\circ$ by $g \circ h = gh$, one may check that $\mathbf{C}_G$ is a groupoid. Therefore, one may define that group is a groupoid with a single object together with its composition informally.

4.1.1 EXAMPLE. $\langle \mathbf{Z}, + \rangle, \langle \mathbf{Q}, + \rangle, \langle \mathbf{R}, + \rangle$ and $\langle \mathbf{C}, + \rangle$ all are groups. But $\langle \mathbf{N}, + \rangle$ is not a group, and $\mathbf{Z}, \mathbf{Q}, \mathbf{R}, \mathbf{C}$ are not groups under multiplication, since 0 has no inverse. However $\mathbf{Q} - \{0\}, \mathbf{R} - \{0\}$ and $\mathbf{C} - \{0\}$ all are groups under multiplication.

4.1.2 EXAMPLE. $M_n(\mathbf{R})$ of all $n \times n$ matrices with entries in $\mathbf{R}$ is a group under addition, but not under multiplication; However, invertible $n \times n$ matrices constitute a group under multiplication. And the group of invertible $n \times n$ matrices with entries in $\mathbf{R}$ is denoted $\text{GL}_n(\mathbf{R})$.

In fact, in the process of constructing groups step by step, we have already derived many important basic properties of groups. However, it is not a bad idea to list them again here, and we will also supplement some fundamental properties that groups satisfy.

4.1.3 PROPOSITION. Let $\langle G, \cdot, 1 \rangle$ be a group.

1. the identity 1 is unique;
2. the inverse is also unique;
3. $xy_1 = xy_2y_1 = y_2$;

4. $x_1y = x_2yx_1 = x_2$;
5. $(x^{-1})^{-1} = x$;
6. $(x_1x_2 \cdots x_n)^{-1} = x_n^{-1}x_{n-1}^{-1} \cdots x_1^{-1}$.

Proof. We are already familiar with the first four propositions; now we prove only the latter two. In a group, $uv = 1$ implies $v = 1v = u^{-1}uv = u^{-1}$. Hence $x^{-1}x = 1$ implies $x = (x^{-1})^{-1}$. The proof of the sixth proposition can be completed by induction and is omitted here. $\square$

4.1.6 DEFINITION. Let $\langle G, \cdot, 1 \rangle$ be a group. Let $a \in G$ and let n be an arbitrary integer. The n th power a^n of a is defined as follows:

1. If $n \geq 0$, then a^n is the product $x_1x_2 \cdots x_n$ in which $x_1 = x_2 = \cdots = x_n = a$;
2. If $n \leq 0$, then $n = -m$ with $m \geq 0$, then $a^n = (a^m)^{-1}$.

4.1.4 PROPOSITION. In a group G , the following properties hold for all $a, b \in G$:

1. $a^0 = 1, a^1 = a$;
2. $a^m a^n = a^{m+n}$;
3. $(a^m)^n = a^{mn}$;
4. $(a^n)^{-1} = a^{-n} = (a^{-1})^n$.

In this section, we introduce a very important class of groups: commutative groups, more often called *abelian groups*. For such groups, we first establish a notational convention: in most cases, we do not use multiplicative notation but rather additive notation. In this case, the n th power of $a \in G$ becomes the integer multiple na :

1. if $n \geq 0$, then na is the sum $x_1 + \cdots + x_n$ in which $x_1 = \cdots = x_n = a$;
2. if $n \leq 0$, then $n = -m$ with $m \geq 0$, then $na = -(ma)$.

4.1.5 PROPOSITION. In an abelian group G , the following properties hold for all $a, b \in G$ and all integers m, n :

1. $ma + na = (m + n)a$;
2. $m(na) = (mn)a$;
3. $0a = 0 = n0$;
4. $-(na) = (-n)a = n(-a)$;
5. $n(a + b) = na + nb$.

4.1.7 DEFINITION. Let A be a nonempty set. A *permutation* of A is an isomorphism $A \rightarrow A$, i.e., an element of $\text{Aut}_{\text{Set}}(A)$. The *symmetric group* of A , denoted S_A is the group $\text{Aut}_{\text{Set}}(A)$ (we have shown that $\text{Aut}_{\text{C}}(A)$ is a group for any category C). In particular, the group of permutations of the set $\{1, 2, \dots, n\}$ is denoted by S_n .

4.1.3 EXAMPLE. The *dihedral group* D_{2n} of a regular polygon with $n \geq 2$ vertices is the group of rotations and symmetries of that polygon.

4.1.4 EXAMPLE. The *quaternion group* Q_8 is defined by $Q_8 = \{1, -1, i, -i, j, i - j, k, -k\}$ with product $\cdot$ computed as follows

$$\begin{aligned}
 1 \cdot a &= a \cdot 1 = a, & \forall a \in Q_8 \\
 (-1) \cdot (-1) &= 1, & (-1) \cdot a = a \cdot (-1) \cdot a = -a, & \forall a \in Q_8 \\
 i \cdot i &= j \cdot j = k \cdot k = -1 \\
 i \cdot j &= k, & j \cdot i &= -k \\
 j \cdot k &= i, & k \cdot j &= -i \\
 k \cdot i &= j, & i \cdot k &= -j
 \end{aligned}$$

Homomorphisms

For two semigroups $(S_1, \cdot_G)$ and $(S_2, \cdot_H)$, a *group homomorphism* $\varphi : (G, \cdot_G) \rightarrow (H, \cdot_H)$ is first of all a function between the sets S_1 and S_2 ; but a function considered purely at the set level may completely ignore the multiplication structure of S_1 and S_2 . In other words, φ must be “compatible” with the two semigroups operations. Noth that, for the function $\varphi : S_1 \rightarrow S_2$, there exists a unique function (morphism) $\varphi \times \varphi : S_1 \times S_1 \rightarrow S_2 \times S_2$ such that the following diagram commutes by the universal property of the product of S_2 and S_2 :

$$\begin{array}{ccc} S_1 & \xrightarrow{\varphi} & S_2 \\ \pi_{S_1} \uparrow & & \uparrow \pi_{S_2} \\ S_1 \times S_1 & \xrightarrow{\varphi \times \varphi} & S_2 \times S_2 \\ \pi_{S_1} \downarrow & & \downarrow \pi_{S_2} \\ S_1 & \xrightarrow{\varphi} & S_2 \end{array} ,$$

In other words, give a set-function $\varphi : S_1 \rightarrow S_2$, it uniquely determines a function $\varphi \times \varphi : S_1 \times S_1 \rightarrow S_2 \times S_2$ which maps (a, b) to $(\varphi(a), \varphi(b))$ for every $(a, b) \in S_1 \times S_1$. Now, we agree that the chosen set-function $\varphi : S_1 \rightarrow S_2$ should “take into account” the group structures of S_1 and S_2 . Therefore, we need to consider the operations $\cdot_{S_1} : S_1 \times S_1 \rightarrow S_1$ and $\cdot_{S_2} : S_2 \times S_2 \rightarrow S_2$; These lead us to consider the following diagram:

$$\begin{array}{ccc} S_1 \times S_1 & \xrightarrow{\varphi \times \varphi} & S_2 \times S_2 \\ \downarrow \cdot_{S_1} & & \downarrow \cdot_{S_2} \\ S_1 & \xrightarrow{\varphi} & S_2 \end{array}$$

What requirement could be more natural than asking that this diagram commute ?

4.1.8 DEFINITION. Let $\langle S_1, \cdot_{S_1} \rangle$ and $\langle S_2, \cdot_{S_2} \rangle$ be two groups. A function $\varphi : S_1 \rightarrow S_2$ defines a *semigroup homomorphism* if the diagram displayed above commutes. In other words, the function is a semigroup homomorphism if it satisfies

$$\forall a, b \in S_1, \varphi(a \cdot_{S_1} b) = \varphi(a) \cdot_{S_2} \varphi(b)$$

In summary, a semigroup homomorphism is a function that preserves the semigroup operation. Nevertheless, many of the structures we encounter in practice are monoids, rather than merely semigroups. Besides the operation, a monoid also includes an identity element. Therefore, it is advisable to impose an additional condition on monoid homomorphisms: preserving the identity. In summary, we define a monoid homomorphism as follows:

4.1.9 DEFINITION. Let $\langle M_1, \cdot, 1_{M_1} \rangle$ and $\langle M_2, \cdot, 1_{M_2} \rangle$ be two monoids. A function $\varphi : M_1 \rightarrow M_2$ defines a *monoid homomorphism* if

$$\forall a, b \in M_1, \varphi(a \cdot b) = \varphi(a) \cdot \varphi(b) \text{ and } \varphi(1_{M_1}) = 1_{M_2}.$$

4.1.10 DEFINITION. A **U-monoid** is a monoid $\langle M, \cdot, 1 \rangle$ whose underlying set M in the usual sense is a **U**-set. (**U** is the Grothendieck universe we fixed)

Obviously, the set consisting of all **U**-monoids is a **U**-class. Thus we can attempt to define a **U**-category, whose object set $\text{Ob}(\text{Mon})$ is the **U**-class consisting of all **U**-monoids. For a Hom-set $\text{Hom}(M_1, M_2)$, it consists of all monoid homomorphisms from M_1 to M_2 . Since $\text{Hom}(M_1, M_2)$ is a subset of $\text{Hom}_{\text{Set}}(M_1, M_2)$, and $\text{Hom}_{\text{Set}}(M_1, M_2)$ is a **U**-set (by the closure properties of the universe), it follows that $\text{Hom}(M_1, M_2)$ is naturally a **U**-set. Consequently, $\text{Mor}(\text{Mon})$ is a **U**-class.

Therefore, if we can define the composition of morphisms, then **Mon** is indeed a **U**-category. Having clarified the basic constitution of the category **Grp**, we will no longer use the notation for the universe.

If M_1, M_2, M_3 are monoids and $\varphi : M_1 \rightarrow M_2$ and $\psi : M_2 \rightarrow M_3$ are two monoid homomorphisms (i.e., they preserve both multiplication and the identity). We show that the composition $\psi \circ \varphi$ in the function sense is a monoid homomorphism by the diagram:

$$\begin{array}{ccccc}
 & & (\psi \circ \varphi) \times (\psi \circ \varphi) & & \\
 & \nearrow & & \searrow & \\
 M_1 \times M_1 & \xrightarrow{\varphi \times \varphi} & M_2 \times M_2 & \xrightarrow{\psi \times \psi} & M_3 \times M_3 \\
 \downarrow \cdot_{M_1} & & \downarrow \cdot_{M_2} & & \downarrow \cdot_{M_3} \\
 M_1 & \xrightarrow{\varphi} & M_2 & \xrightarrow{\psi} & M_3 \\
 & \searrow & \psi \circ \varphi & \nearrow &
 \end{array}$$

This diagram commutes because $(\psi \circ \varphi)(a \cdot b) = \psi(\varphi(a \cdot b)) = \psi(\varphi(a) \cdot \varphi(b)) = \psi(\varphi(a)) \cdot \psi(\varphi(b)) = (\psi \circ \varphi)(a) \cdot (\psi \circ \varphi)(b)$. Moreover, $(\psi \circ \varphi)(1_{M_1}) = \psi(\varphi(1_{M_1})) = \psi(1_{M_2}) = 1_{M_3}$, so the identity is preserved.

Furthermore, it is clear that composition is associative because it is composition of functions, and the identity function $1_{M_1} : M_1 \rightarrow M_1$ is a monoid homomorphism. Therefore, **Mon** (the category of monoids) is indeed a category.

We can similarly proceed to define group homomorphisms. Following the previous approach, we should choose monoid homomorphisms that preserve inverses. However, we will prove that it suffices to require the preservation of the binary operation, as this already guarantees the preservation of the identity and inverses.

4.1.6 PROPOSITION. *Let G_1, G_2 be two groups, then the semigroup homomorphism $\varphi : G_1 \rightarrow G_2$ satisfies the following properties. Therefore, to define a group homomorphism φ , it suffices to require that the function $\varphi : G_1 \rightarrow G_2$ satisfies $\varphi(ab) = \varphi(a)\varphi(b)$.*

1. $\varphi(1_G) = 1_H$;
2. $\varphi(a^{-1}) = \varphi(a)^{-1}$ for every $a \in G$.

Proof. For the first proposition, since $1_H \cdot \varphi(1_G) = \varphi(1_G) = \varphi(1_G \cdot 1_G) = \varphi(1_G)\varphi(1_G)$, $1_H = \varphi(1_G)$ be the cancellation law. For the second proposition, $\varphi(a^{-1})\varphi(a) = \varphi(a^{-1}a) = \varphi(1_G) = 1_H = \varphi(a)^{-1}\varphi(a)$, which implies $\varphi(a^{-1}) = \varphi(a)^{-1}$ by cancellation. $\square$

Similarly, we can define the category **Grp**, whose object class $\text{Ob}(\mathbf{Grp})$ is the class of all groups, and for which a Hom-set $\text{Hom}(G_1, G_2)$ in the morphism class consists of all group homomorphisms from G_1 to G_2 .

4.1.3 REMARK. Obviously, **Mon**, **Grp** are concrete categories. The fundamental reason is that the entire constructions of **Mon** and **Grp** proceed from **Set**, whether it be its objects, morphisms, or composition. We will encounter such a process frequently later on.

4.2 Direct Products

So far, we have defined the categories that we will use very often, **Mon** and **Grp**, in which we can try to define concepts such as products, coproducts, and quotient objects. First, we discuss products.

Let M_1, M_2 be two monoids. We already know that the set $M_1 \times M_2$ is a product in the category **Set**. As for monoids (where we need to define a suitable operation on $M_1 \times M_2$), is it also a product in the category **Mon**? In other words, is the product of M_1 and M_2 in **Mon** supported on the product $M_1 \times M_2$ of the underlying sets? First, let us define an operation on the set $M_1 \times M_2$; we call the operation $\cdot$ *componentwise multiplication*:

$$(a_1, b_1) \cdot (a_2, b_2) := (a_1 a_2, b_1 b_2) \text{ for any } a_1, a_2 \in M_1, b_1, b_2 \in M_2.$$

4.2.1 PROPOSITION. $\langle M_1 \times M_2, \cdot \rangle$ is a monoid and is a product in **Mon**. Furthermore, let G_1 and G_2 be two groups, then $\langle G_1 \times G_2, \cdot \rangle$ is a group and a product in **Grp**.

Proof. There exists a unique function $\varphi_1 \times \varphi_2$ such that the diagram commutes because the set $M_1 \times M_2$ is a product of M_1 and M_2 in **Set**. So we only need to check that $\varphi_1 \times \varphi_2$ is a monoid homomorphism.

$$\begin{array}{ccc}
 & \varphi_1 & \\
 & \nearrow & \\
 Z & \xrightarrow{\varphi_1 \times \varphi_2} & M_1 \times M_2 \\
 & \searrow & \\
 & \varphi_2 & \\
 & \searrow & \\
 & & M_1 \\
 & \pi_1 & \\
 & \searrow & \\
 & & M_2 \\
 & \pi_2 &
 \end{array}$$

□

4.2.1 DEFINITION. Let I be a set and let $\{M_i\}_{i \in I}$ be an I -indexed set of monoids. Define the following operation on the product $\prod_{i \in I} M_i$ (represent elements of $\prod_{i \in I} M_i$ as $(x_i)_{i \in I}$) of the underlying sets:

$$(x_i)_{i \in I} \cdot (y_i)_{i \in I} = (x_i y_i)_{i \in I}$$

One can easily show that $\prod M_i$ is a monoid (the identity element is $(1)_{i \in I}$; if each x_i in $(x_i)_{i \in I}$ is invertible, then $(x_i)_{i \in I}^{-1} = (x_i^{-1})_{i \in I}$). Moreover, if each M_i is a group, then $\prod_{i \in I} M_i$ is also a group. This is called the *direct product* of $\{M_i\}_{i \in I}$. For each $j \in I$, define the projection homomorphism

$$\pi_j : \prod_{i \in I} M_i \rightarrow M_j : (x_i)_{i \in I} \mapsto x_j.$$

The universal property of products can also be extended to a more general form, which we shall not elaborate further here (The same holds for the case of coproducts): let $\{M_i\}_{i \in I}$ be a set of monoids (resp. groups), then $\prod_{i \in I} M_i$ together with the canonical projections $\{\pi_i\}_{i \in I}$ is universal with respect to the property: for any $Z \in \mathbf{Grp}$ and $\{\varphi_i : Z \rightarrow M_i\}$, there is a unique homomorphism $\tilde{\varphi}$ such that $\pi_i \circ \tilde{\varphi} = \varphi_i$ for all $i \in I$.

Using the same approach, can we define coproducts? That is, let M_1 and M_2 be two monoids. Can we define a suitable operation on the disjoint union $M_1 \amalg M_2$ of their underlying sets so that it becomes a monoid, and then prove that it satisfies the universal property of the coproduct?

This approach does not work. Specifically, if the above disjoint union were a coproduct, then by its universal property it would necessarily come with two homomorphisms $\iota_1 : M_1 \rightarrow M_1 \amalg M_2$ and $\iota_2 : M_2 \rightarrow M_1 \amalg M_2$. Thus $\iota_1(1_{M_1}) = (0, 1_{M_1})$ and $\iota_2(1_{M_2}) = (1, 1_{M_2})$ would both be identity elements in $M_1 \amalg M_2$, contradicting the uniqueness of the identity element.

In summary, defining coproducts in non-concrete categories is rather involved, and we will devote the next two sections to gradually constructing the coproduct.

4.3 Free Groups and Free Products

Free Groups

Free objects are another type of object we focus on in these notes, in addition to product objects, coproduct objects, and quotient objects. In this section, we will define free objects using a universal property. As the name suggests, a free object is “free” in the category, though of course we cannot describe mathematical objects in such vague language. To gain a deep understanding of what “free” means, we start with free monoids. (The reason

we directly consider monoids rather than general semigroups is that, in our later studies, we will mostly encounter the two structures: monoids and groups.)

First, a free monoid is generated by a set A , and we denote it by $M(A)$. In other words, to construct a “free” monoid from the set A , the most natural approach is to ensure that the elements of this monoid are subject to no “restrictions” other than those necessary to support the monoid structure. Concretely, we informally define a relation as a relation as an equality between products of elements of A . The relations required for the monoid structure are those of the form $(xy)z = x(yz)$ and $x1 = x = 1x$. A free monoid is then a monoid with no extra relations beyond these.

The first problem we face is: how to construct a monoid out of nothing but a set? To this, we playfully come up with a trick: we “formally” regard the concatenation of two elements from left to right as their product and we “formally” introduce an new element which is not in A as the identity element. The precise mathematical construction proceeds as follows:

Let A be a set. We call A the *alphabet*. A *word* ω on the set A is a finite sequence $\omega = (a_1, a_2, \dots, a_n)$ where $a_i \in A$. The elements of $M(A)$ are all words on the set A . Two words $(a_1, \dots, a_n)$ and $(b_1, \dots, b_m)$ in $M(A)$ are equal if and only if $m = n$ and $a_i = b_i$ for all $i = 1, \dots, n$. And we define the multiplication on $M(A)$ as the *concatenation*:

$$(a_1, a_2, \dots, a_n) \cdot (b_1, b_2, \dots, b_m) = (a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_m).$$

It is immediate that concatenation is associative. The *empty sequence* $()$ is an identity element. Thus, $\langle M(A), \cdot, () \rangle$ is a monoid. (Moreover, apart from the necessary associativity and identity laws, there are no other relations. Concretely, two words are equal if and only if they have the same length and the same letters in each position.) Moreover, every sequence can be written uniquely as a product of one-term sequences:

$$(a_1, a_2, \dots, a_n) = (a_1)(a_2) \cdots (a_n).$$

If every element a of A is identified with the corresponding one-term sequence (a) , then $A \subset M(A)$ and every element of $M(A)$ can be written uniquely as a product of elements of A . The usual notation makes this identification transparent by writing every sequence $(a_1, a_2, \dots, a_n)$ as a product $a_1 a_2 \cdots a_n$ in the alphabet A .

Consider the free group. The situation now becomes more complicated, because a group requires not only the associative law and the identity law, but also the inverse relations $xx^{-1} = x^{-1}x = 1$. The ensuing problem is: how to define inverses? We can still play a small trick: simply take a set that is disjoint from A (somewhere in the universe) and equipped with a bijection to A , and declare the element corresponding to $a \in A$ to be its inverse. Concretely, we proceed as follows:

Let A be a set. Let A^{-1} be a set that is disjoint from A and comes with a bijection $a \mapsto a^{-1}$ of A onto A^{-1} . Then we obtain a new alphabet $A \cup A^{-1}$. (Once our free group is constructed, a^{-1} will be the inverse of a .) It is convenient to denote the inverse bijection $A^{-1} \rightarrow A$ by $b \rightarrow b^{-1}$, so that $(a^{-1})^{-1} = a$ for all $a \in A$. Denote the free monoid on $A \cup A^{-1}$ the set $W(A)$ of all words on $A \cup A^{-1}$.

4.3.1 DEFINITION. Let A be a set. A word $\omega = (a_1, \dots, a_n) \in W(A)$ is *reduced* if $a_{i+1} \neq a_i^{-1}$ for all $1 \leq i < n$.

For example, the empty word and all one-letter words are reduced, for want of consecutive letters. If $A = \{a, b, c, \dots\}$, then (a, b, c) and (a, a, a) are reduced, but (a, b, b^{-1}, c) is not reduced.

4.3.2 DEFINITION. In $W(A)$, we write $a \xrightarrow{1} b$ when $a = (a_1, a_2, \dots, a_n)$, $a_{i+1} = a_i^{-1}$ and $b = (a_1, \dots, a_{i-1}, a_{i+2}, \dots, a_n)$ for some $1 \leq i < n$; we write $a \xrightarrow{k} b$ when $k \geq 0$ and $a \xrightarrow{1} b_1 \xrightarrow{1} b_2 \xrightarrow{1} \cdots \xrightarrow{1} b_k = b$ for some $b_1, \dots, b_k \in W(A)$.

4.3.1 LEMMA. For any word $a \in W(A)$, there is a reduction $a \rightarrow b$ to a reduced b .

Proof. By induction on the length of a . If a is reduced, then $b = a$ serves. Otherwise, $a \xrightarrow{1} c$ for some $c \in W$, $c \rightarrow b$ for some reduced $b \in W$ since c is shorter than a , and then $a \rightarrow b$. $\square$

4.3.2 LEMMA. *If $a \rightarrow b$ and $a \rightarrow c$, then $b \rightarrow d$ and $c \rightarrow d$ for some d . In other words, for every word $a \in W$, there is a unique reduced word b such that $a \rightarrow b$.*

Proof. Say $a \xrightarrow{k} b$ and $a \xrightarrow{l} c$. The result is trivial if $k = 0$ or $l = 0$. We first prove this statement when $l = 1$ by induction on k . We have $a \xrightarrow{1} c$. If $k = 1$. Let $a = (a_1, \dots, a_n)$. We have $a_{i+1} = a_i^{-1}$ and $b = (a_1, \dots, a_{i-1}, a_{i+2}, \dots, a_n)$, for some $1 \leq i < n$; also $a_{j+1} = a_j^{-1}$ and $c = (a_1, \dots, a_{j-1}, a_{j+2}, \dots, a_n)$, for some $1 \leq j < n$. Since $b \neq c$, we have $i \neq j$ and may assume $i < j$. If $j = i + 1$, then $a_i = a_{i+1}^{-1} = a_{j+1} = a_{i+2}$, $(a_{i-1}, a_{i+2}, a_{i+3}) = (a_{j-2}, a_{j-1}, a_{j+2})$, and $b = c$; hence $j \geq i + 2$. Then a_i and a_{i+1} are consecutive letters of c and a_{j+1} are consecutive letters of b , and

$$d = (a_1, \dots, a_{i-1}, a_{i+2}, \dots, a_{j-1}, \dots, a_{j+2}, \dots, a_n)$$

serves. Now, let $k > 1$, so that $a \xrightarrow{1} u \xrightarrow{k-1} b$ for some u . If $u = c$, then $d = b$ serves. Otherwise, $u \xrightarrow{1} v$ and $c \xrightarrow{1} v$ for some v :

$$\begin{array}{ccccc} a & \xrightarrow{1} & u & \longrightarrow & b \\ \downarrow 1 & & \downarrow 1 & & \downarrow \\ c & \xrightarrow{1} & v & \longrightarrow & d \end{array}$$

The induction hypothesis, applied to $u \xrightarrow{k-1} b$ and $u \xrightarrow{1} v$, then yields $b \rightarrow d$ and $c \xrightarrow{1} v \xrightarrow{d}$ for some d . Now, the statement holds when $l \leq 1$; the general case is proved by induction on l :

$$\begin{array}{ccccc} a & \xrightarrow{1} & u & \longrightarrow & c \\ \downarrow & & \downarrow & & \downarrow \\ b & \longrightarrow & v & \longrightarrow & d \end{array}$$

If $l > 1$, then $a \rightarrow b$ and $a \xrightarrow{1} u \xrightarrow{l-1} c$ for some u . By the case $l = 1$, $b \rightarrow v$ and $u \rightarrow v$ for some v . The induction hypothesis, applied to $u \rightarrow v$ and $u \xrightarrow{l-1} c$, then yields $b \rightarrow v \rightarrow d$ and $c \rightarrow d$ for some d . $\square$

Now, we define the reduction function $R : W(A) \rightarrow W(A)$ by setting $R(\omega)$ the unique reduced word b such that $\omega \rightarrow b$. Let $F(A)$ be the set of all reduced words in $W(A)$, that is, the image of the reduction function R we have just defined. Define the operation on $F(A)$ by concatenation and reduction: for reduced words a, b , define $a \cdot b$ as the reduction of the concatenation of a and b :

$$a \cdot b = R(ab)$$

4.3.3 THEOREM. $\langle F(A), \cdot, () \rangle$ is a group.

Proof. If $a \xrightarrow{1} b$, then $ac \xrightarrow{1} bc$ and $ca \xrightarrow{1} cb$ for all $c \in W(A)$. Hence $a \rightarrow b$ implies $ac \xrightarrow{b} c$ and $ca \rightarrow cb$ for all $c \in W(A)$. If now $a, b, c \in W$ are reduced, then $ab \rightarrow a \cdot b$ and $bc \rightarrow b \cdot c$ yield

$$abc \rightarrow (a \cdot b)c \rightarrow (a \cdot b) \cdot c \text{ and } abc \rightarrow a(b \cdot c) \rightarrow a \cdot (b \cdot c).$$

Hence $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ by the lemma above. The empty word $1 = ()$ is reduced and is the identity element of $F(A)$, since $1 \cdot a = R(1a) = R(a) = a$ and $a \cdot 1 = R(a1) = R(a) = a$ for all $a \in F(A)$. The inverse of a reduced word $a = (a_1, \dots, a_n)$ is clearly $a^{-1} = (a_n^{-1}, \dots, a_1^{-1})$; indeed, a^{-1} is reduced, since $a_i^{-1} \neq (a_{i-1}^{-1})^{-1}$ for all $i > 1$, and $aa^{-1} \rightarrow 1, a^{-1}a \rightarrow 1$. Thus, $F(A)$ is a group. $\square$

The group $F(A)$ should be “generated” by A . Strictly, A is not a subset of $F(A)$. However, there is a *canonical injection* $\eta : A \rightarrow F(A)$ defined by $a \mapsto (a)$, which is conveniently extended to $A \cup A^{-1}$ so that $\eta : a^{-1} \mapsto (a^{-1})$; then $F(A)$ is generated by $\eta(A)$: if $a = (a_1, \dots, a_n)$ is reduced, then $a = (a_1)(a_2) \cdots (a_n) = (a_1) \cdot (a_2) \cdots (a_n) = \eta(a_1) \cdot \eta(a_2) \cdots \eta(a_n)$. Moreover, $\eta(a^{-1}) = \eta(a)^{-1}$ for all $x \in A$. Hence every $a \in F(A)$ is a product of elements of $\eta(A)$ and the inverses of elements of $\eta(A)$.

4.3.4 PROPOSITION. *Let $\eta : A \rightarrow F(A)$ be the canonical injection. Then the group $F(A)$ satisfies the property: for every function $f : A \rightarrow G$, there is a unique homomorphism $\varphi : F(A) \rightarrow G$ such that $f = \varphi \circ \eta$.*

Proof. We first show the uniqueness. Let $\varphi : F(A) \rightarrow G$ be a homomorphism such that $f = \varphi \circ \eta$. Extend f to $A \cup A^{-1}$ so that $f(a^{-1}) = f(a)^{-1}$ for all $a \in A$. For every $a \in A$, we have $\varphi(\eta(a)) = f(a)$ and $\varphi(\eta(a^{-1})) = \varphi(\eta(a)^{-1}) = f(a)^{-1} = f(a^{-1})$. If $a = (a_1, \dots, a_n)$ is reduced, then necessarily

$$\varphi(a) = \varphi(\eta(a_1) \cdots \eta(a_n)) = f(a_1) \cdots f(a_n)$$

since φ is a homomorphism. Hence φ is unique. It remains to show that the function $\varphi : F(A) \rightarrow G$ defined by $\varphi(\omega) = f(a_1) \cdots f(a_n)$ is a group homomorphism. First, we can extend φ to all of $W(A)$ by using the formula above for every word. Then $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in W$. Moreover, $a \xrightarrow{1} b$ implies $\varphi(a) = \varphi(b)$: indeed, if $a = (a_1, \dots, a_n), a_{i+1} = a_i^{-1}$, and $b = (a_1, \dots, a_{i-1}, a_{i+2}, \dots, a_n)$ for some $1 \leq i < n$, then

$$\begin{aligned} \varphi(a) &= f(a_1) \cdots f(a_{i-1})f(a_i)f(a_{i+1})f(a_{i+2}) \cdots f(a_n) \\ &= f(a_1) \cdots f(a_{i-1})f(a_{i+2}) \cdots f(a_n) = \varphi(b) \end{aligned}$$

since $f(a_{i+1}) = f(a_i)^{-1}$. Therefore $a \rightarrow b$ implies $\varphi(a) = \varphi(b)$. If now a and b are reduced words, then $\varphi(a \cdot b) = \varphi(ab) = \varphi(a)\varphi(b)$. $\square$

4.3.5 COROLLARY. *If the group G is generated by a subset A , then there is a surjective homomorphism from $F(A)$ to G . In other words, every group G is the homomorphism image of a free group.*

Proof. Consider the inclusion function $\iota : A \rightarrow G$. By the universal property of free group, there is a group homomorphism $\varphi : F(A) \rightarrow G$ such that $\varphi \circ \eta$ is the inclusion $\iota : A \rightarrow G$, then $\text{im } \varphi = G$ since $\text{im } \varphi$ contains every generator $a = \varphi(\eta(a))$ of G . $\square$

4.3.6 COROLLARY. *Let G be a group generated by a subset A , i.e., $G = \{a_1^{\epsilon_1} \cdots a_n^{\epsilon_n} : a_i \in A, \epsilon_i = \pm 1\}$. If every reduced word $g = a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}$, where $a_i \in A$ and $\epsilon_i = \pm 1$, is not equal to 1, then G is free on the set A .*

Proof. Let $F(B)$ be the free group on the set B , where B is a set for which there exists a bijection $f : B \rightarrow A$. Now, consider the diagram

$$\begin{array}{ccc} & & H \\ & \nearrow \psi & \uparrow \tilde{\varphi} \\ F(B) & \xrightarrow{\varphi} & G \\ \uparrow \eta & & \uparrow \iota \\ B & \xrightarrow{f} & A \end{array}$$

$\searrow g$

Since $F(B)$ is free on B , there is a unique homomorphism $\varphi : F(B) \rightarrow G$ such that $\varphi(\eta(b)) = \iota(f(b))$ for all $b \in B$. Now φ is surjective since $\varphi(\eta(B)) = \iota(A) = A$ generates G ; moreover, φ is injective, for if $\omega = (b_1^{\epsilon_1}, \dots, b_n^{\epsilon_n})$ is a nontrivial reduced word on B , then $\omega \notin \ker \varphi$ because, by hypothesis, $\varphi(\eta(\omega)) = a_1^{\epsilon_1} \cdots a_n^{\epsilon_n} \neq 1$. Therefore, φ is an isomorphism and G is free on the set A . $\square$

Here we make some notational remarks. In what follows, we will often identify $a \in A$ with $\eta(a) = (a) \in F(A)$, and instead of using sequence notation, we will write a word as an actual word in the usual sense. For example, we will write (b, a, r, r, y) as *barry*. Meanwhile, we will also employ exponential notation. For example, instead of writing “barry”, we would write bar^2y ; this causes no ambiguity. With this notation, we sometimes directly regard η as the inclusion ι , and say that $F(A)$ is the free group generated by A .

4.3.3 DEFINITION. Let $\mathbf{C}$ be a concrete category and A a nonempty set. A *free object on the set A* is the object F together with a function $\eta : A \rightarrow F$ satisfying the universal property: for all $Z \in \mathbf{C}$ and function $f : A \rightarrow Z$, there exists a unique morphism $\varphi : F \rightarrow Z$ such that the diagram commutes:

$$\begin{array}{ccc} F & \xrightarrow{\exists! \varphi} & Z \\ & \nwarrow \eta \quad \nearrow f & \\ & A & \end{array}$$

4.3.1 EXAMPLE. The group $F(\{x, y\})$ is a coproduct of $\mathbf{Z}$ and $\mathbf{Z}$ in the category $\mathbf{Grp}$.

Proof. Given the universal property of free group $F(\{x, y\})$ on the set $\{x, y\}$: for all groups G and functions $f : \{x, y\} \rightarrow G$, there exists a unique group homomorphism $\varphi : F(\{x, y\}) \rightarrow G$ such that the following diagram commutes:

$$\begin{array}{ccc} F(\{x, y\}) & \xrightarrow{\varphi} & G \\ \uparrow \iota & \nearrow f & \\ \{x, y\} & & \end{array}$$

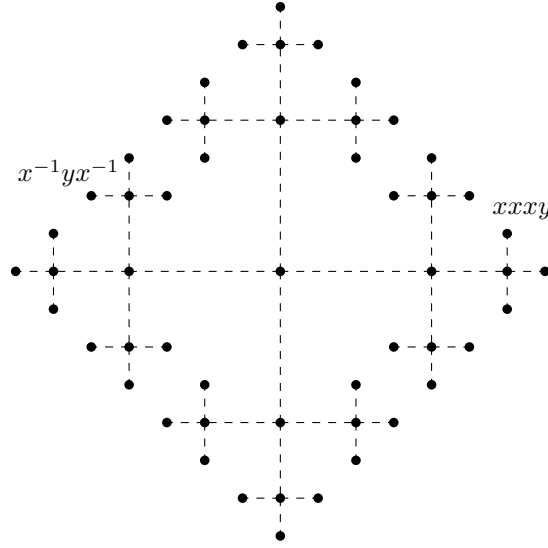
To check that it is a coproduct, we have to check that for any group G and morphisms $f : \mathbf{Z} \rightarrow G, g : \mathbf{Z} \rightarrow G$ there exists a unique morphism $\varphi : F(\{x, y\}) \rightarrow G$ such that the diagram commutes:

$$\begin{array}{ccccc} \mathbf{Z} & & & & \\ & \searrow \iota_1 & & \nearrow f & \\ & & F(\{x, y\}) & \xrightarrow{\varphi} & G \\ & \nearrow \iota_2 & & \nwarrow g & \\ \mathbf{Z} & & & & \end{array}$$

where $\iota_1 : \mathbf{Z} \rightarrow F(\{x, y\})$ and $\iota_2 : \mathbf{Z} \rightarrow F(\{x, y\})$ are homomorphisms defined by $n \mapsto x^n$ and $m \mapsto y^m$, respectively. Define a function $f_0 : \{x, y\} \rightarrow G$ by $f_0(x) = f(1)$ and $f_0(y) = g(1)$. By the universal property of the free group $F(\{x, y\})$, there exists a unique homomorphism $\varphi : F(\{x, y\}) \rightarrow G$ such that $\varphi \circ \iota = f_0$, i.e., $\varphi(x) = f(1)$ and $\varphi(y) = g(1)$.

Now we check the coproduct diagram: for ι_1 , we have $\varphi \circ \iota_1(n) = \varphi(x^n) = \varphi(x)^n = f(1)^n = f(n)$, since $f : \mathbf{Z} \rightarrow G$ is a group homomorphism determined by $f(1)$. Similarly, for ι_2 , $\varphi \circ \iota_2(m) = g(m)$. Thus $\varphi \circ \iota_1 = f$ and $\varphi \circ \iota_2 = g$, so the diagram commutes. It is clear that φ is unique since $\{x, y\}$ generates $F(\{x, y\})$ and $\varphi(x) = f(1), \varphi(y) = g(1)$. Therefore $F(\{x, y\})$ is the coproduct $\mathbf{Z} \coprod \mathbf{Z}$ in $\mathbf{Grp}$. $\square$

Next, we take an intuitive approach to understand why free groups are called “free”. The figure below shows the Cayley graph of the free group $F(\{x, y\})$ (in a Cayley graph, vertices represent group elements, and edges represent “multiplication by” a generator). Here, rightward dashed edges represent multiplication by x , leftward dashed edges represent multiplication by x^{-1} , upward dashed edges represent multiplication by y , and downward dashed edges represent multiplication by y^{-1} . Starting from the identity element, we obtain the following diagram:


 Figure 4.1: the Cayley graph of the free group $F(x, y)$.

Free Products

This section will answer our earlier question: how to construct coproducts of objects in **Grp** and **Mon**? The free product is a group “generated” by two given groups. As in the previous section’s discussion of free groups, the free product contains no unnecessary relations. Similarly, the operations of the two groups here are different, so we have to construct it using an alphabet. However, unlike before, for a word itself, we can use the existing operations to reduce the word, without needing to introduce a hypothetical set. More specifically:

4.3.4 DEFINITION. Let A and B be two groups such that $A \cap B = \{1\}$. A word $x = (x_1, x_2, \dots, x_n) \in W(A \cup B)$ in the alphabet $A \cup B$ is *reduced* when it does not contain consecutive letters x_i, x_{i+1} such that $x_i, x_{i+1} \in A$ or $x_i, x_{i+1} \in B$.

For example, the empty word, and all one-letter words are reduced. If $a_1, a_2 \in A$ and $b \in B$, then a_1ba_2 is reduced, as long as $a_1, a_2, b \neq 1$, but a_1a_2b is not reduced. In general, when $x = (x_1, \dots, x_n)$ is reduced and $n > 1$, then $x_1, \dots, x_n \neq 1$, and elements of A alternate with elements of B in the sequence $x_1, \dots, x_n$.

4.3.5 DEFINITION. In $W(A \cup B)$ with $A \cap B = \{1\}$, we write $x \xrightarrow{1} y$ when $x = (x_1, \dots, x_n), x_i, x_{i+1} \in A$ or $x_i, x_{i+1} \in B$, and $y = (x_1, \dots, x_{i-1}, x_i x_{i+1}, x_{i+2}, \dots, x_n)$ for some $1 \leq i < n$; we write $x \xrightarrow{k} y$ when $k \geq 0$ and $x \xrightarrow{1} y_1 \xrightarrow{1} y_2 \xrightarrow{1} \dots \xrightarrow{1} y_k = y$ for some $y_1, \dots, y_k \in W(A \cup B)$.

There is a condition here: $A \cap B = \{1\}$. In fact, if $A \cap B \neq \{1\}$, then extra relations arise. For example, if $A \cap B = \{1, x\}$, where x satisfies $x^2 = y$ in A and $x^4 = z$ in B , then after reduction the word (x, x, x, x) can be expressed as both (y^2) and (z) , i.e., $(y^2) = (z)$ or $(y^2, z^{-1}) = 1$, which is a relation.

Analogous to the reduced words in the previous section, we have exactly the same conclusion: the reduced form of a word exists and is unique (lemma 3.3.1 and lemma 3.3.2). We will not repeat the proof here.

Now, we define the reduction function $R : W(A \cup B) \rightarrow W(A \cup B)$ by setting $R(x)$ the unique reduced word y such that $x \rightarrow y$. Let $A \amalg B$ be the set of all reduced words in $W(A \cup B)$, that is, the image of the reduction function R . Also, define the operation on $A \amalg B$ by

$$x \cdot y = R(xy).$$

4.3.7 THEOREM. $\langle A \amalg B, \cdot, () \rangle$ is a group.

Proof. If $x \xrightarrow{1} y$, then $xz \xrightarrow{1} yz$ and $zx \xrightarrow{1} zy$ for all $z \in W(A \cup B)$. Hence $x \rightarrow y$ implies $xz \xrightarrow{y} z$ and $zx \rightarrow zy$ for all $z \in W(A \cup B)$. If now $x, y, z \in W(A \cup B)$ are reduced, then $xy \rightarrow x \cdot y$ and $yz \rightarrow y \cdot z$ yield

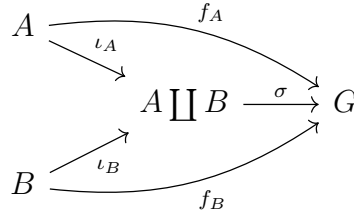
$$xyz \rightarrow (x \cdot y)z \rightarrow (x \cdot y) \cdot z \text{ and } xyz \rightarrow x(y \cdot z) \rightarrow x \cdot (y \cdot z).$$

Hence $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ by the lemma above. The empty word $1 = ()$ is reduced and is the identity element of $A \amalg B$, since $1 \cdot x = R(1x) = R(x) = x$ and $x \cdot 1 = R(x1) = R(x) = x$ for all $x \in A \amalg B$. The inverse of a reduced word $x = (x_1, \dots, x_n)$ is clearly $x^{-1} = (x_n^{-1}, \dots, x_1^{-1})$; indeed, x^{-1} is reduced, since, like x , it does not contain consecutive letters from the same group. Thus, $A \amalg B$ is a group. $\square$

Let A and B be groups such that $A \cap B = \{1\}$. The *free product* of A and B is the group $A \amalg B$ in the theorem together with two functions $\iota_A : A \rightarrow A \amalg B$ and $\iota_B : B \rightarrow A \amalg B$ defined by $\iota_A(x) = (x)$ and $\iota_B(y) = (y)$ for all $x \in A, y \in B$. It is clear that ι_A and ι_B are group homomorphisms and injective as the functions.

If $A \cap B \neq \{1\}$ we replace A and B by isomorphic groups A' and B' such that $A' \cap B' = \{1\}$: for instance, $A' = \{1\} \cup ((A - \{1\}) \times \{0\})$ and $B' = \{1\} \cup ((B - \{1\}) \times \{1\})$, with operations carried from A and B by the bijections $\theta : A \rightarrow A'$ and $\zeta : B \rightarrow B'$: $xy = \theta(\theta^{-1}(x)\theta^{-1}(y))$ for all $x, y \in A'$ and similar for B' ; then $A' \cong A, B' \cong B$ and $A' \cap B' = \{1\}$. Then the free product of A and B is defined as the free product of any copies $A' \cong A$ and $B' \cong B$ such that $A' \cap B' = \{1\}$, with the injections $A \rightarrow A' \rightarrow A' \amalg B'$ and $B \rightarrow B' \rightarrow A' \amalg B'$.

4.3.8 PROPOSITION. *Let $\iota_A : A \rightarrow A \amalg B$ and $\iota_B : B \rightarrow A \amalg B$ be the canonical injections. Then the group $A \amalg B$ satisfies the property: for any group G and homomorphisms $f_A : A \rightarrow G$ and $f_B : B \rightarrow G$, there is a unique homomorphism $\sigma : A \amalg B \rightarrow G$ such that $\sigma \circ \iota_A = f_A$ and $\sigma \circ \iota_B = f_B$:*



Proof. Let A and B be two groups. We may assume $A \cap B = \{1\}$. Then it is convenient to combine ι_A and ι_B into a single function $\lambda : A \cup B \rightarrow A \amalg B$, and to combine f_A and f_B into a single function $\omega : A \cup B \rightarrow G$:

$$\lambda(x) = \begin{cases} \iota_A(x), & x \in A \\ \iota_B(x), & x \in B \end{cases}, \quad \omega(x) = \begin{cases} f_A(x), & x \in A \\ f_B(x), & x \in B \end{cases}$$

Now, every reduced word $x = (x_1, \dots, x_n)$ is a product of some one-letter words $x = \lambda(x_1)\lambda(x_2)\cdots\lambda(x_n)$. If $\sigma \circ \iota_A = f_A$ and $\sigma \circ \iota_B = f_B$, equivalently $\sigma \circ \lambda = \omega$, then $\sigma(x) = \omega(x_1)\omega(x_2)\cdots\omega(x_n)$. Hence σ is unique.

It remains to show the function $\sigma : W(A \cup B) \rightarrow G$ defined by $\sigma(x) = \omega(x_1)\cdots\omega(x_n)$ is a homomorphism. First, we extend σ to all of $W(A \cup B)$ by using the formula for every word. If $x, y \in W(A \cup B)$, then $\sigma(xy) = \sigma(x)\sigma(y)$ for all $x, y \in W(A \cup B)$. Moreover, $x \xrightarrow{1} y$ implies $\sigma(x) = \sigma(y)$: indeed, if $x = (x_1, \dots, x_n)$ and $y = (x_1, \dots, x_{i-1}, x_i x_{i+1}, x_{i+2}, \dots, x_n)$, then $\omega(x_i x_{i+1}) = \omega(x_i)\omega(x_{i+1})$ since f_A is a homomorphism, hence $\sigma(x) = \sigma(y)$. Therefore, $x \rightarrow y$ implies $\sigma(x) = \sigma(y)$. If now, x and y are reduced, then $\sigma(x \cdot y) = \sigma(xy) = \sigma(x)\sigma(y)$. $\square$

In other words, the group $A \amalg B$ together with ι_A, ι_B satisfies the universal property of the coproduct. Thus, we have successfully defined a coproduct structure in the category of groups, namely the free product.

4.4 Subgroups

So far, we have introduced products and coproducts in the category of groups. Next, we will introduce another important object, the quotient group, which will be covered step by step over the next two sections.

4.4.1 DEFINITION. A *submonoid* of a monoid M is a subset N of M that preserves the structure, that is: (i) : $\forall a, b \in N, ab \in N$; (ii) : $1 \in N$. (Associativity holds naturally in any subset.) A *subgroup* of a group G is a submonoid H of G and (iii) : $a \in H \rightarrow a^{-1} \in H$. We denote the relation “ H is a subgroup of G ” by $H \leq G$.

If H is a subgroup of a group G , then the inclusion function $\iota : H \hookrightarrow G$ defined by $\iota(a) = a$ for all $a \in H$, is the group homomorphism of H into G . Conversely, if the inclusion $\iota : H \hookrightarrow G$ is a group homomorphism, then H is a subgroup of G . Besides this, the following proposition provides a more common method for determining subgroups.

4.4.1 PROPOSITION. A subset H of a group G is a subgroup if and only if $H \neq \emptyset$ and for all $a, b \in H$, $ab^{-1} \in H$.

Proof. It is clear that if H is a subgroup, then the stated condition holds. Conversely, assume that the stated condition holds, we have to check that H is closed under the operation of G : Since H is nonempty, we can find an element $h \in H$. Choosing $a = b = h$, we see that

$$1 = hh^{-1} = ab^{-1} \in H,$$

thus H contains the identity. Given any $h \in H$, choosing $a = 1$ and $b = h$ shows that

$$h^{-1} = 1h^{-1} = ab^{-1} \in H$$

thus H contains the inverse of any of its elements. Given any $h_1, h_2 \in H$, choose $a = h_1, b = h_2^{-1}$, the stated condition says that

$$h_1h_2 = h_1((h_2)^{-1})^{-1} = ab^{-1} \in H,$$

proving that H is closed under the operation; finally, the operation is associative in G implies that the induced operation is associative in H . $\square$

4.4.2 PROPOSITION. Let G be a group and A be a subset of G . The set of all products in G of elements of A and inverses of elements of A is a subgroup of G .

Proof. Let $H \subset G$ be a set of all products of elements of A and inverses of elements of A . Then H contains the empty product 1; $h \in H$ implies $h^{-1} \in H$; and $h, k \in H$ implies $hk \in H$, since the product of two products of elements of A and inverses of elements of X is another such product. Thus H is a subgroup of G . Also H contains all the elements of A , which are one-term products of elements of A . $\square$

4.4.2 DEFINITION. The subgroup $\langle A \rangle$ of a group generated by a subset A of G is the set of all products in G of elements of A and inverses of elements of A , i.e., $\langle A \rangle = \{a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n} : n \geq 0, a_i \in A, \epsilon_i = \pm 1\}$. A group G is generated by a subset A if $G = \langle A \rangle$.

4.4.3 PROPOSITION. If G is a group and $\{H_i\}_I$ is a nonempty set of subgroups of G , then $\bigcap_{i \in I} H_i$ is a subgroup of G .

Proof. Let $K = \bigcap_{i \in I} H_i$. Since every H_i is a subgroup, $1 \in H_i$, so $1 \in K$; if $h \in K$, then $h^{-1} \in H_i$ for all $i \in I$, which implies $h^{-1} \in K$; if $h, k \in K$, then $hk \in H_i$ for all $i \in I$, which implies $hk \in K$. Thus, K is a subgroup of G . $\square$

4.4.4 PROPOSITION. $\langle A \rangle = \bigcap_{A \subset H, H \leq G} H$.

Proof. Since each $A \subset \langle A \rangle$, $\bigcap H \subset \langle A \rangle$. But $\bigcap H$ is a group containing A and, since it is closed under the group operation and the process of taking inverses, $\bigcap H$ contains all the products of element of A and the inverses of elements of A , that is, $\langle A \rangle \subset \bigcap H$. $\square$

4.4.5 PROPOSITION. *Let G be a group and let $a \in G$. The set of all powers of a is a subgroup of G ; in fact, it is the subgroup generated by $\{a\}$.*

Proof. That the powers of a constitute a subgroup of G follows from the parts $a^0 = 1$, $(a^n)^{-1} = a^{-n}$, $a^m a^n = a^{m+n}$. Also, nonnegative powers of a are products of a , and negative powers of a are products of a^{-1} . $\square$

4.4.3 DEFINITION. A group or subgroup G is *cyclic* if it can be generated by a single element, i.e., there is some element a in G such that $G = \langle a \rangle$.

In additive notation, G is cyclic if $G = \langle a \rangle = \{na : n \in \mathbf{Z}\}$. A cyclic group may have more than one generator. For example if $G = \langle a \rangle$, then also $G = \langle a^{-1} \rangle$.

4.4.1 EXAMPLE. Every subgroup of $\mathbf{Z}$ is cyclic.

Proof. The proof uses integer division. Let H be a subgroup of (the additive group) $\mathbf{Z}$. If $H = \{0\}$, then H is cyclic, generated by 0. Now assume that $H \neq \{0\}$, so that H contains an integer $m \neq 0$. If $m < 0$, then $-m \in H$; hence H contains a positive integer. Let n be the smallest positive integer that belongs to H . Every integer multiple of n belongs to H . Conversely, let $m \in H$. Then

$$m = nq + r \quad \text{for some } q, r \in \mathbf{Z}, 0 \leq r < n.$$

Since H is a subgroup, $qn \in H$ and $r = m - qn \in H$. Now, $0 < r < n$ would contradict the choice of n ; therefore $r = 0$, and $m = qn$ is an integer multiple of n . Thus H is the set of all integer multiples of n and is cyclic, generated by $n > 0$; we may denote this cyclic group by $n\mathbf{Z}$ (In particular, $\mathbf{Z}$ itself is generated by 1):

$$n\mathbf{Z} = \langle n \rangle = \{m \in \mathbf{Z} : \exists q \in \mathbf{Z}, m = qn\}.$$

$\square$

4.4.6 PROPOSITION. *Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism. If $H \leq G_1$, then $\varphi(H) = \{\varphi(x) : x \in H\} \leq G_2$. If $J \leq G_2$, then $\varphi^{-1}(J) = \{x \in G_1 : \varphi(x) \in J\} \leq G_1$. (the notation $\varphi^{-1}(J)$ should not be read to imply that φ is an isomorphism, or that φ^{-1} is the image of J under some misbegotten function φ^{-1})*

Proof. For the first part, assume $H \leq G_1$, the set $\varphi(H)$ is nonempty since $1_{G_1} \in H$ gives $\varphi(1_{G_1}) = 1_{G_2} \in \varphi(H)$. For any $a \in \varphi(H)$, there exist $x \in H$ with $\varphi(x) = a$, then $a^{-1} = \varphi(x)^{-1} = \varphi(x^{-1}) \in \varphi(H)$ since $x^{-1} \in H$; For any $a, b \in \varphi(H)$, there exist $x, y \in H$ with $\varphi(x) = a$ and $\varphi(y) = b$, and then $ab = \varphi(xy) \in \varphi(H)$ since $xy \in H$. Hence $\varphi(H) \leq G_2$.

For the second part, assume $J \leq G_2$ and $K = \varphi^{-1}(J)$. Since $\varphi(1_{G_1}) = 1_{G_2} \in J$, we have $1_{G_1} \in K$; for any $a \in K$, $\varphi(a) \in J$, which implies $\varphi(a^{-1}) = \varphi(a)^{-1} \in J$, so $a^{-1} \in K$; for any $a, b \in K$, then $\varphi(a), \varphi(b) \in J$, so that $\varphi(ab) \in J$ and thus $ab \in K$. Thus, $\varphi^{-1}(J) \leq G_1$. $\square$

4.4.4 DEFINITION. Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism. We define

- $\ker \varphi = \{x \in G_1 : \varphi(x) = 1\}$ the *kernel* of φ
- $\text{im} \varphi = \{\varphi(x) : x \in G_1\}$ the *image* of φ .

Since $\{1_{G_2}\}$ is a subgroup of G_2 , the proposition above shows that $\ker \varphi$ is indeed a subgroup of G . Since G_1 is a subgroup of G_1 , $\text{im} \varphi$ is a subgroup of G_2 by the proposition above. Kernels are special subgroups, they satisfy a universal property:

4.4.7 PROPOSITION. *Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism. Then the kernel of φ satisfies the property: for all $K \in \mathbf{Grp}$ and group homomorphism $\alpha : K \rightarrow G_1$ such that $\varphi \circ \alpha = \mathbf{0}$, there exists a unique group homomorphism $\bar{\alpha} : K \rightarrow \ker \varphi$ such that $\alpha = \iota \circ \bar{\alpha}$.*

Proof. Since $\varphi \circ \alpha = 0$, for any $k \in K$, we have $\varphi(\alpha(k)) = 1_{G_2}$, which means $\alpha(k) \in \ker \varphi$. Thus α actually a group homomorphism from K into $\ker \varphi$. Define $\bar{\alpha} : K \rightarrow \ker \varphi$ by $\bar{\alpha}(k) = \alpha(k)$ for all $k \in K$. Clearly, $\bar{\alpha}$ is a group homomorphism and $\alpha = \iota \circ \bar{\alpha}$. $\square$

In fact, we may consider the proposition in the category $\mathbf{C}$, in which $\text{Ob}(\mathbf{C}) = \{\alpha \in \mathbf{Grp}^{G_1} : \varphi \circ \alpha = \mathbf{0}\}$, and $\text{Mor}(\mathbf{C}) = \text{Mor}(\mathbf{Grp}^{G_1})$. That is, $\iota : \ker \varphi \hookrightarrow G_1$ is universal with respect to the property: for all $\alpha \in \mathbf{C}$, there exists a unique group homomorphism $\bar{\alpha}$ such that the diagram commutes:

$$\begin{array}{ccccc} & & \mathbf{0} & & \\ & \nearrow & & \searrow & \\ K & \xrightarrow{\alpha} & G_1 & \xrightarrow{\varphi} & G_2 \\ & \searrow \exists! \bar{\alpha} & \uparrow & & \\ & & \ker \varphi & & \end{array}$$

4.4.8 PROPOSITION. *Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism in the category $\mathbf{Grp}$. The following are equivalent:*

1. φ is a monomorphism;
2. $\ker \varphi = \{1\}$;
3. φ is an injection in the function sense.

Proof. (1) $\rightarrow$ (2) : consider the two parallel homomorphisms:

$$\begin{array}{ccccc} \ker \varphi & \xrightarrow{\iota} & G_1 & \xrightarrow{\varphi} & G_2 \\ \ker \varphi & \xrightarrow{e} & G_1 & \xrightarrow{\varphi} & G_2 \end{array}$$

in which $\iota : \ker \varphi \rightarrow G_1$ is the inclusion and $e : \ker \varphi \rightarrow G_1$ is the trivial homomorphism sending every element of $\ker \varphi$ to 1. Now, $\varphi \circ \iota$ is the zero homomorphism since element of $\ker \varphi$ map to 1_{G_1} under φ ; also $\varphi \circ e$ is the zero homomorphism since e sends every element to 1 which φ then sends to 1_{G_2} . Thus, $\varphi \circ \iota = \varphi \circ e$. Since φ is a monomorphism, $\iota = e$, so that $\ker \varphi = \{1\}$.

(2) $\rightarrow$ (3) : given $\ker \varphi = \{1\}$, suppose $\varphi(a) = \varphi(b)$. Then $\varphi(a)\varphi(b)^{-1} = 1$, which implies $\varphi(ab^{-1}) = 1$. Hence $ab^{-1} \in \ker \varphi = \{1\}$, so that $a = b$. Thus, φ is injective.

(3) $\rightarrow$ (1) : if φ is injective in the function sense, then in the category $\mathbf{Set}$ of sets, it has the left cancellation property: for any set Z and any functions $\alpha_1, \alpha_2 : Z \rightarrow G_1$, $\varphi \circ \alpha_1 = \varphi \circ \alpha_2 \rightarrow \alpha_1 = \alpha_2$. Now restrict Z to be a group and α_1, α_2 to be group homomorphisms. The same property holds, so φ is a monomorphism in $\mathbf{Grp}$. $\square$

4.5 Quotient Groups

4.5.1 DEFINITION. A subgroup N of a group G is *normal* if for every $g \in G, n \in N$

$$gng^{-1} \in N.$$

4.5.1 PROPOSITION. *Let $N \leq G$. The following are equivalent.*

1. N is a normal subgroup of G .
2. $gNg^{-1} \subset N$ for every $g \in G$.

3. $gNg^{-1} = N$ for every $g \in G$.
4. $gN \subset Ng$ for every $g \in G$.
5. $gN = Ng$ for every $g \in G$.

The purpose of this section is to study quotient groups, and we can define the quotient group from the perspective of universal properties. Let M be a monoid. Define an equivalence relation $\sim$ on the underlying set M , then we obtain a partition P induced by $\sim$. A quotient of M by P is the monoid $\overline{M}$ together with a monoid homomorphism $\pi : M \rightarrow \overline{M}$ which satisfies $(a \sim b \rightarrow \pi(a) = \pi(b))$ satisfying the universal property: for all $Z \in \mathbf{Mon}$ and monoid homomorphisms $\varphi : M \rightarrow Z$ such that $(a \sim b \rightarrow \varphi(a) = \varphi(b))$, there exists a unique homomorphism $\overline{M} \rightarrow Z$ such that the diagram commutes:

$$\begin{array}{ccc} \overline{M} & \xrightarrow{\exists! \bar{\varphi}} & Z \\ & \swarrow \pi \quad \searrow \varphi & \\ & M & \end{array}$$

Just as in the process of studying homomorphisms, we first study them in the category $\mathbf{Set}$, then make choices and corrections. Similarly, we can first study the quotient set of the underlying set M :

We already know that in the category $\mathbf{Set}$, the quotient set $M/\sim$ of the set M with respect to an equivalence relation $\sim$ satisfies the universal property: for any set Z and any function $\varphi : M \rightarrow Z$ such that $a \sim b \Rightarrow \varphi(a) = \varphi(b)$, there exists a unique function $\bar{\varphi} : M/\sim \rightarrow Z$ such that the diagram commute:

$$\begin{array}{ccc} M/\sim & \xrightarrow{\exists! \bar{\varphi}} & Z \\ & \swarrow \pi \quad \searrow \varphi & \\ & M & \end{array}$$

We should try to construct the monoid $M/\sim$ by defining an operation $\bullet$ on the set $M/\sim$. If $\langle M/\sim, \bullet \rangle$ is a monoid and the canonical projection $\pi : M \rightarrow M/\sim$ is a monoid homomorphism: for $[a] = \pi(a), [b] = \pi(b) \in M/\sim$, the homomorphism condition forces

$$[a] \bullet [b] = \pi(a) \bullet \pi(b) = \pi(ab) = [ab] \text{ and } [1] = \pi(1) \text{ is the identity}$$

To check that $\langle M/\sim, \bullet \rangle$ is indeed a monoid, the first thing we need to check is whether the definition of $\bullet$ is well-defined. In other words, does $[a] = [a']$ implies $[a] \bullet [b] = [a'] \bullet [b]$ and $[b] \bullet [a] = [b] \bullet [a']$? That is, does $a \sim a'$ implies $ab \sim a'b$ and $ba \sim ba'$ for every $b \in M$? It can be verified that this is indeed the case:

4.5.2 PROPOSITION. $\langle M/\sim, \bullet, [1] \rangle$ is a monoid iff $a \sim a' \rightarrow ba \sim ba' \wedge ab \sim a'b$. In this case, the canonical projection $\pi : M \rightarrow M/\sim$ may be a monoid homomorphism.

Proof. Assume that $a \sim a' \rightarrow ba \sim ba'$ and $ab \sim a'b$. Then the operation $[a] \bullet [b] := [ab]$ is well-defined. Moreover, the associativity of $\bullet$ is inherited from the associativity of M :

$$([a] \bullet [b]) \bullet [c] = [ab] \bullet [c] = [(ab)c] = [a(bc)] = [a] \bullet ([b] \bullet [c]).$$

The class $[1]$ is an identity: $[a] \bullet [1] = [a1] = [a] = [1a] = [1] \bullet [a]$. This shows that $M/\sim$ is indeed a monoid. Now, assume that $\langle M/\sim, \bullet \rangle$ is a monoid and $a \sim a'$, then $[ba] = [b] \bullet [a] = [b] \bullet [a'] = [ba']$, which means $ba \sim ba'$. Similarly, $ab \sim a'b$. $\square$

We say that $\sim$ is *compatible* with the monoid structure of M if the condition $a \sim a' \rightarrow ba \sim ba', ab \sim a'b$ for every $b \in M$ simultaneously. In this case, the operation $\bullet$ on the quotient set $M/\sim$ is uniquely determined by the operation on M , therefore, we can omit explicit mention of $\bullet$ when there is no risk of confusion.

4.5.3 PROPOSITION. *Let M be a monoid and $\pi : M \rightarrow M/\sim$ is the canonical projection with respect to the underlying sets. Then π is a monoid homomorphism when we consider the whole monoid structures. And $(M/\sim, \pi)$ satisfies the universal property of the quotient.*

Proof. It is clear that π is a monoid homomorphism. To show that $M/\sim$ satisfies the universal property, assume that $\varphi : M \rightarrow Z$ is a monoid homomorphism such that $a \sim a' \rightarrow \varphi(a) = \varphi(a')$. Since the set $M/\sim$ satisfies the corresponding universal property in the category **Set**, there exists a unique function $\tilde{\varphi} : M/\sim \rightarrow Z$ defined by $\tilde{\varphi}([a]) := \varphi(a)$. So we only need to check that this function $\tilde{\varphi}$ is in fact a monoid homomorphism, and this is immediate:

$$\begin{aligned}\tilde{\varphi}([a] \bullet [b]) &= \tilde{\varphi}([ab]) = \varphi(ab) = \varphi(a)\varphi(b) = \tilde{\varphi}([a])\tilde{\varphi}([b]); \\ \tilde{\varphi}([1]) &= \tilde{\varphi}(\pi(1)) = \varphi(1) = 1.\end{aligned}$$

□

4.5.1 EXAMPLE. Let $\langle M, + \rangle$ be a commutative monoid. One may define the monoid operation on the product $M \times M$ by $(x, y) + (x', y') = (x + x', y + y')$, we have shown that $\langle M \times M, + \rangle$ is a monoid. Define a relation $\sim$ on the set $M \times M$ by $(x, y) \sim (x', y') \leftrightarrow \exists z \in M, x + y' + z = x' + y + z$. One may show that $\sim$ is compatible. Therefore, $\langle M \times M/\sim, + \rangle$ is a monoid in which $+$ is defined by $[x, y] + [x', y'] = [x + x', y + y']$. Furthermore, we may show that $K(M) = M \times M/\sim$ is in fact an abelian group. We denote by $K(M)$ the *Grothendieck group* associated with M . It has the following property.

Given a function $\varphi : M \rightarrow K(M)$ defined by $\varphi(x) = [x, 0]$, then φ is a monoid homomorphism. In fact, $K(M)$ together with the homomorphism φ is universal with respect to the property: for all $A \in \mathbf{Ab}$ and monoid homomorphism $f : M \rightarrow A$, there exists a unique homomorphism $\bar{\varphi} : K(M) \rightarrow A$ such that the diagram commutes:

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & K(M) \\ & \searrow f & \downarrow \exists! \bar{\varphi} \\ & & A \end{array}$$

Proof. Suppose that $\bar{\varphi} : K(M) \rightarrow A$ is a homomorphism satisfies $\bar{\varphi} \circ \varphi = f$. Then $\bar{\varphi}([x, y]) = \bar{\varphi}([x, 0]) - \bar{\varphi}([y, 0]) = f(x) - f(y)$. We have to check that this definition is well-defined: if $[x, y] = [x', y']$, then there exists $z \in M$ such that $x + y' + z = x' + y + z$. Apply f : $f(x) + f(y') + f(z) = f(x') + f(y) + f(z)$. So that $f(x) - f(y) = f(x') - f(y')$. Thus $\bar{\varphi}$ is independent of the choice of representative. □

For the quotient groups, G is a special monoid ($G^\times = G$), then $\langle G/\sim, \bullet, [1] \rangle$ is a monoid. Moreover, we can further prove that $\langle G/\sim, \bullet, [1] \rangle$ is also a group; this is only requires a supplement to proposition 3.4.2: the class $[g^{-1}]$ is the inverse of $[g]$, since $[g] \bullet [g^{-1}] = [1] = [g^{-1}g] = [g^{-1}][g]$. At the same time, Propositions 3.4.3 and 3.4.4 can be directly adapted to the version for groups, without any further explanation.

From this proposition, we can call $G/\sim$ (resp. $M/\sim$) the quotient group (resp. quotient monoid) of the group G (resp. monoid M) with respect to $\sim$. Let us reconsider: this equivalence relation is not chosen arbitrary, it must be compatible with the structure of G :

$$a \sim b \rightarrow ga \sim gb \text{ for every } g \in G; \tag{1}$$

$$a \sim b \rightarrow ag \sim bg \text{ for every } g \in G. \tag{2}$$

4.5.4 PROPOSITION. *Let $\sim$ be a equivalence relation on the underlying set G with $a \sim b \rightarrow ga \sim gb$ for every $g \in G$, then*

1. $[1] = H$ is a subgroup of G , i.e., $[1] \leq G$;

2. $a \sim b \Leftrightarrow a^{-1}b \in H \Leftrightarrow aH = bH$.

Proof. Let H be the equivalence class of 1. Then $H \neq \emptyset$ since $1 \in H$; for any $a, b \in H$, we have $1 \sim b$, then $b^{-1} \sim 1$, hence $ab^{-1} \sim a$, and hence $ab^{-1} \sim a \sim 1$ by the transitivity of $\sim$ and since $a \in H$. This shows that $ab^{-1} \in H$ proving that H is a subgroup. Next, assume that $a, b \in G$ and $a \sim b$, then $1 \sim a^{-1}b$, that is $a^{-1}b \in H$. Since H is closed under the operation, this implies that $a^{-1}bH \subset H$, hence $bH \subset aH$. Since $\sim$ is symmetric, the same reasoning gives that $aH \subset bH$, and hence $aH = bH$. Finally, assume that $aH = bH$. Then $a = a1 \in bH$, and hence $a^{-1}b \in H$, this means $1 \sim a^{-1}b$, then $a \sim b$. $\square$

For condition (2), we can also draw a similar conclusion: let $\sim$ be an equivalence relation on a group G with $a \sim b \Rightarrow ag \sim bg$, then $[1]$ is a subgroup H of G ; and $a \sim b \Leftrightarrow ab^{-1} \in H \Leftrightarrow Ha = Hb$. The proof almost the same reasoning.

The proposition above shows that the equivalence classes of an equivalence relation satisfying (1), (2) are in fact all of the form aH, Ha ($[a] = \{b \in G : b \sim_L a\} = \{b : a^{-1}b \in H\} = \{b : b \in aH\} = aH$), respectively for a fixed subgroup H .

4.5.2 DEFINITION. The *left-cosets* of a subgroup H in a group G are the sets aH for $a \in G$. The *right-cosets* of a subgroup H in a group G are the sets Ha for $a \in G$.

4.5.5 PROPOSITION. If H is a subgroup of a group G , then the relation $\sim_L$ (resp. $\sim_R$) defined by $a \sim_L ba^{-1}b \in HaH = bH$ (resp. $a \sim_R bab^{-1} \in HHa = Hb$) is an equivalence relation for which $a \sim_L b \rightarrow ga \sim_L gb$ (resp. $a \sim_R b \rightarrow ag \sim_R bg$).

Proof. For every $g \in G$, $g^{-1}g = 1 \in H$, so it is always true that $g \sim_L g$; for any $g, h \in G$ in which $g \sim_L h$, we have $g^{-1}h \in H$, and its inverse which is $h^{-1}g$ must also be in H , which implies $h \sim_L g$; for $a, b, c \in G$ in which $a \sim_L b, b \sim_L c$, we have $a^{-1}b \in H, b^{-1}c \in H$, which implies that $a^{-1}c \in H$, so $a \sim_L c$. To show that the relation satisfies the statement, note that

$$a \sim_L b \rightarrow a^{-1}b \in H \rightarrow a^{-1}(g^{-1}g)b \in H \rightarrow (ga)^{-1}(gb) \in H \Rightarrow ga \sim_L gb.$$

$\square$

4.5.6 PROPOSITION. In summary, the two propositions above show that: there is a bijection between subgroups of G and equivalence relations on G satisfying (1) (resp. (2)). In other words, for the relations $\sim_L$ corresponding to a subgroup H of G ,

$$G/\sim_L = \{aH : a \in G\} \text{ (resp. } G/\sim_R = \{Ha : a \in G\})$$

4.5.7 PROPOSITION. The relations $\sim_L, \sim_R$ corresponding to a subgroup H coincide ($G/\sim_L = G/\sim_R$) iff H is a normal subgroup of the group G .

Proof. Assume H is normal, so $gH = Hg$ for all $g \in G$. Hence $aH = Ha$ for all $a \in G$, i.e., $\sim_L = \sim_R$. Conversely, if H is the subgroup corresponding to $\sim_R$ and $\sim_L$, then for all $a \in G$, $a \sim_L a$ implies that $aH = Ha$ naturally. $\square$

This proposition tells us that if H is a normal subgroup, then the equivalence relation $\sim = \sim_L = \sim_R$ corresponding to H satisfies (1), (2) simultaneously, and hence, the quotient

$$G/\sim = \{aH : a \in G\} = \{Ha : a \in G\}$$

has a natural group structure. Since a normal subgroup uniquely determines an equivalence relation, it is sometimes reasonable to avoid discussing the relation; and define the quotient group directly using the normal subgroup itself instead of seeking the explicit definition of the relation:

4.5.3 DEFINITION. Let H be normal subgroup of the group G . The *quotient group of G modulo H* , denoted G/H , is the group $G/\sim$ obtained from the relation $\sim$ defined above. The product in G/H is defined by

$$(aH)(bH) := (ab)H.$$

Then the quotient function $\pi : G \rightarrow G/H$ mapping $g \in G$ to $gH = Hg$ is a group homomorphism and the pair $(G/H, \pi)$ satisfies the universal property: for any group G' and homomorphism $\varphi : G \rightarrow G'$ such that $aH = bH \Rightarrow \varphi(a) = \varphi(b)$, there exists a unique group homomorphism $\tilde{\varphi} : G/H \rightarrow G'$ such that $\varphi = \tilde{\varphi} \circ \pi$:

By proposition 2.6.3., the canonical projection $\pi : G \rightarrow G/H$ sending $g \in G$ to $gH = Hg$ is a group homomorphism and satisfies the universal property: for any homomorphism $\varphi : G \rightarrow G'$ such that $aH = bH \Rightarrow \varphi(a) = \varphi(b)$, there exists a unique homomorphism $\tilde{\varphi} : G/H \rightarrow G'$ such that $\varphi = \tilde{\varphi} \circ \pi$:

4.5.8 PROPOSITION. *Let H be a normal subgroup of a group G . Then for every group homomorphism $\varphi : G \rightarrow G'$ such that $H \subset \ker \varphi$, there exists a unique group homomorphism $\tilde{\varphi} : G/H \rightarrow G'$ such that the diagram commutes:*

$$\begin{array}{ccc} G/H & \xrightarrow{\tilde{\varphi}} & G' \\ \pi \swarrow & & \nearrow \varphi \\ & G & \end{array}$$

Proof. $H \subset \ker \varphi$ means that if $ab^{-1} \in H$ for $a, b \in G$, then $\varphi(a) = \varphi(b)$. Moreover, the relation $\sim$ corresponding to H is defined: $a \sim b \Rightarrow \varphi(a) = \varphi(b)$, the condition giving the universal property in proposition 2.6.3. $\square$

4.5.2 EXAMPLE. Consider the abelian group $\langle \mathbf{Z}, + \rangle$. We have already shown that every subgroup of $\mathbf{Z}$ is a cyclic group of the form $n\mathbf{Z} = \langle n \rangle$. Since $\mathbf{Z}$ is abelian, $n\mathbf{Z}$ is normal naturally. We may consider the quotient group $\mathbf{Z}/n\mathbf{Z} = \{a + \langle n \rangle : a \in \mathbf{Z}\}$. In fact, the equivalence relation $\sim$ induced by the normal subgroup $n\mathbf{Z}$ is defined by

$$a \sim b \Leftrightarrow b - a \in \langle n \rangle \Leftrightarrow n \mid (b - a).$$

We will denote the equivalence class of a by $[a]_n$, i.e., $[a]_n = a + \langle n \rangle$. Now, we show that an analogous result: every subgroup of $\mathbf{Z}/n\mathbf{Z}$ is a cyclic group.

Proof. Let G be a subgroup of $\mathbf{Z}/n\mathbf{Z}$. Consider the canonical projection $\pi : \mathbf{Z} \rightarrow \mathbf{Z}/n\mathbf{Z}$ and $G' = \pi^{-1}(G)$. Then G' is a subgroup of $\mathbf{Z}$. We have already shown that G' is a cyclic subgroup of $\mathbf{Z}$ generated by a nonnegative integer m . It follows that

$$G = \pi(G') = \pi(\langle m \rangle) = \{\pi(km) \mid k \in \mathbf{Z}\} = \{km + \langle n \rangle \mid k \in \mathbf{Z}\},$$

hence $G = \{k(m + \langle n \rangle) : k \in \mathbf{Z}\} = \langle m + \langle n \rangle \rangle$, thus G is indeed a cyclic subgroup of $\mathbf{Z}/n\mathbf{Z}$ generated by a $[m]_n = m + \langle n \rangle$. Furthermore, since $n \in G'$ (because $\pi(n) = n + \langle n \rangle = \langle n \rangle \in G$) and $G' = \langle m \rangle$, we see that $m \mid n$. $\square$

4.5.9 PROPOSITION. *Let H be a normal subgroup of a group G . Every subgroup of G/H is the quotient N/H of a unique subgroup N of G that contains H . That is, the function $u : \mathcal{A} \rightarrow \mathcal{B}$ defined by $u(N) = N/H$ is a bijection where $\mathcal{A} = \{N : N \leq G \wedge H \subset N\}$, $\mathcal{B} = \{\bar{N} : \bar{N} \leq G/H\}$*

Proof. Let $\pi : G \rightarrow G/H$ be the canonical projection and let B be a subgroup of G/H . Since π is a homomorphism and $B \leq G/H$, $A = \pi^{-1}(B) = \{a \in G : aH \in B\}$ is a subgroup of G and contains $\pi^{-1}(1) = \ker \pi = H$. Now H is a subgroup of A and is a normal subgroup of A since $aH = Ha$ for all $a \in A$. The elements aH of A/H all belong to B by definition of A . Conversely, if $xH \in B$, then $x \in A$ and $xH \in A/H$. Thus, $B = A/H$.

For uniqueness, assume that $B = H/H$ where $H \leq G$ contains H . If $h \in H$, then $hH \in H/H = B$ and $h \in A$. Conversely, if $a \in A$, then $aH \in B = H/H$, $aH = hH$ for some $h \in H$, and $a \in hH \subset H$. Thus $H = A$. $\square$

4.5.10 PROPOSITION. *Let H be a normal subgroup of a group G and let N be a subgroup of G containing H . Then the bijection u defined above preserves inclusion and normality ($N/H \trianglelefteq G/H$ if and only if $N \trianglelefteq G$.)*

Proof. By the proposition above, $\pi : G \rightarrow G/H$ induces a bijection $A \mapsto A/H$ from $\mathcal{A}$ onto $\mathcal{B}$, and the inverse bijection is $B \mapsto \pi^{-1}(B)$, since $B = A/H$ if and only if $A = \pi^{-1}(B)$. Both bijections preserve inclusions.

For normality, if N is normal, consider the projection $\varphi : G \rightarrow G/N$: the subgroup H is contained in N , which is the kernel of this homomorphism ($H \subset \ker \varphi$), so φ induces a homomorphism $\tilde{\varphi} : G/H \rightarrow G/N$ by the universal property of quotients. The subgroup N/H of G/H is the kernel of this homomorphism, therefore it is normal. Conversely, if N/H is normal in G/H , consider the composition of homomorphisms

$$G \twoheadrightarrow G/H \twoheadrightarrow (G/H)/(N/H).$$

The kernel of this homomorphism is N , therefore N is normal, □

We also may use the notation $(G/H)_L$ to denote $G/\sim_L$ and $(G/H)_R$ to denote $G/\sim_R$ to avoid discussing the relations. Although aH may not equal Ha at this point, the cardinality of the left cosets of H equals the cardinality of its right cosets. That is the proposition:

4.5.11 PROPOSITION. *Let H be a subgroup of a group G . Then $|(G/H)_L| = |(G/H)_R|$.*

Proof. Let $a \in G$. If $y \in aH$, then $y = ah$ for some $h \in H$ and $y^{-1} = h^{-1}a^{-1} \in Ha^{-1}$. Conversely, if $y^{-1} \in Ha^{-1}$, then $y^{-1} = h'a^{-1}$ for some $h' \in H$ and $y = ah'^{-1} \in aH$. Thus, when $A = aH$ is a left coset of H , the set $A' = \{y^{-1} : y \in A\}$ is a right coset of H and $A' = Ha^{-1}$. Similarly, when $B = Hb$ is a right coset of H , then $B' = \{x^{-1} : x \in B\}$ is a left coset of H . This correspondence is bijective. □

4.5.4 DEFINITION. Let H be a subgroup of a group G . The **index** of H in G , denoted $[G : H]$, is the cardinality of the set of left (or right) cosets of H in G .

4.5.12 THEOREM. *Let K, H, G be groups with $K \leq H \leq G$, then $[G : K] = [G : H][H : K]$. If any two of these indices are finite, then so is the third.*

Proof. It is clear that $G = \bigcup_{i \in I} a_i H$ with $a_i \in G$, $|I| = [G : H]$, and the cosets $a_i H$ are mutually disjoint (that is, $a_i H = a_j H \leftrightarrow i = j$). Similarly, $H = \bigcup_{j \in J} b_j K$ with $b_j \in H$, $|J| = [H : K]$, and the cosets Kb_j are mutually disjoint. Therefore

$$G = \bigcup_{i \in I} a_i H = \bigcup_{i \in I} a_i \left(\bigcup_{j \in J} b_j K \right) = \bigcup_{(i,j) \in I \times J} (a_i b_j) K.$$

It suffices to show that the cosets $a_i b_j K$ are mutually disjoint. By the definition, we must have $[G : K] = |I \times J|$, whence

$$[G : K] = |I \times J| = |I| \cdot |J| = [G : H][H : K].$$

Now suppose $(a_i b_j) K = (a_t b_r) K$. Then there exists $k \in K$ such that $a_i b_j = a_t b_r k$. Since $b_j, b_r, k \in H$, we can multiply on the left by a_t^{-1} : $a_t^{-1} a_i b_j = b_r k \in H$. Hence $a_t^{-1} a_i \in H$ (because $b_j \in H$ and H is a subgroup), so $a_i H = a_t H$. By disjointness of the $a_i H$, we obtain $i = t$. Substituting $i = t$ back gives $a_i b_j = a_i b_r k$. Left multiplying by a_i^{-1} yields $b_j = b_r k \in b_r K$. Thus $b_j K = b_r K$, and by disjointness of the $b_j K$ in H , we get $j = r$. Therefore the left cosets $(a_i b_j) K$ are mutually disjoint, completing the proof. □

4.5.13 COROLLARY (Lagrange's Theorem). *Let H be a subgroup of the group G . Then $|G| = [G : H]|H|$ (the multiplication of cardinality). In particular, if G is finite, the order of H divides the order of G .*

For example, a group of order 9 has no subgroup of order 2. A group G whose order is a prime number has only two subgroups, G itself and $\{1\}$. The original version of Lagrange's theorem applied to functions $f(x_1, \dots, x_n)$ whose arguments are permuted: when $x_1, \dots, x_n$ are permuted in all possible ways, the number of different values of $f(x_1, \dots, x_n)$ is a divisor of n .

4.5.14 COROLLARY. *If p is a prime and $|G| = p$, then G is a cyclic group.*

Proof. Take $g \in G$ with $g \neq 1$. Then the cyclic subgroup $\langle g \rangle$ has more than one element (it contains g and 1), and its order $|\langle g \rangle| > 1$ is a divisor of p . Since p is prime, $|\langle g \rangle| = p = |G|$, and so $\langle g \rangle = G$. $\square$

At this point, it is not clear whether, conversely, a divisor of $|G|$ is necessarily the order of a subgroup of G . Interesting partial answers to this question await us in the next chapter.

Let H, K be subgroups of a group G . Define the set $HK = \{hk : h \in H, k \in K\}$. A right or left coset of a subgroup is a special case. It is worth bearing in mind that HK is not necessarily a group, even if H, K are groups.

4.5.15 PROPOSITION. *If H, K are subgroups of a group G , HK is a subgroup iff $HK = KH$. In particular, if either subgroup is normal, then HK is a subgroup.*

Proof. Assume that $HK = KH$ and let $a, b \in HK$. We prove that $ab^{-1} \in HK$ so HK is a subgroup. Let $a = h_1k_1$ and $b = h_2k_2$ for some $h_1, h_2 \in H, k_1, k_2 \in K$. Thus, $b^{-1} = k_2^{-1}h_2^{-1}$, so $ab^{-1} = h_1k_1k_2^{-1}h_2^{-1}$. Let $k_3 = k_1k_2^{-1} \in K$ and $h_3 = h_2^{-1}$. Thus $ab^{-1} = h_1k_3h_3$. Since $HK = KH$, $k_3h_3 = h_4k_4$ for some $h_3 \in H, k_4 \in K$. Thus $ab^{-1} = h_1h_4k_4 \in HK$.

Conversely, assume that HK is a subgroup of G . Since $K \leq HK$ and $H \leq HK$, then $KH \subset HK$. Let $hk \in HK$. Since HK is a subgroup, write $hk = a^{-1}$ for some $a \in HK$. If $a = h_1k_1$, then $hk = k_1^{-1}h_1^{-1} \in KH$, that is, $HK \subset KH$. $\square$

4.5.16 PROPOSITION. *Let G be a group and let H, K be any two finite subgroups of G . Then we have $|HK| = |H||K|/|H \cap K|$.*

Proof. Note that HK is the union of left cosets of K , that is $HK = \bigcup_{h \in H} hK$. Since each coset of K has $|K|$ elements ($hK = K$ for every $h \in H$), it suffices to find the number of distinct left cosets of K . If $h_1K = h_2K$ for $h_1, h_2 \in H$, then $h_1K = h_2K \Leftrightarrow h_2^{-1}h_1 \in K \Leftrightarrow h_2^{-1}h_1 \in H \cap K \Leftrightarrow h_1(H \cap K) = h_2(H \cap K)$. Which implies that the number of distinct cosets hK of K is the number of distinct cosets $h(H \cap K)$ for $h \in H$. By the Lagrange's theorem, the latter number equals $|H|/|H \cap K|$, as desired. $\square$

At the end of the previous section, we introduced an important subgroup, namely the kernel of a homomorphism. An essential property related to them is that monomorphisms in $\mathbf{Grp}$ are equivalent to injective homomorphisms. Here we introduce another subgroup, the cokernel. The corresponding essential property naturally concerns epimorphisms.

4.5.5 DEFINITION. Let $\varphi : G_1 \rightarrow G_2$ be a homomorphism of abelian groups. The *cokernel* of φ is a group $\text{coker } \varphi$ together with a homomorphism $\pi : G_2 \rightarrow \text{coker } \varphi$ satisfying the universal property: for any $L \in \mathbf{Ab}$ and homomorphism $\alpha : G_2 \rightarrow L$ such that $\alpha \circ \varphi = \mathbf{0}$, there exists a unique homomorphism $\tilde{\varphi} : \text{coker } \varphi \rightarrow L$ such that the diagram commutes:

$$\begin{array}{ccccc} & & 0 & & \\ & \nearrow \varphi & & \searrow \alpha & \\ G_1 & \xrightarrow{\quad} & G_2 & \xrightarrow{\quad} & L \\ & \searrow \pi & \nearrow \exists! \tilde{\varphi} & & \\ & & \text{coker } \varphi & & \end{array}$$

Cokernels exist in $\mathbf{Ab}$: because the image of φ is a subgroup of G_2 , hence a normal subgroup of G_2 since G_2 is abelian; the condition that $\alpha \circ \varphi$ is trivial says that $\text{im } \varphi \subset \ker \alpha$, and hence

$$G_2/\text{im } \varphi \cong \text{coker } \varphi$$

satisfies the universal property of cokernels by the universal property of quotients: The problem in $\mathbf{Grp}$ is that $\text{im } \varphi$ is not guaranteed to be normal in G_2 .

4.5.17 PROPOSITION. *Let $\varphi : G_1 \rightarrow G_2$ be a homomorphism of abelian groups in $\mathbf{Ab}$. The following are equivalent*

1. φ is an epimorphism;
2. $\text{coker}\varphi = \{0\}$;
3. φ is an surjection in the function sense.

Proof. (1) $\rightarrow$ (2) : consider the two parallel homomorphisms:

$$\begin{array}{ccccc} G_1 & \xrightarrow{\varphi} & G_2 & \xrightarrow{\pi} & \text{coker}\varphi \\ G_1 & \xrightarrow{\varphi} & G_2 & \xrightarrow{e} & \text{coker}\varphi \end{array}$$

where π is the canonical projection and e is the trivial homomorphism, sending every element of G_2 to 0. Now, $\pi \circ \varphi$ is the trivial homomorphism since $\pi(\varphi(g)) = \varphi(g) + \text{im}\varphi = \text{im}\varphi$; also $e(\varphi(g)) = \text{im}\varphi$. Thus, both $\pi \circ \varphi$ and $e \circ \varphi$ are the trivial homomorphisms. Since φ is an epimorphism, this implies $\pi = e$.

(2) $\rightarrow$ (3) : assume that $\text{coker}\varphi = G_2/\text{im}\varphi = \{g + \text{im}\varphi : g \in G_2\} = \{0\} = \{\text{im}\varphi\}$, it follows that $G_2 \subset \text{im}\varphi$, thus $\text{im}\varphi = G_2$, meaning that φ is surjective in the function sense.

(3) $\rightarrow$ (1) : if φ is surjective in the function sense, then in the category $\mathbf{Set}$ of sets, it has the right cancellation property: for any set Z and any functions $\alpha_1, \alpha_2 : G_2 \rightarrow Z$, $\alpha_1 \circ \varphi = \alpha_2 \circ \varphi \rightarrow \alpha_1 = \alpha_2$. Now restrict Z to be an abelian group and α_1, α_2 to be group homomorphisms. The same property holds, so φ is a monomorphism in $\mathbf{Ab}$. $\square$

4.6 Presentations

In the last section, we introduced the universal property and standard construction of free groups. The key feature of a free group lies in its “free”, meaning its construction is not constrained by any unnecessary relations. This section, we will impose “restrictions” on free groups, i.e., the relations, and we refer to a free group with such relations added as a presentation. This will serve as a direct application of quotient groups.

4.6.1 THEOREM (First Isomorphism Theorem). *If $\varphi : G_1 \rightarrow G_2$ is a group homomorphism, then $G_1/\ker\varphi \cong \text{im}\varphi$. In fact, there exists a unique isomorphism $\tilde{\varphi} : G_1/\ker\varphi \rightarrow \text{im}\varphi$ such that $\varphi = \iota \circ \tilde{\varphi} \circ \pi$, where $\iota : \text{im}\varphi \hookrightarrow G_2$ is the inclusion homomorphism and $\pi : G_1 \twoheadrightarrow G_1/\ker\varphi$ is the canonical projection:*

$$\begin{array}{ccc} G_1 & \xrightarrow{\varphi} & G_2 \\ \downarrow \pi & & \uparrow \iota \\ G_1/\ker\varphi & \xrightarrow[\sim]{\tilde{\varphi}} & \text{im}\varphi \end{array}$$

Proof. Let $\psi : G_1 \rightarrow \text{im}\varphi$ be the same function as φ but viewed as a homomorphism of G_1 onto $\text{im}\varphi$. Then $\ker\psi = \ker\varphi$. By the universal property of quotient groups, there is a unique homomorphism $\tilde{\varphi} : G_1/\ker\varphi \rightarrow \text{im}\varphi$ such that $\psi = \tilde{\varphi} \circ \pi$. Then

$$\tilde{\varphi}(x \ker\varphi) = \psi(x) = \varphi(x), \quad \forall x \in G_1.$$

Moreover, $\varphi = \iota \circ \tilde{\varphi} \circ \pi$ since $\varphi(x) = \psi(x) = \tilde{\varphi}(\pi(x)) = \iota(\tilde{\varphi}(\pi(x)))$ for all $x \in G_1$. It suffices to show that $\tilde{\varphi}$ is bijective in the sense of function. Since ψ is surjective, so is $\tilde{\varphi}$; $\tilde{\varphi}$ is injective since $\tilde{\varphi}(x \ker\varphi) = 1$ implies $\varphi(x) = 1$, $x \in \ker\varphi$, and $x \ker\varphi = 1$ in $G_1/\ker\varphi$. For the uniqueness, if $\tilde{\varphi}' : G_1/\ker\varphi \rightarrow \text{im}\varphi$ is another isomorphism such that $\varphi = \iota \circ \tilde{\varphi}' \circ \pi$, then

$$\tilde{\varphi}'(x \ker\varphi) = \iota(\tilde{\varphi}'(\pi(x))) = \varphi(x) = \iota(\tilde{\varphi}(\pi(x))) = \tilde{\varphi}(x \ker\varphi).$$

$\square$

The homomorphism theorem implies that every homomorphism is a composition of three basic types of homomorphisms: *inclusion homomorphism of subgroups*; *isomorphisms*; and *canonical projections to quotient groups*.

4.6.2 COROLLARY. *Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism. If φ is injective in the sense of function, then $G_1 \cong \text{im}\varphi$; if φ is surjective in the sense of function, then $G_2 \cong G_1/\ker \varphi$.*

4.6.1 DEFINITION. Let A be a set. A *relation* on A is an ordered pair (a, b) of two reduced words a, b . We say the relation (a, b) holds in a group G via the function $f : A \rightarrow G$ if $\varphi(a) = \varphi(b)$ where $\varphi : F(A) \rightarrow G$ is the unique homomorphism that extends f .

Relations (a, b) are normally written as $(ab^{-1}, 1)$: indeed, (a, b) holds in a group G via f if and only if $\varphi(a) = \varphi(b)$ if and only if $\varphi(ab^{-1}) = 1 = \varphi(1)$ if and only if $(ab^{-1}, 1)$ holds in a group G via f . Now, given a set of reduced words R , the relations of R is the set $R^* = \{(r, 1) : r \in R\}$.

An immediate consequence of this corollary and the corollary 3.3.5 (every group is the image of a surjective homomorphism of a free group) is that any group G is isomorphic to a quotient group $F(A)/N$ where $G = \langle A \rangle$ and N is the kernel of the surjective homomorphism $\varphi : F(A) \rightarrow G$:

$$G \cong F(A)/\ker \varphi.$$

Therefore, in order to describe G up to isomorphism we need only specify $A, F(A)$ and N . $F(A)$ is determined up to isomorphism by A and N is determined by some subset that generates it as a subgroup of $F(A)$. If a subset R is a set of generators of $N = \ker \varphi$ and $r = a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n} \in F(A)$ is such a generator of $N = \ker \varphi$, then under the surjective homomorphism $\varphi : F(A) \rightarrow G$, $\varphi(r) = a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n} = 1 \in G$. Then the relations $(r, 1)$ holds in G via the inclusion $\iota : A \rightarrow G$. And these reduced words $a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n}$ in $N = \ker \varphi$ identify all the elements of $N = \ker \varphi$:

$$\ker \varphi = \langle R \rangle$$

Therefore, we can directly assert that, up to isomorphism, any group G is completely determined by a set A of generators of G and a suitable subset R .

Conversly, suppose we are given a set A and a set R of reduced words of $F(A)$. Does there exist a group G such that G can be generated by the set A and all the relations $R^* = \{(r, 1) : r \in R\}$ are valid? The answer is yes! The most natural idea is to construct it through “quotient sets”: A generates the free group $F(A)$ and R generates a subgroup N . However, a subgroup generated by R may not necessarily be a normal subgroup, which requires a technical adjustment here:

4.6.2 DEFINITION. Clearly, the intersection of normal subgroups of a group is again a normal subgroup, so we have the definition: Let G be a group and $R \subset G$, we denote the smallest normal subgroup containing R by $\langle\langle R \rangle\rangle$, that is, $\langle\langle R \rangle\rangle = \bigcap_{R \subset H, H \trianglelefteq G} H$.

4.6.3 PROPOSITION. *Let G be a group and $R \subset G$. Then*

$$\langle\langle R \rangle\rangle = \langle \{ \omega^{-1} r^\epsilon \omega : \omega \in G, r \in R, \epsilon = \pm 1 \} \rangle$$

Proof. Let $N = \langle \{ \omega^{-1} r^\epsilon \omega : \omega \in G, r \in R, \epsilon = \pm 1 \} \rangle$. Since $H \trianglelefteq G$ contains R , $\langle\langle R \rangle\rangle$ contains R . Moreover, the intersection of normal subgroups is a normal subgroup, therefore $\langle\langle R \rangle\rangle$ is itself a normal subgroup containing R . Since $\langle\langle R \rangle\rangle$ is itself a subgroup, the inverse of each $r \in R$ denoted $r^{-1} \in \langle\langle R \rangle\rangle$, and by the closure under conjugation, each $\omega^{-1} r^\epsilon \omega \in \langle\langle R \rangle\rangle$. Therefore, $\langle\langle R \rangle\rangle$ contains all the generators of K , which implies that $K \subset \langle\langle R \rangle\rangle$ by the closure under products of these generators. To show that $\langle\langle R \rangle\rangle \subset K$, let $r \in R$. Then $r = 1^{-1} r 1 \in K$, so $R \subset K$. It is clear that K is a subgroup of G , and we

show that K is normal: let $x \in K$ written as $x = \omega_1^{-1}r_1^{\epsilon_1}\omega_1 \cdots \omega_n^{-1}r_n^{\epsilon_n}\omega_n$, taking $\omega \in G$ yields

$$\begin{aligned}\omega^{-1}x\omega &= \omega^{-1}(\omega_1^{-1}r_1^{\epsilon_1}\omega_1)(\omega_2^{-1}r_2^{\epsilon_2}\omega_2) \cdots (\omega_n^{-1}r_n^{\epsilon_n}\omega_n)\omega \\ &= (\omega^{-1}\omega_1^{-1}r_1^{\epsilon_1}\omega_1)(\omega^{-1}\omega_2^{-1}r_2^{\epsilon_2}\omega_2) \cdots (\omega^{-1}\omega_n^{-1}r_n^{\epsilon_n}\omega_n)\omega \\ &= ((\omega_1\omega)^{-1}r_1^{\epsilon_1}(\omega_1\omega))((\omega_2\omega)^{-1}r_2^{\epsilon_2}(\omega_2\omega)) \cdots ((\omega_n\omega)^{-1}r_n^{\epsilon_n}(\omega_n\omega))\end{aligned}$$

Since $\omega_1\omega \in G$, it follows that $\omega^{-1}x\omega \in K$. Therefore, $\omega^{-1}x\omega$ can be written as a product of some conjugation of R , i.e., K is normal. Then $R \subset K \trianglelefteq G$. It follows that $\langle\langle R \rangle\rangle \subset K$. $\square$

Group G generated by A and satisfies the relations in $R^* = \{(r, 1) : r \in R\}$ is constructed as follows: Let $F(A)$ be the free group generated by A and let N be the normal subgroup of $F(A)$ generated by the set R . Any relation $(r, 1)$ in R^* holds in the group G via the canonical projection $\pi : F(A) \rightarrow G = F(A)/N$ because, for $r = a_1^{\epsilon_1} \cdots a_n^{\epsilon_n} \in R$, we have $rN = N$ since $r \in R \subset N$. Since N is the identify element in $F(A)/N$, we have in group G that $r = 1$ where 1 denote the identify in $G = F(A)/N$.

4.6.3 DEFINITION. Let A be a set and R a set of words in $F(A)$. The group $\langle A \mid R \rangle$ is the quotient of the free group $F(A)$ by the smallest normal subgroup N of $F(A)$ that contains R :

$$\langle A \mid R \rangle := F(A)/N = F(A)/\langle\langle R \rangle\rangle$$

We will call a pair $\langle A \mid R \rangle$ a presentation. The elements of A is called the generators and the words in R are called *relators*. The following proposition reveals the universal property of a presentation.

4.6.4 PROPOSITION. Let A be a set and $R \subset F(A)$. Let $\pi : F(A) \rightarrow \langle A \mid R \rangle$ denote the canonical projection of $F(A)$. Then the group $\langle A \mid R \rangle$ together with the inclusion $\iota : A \rightarrow \langle A \mid R \rangle$ is universal with respect to the property: for any group G and function $f : A \rightarrow G$ such that every relation in R^* holds in G via f , there exists a unique homomorphism $\psi : \langle A \mid R \rangle \rightarrow G$ such that the diagram commute:

$$\begin{array}{ccc} \langle A \mid R \rangle & \xrightarrow{\varphi} & G \\ \uparrow \iota & \nearrow f & \\ A & & \end{array}$$

Proof. Applying the universal property for free group to $F(A)$ induces a unique homomorphism $\varphi : F(A) \rightarrow G$ extending $f : A \rightarrow G$. Since every $r \in R$ holds in G via f , i.e., $\varphi(r) = \varphi(1) = 1$, it follows that $R \subset \ker \varphi$ and $\langle\langle R \rangle\rangle \subset \ker \varphi$. Now, we apply the universal property for quotient group to $\langle A \mid R \rangle$ to induce the unique homomorphism $\psi : \langle A \mid R \rangle \rightarrow G$.

$$\begin{array}{ccc} \langle A \mid R \rangle & \xrightarrow{\psi} & G \\ \uparrow \pi & \nearrow \varphi & \\ F(A) & & \\ \uparrow \iota^* & \nearrow f & \\ A & & \end{array}$$

$\square$

If G is generated by $f(A)$, then $\text{im} \psi = G$ since $\text{im} \psi$ contains every generator $f(a) = \psi(\iota(a))$ of G . In particular, when G is generated by A , and every relation $(r, 1) \in R^*$ holds in G via the inclusion $A \rightarrow G$, then there is a surjective homomorphism $\langle A \mid R \rangle \rightarrow G$ and G is isomorphic to a quotient group of $\langle A \mid R \rangle$. In this sense, $\langle A \mid R \rangle$ is the largest group generated by A subject to every relation in R^* .

4.6.5 THEOREM. Every group is isomorphic to a presentation.

Proof. Let G be a group and $G = \langle A \rangle$, $F(A)$ is the free group generated by the set A . Then $G \cong F(A)/\ker \varphi$ where $\varphi : F(A) \rightarrow G$ is the unique surjective homomorphism. Let $R = \ker \varphi$, then $\langle R \rangle' = \ker \varphi$ since $\ker \varphi$ is normal. Hence $G \cong \langle A \mid R \rangle$. $\square$

4.6.1 EXAMPLE. $D_{2n} \cong \langle a, b \mid a^n = b^2 = 1, bab = a^{-1} \rangle$.

Proof. Let $G = \langle a, b \mid a^n = b^2 = 1, bab = a^{-1} \rangle$. The elements of G are products of a 's, b 's, a^{-1} 's and b^{-1} 's. Since a and b have finite order, the elements of G are in fact products of a 's and b 's. Using the equality $ba = a^{n-1}b$ every product of a 's and b 's can be written so that all a 's precede all b 's. Since $a^n = b^2 = 1$, every element of G is now a product of fewer than n a 's followed by fewer than 2 b 's; in other words, $G = \{a^i, a^i b : 0 \leq i < n\}$. In particular, G has at most $2n$ elements. In D_{2n} , the equality $r^n = s^n = 1$ and $sr s = r^{-1}$ hold. Consider the function $f : \{a, b\} \rightarrow D_{2n}$ defined by $f(a) = r, f(b) = s$. By the universal property, f induces a surjective homomorphism $\tilde{\varphi} : G \rightarrow D_{2n}$. Hence G has at least $2n$ elements. Therefore, G has exactly $2n$ elements, i.e., $\tilde{\varphi}$ is bijective, so that $\tilde{\varphi}$ is an isomorphism. $\square$

4.6.2 EXAMPLE. $Q_8 \cong \langle a, b \mid a^4 = 1, a^2 b^{-2} = 1, abab^{-1} = 1 \rangle$.

Proof. Let $G = \langle a, b \mid a^4 = 1, a^2 b^{-2} = 1, abab^{-1} = 1 \rangle$. By the same discussion of elements in G as in the previous example, we obtain that $G = \{a^i b^j : i \in \{0, 1, 2, 3\}, j \in \{0, 1\}\}$. Hence G has at most 8 elements. Consider the function $f : \{a, b\} \rightarrow Q_8$ defined by $f(a) = i, f(b) = j$. Then the homomorphism induced by f is surjective. Hence G has exactly 8 elements. Therefore $Q_8 \cong G$. $\square$

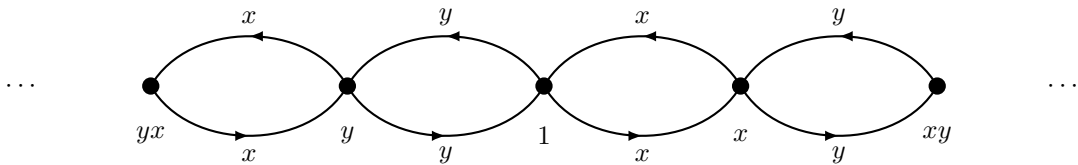
4.6.3 EXAMPLE (Coxeter Groups). Let W be a group generated by a finite set $S = \{s_1, \dots, s_n\}$. W is a *Coxeter group* if it has the presentation

$$\langle S \mid (s_i s_j)^{m_{ij}} = 1 \rangle,$$

where $m_{ii} = 1, m_{ij} = m_{ji}$ and $m_{ij} > 1$ for $i \neq j$. The group W may be determined by its universal property: for all $Z \in \mathbf{Grp}$ and any function $f : S \rightarrow Z$ such that every relation holds in Z via f , there exists a unique group homomorphism $\psi : W \rightarrow Z$ such that the diagram commute. The dihedral group D_{2n} of order $2n$ is a Coxeter group. In fact one may show that $D_{2n} \cong \langle r, s \mid r^2 = s^2 = (rs)^n = 1 \rangle$.

Coxeter groups are a very important class of groups and are one of the main subjects of study in geometric group theory. However, this is not the main focus of these notes. Interested readers may refer to the book Office Hours with a Geometric Group Theorist.

4.6.4 EXAMPLE. Finally, we introduce the Cayley graph of the presentation $\langle x, y \mid x^2 = y^2 = 1 \rangle$. Compared with the Cayley graph of the free group $F(\{x, y\})$, one can intuitively observe that this presentation is no longer “free”.



Intuitively, once relations are imposed on a free group, the result manifests itself in the Cayley graph as the appearance of “cycles”.

4.6.6 THEOREM (Second Isomorphism Theorem). Let G be a group and let H, K be normal subgroups of G . If $K \subset H$, then $K \trianglelefteq H$, $H/K \trianglelefteq G/K$, and

$$G/H \cong (G/K)/(H/K).$$

In fact, there is a unique isomorphism $\varphi : G/H \rightarrow (G/K)/(H/K)$ such that $\varphi \circ \pi_H = \pi_{H/K} \circ \pi_K$ where $\pi_H : G \rightarrow G/H$, $\pi_K : G \rightarrow G/K$ and $\pi_{H/K} : G/K \rightarrow (G/K)/(H/K)$ are the canonical projections:

$$\begin{array}{ccc} G & \xrightarrow{\pi_K} & G/K \\ \pi_H \downarrow & \swarrow \sigma & \downarrow \pi_{H/K} \\ G/H & \xrightarrow{\varphi} & (G/K)/(H/K) \end{array}$$

Proof. Since $K \subset \ker \pi_H$, there exists a unique homomorphism $\sigma : G/K \rightarrow G/H$ such that $\pi_H = \sigma \circ \pi_K$ by the universal property of quotient groups, so that σ is defined by $gK \mapsto gH$. Hence σ is surjective. We show that $\ker \sigma = H/K$:

First, $K \trianglelefteq H$, since $K \trianglelefteq G$. If $hK \in H/K$, where $h \in H$, then $\sigma(hK) = hH = 1$ in G/H . Conversely, if $\sigma(gK) = 1$, then $gH = H$ and $g \in H$. Thus, $\ker \sigma = \{hK : h \in H\} = H/K$; in particular, $H/K \trianglelefteq G/K$. By the first isomorphism theorem, $G/H = \text{im } \sigma \cong (G/K)/\ker \sigma = (G/K)/(H/K)$. In fact, the first isomorphism theorem yields an isomorphism $\varphi : G/K \rightarrow (G/H)/(H/K)$ such that $\varphi \circ \sigma = \pi_{H/K}$, and then $\varphi \circ \pi_H = \pi_{H/K} \circ \pi_K$. $\square$

4.6.7 THEOREM (Third Isomorphism Theorem). *Let G be a group and let $H \leq G$, $K \trianglelefteq G$. Then $HK \trianglelefteq G$, $K \trianglelefteq HK$, $H \cap K \trianglelefteq H$ and*

$$HK/K \cong H/(H \cap K).$$

In fact, there is a unique isomorphism $\varphi : H/(H \cap K) \rightarrow HK/K$ such that $\varphi \circ \pi_{H \cap K} = \pi_K \circ \iota$, where $\pi_{H \cap K}$ and π_K are canonical projections and ι is the inclusion homomorphism:

$$\begin{array}{ccc} H & \xrightarrow{\pi_{H \cap K}} & H/(H \cap K) \\ \iota \downarrow & \searrow \sigma & \downarrow \varphi \\ HK & \xrightarrow{\pi_K} & HK/K \end{array}$$

Proof. We show that $HK \leq G$. First, $1 \in HK$. Since $K \trianglelefteq G$, $HK = KH$; hence $hk \in HK$ implies $(hk)^{-1} = k^{-1}h^{-1} \in KH = HK$. Finally, $HKHK = HHHK = HK$.

Now, $K \trianglelefteq HK$. Let $\sigma = \pi_K \circ \iota$. Then $\sigma(h) = hK \in HK/K$ for all $h \in H$, and σ is surjective. Moreover, $\sigma(h) = 1$ if and only if $h \in K$, so that $\ker \sigma = H \cap K$; in particular, $H \cap K \trianglelefteq H$. By the first isomorphism theorem, $HK/K = \text{im } \sigma \cong H/\ker \sigma = H/(H \cap K)$; in fact, there is a unique isomorphism $\varphi : H/(H \cap K) \rightarrow HK/K$ such that $\varphi \circ \pi_{H \cap K} = \sigma = \pi_K \circ \iota$. $\square$

This theorem is actually quite easy to understand. Intuitively, HK/K is obtained by “modding out” the K -part of HK . However, the result is not directly H , since H and K may overlap—that is, elements in $H \cap K$ are identified with the identity in K . Hence, what remains is essentially H modulo its intersection with K , namely $H/(H \cap K)$.

4.7 Composition Series

Jordan-Hölder Theorem

4.7.1 DEFINITION. A *series* of subgroups G_i of a group G is a strictly decreasing sequence of subgroups starting from G :

$$G = G_0 \supset G_1 \supset G_2 \supset \cdots \supset G_n \supset \cdots$$

We are more interested in series that end with $\{1\}$ (some textbooks simply define series this way). In this case, we can define the notion of length. The *length* of the series is the number of strict inclusions. A series is *normal* if $G_{i+1} \trianglelefteq G_i$ for all i . The *factors* of a normal series are the quotient groups G_i/G_{i+1} .

All non-strict inclusions in a series are of little significance and are merely redundant repetitions. Therefore, for convenience of discussion, we shall directly consider series that are entirely strict inclusions:

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \cdots \supsetneq G_n = \{1\}.$$

We can naturally perform a so-called “refinement” on a normal series. Intuitively, the process of obtaining a new normal series by repeatedly inserting new normal subgroups via expressions of the form

$$[\cdots G_i \supsetneq G_{i+1} \supsetneq \cdots] \rightsquigarrow [\cdots \supsetneq G_1 \supsetneq G' \supsetneq G_{i+1} \supsetneq \cdots]$$

is called a refinement. The refinement process naturally increases the length of a normal series, which leads us to be interested in the maximal possible length. We denote the maximal length of a normal series of a group G by $\ell(G)$.

It is evident that if $\ell(G) = 0$, then G must be the trivial group $\{1\}$; if $\ell(G) = 1$, then every normal series of G contains at most one strict inclusion. If $\ell(G) = 1$, then the normal subgroups of the group G can only be trivial, namely G itself and $\{1\}$. We call such groups *simple* groups.

We are very interested in simple groups. To some extent, $\ell(G)$ reflects how “simple” the group G is; the larger it is, the less simple it is. More concretely, we need to consider the process starting from $G \supsetneq \{1\}$ and refining it until no further refinement is possible. How can we tell whether a refinement is still possible? We should naturally examine the quotient groups G_i/G_{i+1} . By Theorem 4.6.9, the preimage of any normal subgroup of G_i/G_{i+1} is a normal subgroup of G_i containing G_{i+1} , which naturally allows for a refinement. Hence, when all the factors G_i/G_{i+1} are simple groups, no further refinement is possible. Then, we give the following definition.

4.7.2 DEFINITION. A *composition series* for a group G is a normal series $G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{1\}$ such that the factors G_i/G_{i+1} are simple.

4.7.1 REMARK. Every finite groups have composition series, while infinite groups do not necessarily have one. One may consider the group $\mathbf{Z}$.

It is clear that if a normal series has the maximal length $\ell(G)$, then it is a composition series. We expect that every composition series (that is, a normal series which admits no further refinement) attains the maximal length $\ell(G)$. However, it is not a priori clear whether the length of a composition series truly equals $\ell(G)$, since there might conceivably be an accidental refinement process that produces a composition series whose length falls short of $\ell(G)$. Fortunately, such a situation never happens. This is guaranteed directly by the Jordan-Hölder theorem. However, prior to proving this theorem, we need to establish several lemmas.

4.7.3 DEFINITION. Let $G = G_0 \supsetneq G_1 \supsetneq \cdots \supsetneq G_r = \{1\}$ be normal series. We regard its subquotients $(G_i/G_{i+1})_{0 \leq i \leq r-1}$ as a multiset (i.e., a set counted with multiplicity, disregarding order). If two normal series have the same length and their subquotients are equal in the above sense, then the two normal series are said to be *equivalent*.

4.7.1 LEMMA (Zassenhaus). *Let G be a group. Consider subgroups U, V and their respectively normal subgroups $u \trianglelefteq U, v \trianglelefteq V$. Then $u(U \cap v), u(U \cap V), (u \cap V)v$ and $(U \cap V)v$ are subgroups of G ; and*

$$\begin{aligned} u(U \cap v) &\trianglelefteq u(U \cap V), \\ (u \cap V)v &\trianglelefteq (U \cap V)v, \\ \frac{u(U \cap V)}{u(U \cap v)} &\cong \frac{(U \cap V)v}{(u \cap V)v}. \end{aligned}$$

Proof. Draw the lattice as following. Vividly, this theorem is therefore called the Butterfly Lemma. If one asks which theorem in group theory is the most “beautiful,” then this one surely deserves the title. We now verify it step by step from top to bottom on the left wing (and similarly for the right wing).

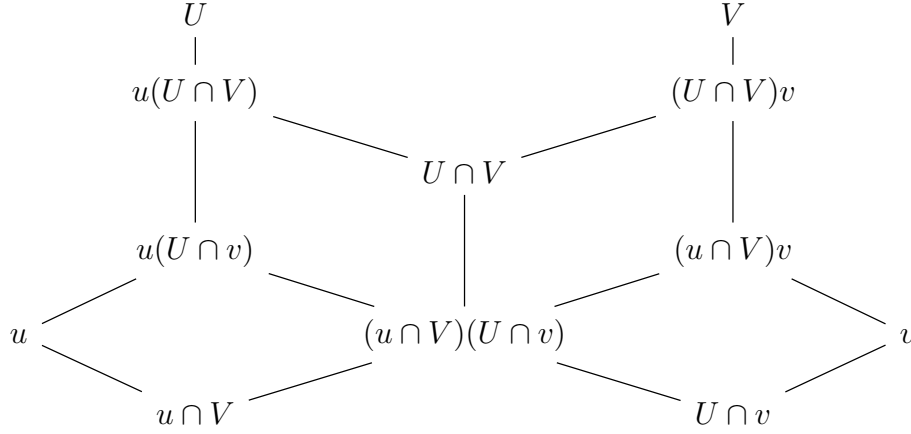
$u(U \cap v)$ and $u(U \cap V)$ are subgroups of U , since $u \trianglelefteq U$ and $U \cap V, U \cap v \leq U$. Also $U \cap v \trianglelefteq U \cap V$ since $v \trianglelefteq V$. Let $x = ab \in u(U \cap V)$ and $y = cd \in u(U \cap v)$ with $a, c \in u, b \in U \cap V, d \in U \cap v$. Then $xcx^{-1} \in u$, since $x \in U$ and $u \trianglelefteq U$; $bdb^{-1} \in U \cap v$, since $b \in U \cap V$ and $U \cap v \trianglelefteq U \cap V$; $xdx^{-1} = abdb^{-1}a^{-1} \in u(U \cap v)u = u(U \cap v)$, since $u \trianglelefteq U$; and $xyx^{-1} = xcx^{-1}xdx^{-1} \in u(U \cap v)$. Thus $u(U \cap v) \trianglelefteq u(U \cap V)$.

Let $S = U \cap V$, $T = u(U \cap v)$, and $M = u(U \cap V)$. Then $S \leq M, T \leq M$ and $ST = TS = u(U \cap v)(U \cap V) = u(U \cap V) = M$.

Now we claim $S \cap T = (u \cap V)(U \cap v)$. First, $u \cap V \subset S, u \cap V \subset u \subset T, U \cap v \subset S$, and $U \cap v \subset T$, so that $(u \cap V)(U \cap v) \subset S \cap T$. Conversely, if $t \in S \cap T$, then $t \in U \cap V$ and $t = ab$ for some $a \in u$ and $b \in U \cap v$; then $b \in V$, $a = tb^{-1} \in V$, and $s = ab \in (u \cap V)(U \cap v)$. Thus $S \cap T = (u \cap V)(U \cap v)$.

By the second isomorphism theorem, $T \triangleleft ST, S \cap T \triangleleft S$, and $ST/T \cong S/(S \cap T)$. Hence $(u \cap V)(U \cap v) = S \cap T \triangleleft S = U \cap V$ and

$$\frac{u(U \cap V)}{u(U \cap v)} = ST/T \cong S/(S \cap T) = \frac{(U \cap V)}{(u \cap V)(U \cap v)}.$$



□

4.7.2 THEOREM (Schreier’s Theorem). *Let*

$$\begin{aligned} G &= G_0 \supsetneq \cdots \supsetneq G_r \supsetneq G_{r+1} = \{1\}; \\ G &= H_0 \supsetneq \cdots \supsetneq H_s \supsetneq H_{s+1} = \{1\}, \end{aligned}$$

be two normal series of G . Then the two series have equivalent refinements.

Proof. For each $0 \leq i \leq r$ and $0 \leq j \leq s$, define

$$G_{i,j} := G_{i+1}(H_j \cap G_i), \quad H_{j,i} := (G_i \cap H_j)H_{j+1}.$$

First consider $G_{i,j}$. Since $G_{i+1} \triangleleft G_i$, it is a subgroup. The inclusion $G_{i,j+1} \trianglelefteq G_{i,j}$ holds, and moreover, $G_{i,0} = G_{i+1}(G \cap G_i) = G_i$, $G_{i,s+1} = G_{i+1}$. Thus we obtain a refinement of $(G_i)_{i=0}^r$ (Since the resulting $G_{i,j+1}$ and $G_{i,j}$ may be identical, this does not strictly count as a refinement in the proper sense. However, for convenience of exposition, we shall also call it a refinement.):

$$\mathcal{G} := [\cdots \supsetneq G_i = G_{i,0} \supsetneq G_{i,1} \supsetneq \cdots \supsetneq G_{i,s} \supsetneq G_{i,s+1} = G_{i+1} \supsetneq \cdots].$$

Similarly, $H_{j,i}$ gives a refinement of $(H_j)_{j=0}^s$, denoted by $\mathcal{H}$. Applying the Lemma with $u := G_{i+1}$, $U := G_i$, and $v := H_{j+1}$ and $V := H_j$, we obtain

$$\frac{G_{i,j}}{G_{i,j+1}} = \frac{u(U \cap V)}{u(U \cap v)} \cong \frac{(U \cap V)v}{(u \cap V)v} = \frac{H_{j,i}}{H_{j,i+1}}.$$

As (i, j) ranges over all possible pairs, each subquotient of the normal series $\mathcal{G}$ and $\mathcal{H}$ appears exactly once on each side up to isomorphism. (Now we perform the process of deleting redundant identical subgroups. From the above formula, we have that $G_{i,j+1} = G_{i,j}$ if and only if $H_{j,i+1} = H_{j,i}$. Therefore, in the end, the number of redundant identical subgroups deleted from $\mathcal{G}$ and $\mathcal{H}$ is exactly the same.) Hence the theorem follows. $\square$

4.7.3 COROLLARY (Jordan-Hölder Theorem). *Any two composition series of a group G are equivalent. That is, any two composition series have the same length and their subquotients are identical as multisets.*

Therefore, once a group has a composition series, its subquotients are independent of the choice of the composition series in the sense of equivalence. Hence we can give the following definition.

4.7.4 DEFINITION. Assume that a group G has a composition series. Define its set of composition factors, or Jordan-Hölder factors, denoted by $\text{JH}(G)$, to be the multiset of all subquotients of any composition series of G .

Roughly speaking, the set of composition factors $\text{JH}(G)$ is an invariant of the group G . However, it does not completely characterize the group; for example, the groups $\mathbf{Z}/4\mathbf{Z}$ and $\mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}$ both have the same composition factor set $\{\mathbf{Z}/2\mathbf{Z}, \mathbf{Z}/2\mathbf{Z}\}$. Nevertheless, this still provides us with an important tool for understanding the structure of groups, and it is one of the reasons why we study composition series.

4.7.4 PROPOSITION. *Let G be a group and let N be a normal subgroup of G . Then G has a composition series if and only if both N and G/N have composition series. Further, if this is the case, then*

$$\ell(G) = \ell(N) + \ell(G/N) \text{ and } \text{JH}(G) = \text{JH}(N) \sqcup \text{JH}(G/N).$$

Proof. Suppose that G/N has a composition series $G/N = \overline{G}_0 \supsetneq \overline{G}_1 \supsetneq \cdots \supsetneq \overline{G}_r = \{1N\}$. By the theorem 4.6.9, each $\overline{G}_i$ is of the form G_i/N , where $G_i \leq G$ containing N ; and $\overline{G}_i/\overline{G}_{i+1} \cong G_i/G_{i+1}$ by the third isomorphism theorem. Moreover $G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_r = N$ is a composition series by the theorem 4.6.10. Suppose N has a compatible series $N = N_0 \supsetneq N_1 \supsetneq \cdots \supsetneq N_s = \{1\}$. Juxtaposing the two chains yields

$$G = G_0 \supsetneq G_1 \supsetneq \cdots \supsetneq G_r = N = N_0 \supsetneq N_1 \supsetneq \cdots \supsetneq N_s = \{1\}.$$

Conversly, suppose that G has a composition series $G = G_0 \supsetneq G_1 \supsetneq \cdots \supsetneq G_n = \{1\}$ and that N is a normal subgroup of G . Intersecting the series with N gives a sequence of subgroups of the latter:

$$N = G \cap N \supsetneq G_1 \cap N \supsetneq \cdots \supsetneq \{1\} \cap N = \{1\},$$

such that $G_{i+1} \cap N$ is normal in $G_i \cap N$ for all i . We claim that this is a composition series for N once repetitions are eliminated: indeed, this follows once we establish that

$$\frac{G_i \cap N}{G_{i+1} \cap N}$$

is either trivial (so that $G_{i+1} \cap N = G_i \cap N$) or isomorphic to G_i/G_{i+1} . To see this, consider the homomorphism $G_i \cap N \hookrightarrow G_i \twoheadrightarrow G_i/G_{i+1}$ whose kernel is clearly $G_{i+1} \cap N$. Therefore (by the first isomorphism theorem) we have an injective homomorphism

$$\frac{G_i \cap N}{G_{i+1} \cap N} \hookrightarrow G_i/G_{i+1}$$

identifying $(G_i \cap N)/(G_{i+1} \cap N)$ with a subgroup of G_i/G_{i+1} . Now, this subgroup is normal and G_i/G_{i+1} is simple; our claim follows.

As for G/N , obtain a sequence of subgroups from a composition series for G : $G/N \supset (G_1N)/N \supset (G_2N)/N \supset \cdots \supset (\{1\}N)/N$, such that $(G_{i+1}N)/N$ is normal in $(G_iN)/N$. As above, we should check that

$$\frac{(G_iN)/N}{(G_{i+1}N)/N}$$

is either trivial or isomorphic to G_i/G_{i+1} . By the third isomorphism theorem, this quotient is isomorphic to $(G_iN)/(G_{i+1}N)$. Now, consider the homomorphism $G_i \hookrightarrow G_iN \twoheadrightarrow (G_iN)/(G_{i+1}N)$. One may check that this is a surjective homomorphism and the subgroup G_{i+1} of the source is sent to the identity element in the target; hence by the universal property of quotient group there is an onto homomorphism

$$G_i/G_{i+1} \twoheadrightarrow (G_iN)/(G_{i+1}N).$$

Since G_i/G_{i+1} is simple, it follows that $(G_iN)/(G_{i+1}N)$ is either trivial or isomorphic to it (by the first isomorphism theorem $(G_i/G_{i+1})/\ker \varphi \cong (G_iN)/(G_{i+1}N)$). But G_i/G_{i+1} is simple, the only possibility of $(G_i/G_{i+1})/\ker \varphi$ are G_i/G_{i+1} or $\{1\}$, as desired. $\square$

Solvable Groups

4.7.5 DEFINITION. Let G be a group, the *commutator* of $g \in G$ and $h \in G$ is the element $ghg^{-1}h^{-1}$; and is often denoted $[g, h]$. For arbitrary sets A and B , define $[A, B] = \langle [a, b] : a \in A, b \in B \rangle$. In particular, we are more concerned with $[G, G]$. At the same time, we may define it recursively:

$$\begin{cases} \mathcal{D}^{i+1}G &:= \mathcal{D}(\mathcal{D}^iG) = [\mathcal{D}^iG, \mathcal{D}^iG] & i \geq 0 \\ \mathcal{D}^0G &:= G \end{cases}$$

4.7.5 PROPOSITION. Let G_1, G_2 be groups and $\varphi : G_1 \rightarrow G_2$ a homomorphism.

1. $\varphi([g, h]) = [\varphi(g), \varphi(h)]$ for any $g, h \in G_1$;
2. $\mathcal{D}^iG_1 \trianglelefteq G_1$ for every $i \geq 0$.

In particular, we call $G_{\text{der}} := \mathcal{D}^1G$ the *derived subgroup* of G , and $G_{\text{ab}} := G/G_{\text{der}}$ the *abelianization* of G . Let $g, h \in G$, then $gh = [g, h]hg$, so that $gh = hg$ if and only if $[g, h] = 1$. We call $[g, h]$ the non-commutative deviation of the elements $g, h \in G$ (this is not a mathematical definition, but merely for intuitive understanding, since we cannot truly regard the commutator as something like a metric or a norm). The derived subgroup $[G, G]$ is the normal subgroup generated by these non-commutative deviations. When we choose to mod out the derived subgroup, informally speaking, the non-commutative deviations of two elements are absorbed, so they commute. Specifically, we have the following theorem:

4.7.6 PROPOSITION. G_{ab} is abelian. And G_{ab} together with the canonical projection $\pi : G \rightarrow G_{\text{ab}}$ satisfies the universal property: for any abelian group A and homomorphism $\varphi : G \rightarrow A$, there exists a unique homomorphism $\tilde{\varphi} : G_{\text{ab}} \rightarrow A$ such that the diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{\pi} & G_{\text{ab}} \\ & \searrow \varphi & \swarrow \tilde{\varphi} \\ & A & \end{array}$$

In particular, G/N is abelian if and only if $G_{\text{der}} = \mathcal{D}G \subset N$. (Consider the canonical projection $\pi_N : G \rightarrow G/N$ whose kernel $\ker \pi_N = N$)

Proof. For the first statement, the commutator of two coset $gG_{\text{der}}, hG_{\text{der}}$ is the coset $ghg^{-1}h^{-1}G_{\text{der}}$, i.e., the coset of the commutator $[g, h]$. Hence it is the identity G_{der} in G/G_{der} . It follows that G/G_{der} is abelian.

For any $[a, b] \in G_{\text{der}}$, $\varphi([a, b]) = [\varphi(a), \varphi(b)] = 1$ since A is abelian. So that $G_{\text{der}} \subset \ker \varphi$. By the universal property of quotient group, there exists a unique homomorphism $\tilde{\varphi} : G/G_{\text{der}} \rightarrow A$ such that the diagram commutes. $\square$

Clearly, these derived subgroups form a descending series (if not strictly contained, it is regarded as the same as the previous subgroup), and it is a normal series. We call this series the derived series:

$$G = \mathcal{D}^0 G \supseteq \mathcal{D}^1 G \supseteq \mathcal{D}^2(G) \supseteq \cdots \supseteq \mathcal{D}^n G \supseteq \cdots$$

By the same analysis as above, $\mathcal{D}^2 G = [\mathcal{D}G, \mathcal{D}G]$ is again the non-commutative deviation of the non-commutative deviations; we may call it the second-order non-commutative deviation (and correspondingly there are i -th order deviations). Similarly, $\mathcal{D}G/\mathcal{D}^2 G$ is also an abelian group. In general, each $\mathcal{D}^i G/\mathcal{D}^{i+1} G$ is abelian.

Let us explain this process in a figurative and informal language. $\mathcal{D}G$ strips away the non-commutativity of G , and $\mathcal{D}^2 G$ strips away the non-commutativity of $\mathcal{D}G$, which is already contained in $\mathcal{D}G$. The stripping of non-commutativity of $\mathcal{D}G$ by $\mathcal{D}^2 G$ can be regarded as part of the total non-commutativity of G (hence $G^{i+1} \subset G^i$). In other words, $\mathcal{D}^2 G$ finely strips away a smaller portion of non-commutativity than what $\mathcal{D}G$ strips away. Continuing this process, we repeatedly strip away finer and finer layers. At each step of this process, a quotient factor, which is an abelian group, is produced. When we can no longer strip anything further, this indicates that the non-commutative part stripped out in the previous step is completely abelian, and we can no longer refine the stripping process (any finer and even the identity 1 would be gone!!!). In a word, the derived series is like searching through a garbage heap for garbage within garbage that meets stricter and stricter screening criteria, and repeating this process over and over again:

Suppose G is all the things in my house (including both what I consider garbage and what I consider non-garbage). Then $\mathcal{D}G$ represents the garbage that I throw away, i.e., what I consider to be trash. $\mathcal{D}^2 G$ represents what the scavengers discard from this pile of garbage as the garbage they consider worthless—for example, they would keep plastic bottles and cardboard to exchange for money. $\mathcal{D}^3 G$ represents what the little dog discards from the remaining garbage left by the scavengers as the garbage it considers worthless—for example, the little dog would keep the expired bread inside. $\mathcal{D}^4 G$ represents what the microorganisms discard from the remaining garbage left by the little dog as the garbage they consider worthless, but they can decompose anything, so at this point there is no more garbage left, i.e., $\mathcal{D}^4 G = \{1\}$. And the things that are retained throughout this process are relatively not garbage, which corresponds precisely to what we call abelian groups. (This is just an informal description of the meaning of the derived series; we may delve deeper into this process later.)

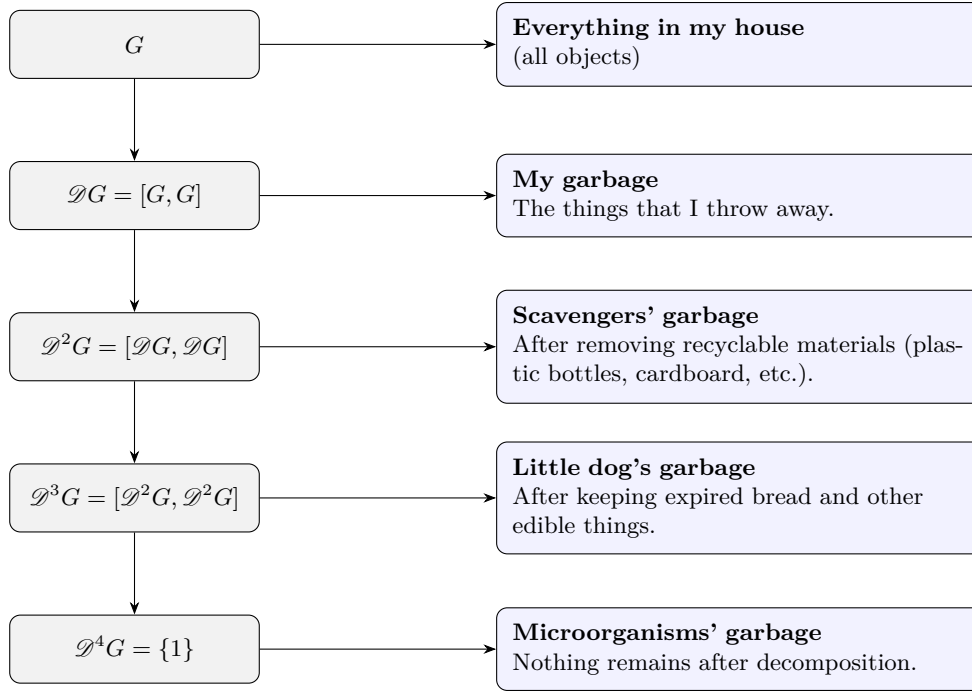


Figure 4.2: An informal interpretation of the derived series.

The derived series need not necessarily end at the trivial subgroup $\{1\}$. For example, when G is a non-abelian simple group, we have $G = \mathcal{D}^1G = \mathcal{D}^2G = \cdots$. Of course, if G itself is abelian, then naturally we have $G \supseteq \mathcal{D}^1G = \{1\}$. We are interested in such terminating derived series, so we have the following definition.

4.7.6 DEFINITION. A group G is *solvable* if its derived series terminates with the identity. That is, there exists $n \geq 0$ such that $\mathcal{D}^nG = \{1\}$.

4.7.7 PROPOSITION. Let G and K be groups, let H be a subgroup of G and let $\varphi : G \rightarrow K$ be a surjective homomorphism.

1. $\mathcal{D}^iH \leq \mathcal{D}^iG$ for all $i \geq 0$. In particular, if G is solvable, then so is H , in other words, subgroups of solvable groups are solvable.
2. $\varphi(\mathcal{D}^iG) = \mathcal{D}^iK$. In particular, consider the canonical projection $\pi : G \rightarrow G/N$, every quotient group of a solvable group is solvable;
3. If $N \trianglelefteq G$ and G/N are solvable, then G is solvable.

Proof. For (1), assume that G is solvable and $H \leq G$, then $\mathcal{D}H \subset \mathcal{D}G$, $\mathcal{D}^iH \subset \mathcal{D}^i(G)$ for each i , so that $\mathcal{D}^nG = \{1\}$ implies $\mathcal{D}^nH = \{1\}$.

For (2), we prove it by induction on i . For $i = 0$, $\varphi(G) = K$. Assume that $\varphi(\mathcal{D}^iG) = \mathcal{D}^iK$. We first show that $\varphi(\mathcal{D}^{i+1}G) \leq \mathcal{D}^{i+1}K$: for any $x, y \in \mathcal{D}^iG$, $\varphi([x, y]) = [\varphi(x), \varphi(y)] \in [\mathcal{D}^iG, \mathcal{D}^iG] = [\mathcal{D}^iK, \mathcal{D}^iK] = \mathcal{D}^{i+1}K$. So that $\varphi(\mathcal{D}^{i+1}G) \leq \mathcal{D}^{i+1}K$. To show the converse, Since φ is surjective, every commutator in K is the image of a commutator in G , for any $u, v \in \mathcal{D}^iK = \varphi(\mathcal{D}^iG)$, there exists $x, y \in \mathcal{D}^iG$ such that $\varphi(x) = u, \varphi(y) = v$. Then

$$[u, v] = [\varphi(x), \varphi(y)] = \varphi([x, y]) \in \varphi(\mathcal{D}^{i+1}G).$$

If G is solvable, then $\mathcal{D}^nG = \{1\}$, and $\mathcal{D}^nK = \varphi(\mathcal{D}^nG) = \varphi(\{1\}) = \{1\}$. In particular, $\pi(\mathcal{D}^nG) = \pi(\{1\}) = N$.

For (3), if G/N and N are solvable, then by (2) applied to the canonical projection $\pi : G \rightarrow G/N$ we obtain that $\pi(\mathcal{D}^nG) = N$, i.e., $\mathcal{D}^nG \leq N$. If $\mathcal{D}^mN = \{1\}$, then $\mathcal{D}^{n+m}G = \mathcal{D}^m(\mathcal{D}^nG) = \{1\}$. It follows that G is solvable, as desired. $\square$

4.7.8 PROPOSITION. G is solvable if and only if there exists a normal series $G = G_0 \supsetneq G_1 \supsetneq \cdots \supsetneq G_r = \{1\}$ such that every factor G_i/G_{i+1} is abelian.

Proof. If $\mathcal{D}^n G = \{1\}$, then $G \supseteq \mathcal{D}^1 G \supseteq \mathcal{D}^2 G \supseteq \cdots \supseteq \mathcal{D}^n G = \{1\}$ is a normal series and each factor $\mathcal{D}^i G / \mathcal{D}^{i+1} G$ is abelian. Conversely, assume that G has a normal series $G \supsetneq G_1 \supsetneq \cdots \supsetneq G_n = \{1\}$ whose factors G_i / G_{i+1} are abelian. Then we claim that $\mathcal{D}^i G \subset G_i$ for all i : this can be verified by induction. For $i = 1$, G / G_1 is abelian, thus $\mathcal{D}^1 G \subset G_1$. Assume that $\mathcal{D}^i G \subset G_i$. Since G_i / G_{i+1} is abelian, $\mathcal{D}^i G \subset G_{i+1}$, and hence

$$\mathcal{D}^{i+1} G = \mathcal{D}(\mathcal{D}^i G) \subset \mathcal{D}G_{i+1} \subset G_{i+1}.$$

□

4.7.9 THEOREM (Feit-Thompson). *If $|G|$ is odd, then G is solvable.*

The original proof of this theorem runs to 255 pages, and this length alone suffices to show how highly intricate and technical it is. For the average practitioner, it is both completely impractical and pointless to devote oneself to studying that proof in detail.

In fact, even some of the theorem proofs in these very notes are “pointless” in the same sense, for instance, the proofs of the Butterfly Lemma and the Schreier Refinement Theorem. Rather than spending a great deal of time on the tedious derivations of these theorems, it is better to focus on building one’s own understanding of the core ideas behind them.

Nilpotent Groups

Each time we take the derived subgroup, the resulting set extracts the “non-commutative information” of the target set itself. Now consider a more aggressive extraction method: we extract the “non-commutative information” between this set and the original set. Specifically, we consider the following recursive form:

4.7.7 DEFINITION.

$$\begin{cases} \mathcal{C}^{i+1} G &:= \mathcal{C}(\mathcal{C}^i G) = [\mathcal{C}^i G, G] & i \geq 0 \\ \mathcal{C}^0 G &:= G \end{cases}$$

It is clear that $\mathcal{D}^i G \subset \mathcal{C}^i G$ and $\mathcal{C}^i G \trianglelefteq G$ for each i . Similarly, we will obtain the following normal series, called the lower central series.

$$G = \mathcal{C}^0 G \supseteq \mathcal{C}^1 G \supseteq \mathcal{C}^2 G \supseteq \cdots \supseteq \mathcal{C}^n G \supseteq \cdots$$

We will not go into too much detail about the specific recursive process; here we still use the garbage-throwing analogy to intuitively understand the descending central series:

Initially, the garbage I throw away is $\mathcal{C}(G) = [G, G]$. The scavengers outside are very cautious and may think that some bottles and cardboard are cursed. So they come to my house, observe the placement and order of all my furniture and belongings, and determine that among the garbage I threw away, some bottles and cardboard that they would not normally discard are actually cursed. They become very worried and decide to discard some of the bottles and cardboard they would otherwise have kept. Together with the garbage they would have discarded anyway, they obtain $\mathcal{C}^2 G = [\mathcal{C}G, G]$. Then the little dog, being a very clever dog, also thinks that some of this garbage might have been contaminated with chocolate (assuming this dog cannot smell chocolate and can only rely on sight). So the dog also comes to my house, observes all my things, and decides that some expired bread might have been touched by chocolate, so it decides to treat these as garbage as well. Combining this with the garbage the dog had previously discarded, we get $\mathcal{C}^3 G = [\mathcal{C}^2 G, G]$.

Finally, we have the microorganisms, which are somewhat intelligent” microorganisms. They might think that I have sprayed



Figure 4.3: The Clever Dog, Mr. Peabody (from the animated film *Mr. Peabody & Sherman*, DreamWorks, 2014).

something on these garbage items to inhibit microbial growth, so they come to my house and pretend to investigate. But I am far too busy to deal with such things, so luckily for them, they can eat all the garbage, giving $\mathcal{C}^4 G = [\mathcal{C}^3 G, G] = \{1\}$. The clever dog and the intelligent bacteria are both “real” —the author has a basis in reality for them.



Figure 4.4: *Physarum polycephalum* (slime mold) in its plasmodial stage. Image by frankenstoan, via Flickr (CC BY 2.5), from Wikimedia Commons.

Likewise, the lower central series need not end at the trivial subgroup. For example, $S_3 \supsetneq A_3 = A_3 = \dots$. In addition, we introduce another such example, the infinite dihedral group D_∞ .

4.7.1 EXAMPLE.

$$D_\infty = \langle r, s \mid s^2 = 1, srs = r^{-1} \rangle$$

One may show that $\mathcal{C} D_\infty = \langle r^2 \rangle$, $\mathcal{C}^2 D_\infty = \dots = \mathcal{C}^n D_\infty = \langle r^2 \rangle$. However, it is solvable, since $[\langle r^2 \rangle, \langle r^2 \rangle] = \{1\}$ (or $D_\infty / \langle r^2 \rangle \cong \mathbf{Z}/2\mathbf{Z}$ and $\langle r^2 \rangle \cong \mathbf{Z}$ where $\mathbf{Z}/2\mathbf{Z}$ and $\mathbf{Z}$ are abelian (so is simple)).

4.7.8 DEFINITION. A group G is *nilpotent* if its lower central series terminates with the trivial group $\{1\}$. That is, there exists $n \geq 0$ such that $\mathcal{C}^n G = \{1\}$.

According to this definition, it is easy to see that every nilpotent group is solvable (since $\mathcal{C}^i G \supset \mathcal{D}^i G$). The converse is not true, as shown by the previous example of the infinite dihedral group, which is solvable but not nilpotent.

4.7.10 PROPOSITION. Let G and K be groups, let H be a subgroup of G and let $\varphi : G \rightarrow K$ be a surjective homomorphism.

1. $\mathcal{C}^i H \leq \mathcal{C}^i G$ for all $i \geq 0$. In particular, if G is nilpotent, then so is H , in other words, subgroups of nilpotent groups are nilpotent.
2. $\varphi(\mathcal{D}^i G) = \mathcal{D}^i K$. In particular, consider the canonical projection $\pi : G \rightarrow G/N$, every quotient group of a nilpotent group is nilpotent;
3. If $N \subset Z(G)$, and G/N is nilpotent, then G is nilpotent.

Proof. □

4.7.11 PROPOSITION. A group G is nilpotent if and only if there exists a normal series $G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \dots \supsetneq G_r = \{1\}$ such that $G_i \trianglelefteq G$ and $G_i/G_{i+1} \subset Z_{G/G_{i+1}}$ for each i (this is equivalent to $[G_i, G] \subset G_{i+1}$ or $[G_i/G_{i+1}, G/G_{i+1}] = \{1\}$).

Proof. Suppose that G is nilpotent, and $G \supset \mathcal{C} G \supset \mathcal{C}^2 G \supset \dots \supset \mathcal{C}^n G = \{1\}$. Let $\overline{G} := G/\mathcal{C}^{i+1} G$, clearly $[\mathcal{C}^i/\mathcal{C}^{i+1} G, \overline{G}] = \{1\}$, so that $\mathcal{C}^i G/\mathcal{C}^{i+1} G \subset Z_{G/\mathcal{C}^{i+1} G}$, as desired. Conversely, assume that G has a normal series $G \supsetneq G_1 \supsetneq \dots \supsetneq G_r = \{1\}$ such that $G_i \trianglelefteq G$ and $G_i/G_{i+1} \subset Z_{G/G_{i+1}}$. We claim that $\mathcal{C}^i G \subset G_i$ for each i ; this can be verified by induction on i . For $i = 0$, $G \subset G$ is clear. Then $\mathcal{C}^{i+1} G = \mathcal{C}(\mathcal{C}^i G) \subset \mathcal{C}(G_i) \subset G_{i+1}$ (since $G_i/G_{i+1} \subset Z_{G/G_{i+1}}$ implies $[G_i, G] \subset G_{i+1}$). □

4.7.2 EXAMPLE. (This example and proof follow 李文威, 代数学方法 (卷一: 基础架构), p. 125.) Let $\mathbf{K}$ be any field and let $\mathrm{GL}_n(\mathbf{K})$ be the group of invertible $n \times n$ matrices over $\mathbf{K}$. Consider the subgroup of $\mathrm{GL}_n(\mathbf{K})$:

$$U := \left\{ \begin{pmatrix} 1 & * & \dots & * \\ 0 & 1 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \right\},$$

where the diagonal entries of U are all 1. Then U is nilpotent. These facts can be verified by matrix computations, but it is perhaps more illuminating to adopt an abstract viewpoint. Let V be an n -dimensional $\mathbf{K}$ -vector space. Consider a series of subspaces

$$V = V_0 \supseteq V_1 \supseteq \dots \supseteq V_n = \{0\}, \quad \dim_{\mathbf{K}} V_i/V_{i+1} = 1 \quad (0 \leq i < n).$$

For convenience, set $V_m = \{0\}$ for $m > n$. Let $\text{GL}(V)$ denote the group of linear automorphisms of V . Define

$$U_r := \{x \in \text{GL}(V) : \forall i, (x-1)(V_i) \subset V_{i+r}\}, \quad r \in \mathbb{Z}_{\geq 1}.$$

Clearly, $U_1 \supset U_2 \supset \cdots \supset U_n = \{1\}$. Note that the definition of U_r implies $x(V_i) \subset V_i$ for all i ($(x-1)(V_i) \subset V_{i+r} \subset V_i$). We claim that each U_r is a subgroup and that

$$[U_r, U_s] \subset U_{r+s} \text{ for all } r, s \geq 1.$$

Indeed, if $x, x' \in U_r$, then $xx' - 1 = (x-1)x' + (x'-1)$ maps each V_i into V_{i+r} , so U_r is closed under multiplication. If $x = 1 - X \in U_r$, then X is nilpotent ($X^n = 0$, since $X(V_i) = -(x-1)(V_i) \subset V_{i+r}$, for any $v \in V$, $X(v) \in V_r$, $X^2(v) \in V_{2r}, \dots$), and

$$(1 - X)^{-1} = \sum_{k=0}^{n-1} X^k = 1 + X \sum_{k=1}^{n-1} X^{k-1} \in U_r,$$

so U_r is closed under inverses. Now let $x = 1 - X \in U_r$ and $y = 1 - Y \in U_s$. Expand the commutator $[y, x] = yxy^{-1}x^{-1}$ in $\text{End}_{\mathbf{K}}(V)$. Introducing formal variables q, v and writing qX, vY in place of X, Y , we have

$$(1 - vY)(1 - qX)(1 - vY)^{-1}(1 - qX)^{-1} = (1 - vY)(1 - qX) \sum_{h=0}^{n-1} v^h Y^h \sum_{k=0}^{n-1} q^k X^k$$

The right-hand side splits into: constant term (equal to 1), terms involving only q (which vanish upon setting $v = 0$), terms involving only v (which vanish upon setting $q = 0$), terms involving both q and v . Substituting $q = v = 1$ gives $[y, x]$.

Any term containing both X and Y maps V_i into V_{i+r+s} , hence lies in U_{r+s} . Therefore $[y, x] \in U_{r+s}$, and since commutators generate $[U_r, U_s]$, we obtain $[U_r, U_s] \subset U_{r+s}$. As a consequence, $[x, y]^{-1} = [y, x] \in U_{r+s}$, and we have shown $[U_r, U_s] \subset U_{r+s}$. In particular, since $xy^{-1} = [x, y]y$, we have for $r \geq s$ that $U_r \triangleleft U_s$.

Now set $U := U_1$. Then $U_r \triangleleft U$ and $[U_r, U] \subset U_{r+1}$. This shows that $(U_r)_{1 \leq r \leq n}$ is a central series for U , hence U is nilpotent.

5

Group II: Two Basic Decompositions in Group Theory

In the previous chapter, we introduced three basic constructions of groups: direct products, free products, and quotient groups. These three constructions correspond respectively to the intuitive ideas of “parallel combination,” “free amalgamation,” and “compression/collapsing.” Naturally, we may now ask the reverse question: given a group that has already been built, in what ways can it be decomposed back into those basic building blocks? In other words, can we recognize from a given group whether it is composed of simpler groups via the above constructions?

Since the decomposition problem for free products, namely, the uniqueness of free products or the so-called free decomposition of groups — is considerably more involved and far from trivial in general, we shall focus primarily on the two directions arising from direct products and quotient groups. More precisely, we shall discuss direct product decompositions of groups, i.e., when a group can be written as a direct product of two subgroups; and the group extension problem, i.e., when a group can be written as an “extension” of one of its quotient groups (the precise meaning of extension will be formally introduced later).

$$\boxed{N \trianglelefteq G, \quad Q \leq G, \quad NQ = G, \quad N \cap Q = 1}$$

The first three sections will cover the basic theory of group actions, the structure of symmetric groups, and Sylow’s theorems, which together form the preparatory foundation for this part.

As for why we need group actions, the deeper reason will gradually unfold in our discussion of group extensions. The reader will come to appreciate that direct sum decompositions of groups are essentially a special class of group extensions whether the conjugation action of a normal subgroup on its complement is trivial determines whether the extension degenerates into a direct product or retains a nontrivial semidirect product structure. Thus, group actions serve not only as the language for understanding extensions, but also as the bridge that unifies direct product decompositions and extension problems.

5.1 Group Actions

A group, unless it is abelian, can generally be very complicated in structure. We first focus on the extent to which elements in a group commute with each other. The higher the degree of commutativity, the relatively simpler the structure; the lower the degree of commutativity, the more complex the group is perceived to be. How can we define the degree of commutativity? To understand this, we can approach it from the perspectives of subgroups and group actions. Since we wish to clarify these terms first, we begin by discussing subgroups.

Center, Centralizer, Normalizer

For a group with many elements, it is natural to first understand its subgroups that contain fewer elements. Naturally, we collect all elements that commute with every element of the group; this collection is called the center of the group. For a group G , its center is denoted by $Z(G)$. In other words,

$$Z(G) := \{g \in G : \forall x \in G, gx = xg\}.$$

Note that the elements g collected here are those that commute with every element in G , not just with a specific x . It is easy to see that a larger $Z(G)$ indicates more commuting elements, so it is used to measure the degree to which a group is abelian. Moreover, $Z(G)$ is a subgroup of G , and since every element of $Z(G)$ commutes with all elements of G , $Z(G)$ is itself an abelian group.

5.1.1 PROPOSITION. *Let G be a finite group and $G/Z(G)$ cyclic, then G is abelian.*

Proof. Since $G/Z(G)$ is cyclic, there exists an element $g \in G$ such that the class $gZ(G)$ generates $G/Z(G)$. Then for every $a \in G$, $aZ(G) = (gZ(G))^r$ for some $r \in \mathbf{Z}$; that is, there is an element $z \in Z(G)$ of the center such that $a = g^r z$.

Let $a, b \in G$, and $a = g^r z, b = g^s w$ for some $r, s \in \mathbf{Z}$ and $z, w \in Z(G)$; then $ab = (g^r z)(g^s w) = g^{r+s} zw = (g^s w)(g^r z) = ba$ (since z and w commute with every element of G .) This proves that G is abelian. $\square$

Sometimes the commutativity of G is so poor that $Z(G)$ contains almost nothing. In such cases, we compromise by collecting the elements that commute with all elements of a subgroup H of G (since H has fewer elements than G , there should be more elements that commute with all of H). This is called the centralizer of H in G and is denoted by $C_G(H)$. In other words,

$$C_G(H) := \{g \in G : \forall x \in H, gx = xg\}$$

Obviously, when $H = G$, we have $C_G(G) = Z(G)$. However, do not mistakenly think that $C_G(H)$ is the center of H . By definition, the center of H must lie inside H , but $C_G(H)$ may contain elements outside H , so $C_G(H)$ is not necessarily a subgroup of H . On the other hand, every element of $C_G(H)$ commutes with all elements of H , but it is not necessarily true that elements of $C_G(H)$ commute among themselves (so $C_G(H)$ need not be abelian). Therefore, $C_G(H)$ is not necessarily the center of H either. However, if we restrict $C_G(H)$ to H , that is, consider $H \cap C_G(H)$, then this is precisely the center of H . Because:

$$H \cap C_G(H) = \{g \in H : \forall x \in H, gx = xg\} = Z(H).$$

From the proof that $C_G(H)$ is still a subgroup of G (though not necessarily abelian), it is not difficult to see that we do not actually need H to be a subgroup of G . That is, for any nonempty subset S of G , consider the set $C_G(S) := \{g \in G : \forall x \in S, gx = xg\}$ will still be a subgroup of G . We still call it the centralizer of S in G .

5.1.2 PROPOSITION. *If $H \leq G$ and $H = \langle S \rangle$, then $C_G(H) = C_G(S)$.*

Proof. It is clear that $C_G(H) \subset C_G(S)$. Now, suppose that $g \in C_G(S)$, so $gs = sg$ for all $s \in S$. Any $h \in H$ can be written as a finite product of elements of S and their inverses: $h = s_1^{\epsilon_1} s_2^{\epsilon_2} \cdots s_n^{\epsilon_n}$ with $s_i \in S$ and $\epsilon_i = \pm 1$. If g commutes with $s \in S$, it also commutes with s^{-1} since $gs = sg$ implies $s^{-1}g = gs^{-1}$; if g commutes with a and b , then it commutes with ab since $g(ab) = (ga)b = (ag)b = a(bg) = (ab)g$. Thus, g commutes with any finite product of elements in S and the inverses, so that $g \in C_G(H)$. $\square$

In general, we prefer to use $C_G(S)$ to find $C_G(H)$. Since the number of elements in S is often much smaller than that in $\langle S \rangle = H$, checking whether an element commutes with S is naturally much easier than checking whether it commutes with H . For example,

when S consists of a single element a , the smallest subgroup of G containing a , namely $\langle a \rangle$. When $S = \{a\}$ has only one element, we denote its centralizer by $C_G(a)$, i.e.,

$$C_G(a) := \{g \in G : ga = ag\}$$

and we directly call $C_G(a)$ the centralizer of a in G . Since $C_G(a) = C_G(\langle a \rangle)$, to find the elements in it, we only need to check whether they commute with a ; this is certainly much more convenient than checking the entire cyclic group $\langle a \rangle$.

In fact, even if we only consider the elements of a group that commute with all elements of a given subset, this condition is still rather restrictive. To address this, we may appropriately relax the commutativity requirement: instead of requiring that $gx = xg$ for all $x \in H$ (i.e., that g commutes with every element of the subgroup H), we may ask that for every $x \in H$, there exists some $x' \in H$ such that $gx = x'g$. In particular, if SH is a normal subgroup, then for every $g \in G$ we have $gH = Hg$. In general, we may define the *normalizer* of S in G by:

$$N_G(H) := \{g \in G : gH = Hg\}$$

In fact, $N_G(H)$ is a subgroup of G , but note that it is not necessarily a normal subgroup of G ; however, by definition, H is a normal subgroup of $N_G(H)$. Moreover, $N_G(H)$ is the largest subgroup of G in which H is normal. This is precisely the reason why it is called the normalizer.

Group Action

Group actions are an important tool for studying the structure of groups themselves. Roughly speaking, a group action allows one to obtain the structure and properties of a group indirectly by observing how the group “moves” on some object. (Of course, this requires choosing suitable group actions.) In addition to this, one can also study certain properties of that object along the way. The suitable group action we mentioned is precisely the one obtained via conjugation. Now let us first consider the general case.

Formally, an *action* of a group G on an object A of a category $\mathbf{C}$ is a homomorphism (we have shown that $\text{Aut}_{\mathbf{C}}(A)$ is a group with the composition of morphisms)

$$\sigma : G \rightarrow \text{Aut}_{\mathbf{C}}(A)$$

Hence, each element of G corresponds to an isomorphism of A in $\mathbf{C}$. As we stated at the beginning, group actions are an important tool for studying groups themselves. Therefore, up to this point, we very much hope that the above σ is an isomorphism, or at least a monomorphism:

5.1.1 DEFINITION. An action of a group G on an object A of a category $\mathbf{C}$ is *faithful* if the corresponding $\sigma : G \rightarrow \text{Aut}_{\mathbf{C}}(A)$ is a monomorphism (injective homomorphism).

If G acts on A , then $\sigma(1) = \text{id}_A$; the action of a group G on an object A is faithful if and only if $\ker \sigma = \{1\}$. In particular, if $\mathbf{C} = \mathbf{Set}$, then σ is faithful if and only if 1 is the only element g of G such that $\sigma(g)(a) = \text{id}_A(a) = a$ for all $a \in A$.

This note focuses more on group actions on sets, and we have the following discussion. Now, given a group homomorphism $\sigma : G \rightarrow \text{Aut}_{\mathbf{C}}(A)$, define a function $\rho : G \times A \rightarrow A$ from the underlying set G to a set A by $\rho(g, a) = \sigma(g)(a)$. Then $\rho(1, a) = \sigma(1)(a) = a$ for all $a \in A$; and for any $g_1, g_2 \in G$, $\rho(g_1 g_2, a) = \sigma(g_1 g_2)(a) = (\sigma(g_1) \sigma(g_2))(a) = \sigma(g_1)(\sigma(g_2)(a)) = \sigma(g_1) \rho(g_2, a) = \rho(g_1, \rho(g_2, a))$.

Conversely, given a function $\rho : G \times A \rightarrow A$ satisfying the conditions: we may define a function $\sigma : G \rightarrow \text{Hom}_{\mathbf{Set}}(A, A)$ by $\sigma(g)(a) = \rho(g, a)$. This function preserves the operation, because

$$\begin{aligned} \sigma(g_1 g_2)(a) &= \rho(g_1 g_2, a) = \rho(g_1, \rho(g_2, a)) = \sigma(g_1)(\rho(g_2, a)) \\ &= \sigma(g_1)(\sigma(g_2)(a)) = \sigma(g_1) \circ \sigma(g_2)(a) \end{aligned}$$

Moreover, this verifies that $\sigma(g^{-1})$ is the inverse of $\sigma(g)$ because $\sigma(g^{-1}) \circ \sigma(g)(a) = \sigma(g^{-1}g)(a) = \sigma(1)(a) = a$ for all $a \in A$. Thus, the image of σ consists of isomorphisms. In other words, the map σ we defined can be written as $\sigma : G \rightarrow \text{Aut}_C(A)$.

In other words, the function $\sigma \mapsto \rho$ is a bijection. Therefore, when discussing group actions, we also use ρ to denote a group action. The latter is more commonly used due to its more intuitive form. Hence, we adopt a more frequently used definition specifically for the action of a group G on a set A :

5.1.2 DEFINITION. An *action* of a group G on a set A is a function $\rho : G \times A \rightarrow A$ such that $\rho(1, a) = a$ for all $a \in A$ and $\rho(g_1g_2, a) = \rho(g_1, \rho(g_2, a))$ for any $g_1, g_2 \in G$ and $a \in A$.

In practice, $\rho(g, a)$ is written simply $g \cdot a$ or ga . This is *not* an actual multiplication of elements from two possibly different sets G and A . It is just the notation for the effect of g on the element a that gives us an element ga in A . In this notation, the two conditions for a group action in the definition take the following form:

1. $1a = a$ for all $a \in A$;
2. $g_1(g_2a) = (g_1g_2)a$ for any $g_1, g_2 \in G$ and $a \in A$.

If we revisit the two forms of group actions, we can derive another function $\varrho : A \times G \rightarrow A$ corresponding to the group action $\sigma : G \rightarrow \text{Aut}_{\text{Set}}(A)$ using the same method. Specifically, given a group action σ , define a function $\varrho : A \times G \rightarrow A$ by $\varrho(a, g) = \sigma(g^{-1})(a)$ (this definition is well-defined since the inverse of an element is unique). One may also show that $\varrho(a, 1) = a$ and $\varrho(a, g_1g_2) = \varrho(\varrho(a, g_1), g_2)$. Conversely, a function ϱ satisfying these two conditions also defines a homomorphism σ .

To distinguish between them, we call the group action defined in Definition 5.1.2 a *left* action, and the one described above a *right* action. Similarly, we can abbreviate the definition of a right action as follows:

1. $a1 = a$ for all $a \in A$;
2. $(ag_1)g_2 = a(g_1g_2)$ for all $g_1, g_2 \in G$ and $a \in A$

Every right action may be turned into a left action since $\varrho(a, g) = \sigma(g^{-1})(a) = \rho(g^{-1}, a)$. Therefore, right actions will not appear frequently in our subsequent study. In any case, when we refer to a group action, we shall assume by default that it is a left action.

5.1.1 EXAMPLE. Let G be a group and $H \leq G$. An action of the group H on the set underlying G is given by $(h, x) \mapsto hx$, where hx is the product in G . The action of $h \in H$ on G is called a *translation*. If K is another subgroup of G , then H acts on $(G/K)_L$ by translation $(h, xK) \mapsto hxK$.

5.1.3 THEOREM (Cayley's Theorem). *Every group acts faithfully on some set.*

Proof. Let G acts on itself by left multiplication: $\rho : G \times G \rightarrow G$ defined by $\rho(g, a) = ga$. Then the corresponding homomorphism $\sigma : G \rightarrow \text{Aut}_{\text{Set}}(G)$ is injective: if $\sigma(g) = \text{id}_G$, then $ga = \sigma(g)(a) = a$ for all $a \in G$ and $g = 1$. And $G \cong \text{im } \sigma \leq \text{Aut}_{\text{Set}}(G)$. $\square$

As we mentioned at the beginning, once we have a group action, we can obtain some information about A through the action. Specifically, we can partition the set A using the action: fix $a \in A$. If for $b \in A$ there exists $g \in G$ such that $ga = b$, then we say that a and b are related:

5.1.4 PROPOSITION. *Let G be a group acting on the set A . Then the relation on A defined by $a \sim b \leftrightarrow a = gb$ for some $g \in G$ is an equivalence relation.*

Proof. $a = 1a$ implies that $\sim$ is reflexive. If $a \sim b$, then $a = gb$ for some $g \in G$ so that $g^{-1}a = g^{-1}(gb) = 1b = b$, that is $b \sim a$ and the relation is symmetric. Finally, if $a \sim b$ and $b \sim c$, then $a = gb$ and $b = hc$ for some $g, h \in G$ so $a = gb = g(hc) = (gh)c$, hence $a \sim c$ and the relation is transitive. $\square$

The *orbit* of $a \in A$ under an action of a group G is the set $O_G(a) := \{ga : g \in G\}$. Hence the orbits of an action of a group G on a set A form a partition of A . For instance, if a subgroup H of G acts on G by left multiplication, then the orbit of an element x of G is its right coset Hx since $a \sim b \leftrightarrow a = hb \leftrightarrow ab^{-1} \in H \leftrightarrow Ha = Hb$.

However, not every $g \in G$ can help us find another element in A that is equivalent to a . It is possible that ga is itself, i.e., $ga = a$. Therefore, we single out such g and consider:

5.1.3 DEFINITION. In a group action of a group G on a set A , the *stabilizer* of $a \in A$ is the subgroup $\text{Stab}_G(a) := \{g \in G : ga = a\}$ of G . ($\text{Stab}_G(a)$ is a subgroup since $1a = a$, $ga = a$ implies $a = g^{-1}(ga) = g^{-1}a$, and $ga = ha = a$ implies $(gh)a = g(ha) = ga = a$).

5.1.5 PROPOSITION. If a group G acts on a set A . Then the cardinal number of the orbit of $a \in A$ is the index $[G : \text{Stab}_G(a)]$.

Proof. Let G acts on A . Let $a \in A$. Define a function $\varphi : G \rightarrow O_G(a)$ by $\varphi(g) = ga$. Clearly φ is surjective by the definition of the orbit. Now suppose $\varphi(g_1) = \varphi(g_2)$, i.e., $g_1a = g_2a$. Then $g_2^{-1}g_1a = a$, so $g_2^{-1}g_1 \in \text{Stab}_G(a)$. Hence $g_1 \in g_2\text{Stab}_G(a)$. Conversely, if $g_1 = g_2h$ for some $h \in \text{Stab}_G(a)$, then

$$g_1a = g_2ha = g_2(ha) = g_2a.$$

Thus $\varphi(g_1) = \varphi(g_2)$ if and only if g_1 and g_2 lies in the same left coset of $\text{Stab}_G(a)$. Therefore, φ induces a bijection $\bar{\varphi} : g\text{Stab}_G(a) \mapsto ga$. Hence $[G : \text{Stab}_G(a)] = |O_G(a)|$. $\square$

5.1.1 REMARK. Let K be the kernel of the corresponding homomorphism $\sigma : G \rightarrow \text{Aut}(A)$, then $K = \ker \sigma = \{g \in G : \sigma(g) = \text{id}_A\} = \{g \in G : \sigma(g)(a) = \rho(g, a) = ga = a, \forall a \in A\} = \bigcap_{a \in A} \text{Stab}_G(a)$.

5.1.2 REMARK. Let G acts on a set A . Since $a \sim b$ is an equivalence relation, any two orbits of G are disjoint or equal. Hence A is the disjoint union of distinct orbits, and we can write

$$A = \bigsqcup_{i \in I} O_G(a_i)$$

where I is some index set, and the a_i are elements of distinct orbits. If A is finite, this gives a decomposition of the order of A as a sum of orders of orbits, which we call the *orbit decomposition formula*, namely

$$|A| = \sum_{i \in I} |O_G(a_i)| = \sum_{i \in I} [G : \text{Stab}_G(a_i)].$$

5.1.4 DEFINITION. Let p be a prime number. A *p-group* is a group G whose order is a power of a prime p , i.e., $|G| = p^n$ for some $n \in \mathbf{N}$. If H is a subgroup of a group G' and H is a p -group, H is said to be a *p-subgroup* of G' . In particular, the trivial subgroup $H = \{1\}$ is a p -subgroup of G' for every prime p .

5.1.6 LEMMA. Let G be a p -group acting on a finite set A , and let $A_0 = \{a \in A : ga = a, \forall g \in G\}$. Then $|A_0| \equiv |A| \pmod{p}$.

Proof. An orbit $O_G(a)$ contains exactly one element if and only if $a \in A_0$. Hence A can be written as a disjoint union $A = A_0 \cup O_G(a_1) \cup \cdots \cup O_G(a_n)$, with $|O_G(a_i)| > 1$ for all i . Hence $|A| = |A_0| + \sum_{|O_G(a_i)| > 1} |O_G(a_i)|$. Now $p \mid |O_G(a_i)|$ for each i since $|O_G(a_i)| > 1$ and $|O_G(a_i)| = [G : \text{Stab}_G(a_i)]$ divides $|G| = p^n$. Thus, $|A| \equiv |A_0| \pmod{p}$. $\square$

5.1.7 COROLLARY (Cauchy's Theorem). Let G be a group, and let p be a prime divisor of $|G|$. Then G contains an element x of order p .

Proof. Consider the set A of ordered p -tuples of elements of G such that $a_1 \cdots a_p = \mathbf{1}$: $A = \{(a_1, \dots, a_p) : a_i \in G \wedge a_1 \cdots a_p = \mathbf{1}\}$. We claim that $|A| = |G|^{p-1}$: indeed, once $a_1, \dots, a_{p-1}$ are chosen, then a_p is determined since it is the inverse of $a_1 \cdots a_{p-1}$. Therefore, p divides the order of A since it divides the order of G .

Now we act with the group $\mathbf{Z}/p\mathbf{Z}$ on A : given $[m]$ in $\mathbf{Z}/p\mathbf{Z}$ with $0 \leq m < p$, act by $[m]$ on $(a_1, \dots, a_p)$ by sending it to $(a_{m+1}, \dots, a_p, a_1, \dots, a_m)$ (it is clear that this is still an element of A). The lemma implies that $|A_0| \equiv |A| \equiv 0 \pmod{p}$. The elements of A_0 are of the form

$$(a, \dots, a),$$

and note that $A_0 \neq \emptyset$ since $(1, \dots, 1) \in A_0$. Since $p \geq 2$ and p divides $|A_0|$, we conclude that $|A_0| > 1$; therefore there exists some element in A of the form $(a, \dots, a)$ with $a \neq 1$. This says that there exists an $a \in G, a \neq 1$ such that $a^p = 1$, proving the statement. $\square$

Since a group action of G on A helps us understand A , we may consider the case $A = G$ to understand G . The most common group action of G on G is given by conjugation. Let G be a group. Define an action of the group G on the underlying set G by

$$(g, x) \mapsto gxg^{-1},$$

This is indeed an action: for all $g_1, g_2 \in G$ and $x \in G$, $g_1(g_2xg_2^{-1})g_1^{-1} = (g_1g_2)x(g_1g_2)^{-1}$. This action is called the *conjugation*.

Under this conjugation action, what is the stabilizer of $a \in A = G$? By definition, $ga = a$ means $gag^{-1} = a$, i.e., $ga = ag$. Hence, under this group action,

$$\text{Stab}_G(a) = \{g \in G : gag^{-1} = a\} = \{g \in G : ga = ag\} = C_G(a),$$

so under the conjugation action, the stabilizer of $a \in G$ is precisely the centralizer of a in G . Moreover, $\bigcap_{x \in G} \text{Stab}_G(x) = \{g \in G \mid gxg^{-1} = x, \forall x \in G\} = \{g \in G \mid gx = xg, \forall x \in G\} = Z(G)$. That is, under the conjugation group action, the kernel is precisely the center of G .

5.1.8 THEOREM (The Class Equation). *Let G be a finite group. Then*

$$|G| = \sum |C| = |Z(G)| + \sum_{|C|>1} |C|$$

Proof. First, $|G| = \sum |C|$ since the conjugacy classes (orbits) constitute a partition of G . Now, the conjugacy class of a is trivial ($|C| = 1$) if and only if $gag^{-1} = a$ for all $g \in G$ if and only if $a \in Z(G)$; hence there are $|Z(G)|$ trivial conjugacy classes and

$$|G| = \sum |C| = \sum_{|C|=1} |C| + \sum_{|C|>1} |C| = |Z(G)| + \sum_{|C|>1} |C|.$$

$\square$

5.1.9 COROLLARY. *Every nontrivial p -group has a nontrivial center.*

Proof. By the proposition 5.1.4, $|C|$ divides $|G| = p^n$ for every conjugacy class C . In particular, p divides $|C|$ when $|C| > 1$. In the class equation, p divides $|G|$ and p divides $\sum_{|C|>1} |C|$; hence p divides $|Z(G)|$ and $|Z(G)| \geq p$. $\square$

5.1.2 EXAMPLE. Consider a group G of order 6. If G is abelian, then $G = Z(G)$, so that the class formula tells us that $6 = 6$. If G is not abelian, then its center must be trivial (otherwise, $G/Z(G)$ is cyclic and hence G is abelian which is a contradiction by the lemma); so the class formula is $6 = 1 + \dots$, where $\dots$ collects the sizes of the nontrivial conjugacy classes. But each of these summands must be larger than 1, smaller than 6, and must divide 6; that is, there are no choices:

$$6 = 1 + 2 + 3.$$

5.1.3 REMARK. Another useful observation is that normal subgroups must be unions of conjugacy classes: because if H is a normal subgroup, $a \in H$ and $b = gag^{-1}$ is conjugate to a , then $b \in gHg^{-1} = H$.

We may also consider a group action on S , where S is the set of subgroups of G (i.e., the elements of S are the subgroups of G). In this setting, the most common group action is again conjugation. Here, for $g \in G$ and $H \in S$ (i.e., H is a subgroup of G), we define

$$(g, H) \mapsto gHg^{-1} = \{ghg^{-1} : h \in H\}$$

It is easy to show that gHg^{-1} is also a subgroup of G . This is indeed an action: for all $H \in S$ and $g_1, g_2 \in G$, $g_1(g_2Hg_2^{-1})g_1^{-1} = \{g_1g_2hg_2^{-1}g_1^{-1}\} = \{(g_1g_2)h(g_1g_2)^{-1}\} = (g_1g_2)H(g_1g_2)^{-1}$.

5.1.4 REMARK. Naturally, we have an equivalent definition of a normal subgroup. Let G be a group and let H be a subgroup of G . If in the conjugation action of G on the set of all subgroups of G , the subgroup H is fixed (i.e., has an orbit of size 1), then H is called a *normal subgroup* of G .

Under this conjugation group action, what is the stabilizer of $H \in S$? By definition, $g \cdot H = H$ means $gHg^{-1} = H$, i.e., $gH = Hg$. Hence, under this group action,

$$\text{Stab}_G(H) = \{g \in G : gH = Hg\} = N_G(H),$$

That is, under the conjugation group action, the stabilizer of the subgroup H is precisely the normalizer of H in G . Therefore, the cardinality of the orbit $O_G(H)$ is $[G : N_G(H)]$. If G is finite, this means that the number of subgroups conjugate to H is $[G : N_G(H)]$.

In the conjugation action of a group on its subgroups, the stabilizer of a subgroup is its normalizer. Thus, again this is clear also directly from the definition: a subgroup is normal in G if and only if $N_G(H) = G$.

Ping-Pong Lemma

After much deliberation, the author has finally decided to use the Ping-Pong Lemma as a concrete way to appreciate why we say that “group actions allow us to understand some information about the group itself.”

The Ping-Pong Lemma, sometimes also called the Schottky Lemma or Klein’s criterion, provides a set of conditions under which one can determine that a group is free. Let us begin with a method for testing whether an unknown group is F_2 .

5.1.10 LEMMA (Ping-Pong Lemma for two players). *Suppose that a and b generate a group G that acts on a set X . If X has disjoint nonempty subsets X_a and X_b , and $a^k(X_b) \subset X_a$ and $b^k(X_a) \subset X_b$ for all nonzero powers k , then G is isomorphic to a free group of rank 2.*

Let us first see what this lemma is doing. As its name suggests, this lemma describes an exciting game of ping-pong. Draw some figures and I will explain some details:

Let X be the table tennis table, with the left side as the ball control zone X_b and the right side as X_a . Player A serves from the left toward X_a , and player B serves from the right toward X_b . Both A and B are super athletes; they can quickly perform several tactical simulations to plan their next return route while the ball is moving toward its landing point, and their next attack will surely execute the route they have conceived. Each tactical simulation in A ’s mind is denoted by a and a^{-1} (with a denoting an offensive tactic and a^{-1} a defensive tactic), and for B , by b and b^{-1} . Suppose the match begins with A serving from point p in region X_b toward region X_a . Taking into account B ’s positioning and usual playing style, A decides to adopt a defensive tactic and, after simulating 15 defensive routes, finally serves with a^{-15} . While the ball is in motion, B

simulates 8 offensive routes as b^8 to obtain the best return trajectory (which costs B considerable mental effort), catches the ball, and returns it toward region X_b . This ball is as elusive as a ghost, but A knows B 's routes inside out, simulates only 2 offensive routes as a^2 , and delivers a precise and fierce smash to win the point! The element generated by a and b in this exchange, $a^2b^8a^{-15}$, represents a complete sequence of strokes during the match:

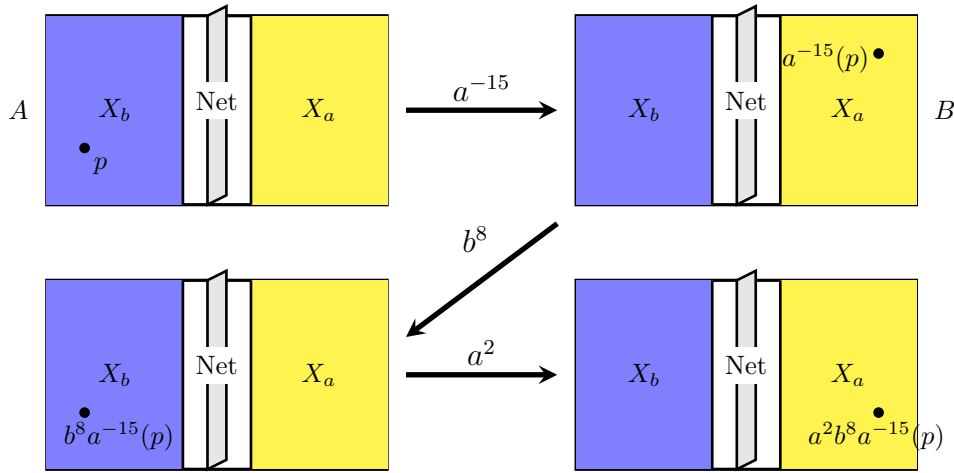


Figure 5.1: Multiple views of table tennis table

At the end, the last simulation a^2 takes the image of p to X_a . Since X_a is disjoint from X_b , we conclude that $a^2b^8a^{-15}(p) \neq p$. In general, no element g of G represented by a word of the form $a^*b^*\cdots b^*a^*$ is the identity since $g(X_b) \subset X_a$. Since any element ω of G is conjugate to one of the form $a^*b^*\cdots b^*a^*$, i.e., $a^*\omega(a^*)^{-1} = a^*b^*\cdots b^*a^*$, and $a^*b^*\cdots b^*a^* \neq 1$, so that $\omega \neq 1$, which implies that G has no relations with respect to the generating set $\{a, b\}$: no nonempty reduced words equal the identity. This means that G is the free group of rank 2.

In fact, by now we have already proved the Ping-Pong Lemma; we only need to prove one statement: every element of G is conjugate to one of the form $a^*b^*a^*\cdots b^*a^*$. This is also very simple: It is necessary to consider whether the beginning and the end end with b^* , which can be divided into three cases (or four). But in any case, by conjugating with a sufficiently large power of a^k , one can always achieve the goal.

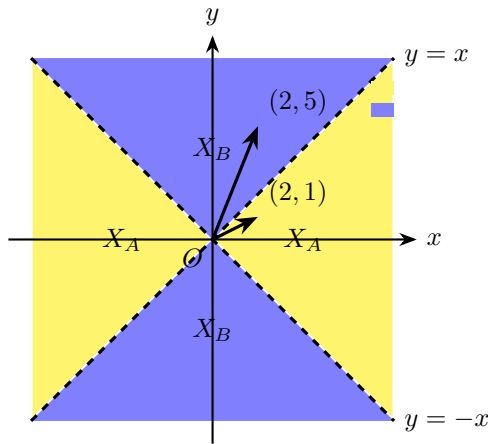
5.1.3 EXAMPLE. Recall that $\text{GL}(2, \mathbf{Z})$ is the group of 2×2 integer matrices with determinant equal to ± 1 . Now, consider the two matrices ($m \geq 2$)

$$A = \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 0 \\ m & 1 \end{pmatrix}$$

One may verify by induction that (if $m = 1$, then $ABA^{-1}BAB^{-1} = I$ is a non-trivial relation between A and B .)

$$A^k = \begin{pmatrix} 1 & km \\ 0 & 1 \end{pmatrix} \text{ and } B^k = \begin{pmatrix} 1 & 0 \\ km & 1 \end{pmatrix}$$

Then, consider the two subsets $X_A = \{(x, y)^T \in \mathbf{R}^2 : |x| > |y|\}$ and $X_B = \{(x, y) \in \mathbf{R}^2 : |x| < |y|\}$, so that the two subsets fulfill the conditions of the ping-pong lemma. ($A^k(X_B) \subset X_A, B^k(X_A) \subset X_B$). The following figure is the (x, y) -plane with region X_A and X_B . If $m = 2$ in the matrix B , then $B(2, 1)^T = (2, 5)^T$, so $(2, 1)^T$ is an example of a vector that B moves from X_A to X_B .



5.1.11 LEMMA (Ping-Pong Lemma for n players). Suppose that $\{g_1, \dots, g_n\}$ generates a group G that acts on a set X . If X has pairwise disjoint nonempty subsets $\{X_1, \dots, X_n\}$ and $g_i^k(X_j) \subset X_i$ for all nonzero powers k and $i \neq j$, then G is a free group of rank n .

5.2 Symmetric Groups

5.2.1 DEFINITION. Let A be a nonempty set. A *permutation* of A is an isomorphism $A \rightarrow A$ in $\text{Aut}_{\text{Set}}(A)$. The *symmetric group* or group of permutations of A , denoted S_A is the group $\text{Aut}_{\text{Set}}(A)$. The group of permutations of the set $\{1, 2, \dots, n\}$ is denoted by S_n .

In many cases, beginners encounter the symmetric groups S_n . Let's first study the elements of S_n . The group S_1 is a trivial group. S_2 consists of the two possible permutations:

$$\left\{ \begin{array}{l} 1 \mapsto 1 \\ 2 \mapsto 2 \end{array} \right. \quad \text{and} \quad \left\{ \begin{array}{l} 1 \mapsto 2 \\ 2 \mapsto 1 \end{array} \right.$$

which we can call $\mathbf{1}$ (identity) and f (flip), with operation $\mathbf{1}\mathbf{1} = ff = \mathbf{1}$, $\mathbf{1}f = f\mathbf{1} = f$. We cannot assign a new name to every permutation in S_n , but we can use relatively simple notation to represent these permutations; we can use a two-rowed notation to denote a permutation: let $\sigma \in S_n$, take

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \cdots & \sigma(n) \end{pmatrix}.$$

Informally, lists and permutations are simply different ways of describing the same thing. And the elements $\mathbf{1}, f$ in S_2 may be denoted by

$$e = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, f = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

Cyclic groups are naturally inseparable from group actions. Specifically, we may consider the group action $S_n \rightarrow \text{Aut}(\{1, 2, \dots, n\}) = S_n$ given by the identity homomorphism, which is clearly faithful. Hence it induces a group action $S_n \times \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ by $(\sigma, i) \mapsto \sigma(i)$. Now restricting this action to any subgroup H of S_n , we obtain a function $H \times \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, which is evidently also a group action. In particular, when we fix a permutation σ and consider the cyclic subgroup $\langle \sigma \rangle$ acting on $\{1, 2, \dots, n\}$, we can partition $\{1, 2, \dots, n\}$ according to the definition of orbits. That is, each orbit is of the form $\{\sigma^r(i) : r \in \mathbf{Z}\}$ for some $i \in \{1, 2, \dots, n\}$. For example, if

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 1 & 2 & 7 & 5 & 3 & 4 & 6 \end{pmatrix}$$

It is easy to see that this action has three orbits: $\{1, 2, 3, 6, 8\}$, $\{4, 7\}$, and $\{5\}$. At the same time, when the group $\langle \sigma \rangle$ acts on a given orbit, by the definition of an orbit, any

element of that orbit can be mapped, after applying σ some number of times, to any other element of the same orbit, and eventually back to the original element, as in the example above:

$$1 \mapsto 8 \mapsto 6 \mapsto 3 \mapsto 2 \mapsto 1, \quad 4 \mapsto 7 \mapsto 4, \quad 5 \mapsto 5.$$

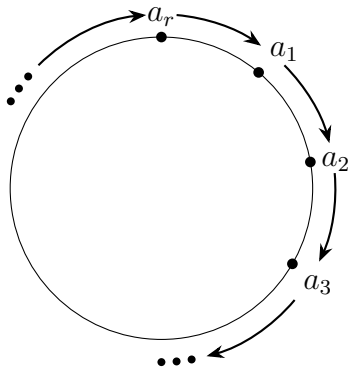
5.2.2 DEFINITION. A (*nontrivial*) *cycle* is an element of S_n with exactly one nontrivial orbit. For instance $a_1, \dots, a_r$ in $\{1, \dots, n\}$, the notation $(a_1 a_2 \dots a_r)$ denotes the cycle σ in S_n with nontrivial orbit $\{a_1, \dots, a_r\}$, acting as

$$a_1 \mapsto a_2 \mapsto \dots \mapsto a_r \mapsto a_1.$$

In this case r is the *length* of the cycle. A cycle of length r is called an r -cycle. The identity is considered a cycle of length 1 in a trivial way and is denoted by (i) for any i .

The term *cycle* from the Greek word for circle (the figure as following). Any a_j can be taken as the “starting point” and so there are r “different” cycle notation for any r -cycle; but for notational convenience, we typically take the smallest number as the “starting point”:

$$(a_1 a_2 \dots a_r) = (a_2 a_3 \dots a_r a_1) = \dots = (a_r a_1 a_2 \dots a_{r-1}).$$



5.2.1 LEMMA. *Disjoint cycles commute.*

Proof.

□

5.2.2 LEMMA. *Every $\sigma \in S_n, \sigma \neq 1$ can be written as a product of disjoint nontrivial cycles, in a unique way up to permutation of the factors.*

Proof. We first prove existence. Consider the action of the cyclic subgroup $\langle \sigma \rangle$ on the set $X = \{1, 2, \dots, n\}$. This action partitions X into disjoint orbits. For each nontrivial orbit $\mathcal{O} = \{a_1, a_2, \dots, a_r\}$, where the elements are arranged so that

$$a_1 \mapsto a_2 \mapsto \dots \mapsto a_r \mapsto a_1,$$

we associate the r -cycle $\gamma_{\mathcal{O}} = (a_1 a_2 \dots a_r)$. Let $\gamma_1, \dots, \gamma_m$ be the cycles corresponding to all nontrivial orbits. These cycles are pairwise disjoint because their underlying sets are disjoint. We claim that

$$\sigma = \gamma_1 \gamma_2 \dots \gamma_m.$$

Indeed, for any $x \in X$, if x lies in a trivial orbit, then $\sigma(x) = x$ and every γ_i fixes x ; if x lies in a nontrivial orbit $\mathcal{O}_j$, then only γ_j moves x , and it sends x to $\sigma(x)$. Thus the product agrees with σ on every element, so equality holds. For uniqueness, suppose

$$\sigma = \delta_1 \delta_2 \dots \delta_t$$

is another decomposition into pairwise disjoint nontrivial cycles. Let A_j be the set of elements moved by δ_j . For any $x \in A_j$, since all other δ_i 's fix x , we have $\sigma(x) = \delta_j(x)$, and more generally $\sigma^m(x) = \delta_j^m(x)$ for all integers m . Hence A_j is exactly an orbit of $\langle \sigma \rangle$ on X , and δ_j is precisely the cycle determined by that orbit. Therefore the sets $A_1, \dots, A_t$ are precisely the nontrivial orbits of σ , and the cycles δ_j are exactly the cycles $\gamma_{\mathcal{O}}$ constructed above. Thus the two decompositions contain the same cycles, differing only in the order of the factors. □

We call the cycle lengths $n_1, n_2, \dots, n_r$ appearing in the cycle decomposition (including cycles of length 1) the *cycle type* of the permutation, where multiplicities are counted but order is ignored. To obtain uniqueness, we may arrange them as $n_1 \geq n_2 \geq \dots \geq n_r$. The cycle type then corresponds to a *partition* of the integer $n = n_1 + \dots + n_r$. And the partition may be denoted by

$$[n_1, n_2, \dots, n_r]$$

5.2.3 LEMMA. *Let $\tau \in S_n$ and let $(a_1 \cdots a_r)$ be a cycle. Then*

$$\tau(a_1 \cdots a_r)\tau^{-1} = (\tau(a_1) \cdots \tau(a_r)).$$

Proof. We show that both permutations act the same on every element x . If $x = \tau(a_i)$ for some i . Then $\tau(a_1 \cdots a_r)\tau^{-1}(x) = \tau(a_1 \cdots a_r)(a_i) = \tau(a_{i+1})$, while $(\tau(a_1) \cdots \tau(a_r))(x) = (\tau(a_1) \cdots \tau(a_r))(\tau(a_i)) = \tau(a_{i+1})$. Hence the two sides agree on such x .

If $x \notin \{\tau(a_1), \dots, \tau(a_r)\}$. Then $\tau^{-1}(x) \notin \{a_1, \dots, a_r\}$, so $(a_1 \cdots a_r)(\tau^{-1}(x)) = \tau^{-1}(x)$. Therefore $\tau(a_1 \cdots a_r)\tau^{-1}(x) = x$, and the cycle $(\tau(a_1) \cdots \tau(a_r))$ also fixes x . Thus the two sides agree on all such x . Since the two permutations agree on every element of the set, they are equal. $\square$

By cleverly inserting identity factors of the form $\tau^{-1}\tau$ at appropriate positions, a standard technique, this conjugation formula can be immediately generalized to arbitrary products of cycles:

$$\tau(a_1 \cdots a_r)(b_1 \cdots b_s) \cdots (c_1 \cdots c_k)\tau^{-1} = (\tau(a_1) \cdots \tau(a_r))(\tau(b_1) \cdots \tau(b_s))(\tau(c_1) \cdots \tau(c_k)).$$

This holds whether the cycles are disjoint or not. However, since τ is a bijection, disjoint cycles remain disjoint after conjugation. This is essentially all there is to the following observation:

5.2.4 PROPOSITION. *Two elements of S_n are conjugate if and only if they have the same type. The number of conjugacy classes in S_n equals the number of partitions of n .*

Proof. Let σ be a product of disjoint cycles $\sigma = \gamma_1 \cdots \gamma_r$. Each γ_i is a k_i -cycle for some $k_i \geq 2$. By lemma 5.2.1, we may assume that $k_1 \geq k_2 \geq \dots \geq k_r$, and then the cycle type of σ is $[k_1, k_2, \dots, k_r]$. By the lemma 5.2.3, $\tau\gamma_i\tau^{-1}$ is a k_i -cycle like γ_i and

$$\tau\sigma\tau^{-1} = (\tau\gamma_1\tau^{-1}) \cdots (\tau\gamma_r\tau^{-1})$$

is a product of disjoint cycles. Therefore the cycle type of $\tau\sigma\tau^{-1}$ is $[k_1, \dots, k_r]$, the same as that of σ .

Conversely, let σ and ρ have the same type $[k_1, \dots, k_r]$. Then σ and ρ are products of r pairwise disjoint cycles

$$\sigma = \gamma_1\gamma_2 \cdots \gamma_r \text{ and } \rho = \delta_1\delta_2 \cdots \delta_r,$$

in which γ_i and δ_i are k_i -cycles, $\gamma_i = (a_1 a_2 \cdots a_{k_i})$, $\delta_i = (b_1 b_2 \cdots b_{k_i})$. Let τ_i be the bijection of $\{a_1, a_2, \dots, a_{k_i}\}$ onto $\{b_1, b_2, \dots, b_{k_i}\}$ that sends a_t to b_t . The permutation σ and ρ have $n - k_1 - k_2 - \dots - k_r$ fixed points; let τ_0 be any bijection of the set of fixed points of σ onto that of ρ . Then $\tau_0, \tau_1, \dots, \tau_r$ can be pasted together into a bijection τ of $\{1, \dots, n\}$ onto $\{1, \dots, n\}$. By lemma 5.2.3, $\tau\sigma_i\tau^{-1} = (\tau_i(a_1) \cdots \tau_i(a_{k_i})) = (b_1 \cdots b_{k_i}) = \delta_i$ for all i .

So there is a bijection between the conjugacy classes of S_n and the permissible cycle types and each type for a permutation in S_n is a partition of n , the second assertion of the proposition follows, completing the proof. $\square$

It is also reasonably straightforward to compute the number of elements in each conjugacy class, in terms of the type. For example, in order to count the number of permutations of type $[2, 1, 1]$ in S_5 , note that there are $5! = 120$ ways to write a permutation as a product of two 2-cycles:

$$(a_1 a_2)(a_3 a_4)(a_5);$$

but switching $a_1 \leftrightarrow a_2$ and $a_3 \leftrightarrow a_4$, as well as switching the two cycles, gives the same permutation $((a_1 a_2)(a_3 a_4) = (a_2 a_1)(a_4 a_3) = (a_3 a_4)(a_1 a_2))$. Therefore, there are

$$\frac{120}{2 \times 2 \times 2} = 15$$

permutations of type $[2, 2, 1]$. Performing this computation for all the types gives us the size of each conjugacy class, that is, the class formula for S_5 :

$$120 = 1 + (10 + 15 + 20 + 30 + 24).$$

5.2.1 EXAMPLE. There are no normal subgroups of size 30 in S_5 .

Proof. Normal subgroups are unions of conjugacy classes; since the identity is in every subgroup and $30 - 1 = 29$ cannot be written as a sum of the numbers appearing in the class formula for S_5 , there is no such subgroup. $\square$

Transpositions and Alternating Groups

As we have seen that every permutation in S_n can be written as a product of disjoint cycles in an “essentially” unique fashion. In contrast, every element of S_n can be written in many different ways as a (nondisjoint) product of cycles. For example, the permutation $\sigma = (1\ 2\ 3)$ may be written as

$$(1\ 2\ 3) = (1\ 3)(1\ 2) = (1\ 2)(1\ 3)(1\ 2)(1\ 3) = (1\ 2)(2\ 3)$$

Not requiring the cycles to be disjoint totally destroys the uniqueness of a representation of a permutation as a product of cycles. We can, however, obtain a sort of “parity check” from writing permutations (nonuniquely) as a product of 2-cycles.

5.2.5 PROPOSITION. Let S_n be the symmetric group of $\{1, \dots, n\}$. Let T be a set consists of transpositions σ_i of the form $(i\ i+1)$, i.e., $T = \{\sigma_1, \dots, \sigma_{n-1}\}$. Then $S_n = \langle T \rangle$.

Proof. We first show that S_n is generated by all the transpositions. First we observe that every nontrivial cycle $(a_1 a_2 \dots a_r)$ may be written as

$$(a_1 a_2 \dots a_m) = (a_1 a_2)(a_1 a_3)(a_1 a_4) \dots (a_1 a_{r-1})(a_1 a_r)$$

Now every permutation in S_n may be written as a product of cycles, which implies that every permutation may be written as a product of transpositions. Next, we use induction on n . For $n = 1$, S_n is generated by the empty set. Now suppose the statement is true for S_n , and consider S_{n+1} .

For each n , let $T_n = \{(i\ i+1) : 1 \leq i \leq n-1\}$. We want to show that every permutation in S_{n+1} can be written as a product of members of T_{n+1} . Let $\sigma \in S_{n+1}$ and write σ as a product of transpositions.

For each transposition $(j\ k)$ with $j < k$, there are two cases: first, if $1 \leq j < k \leq n$, then the transposition belongs to S_n and by the induction hypothesis can be written as a product of members of T_n . The remaining case is where $k = n+1$. If $j = n$, then $(j\ k)$ is already in T_{n+1} and we are done, so suppose that $j < n$. Let

$$\tau = (k-1\ k)(j\ k-1)(k-1\ k).$$

Notice that $\tau(j) = k$ and $\tau(k) = j$, with all other points fixed, i.e., $\tau = (j\ k)$. Since $(j\ k-1)$ can, by the induction hypothesis, be written as a product of members of T_n and $(k-1\ k) \in T_{n+1}$, $(j\ k)$ is the product of members of T_{n+1} . $\square$

5.2.1 REMARK. It is clear that these special transpositions satisfy the following properties: $\sigma_i^2 = 1$; $(\sigma_i \sigma_j)^2 = 1$ if $|i - j| > 1$; and $(\sigma_i \sigma_{i+1})^3 = 1$ for each i . In fact, one may give the presentation of S_n :

$$S_n \cong \left\langle s_1, s_2, \dots, s_{n-1} \left| \begin{array}{l} s_i^2 = 1, \quad 1 \leq i \leq n-1, \\ (s_i s_j)^2 = 1, \quad |i - j| > 1, \\ (s_i s_{i+1})^3 = 1, \quad 1 \leq i \leq n-2 \end{array} \right. \right\rangle$$

Proof. Let $G_n = \langle \dots | \dots \rangle$. It suffices to show that there exists a isomorphism $\varphi : S_n \rightarrow G_n$. We already have a natural surjective homomorphism $\varphi : G_n \rightarrow S_n, s_i \mapsto \sigma_i$. Thus it suffices to prove that $|G_n| \leq n!$. We proceed by induction on n . For $n = 1$, the group is trivial, so $|G_1| = 1 = 1!$. For $n = 2$, we have one generator s_1 with $s_1^2 = 1$, hence $|G_2| \leq 2 = 2!$. Assume that for $n - 1$ we have $|G_{n-1}| \leq (n - 1)!$. Consider G_n . Let G_{n-1} act on G_n by right multiplication and consider the orbits $\{gh : \forall h \in G_{n-1}\}$ of this action. We claim that

$$\begin{aligned} G_n &= O(1) \cup O(s_{n-1}) \cup \dots \cup O(s_1 s_2 \dots s_{n-1}) \\ &= 1 \cdot G_{n-1} \cup s_{n-1} \cdot G_{n-1} \cup \dots \cup (s_1 \dots s_{n-1}) \cdot G_{n-1}. \end{aligned}$$

Indeed: it suffices to show that the right-hand side is closed under left multiplication by each generator s_i . Consider $s_i O(s_j \dots s_{n-1})$. When $i = j - 1$ or $i = j$, closure is immediate from $s_j^2 = 1$. When $i < j - 1$, we use $(s_i s_j)^2 = 1$ to move s_i to the right and absorb it into G_{n-1} , again preserving closure. Thus we may assume $i > j$. Using the $(s_i s_j)^2 = 1$, we move s_i to the right until we obtain

$$\dots s_{i-2} s_i s_{i-1} s_i s_{i+1} \dots G_{n-1}.$$

Then applying the third relation yields

$$\dots s_{i-2} s_{i-1} s_i s_{i+1} \dots G_{n-1}.$$

The indicated s_{i-1} can then be moved all the way to the right until it is absorbed into G_{n-1} , proving the desired left-invariance. $\square$

This presentation shows that the symmetric group belongs to a class of constructions known as Coxeter groups, as we mentioned in the example at the end of the last section of the previous chapter.

Again we emphasize that for any $\sigma \in S_n$, there may be many ways of writing σ as a product of transpositions. For fixed σ we now show that the parity (i.e., an odd or even number of terms) is the same for any product of transpositions equaling σ . Let $x_1, \dots, x_n$ be independent variables and let Δ_n be the polynomial

$$\Delta_n = \prod_{1 \leq i < j \leq n} (x_i - x_j)$$

i.e., the product of all the terms $x_i - x_j$ for $i < j$. For example, when $n = 3$, $\Delta_3 = (x_1 - x_2)(x_1 - x_3)(x_2 - x_3)$. Now, let any $\sigma \in S_n$ act on Δ_n by permuting the indices according to σ :

$$\sigma(\Delta_n) := \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)}).$$

For example $(1 \ 2 \ 3)(\Delta_3) = (x_2 - x_3)(x_2 - x_1)(x_3 - x_1)$. Note that Δ_n contains one factor $x_i - x_j$ for all $i < j$, and since σ is bijection of the indices, $\sigma(\Delta_n)$ must contain either $x_i - x_j$ or $x_j - x_i$, but not both, for all $i < j$. If $\sigma(\Delta_n)$ has a factor $x_j - x_i$ where $j > i$, write this term as $-(x_i - x_j)$. Collecting all the changes in sign together we see that Δ_n and $\sigma(\Delta_n)$ have the same facts up to product of -1 's, i.e.,

$$\sigma(\Delta_n) = \pm \Delta_n$$

5.2.3 DEFINITION. The *sign* of a permutation $\sigma \in S_n$, denoted $(-1)^\sigma$, is determined by the action of σ on Δ_n : $\sigma(\Delta_n) = (-1)^\sigma \Delta_n$. We say that a permutation is *even* if its sign is $+1$ and *odd* if its sign is -1 .

We may define a function $\text{sgn} : S_n \rightarrow \{1, -1\}$ by $\text{sgn}(\sigma) = (-1)^\sigma$. In the example with $n = 3$ and $\sigma = (1\ 2\ 3)$, we have $(1\ 2\ 3)(\Delta_3) = -\Delta_3$, so $\text{sgn}(\sigma) = -1$. Note that $\forall \sigma, \tau \in S_n$, we have

$$(\sigma\tau)(\Delta_n) = \sigma(\tau\Delta_n),$$

so that $(-1)^{\sigma\tau} \Delta_n = \sigma((-1)^\tau \Delta_n) = (-1)^\sigma (-1)^\tau \Delta_n$. it follows that $(-1)^{\sigma\tau} = (-1)^\sigma (-1)^\tau$. Viewing $\{-1, 1\}$ as a group under multiplication, we see that the function $\text{sgn} : S_n \rightarrow \{-1, 1\}$ is a group homomorphism.

Consider the sign of every transposition in S_n . It is clear that the sign of a transposition is -1 : indeed, $(i\ j)$ acts on Δ_n by permuting its factors and changing the sign of an odd number of factors (for $i < j$, the factor $(x_i - x_j)$ and the pairs of factors $(x_i - x_k), (x_k - x_j)$ for all $i < k < j$).

5.2.6 PROPOSITION. Let $\sigma \in S_n$, writing $\sigma = \gamma_1 \cdots \gamma_r$ as a product of transpositions. Then r is even if and only if σ is even.

Proof. σ is even if and only if $\text{sgn}(\sigma) = 1$ if and only if $1 = (-1)^r$ (Since $\sigma = \gamma_1 \cdots \gamma_r$ and sgn is a group homomorphism) if and only if r is even. $\square$

Recall that an k -cycle may be written as a product of $k - 1$ transpositions. Thus an k -cycle is odd if and only if k is even. Now, for any permutation σ , let $\sigma = \tau_1 \cdots \tau_m$ be its cycle decomposition. Then $\text{sgn}(\sigma)$ is given by $\text{sgn}(\tau_1) \cdots \text{sgn}(\tau_m)$ and $\text{sgn}(\tau_i) = -1$ if and only if the length of τ_i is even. It follows that for $\text{sgn}(\sigma)$ to be -1 the product of the $\text{sgn}(\tau_i)$'s must contain an odd number of factors of (-1) . In other words, the permutation σ is odd if and only if the number of cycles of even length in its cycle decomposition is odd.

5.2.4 DEFINITION. The *alternating group* on $\{1, 2, \dots, n\}$ denoted A_n consists of all even permutations $\sigma \in S_n$. (the kernel of the homomorphism sgn).

5.2.7 THEOREM. Let A_n be the alternating group. Let $T = \{(i\ j\ k) : 1 \leq i, j, k \leq n\}$. Then $A_n = \langle T \rangle$. That is, A_n is generated by 3-cycles.

Proof. Since every even permutation is a product of an even number of 2-cycles, it suffices to show that every product of two 2-cycles may be written as product of 3-cycles. Therefore, consider a product

$$(a\ b)(c\ d)$$

with $a \neq b, c \neq d$. If $(a\ b) = (c\ d)$, then this product is the identity, and there is nothing to prove. If $\{a, b\}, \{c, d\}$ have exactly one element in common, then we may assume that $c = a$ and observe

$$(a\ b)(a\ d) = (a\ b\ d).$$

If $\{a, b\}, \{c, d\}$ are disjoint, then

$$(a\ b)(c\ d) = (a\ b\ c)(a\ d\ c).$$

$\square$

Denote by $[\sigma]_{S_n} = \{\tau\sigma\tau^{-1} : \tau \in S_n\}$ (resp. $[\sigma]_{A_n}$) the conjugacy class of an even permutation σ in S_n (resp. A_n). It is clear that $[\sigma]_{A_n} \subset [\sigma]_{S_n}$; we proceed to compare these two sets.

Imagine S_n as a collection of mathematicians and physicists, where A_n is the group consisting of mathematicians, and the rest form the set of physicists. These people share knowledge with one another through mutual communication (i.e., conjugation).

Now suppose there is a mathematician σ who wants to switch fields of research. However, this mathematician σ works in algebraic geometry, which is a very broad area, so

when he communicates with some people, he gains no new inspiration, these people are exactly the centralizer of σ .

If the centralizer of σ consists entirely of mathematicians, then he can gain entirely new inspiration from every physicist in the whole physics community, and thus he might become a master of PDEs or string theory (fields that he cannot access within the mathematical circle A_n). If his centralizer contains even one physicist, then σ himself is a mathematician who is also proficient in physics; even though he usually communicates within the mathematical circle, he can still use his physics knowledge to study PDEs and string theory on his own.

5.2.8 LEMMA. *Let $n \geq 2$, and let $\sigma \in A_n$. Then $[\sigma]_{A_n} = [\sigma]_{S_n}$ or the size of $[\sigma]_{A_n}$ is half the size of $[\sigma]_{S_n}$, according to whether the centralizer $C_{S_n}(\sigma)$ is not or is contained in A_n .*

Proof. Note that $C_{A_n}(\sigma) = A_n \cap C_{S_n}(\sigma)$. Recall that the centralizer of σ is its stabilizer under conjugation, and therefore the size of the conjugacy class of σ equals the index of its centralizer. If $C_{S_n}(\sigma) \subset A_n$, then $C_{A_n}(\sigma) = C_{S_n}(\sigma)$, so that

$$[S_n : C_{S_n}(\sigma)] = [S_n : C_{A_n}(\sigma)] = [S_n : A_n][A_n : C_{A_n}(\sigma)] = 2 \cdot [A_n : C_{A_n}(\sigma)];$$

therefore, $[\sigma]_{A_n}$ is half the size of $[\sigma]_{S_n}$ in this case. If $C_{S_n}(\sigma) \not\subset A_n$, then note that $A_n C_{S_n}(\sigma) = S_n$: indeed, $A_n C_{S_n}(\sigma)$ is a subgroup of S_n (since $A_n \trianglelefteq S_n$), and $A_n \subset A_n C_{S_n}(\sigma)$, so it must equal S_n as A_n has index 2 in S_n . Then

$$[A_n : C_{A_n}(\sigma)] = [A_n : A_n \cap C_{S_n}(\sigma)] = [A_n C_{S_n}(\sigma) : C_{S_n}(\sigma)] = [S_n : C_{S_n}(\sigma)].$$

so the classes have the same size. Since $[\sigma]_{A_n} \subset [\sigma]_{S_n}$ in any case, it follows that $[\sigma]_{A_n} = [\sigma]_{S_n}$, completing the proof. $\square$

Therefore, conjugacy classes of even permutations either are preserved from S_n to A_n or they split into two distinct, equal-sized classes. We are now in a position to give precise conditions determining which happens when.

5.2.9 PROPOSITION. *Let $\sigma \in A_n$. Then the conjugacy class of σ in S_n splits into two conjugacy classes in A_n precisely if and only if the type of σ consists of distinct odd numbers.*

Proof. By the lemma, we have to verify that $C_{S_n}(\sigma)$ is contained in A_n , that is we have to show that $\sigma = \tau \sigma \tau^{-1}$ implies τ is even when the type of σ consists of distinct odd numbers. Write σ in cycle notation $\sigma = (a_1 \cdots a_\lambda)(b_1 \cdots b_\mu) \cdots (c_1 \cdots c_\nu)$, and recall that

$$\tau \sigma \tau^{-1} = (\tau(a_1) \cdots \tau(a_\lambda))(\tau(b_1) \cdots \tau(b_\mu)) \cdots (\tau(c_1) \cdots \tau(c_\nu)).$$

Assume that $\lambda, \mu, \dots, \nu$ are odd and distinct. If $\tau \sigma \tau^{-1} = \sigma$, then conjugation by τ must preserve each cycle in σ , since all cycle lengths are disjoint:

$$\tau(a_1 \cdots a_\lambda) \tau^{-1} = (a_1 \cdots a_\lambda), \text{ etc.},$$

that is, $(\tau(a_1) \cdots \tau(a_\lambda)) = (a_1 \cdots a_\lambda)$, etc.. This means that τ acts as a cyclic permutation on $a_1, \dots, a_\lambda$ and therefore in the same way as a power of $(a_1 \cdots a_\lambda)$. It follows that

$$\tau = (a_1 \cdots a_\lambda)^r (b_1 \cdots b_\mu)^s \cdot (c_1 \cdots c_\nu)^t$$

for suitable $r, s, \dots, t$. Since all cycles have odd lengths, each cycle is an even permutation; and τ must then be even as it is a product of even permutations. This proves that $C_{S_n}(\sigma) \subset A_n$ if the type of σ consists of distinct odd numbers.

Conversely, assume that either some of the cycles in the cycle decomposition have even length, or all have odd length but two of the cycles have the same length. In the first case, let τ be an even-length cycle in the cycle decomposition of σ . Note that $\tau \sigma \tau^{-1} = \sigma$: indeed, τ commutes with itself and with all cycles in σ other than τ . Since τ has even length, then it is odd as a permutation: this shows that $C_{S_n}(\sigma) \not\subset A_n$.

In the second case, without loss of generality assume that $\lambda = \mu$, and consider the odd permutation

$$\tau = (a_1 \ b_1)(a_2 \ b_2) \cdots (a_\lambda \ b_\lambda) :$$

conjugating by τ simply interchanges the first two cycles in σ ($\tau(a_1 \ a_2 \cdots a_\lambda)\tau^{-1} = (\tau(a_1) \ \tau(a_2) \cdots \tau(a_\lambda)) = (b_1 \ b_2 \cdots b_\lambda)$); hence $\tau\sigma\tau^{-1} = \sigma$. Since τ is odd, this again shows that $C_{S_n}(\sigma) \not\subseteq A_n$. $\square$

5.2.2 EXAMPLE. Consider the alternating group A_5 . The types of the even permutations in S_5 are $[1, 1, 1, 1, 1]$, $[2, 2, 1]$, $[3, 1, 1]$ and $[5]$. By the proposition above, the conjugacy classes corresponding to the first three types are preserved in A_5 , while the last one splits. Therefore, there are exactly 5 conjugacy classes in A_5 itself, and the class formula for A_5 is

$$60 = 1 + 15 + 20 + 12 + 12.$$

5.2.10 COROLLARY. A_5 is simple.

Proof. A normal subgroup of A_5 is necessarily the union of conjugacy classes contains the identity, and has order equal to a divisor of 60. The divisors of 60 other than 1 and 60 are

$$2, 3, 4, 5, 6, 10, 12, 15, 20, 30;$$

counting the elements other than the identity would give one of

$$1, 2, 3, 4, 5, 9, 11, 14, 19, 29$$

as a sum of numbers $\neq 1$ from the class formula for A_5 (15, 20, 12, 12). But this simply does not happen. So the normal subgroups of A_5 are A_5 and $\{1\}$. $\square$

5.2.3 EXAMPLE. A_n is simple for $n \geq 5$.

Proof. In fact, we have shown that A_5 is simple. Now, the simplicity of A_n when $n > 5$ is proved by induction on n . Let $N \neq \{1\}$ be a normal subgroup of A_n . We show that $N = A_n$.

First, we show that N is transitive: for every $i, j \in \{1, 2, \dots, n\}$, there exists $\sigma \in N$ such that $\sigma(i) = j$. Since $N \neq \{1\}$, we have $\sigma(k) \neq k$ for some $\sigma \in N$ and $k \in \{1, 2, \dots, n\}$. For any $i \in \{1, 2, \dots, n\}$ we can rig an even permutation α such that $\alpha(k) = k$ and $\alpha(\sigma(k)) = i$; then $\alpha\sigma\alpha^{-1} \in N$ and $\alpha\sigma\alpha^{-1}(k) = i$. If now $i, j \in \{1, 2, \dots, n\}$, there exist $\mu, \nu \in N$ such that $\mu(k) = i, \nu(k) = j$, and then $\nu\mu^{-1} \in N$ and $\nu\mu^{-1}(i) = j$.

Next, we show that some $\sigma \in N, \sigma \neq 1$, has a fixed point ($\sigma(k) = k$ for some k). As above, we have $\sigma(k) = j \neq k$ for some $\sigma \in N$. Then $\sigma(j) \neq j$. Let $i \neq j, k, \sigma(j)$. If $\sigma(i) = i$, σ serves. Otherwise, $i, \sigma i, j, \sigma j$ are all different; since $n \geq 6$ we can concoct an even permutation α such that $\alpha(j) = k, \alpha\sigma(j) = j, \alpha(i) = i$, and $\alpha\sigma(i) \neq i, j, k, \sigma(i)$. Then $\mu = \alpha\sigma\alpha^{-1} \in N, \mu(k) = \alpha\sigma(j) = j = \sigma(k)$, and $\mu(i) = \alpha\sigma(i) \neq \sigma(i)$. Hence $\nu = \sigma^{-1}\mu \in N, \nu \neq 1$ since $\nu(i) \neq i$ and $\nu(k) = k$.

Let k be a fixed point of some $\sigma \in N, \sigma \neq 1$. Let $B = \{\alpha \in A_n : \alpha(k) = k\}$. Then $N \cap B \neq \{1\}$ and $N \cap B \trianglelefteq B$. Since $B \cong A_{n-1}$ is simple by our hypothesis, this implies $N \cap B = B$ and $B \subset N$. If $\alpha \in A$, then $\mu(k) = \alpha(k)$ for some $\mu \in N$, since N is transitive, so that $\mu^{-1}\alpha \in B \subset N$ and $\alpha = \mu(\mu^{-1}\alpha) \in N$. Thus $N = A_n$. $\square$

5.2.4 EXAMPLE. S_n is not solvable for $n \geq 5$.

Proof. Since A_n is simple, the series $S_n \supsetneq A_n \supsetneq \{(1)\}$ is a composition series for S_n . It follows that the composition factors of S_n are $\mathbf{Z}/2\mathbf{Z}$ and A_n . However, A_n is non-abelian (for all $n \geq 4$), it follows that S_n is not solvable. $\square$

5.3 Sylow's Theorem

Let p be a prime. A p -Sylow subgroup of a finite group G is a subgroup of order p^r , where $|G| = p^r m$ and $\gcd(p, m) = 1$. That is, $P \subset G$ is a p -Sylow subgroup if it is a p -group and p does not divide $[G : P]$.

5.3.1 EXAMPLE. Every finite p -group is solvable.

Proof. Assume $|G| = p^n$. We prove by induction on n . If $n \leq 2$, then G is abelian, hence solvable. In general, G has a subgroup N of order p^{n-1} , then N and G/N are solvable by our hypothesis, so that G is solvable. $\square$

5.3.2 EXAMPLE. Every group of order $p^n q$ (p, q are primes) is solvable. (This is a special case of Burnside's theorem, which states that all groups of order $p^n q^m$ are solvable. However, we are not yet in a position to prove this theorem.)

Proof. We prove by induction on n . Let S be a p -Sylow subgroup of G . Then G/S and S are solvable since $|G/S| = q$ and $|S| = p^n$.

Assume S is not normal in G . Then $S \subset N_G(S) \subsetneq G$. Since $[G : N_G(S)][N_G(S) : S] = [G : S] = q$ is prime, $[G : N_G(S)] = 1$ or $[G : N_G(S)] = q$. However, if $[G : N_G(S)] = 1$, then $G = N_G(S)$, so that $S \trianglelefteq G$, which is a contradiction. The only case is that $[G : N_G(S)] = q$ and hence $S = N_G(S)$. And hence there are q many p -Sylow subgroups.

If the q many p -Sylow subgroups of G are pairwise disjoint, then G has $q(p^n - q)$ elements whose order is a positive power of p , leaving at most q elements whose order is a power of q . Therefore G has only one q -Sylow subgroup Q and $Q \trianglelefteq G$. And Q is solvable, G/Q is solvable, so that G is solvable.

Now, assume that the q many p -Sylow subgroups of G are not pairwise disjoint. Then there are p -Sylow subgroups S and T such that $S \cap T \neq \{1\}$, and one can choose S and T so that $M = S \cap T$ has the greatest possible number of elements. By the lemma above, $H = N_G(M)$ has more than one p -Sylow subgroup; and M is the intersection of all the p -Sylow subgroups of H ; and every p -Sylow subgroup of H is contained in a unique p -Sylow subgroup of G . Now the number of p -Sylow subgroups of H divides $p^n q$ but is not divisible by p ; hence H has q many p -Sylow subgroups. Since G also has q many p -Sylow subgroups, M is contained in every p -Sylow subgroup of G . Therefore M is the intersection of all the p -Sylow subgroups of G . Since the latter are all conjugate, this implies that $M \trianglelefteq G$. Now, $|M| = p^k$, where $1 \leq k < n$. Hence G/M is solvable, by the induction hypothesis; M is solvable and G is solvable. $\square$

5.3.3 EXAMPLE. Every finite p -group is nilpotent.

Proof. We prove this by induction on n . If $n \leq 2$, then G is abelian, hence nilpotent. If $n > 2$, then G has a nontrivial center (proposition 5.1.9). Consider the quotient group $G/Z(G)$, whose order is p^m and $m < n$. Then $G/Z(G)$ is nilpotent by our induction hypothesis. Since $Z(G) \subset Z(G)$ and $G/Z(G)$ is nilpotent, G is nilpotent. $\square$

5.3.1 PROPOSITION. Let G be a finite group and let p be a prime number. If p^k divides the order of G , then G has a subgroup of order p^k .

Proof. If $k = 0$, there is nothing to prove, so we assume $k \geq 1$, hence $p \mid |G|$. We proceed by induction on $|G|$: if $|G| = p$, then G itself serves. Otherwise, we assume that $|G| > p$: If there exists a proper subgroup H such that $p \nmid [G : H]$, then $p^k \mid |H|$ since $p^k \mid |G| = |H|[G : H]$ and $p \nmid [G : H]$. Since $|H| < |G|$, by the hypothesis $|H|$ contains a subgroup K of order p^k . Then K is also a subgroup of G , and we are done.

If for every proper subgroup H we have $p \mid [G : H]$, we show that the desired subgroup still exists. Since every proper subgroup has index divisible by p , consider the class equation

$$|G| = |Z(G)| + \sum_{g \notin Z(G)} [G : C_G(g)]$$

For each $g \notin Z(G)$, $C_G(g)$ is a proper subgroup of G , so by our assumption $p \mid [G : C_G(g)]$. Hence each term in the sum is divisible by p . Since $p \mid |G|$, the class equation forces $p \mid |Z(G)|$. By Cauchy's theorem, there exists an element $a \in Z(G)$ of order p . Let $N = \langle a \rangle$; then $N \subset Z(G)$, so N is a normal subgroup of G . Consider the quotient group G/N . We have $|G/N| = |G|/p$. Since $p^k \mid |G|$, it follows that $p^{k-1} \mid |G/N|$. Note that $|G/N| < |G|$ because $p \geq 2$. By the induction hypothesis, G/N contains a subgroup of order p^{k-1} .

By the correspondence theorem, this subgroup must be of the form P/N for some subgroup $P \leq G$ containing N . Then $|P| = |P/N||N| = p^k$. Thus P is a subgroup of G of order p^k . This completes the induction. $\square$

5.3.2 THEOREM (First Sylow Theorem). *Every finite group G contains a p -Sylow subgroup for all primes p .*

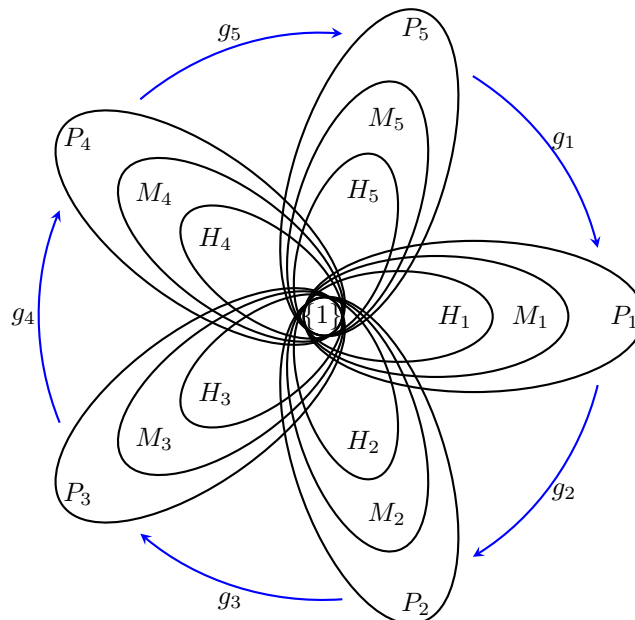
Proof. This theorem is merely a special case of the above proposition. Of course, if there exists a prime p that does not divide the order of G , we consider the trivial subgroup $\{1\}$, which is clearly a p -Sylow subgroup. $\square$

In fact, the first Sylow theorem provides only limited information and leaves further questions: since p -Sylow subgroups exist, how many of them are there? Apart from this obvious remaining question, there is a deeper issue that we will address next: theoretically, a p -Sylow subgroup is the largest p -subgroup in terms of order. Does this mean it is also the largest p -subgroup in terms of inclusion? That is, can every p -subgroup whose order is smaller than that of a p -Sylow subgroup be contained in this p -Sylow subgroup? The second Sylow theorem, which we introduce below, resolves this problem.

5.3.3 THEOREM (Second Sylow Theorem). *Let G be a finite group, let P be a p -Sylow subgroup, and let $H \subset G$ be a p -group. Then H is contained in a conjugate of P : there exists $g \in G$ such that $H \subset gPg^{-1}$. In particular, any two p -Sylow subgroups are conjugate.*

Proof. Let H act on the set $(G/P)_L$ of left cosets of P by left multiplication. Since there are $[G : P]$ cosets and $p \nmid [G : P]$, $(G/P)_{L_0} = \{gP : h(gP) = gP, \forall h \in H\} \neq \emptyset$ by the lemma 5.1.5. Let $gP \in (G/P)_{L_0}$, this means $\forall h \in H, h(gP) = gP$, that is, $g^{-1}hgP = P$ for all $h \in H$; that is, $g^{-1}Hg \subset P$; that is, $H \subset gPg^{-1}$, as desired. $\square$

That is, given a p -Sylow subgroup, an arbitrary p -subgroup is not necessarily contained in it, but it must be contained in another p -Sylow subgroup (note that gPg^{-1} is also a p -Sylow subgroup because conjugation is a group action). The following diagram illustrates this theorem, where P_1 is a given p -Sylow subgroup, and H_i, M_i are p -subgroups:



There is an interesting conjecture (in fact, the author had a momentary lapse of reason): is there a proposition analogous to the second Sylow theorem stating that if P is a p -subgroup of order p^r , where p^r is less than the order of a p -Sylow subgroup, and H is a p -subgroup of order less than p^r , then H must be contained in some conjugate of P ?

Unfortunately, this is false. For a concrete example, consider the dihedral group D_8 . Take $P = \langle r \rangle$, which is a p -subgroup of order 4, and $H = \langle s \rangle$, which is a p -subgroup of order 2. However, P is conjugate only to itself.

Are p -subgroups really so inadequate? The reason is that the conjugates of a non-Sylow p -subgroup P cannot “exhaust” most p -subgroups of the same order. However, this does not mean that none exist; that is, we can always find some other p -subgroup, not necessarily conjugate to the given p -subgroup P but of the same order as P , that contains H :

5.3.4 LEMMA. *Let H be a p -group contained in a finite group G . Then*

$$[N_G(H) : H] \equiv [G : H] \pmod{p}.$$

Proof. If H is trivial, then $N_G(H) = G$ and the two numbers are equal. Assume that H is nontrivial and act with H on $(G/H)_L$ by left multiplication. Then $(G/H)_{L_0} = \{gH \in (G/H)_L : hgH = gH, \forall h \in H\}$, that is, every element gH in this set satisfies $hgH = gH$ for all $h \in H$, so that $g^{-1}hg \in H$ for all $h \in H^{-1}$; and $H \subset gHg^{-1}$. Since $|H| = |gHg^{-1}|$, $H = gHg^{-1}$. This means precisely that $g \in N_G(H)$. Therefore, $(G/H)_{L_0}$ consists of the set of cosets of H in $N_G(H)$. By lemma 5.1.5, $[G : H] \equiv [N_G(H) : H] \pmod{p}$. $\square$

5.3.5 PROPOSITION. *Let H be a p -subgroup of a finite group G , and assume that H is not a p -Sylow subgroup. Then there exists a p -subgroup H' of G containing H such that $[H' : H] = p$ and H is normal in H' .*

Proof. Since H is not a p -Sylow subgroup of G , $p \mid [G : H]$, then $p \mid [N_G(H) : H]$ by lemma 5.2.4. Since H is normal in $N_G(H)$, we may consider the quotient group $N_G(H)/H$, and p divides the order of this group. By Cauchy's theorem, $N_G(H)/H$ has an element of order p ; this generates a subgroup of order p of $N_G(H)/H$, which must be of the form H'/H for a subgroup H' of $N_G(H)$. It is clear that $H \subset H'$ and $|H'| = p \cdot |H|$ with $H \trianglelefteq H'$. $\square$

5.3.1 REMARK. Note that this proposition provides another approach to proving the first Sylow theorem: one simply starts with the trivial subgroup $\{1\}$ and repeatedly extends it.

Therefore, suppose we already have a chain of inclusions of p -subgroups: $\{1\} \trianglelefteq H_1 \trianglelefteq H_2 \trianglelefteq \cdots \trianglelefteq H_k$, where each H_i is a p -subgroup of order p^i . We can always extend this chain further; there exist $H_{k+1}, H_{k+2}, \dots$ until we reach a p -Sylow subgroup, giving the following complete chain:

$$\{1\} \trianglelefteq H_1 \trianglelefteq H_2 \trianglelefteq \cdots \trianglelefteq H_{k+1} \trianglelefteq \cdots \trianglelefteq H_r.$$

Regarding the question left by the first Sylow theorem, there remains the issue of the exact number of p -Sylow subgroups, to which the third Sylow theorem provides a profound answer.

5.3.6 THEOREM (Third Sylow Theorem). *Let p be a prime integer, and let G be a finite group of order $|G| = p^r m$. Assume that p does not divide m . Then the number of p -Sylow subgroups of G divides m and is congruent to 1 modulo p , i.e., $N_p \equiv 1 \pmod{p}$.*

Proof. Let N_p denote the number of p -Sylow subgroups of G . By the second Sylow theorem, the p -Sylow subgroups of G are conjugates of any given p -Sylow subgroup P . Since the number of subgroups conjugate to P equals the index $[G : N_G(P)]$ of the normalizer of P in G , $N_p = [G : N_G(P)]$. Then

$$m = [G : P] = [G : N_G(P)][N_G(P) : P] = N_p \cdot [N_G(P) : P].$$

Now, by lemma 5.2.4, we have $m = [G : P] \equiv [N_G(P) : P] \pmod{p}$. Thus, $mN_p \equiv m \pmod{p}$. Since $p \nmid [G : P] = m$ and p is a prime, it follows that

$$N_p \equiv 1 \pmod{p}.$$

□

5.3.7 COROLLARY. *Let P be a p -Sylow subgroup of a finite group G . Then P is normal in G if and only if P is the unique p -Sylow subgroup of G , i.e., $N_p = 1$.*

Proof. If $P \trianglelefteq G$, and Q is a p -Sylow subgroup of G , then by the first Sylow theorem, there exists $g \in G$ such that $Q = gPg^{-1} = P$. Thus P is the unique p -Sylow subgroup of G . Conversely, if $N_p = 1$, then $gPg^{-1} = P$ for all $g \in G$, hence P is normal in G . □

Now, we wish to explore some applications of the Sylow theorems to combinatorics, arithmetic, and finite group theory, in order to demonstrate their significance.

5.3.4 EXAMPLE (Wilson's Theorem). p is prime if and only if $(p-1)! \equiv -1 \pmod{p}$.

Proof. We will only prove the “only if” part since it uses the Sylow theorems. Consider the symmetric group S_p for p prime. Since $p \mid p!$ and $p^2 \nmid p!$, □

5.3.5 EXAMPLE (Groups of order pq , p and q primes with $p < q$). Suppose that $|G| = pq$ for primes p and q with $p < q$. Let $P \in \text{Syl}_p(G)$ and let $Q \in \text{Syl}_q(G)$. We show that Q is normal in G ; and if $p \nmid q-1$, then G is cyclic.

Proof. By the third Sylow theorem, $N_q = 1 + kq$ for some integer $k \geq 0$ and $N_q \mid p$. However, since $p < q$, the only possible case is $k = 0$, i.e., $N_q = 1$. Hence $Q \trianglelefteq G$.

Since N_p divides the prime q , the only possibilities are $N_p = 1$ or $N_p = q$. In particular, if $p \nmid q-1$, then $N_p = 1 + kp$ cannot equal q , so that $P \trianglelefteq G$. Since P is normal, conjugation gives an action of G on P , that is, we may define the homomorphism $\sigma : G \rightarrow \text{Aut}(P)$ (corresponding the action $\rho : G \times P \rightarrow P, (g, h) \mapsto ghg^{-1}$).

Now, P is cyclic of order p , so that $|\text{Aut}(P)| = p-1$. Then $|\sigma(G)| \mid pq$ by the first isomorphism theorem. Since $\sigma(G)$ is a subgroup of $\text{Aut}(P)$, the order of $\sigma(G)$ must divide $p-1$. So, $\sigma(G) \mid \gcd(pq, p-1)$. Then $\sigma(G)$ is the trivial group, that is, $\sigma(g)(h) = ghg^{-1} = h$ for every $g \in G, h \in P$ so $P \subset Z(G)$.

Now, consider the quotient group G/P , $|G/P| = q$, so that G/P is cyclic. Since $P \subset Z(G)$, $G/Z(G) \cong (G/P)/(Z(G)/P)$. It follows that $G/Z(G)$ is cyclic. Then G is abelian. Finally, an abelian group of order pq with $p < q$ primes is necessarily cyclic: indeed, it must contain elements g, h of order p, q by Cauchy's theorem, and then $|gh| = pq$. □

5.3.6 EXAMPLE ($p = 2$). Let q be an odd prime, and let G be a noncommutative group of order $2q$. Then $G \cong D_{2q}$.

Proof. By Cauchy's theorem, there exists an element $y \in G$ such that y has order q . By the third Sylow theorem, $N_q \mid 2$ and $N_p \equiv 1 \pmod{q}$, it follows that $\langle y \rangle$ is the unique subgroup of order q in G and is therefore normal. Since G is not commutative and in particular it is not cyclic, it has no elements of order $2q$, therefore, every element in the complement of $\langle y \rangle$ has order 2; let x be such element.

The conjugate xyx^{-1} of y by x is an element of order q , so $xyx^{-1} \in \langle y \rangle$. Thus $xyx^{-1} = y^r$ for some r between 0 and $q-1$. Now, observe that

$$(y^r)^r = (xyx^{-1})^r = xy^r x^{-1} = x^2 t (x^{-1})^2 = y,$$

therefore $y^{r^2-1} = 1$, which implies $q \mid (r^2-1) = (r-1)(r+1)$. since q is prime, this says that $q \mid (r-1)$ or $q \mid (r+1)$. Since $0 \leq r \leq q-1$, it follows that $r = 1$ or $r = q-1$.

If $r = 1$, then $xyx^{-1} = x$, that is $xy = yx$. But then the order of xy is $|x||y| = 2q$, and G is cyclic, which is a contradiction. Therefore, $r = q - 1$, and we have established the relations

$$\begin{cases} x^2 = 1, \\ y^q = 1, \\ yx = xy^{q-1} \end{cases}$$

Now, $\langle x, y \rangle = G$ since $q < |\langle x, y \rangle| \leq 2q$, forces $|\langle x, y \rangle| = |2q|$, i.e., $G = \langle x, y \rangle$. Since we have seen that $D_{2q} \cong \{r, s \mid r^q = s^2 = 1, rs = sr^{q-1}\}$, it follows that $G \cong D_{2q}$. $\square$

5.4 Direct Sum

Direct Sum

If G is a product of the groups G_1 and G_2 , then G is isomorphic to $G_1 \times G_2$, i.e., there exists an isomorphism $\varphi : G_1 \times G_2 \rightarrow G$. Next, we give conditions under which a group splits into a direct product.

5.4.1 PROPOSITION. *A group G is a product of G_1 and G_2 if and only if it contains normal subgroups $A \cong G_1$ and $B \cong G_2$ such that $A \cap B = \{1\}$ and $AB = G$.*

Proof. The direct product $G_1 \times G_2$ comes with projections $\pi_1 : G_1 \times G_2 \rightarrow G_1, (x_1, x_2) \mapsto x_1$ and $\pi_2 : G_1 \times G_2 \rightarrow G_2, (x_1, x_2) \mapsto x_2$, which are homomorphisms, since the operation on $G_1 \times G_2$ is componentwise. Hence

$$\begin{aligned} \ker \pi_1 &= \{(x_1, x_2) \in G_1 \times G_2 : x_1 = 1\} \\ \ker \pi_2 &= \{(x_1, x_2) \in G_1 \times G_2 : x_2 = 1\} \end{aligned}$$

are normal subgroups of $G_1 \times G_2$. We see that $(x_1, 1) \mapsto x_1$ is an isomorphism of $\ker \pi_2$ onto G_1 and that $(1, x_2) \mapsto x_2$ is an isomorphism of $\ker \pi_1$ onto G_2 . Moreover, $\ker \pi_1 \cap \ker \pi_2 = \{(1, 1)\} = \{1\}$, and $(\ker \pi_1)(\ker \pi_2) = G_1 \times G_2$, since every $(x_1, x_2) \in G_1 \times G_2$ is the product $(x_1, x_2) = (x_1, 1)(1, x_2)$ of $(x_1, 1) \in \ker \pi_2$ and $(1, x_2) \in \ker \pi_1$.

Now, if G is also a product of $G_1 \times G_2$, and $\varphi : G_1 \times G_2 \rightarrow G$ is an isomorphism, then $A = \varphi(\ker \pi_2)$ and $B = \varphi(\ker \pi_1)$ are normal subgroups of G , and $A \cong \ker \pi_2 \cong G_1, B \cong \ker \pi_1 \cong G_2, A \cap B = \{1\}$ and $AB = G$.

Conversely, assume that $A \trianglelefteq G, B \trianglelefteq G, A \cap B = \{1\}, AB = G$. Then every element g of G is a product $g = ab$ of some $a \in A$ and $b \in B$. Moreover, if $ab = a'b'$ with $a, a' \in A, b, b' \in B$, then $a'^{-1}a = b'b^{-1} \in A \cap B$ yields $a'^{-1}a = b'b^{-1} = 1$ and $a = a', b = b'$. Hence the function $\varphi : A \times B \rightarrow G = AB$ defined by $(a, b) \mapsto ab$ is a bijection. We show that φ is an isomorphism:

For all $a \in A$ and $b \in B$, we have $aba^{-1}b^{-1} = a(ba^{-1}b^{-1}) \in A$ and $aba^{-1}b^{-1} = (aba^{-1})b^{-1} \in B$ since $A, B \trianglelefteq G$. Hence $aba^{-1}b^{-1} = 1$ and $ab = ba$. Therefore

$$\varphi((a, b)(a', b')) = \varphi(aa', bb') = aa'bb' = aba'b' = \varphi(a, b)\varphi(a', b').$$

$\square$

5.4.2 PROPOSITION. *A group G is a product of $G_1, \dots, G_n$ if and only if G contains normal subgroups $A_1 \cong G_1, \dots, A_n \cong G_n$ such that $A_j \cap (A_1 \cdots A_{j-1}A_{j+1} \cdots A_n) = \{1\}$ and $G = A_1 \cdots A_n$.*

In particular, a group G is a product of two subgroups H_1, H_2 if and only if H_1, H_2 are normal subgroups and $H_1 \cap H_2 = \{1\}$ and $H_1H_2 = G$. And the isomorphism $\varphi : H_1 \times H_2 \rightarrow G$ is defined by $(h_1, h_2) \mapsto h_1h_2$. This means that each element g of G can be written uniquely in the form $g = h_1h_2$.

Naturally, we would wonder whether this conclusion can be extended to infinite sets, that is, considering a possibly infinite set K , whether there exists a similar conclusion

as in the finite case. First, we must make clear that in group theory, writing an element as $a_1 a_2 \cdots a_n \cdots$ is not defined, nor do we have a definition for $A_1 A_2 \cdots A_n \cdots$. For this reason, we should discuss the subgroup generated by the union of these subgroups as sets, denoted by $\langle \bigcup_{i \in I} A_i \rangle$, to correspond to the conclusion “ $G = A_1 A_2 \cdots A_n$ ” above. In fact, when all of them are normal subgroups, we have

$$A_1 A_2 \cdots A_n = \left\langle \bigcup_{i=1}^n A_i \right\rangle.$$

Therefore, the question we need to address is whether the following statement holds: “ G is the direct product of its subgroups $\{A_i : i \in I\}$ if and only if $A_i \trianglelefteq G$, $A_j \cap \langle \bigcup_{k \neq j} A_k \rangle = \{1\}$, and $G = \langle \bigcup_{k \in K} A_k \rangle$ ”? The answer, of course, is negative. For example,

From now on, we shall primarily use additive notation, with the convention that AB is written as $A + B$. In particular, the identity element will be denoted by 0. When no confusion may arise, we shall denote the identity element of most groups by 0 or by $\mathbf{0}$ to distinguish it when necessary.

5.4.1 EXAMPLE. Let $G = \prod_{n=1}^{\infty} \mathbf{Z}_2$ be the direct product of countably many copies of $\mathbf{Z}_2$, i.e., the group of all 0 - 1 sequences $(x_1, x_2, x_3, \dots)$, $x_n \in \{0, 1\}$ under componentwise addition modulo 2. For each $n \in \mathbf{N}$, define

$$A_n = \left\{ \mathbf{0}, (0, 0, \dots, 0, \underbrace{1}_{n\text{-th position}}, 0, \dots) \right\}.$$

Equivalently, $A_n = \{\mathbf{0}, \mathbf{e}_n\}$, where $\mathbf{e}_n$ is the sequence with 1 in the n -th coordinate and 0 elsewhere. First, since G is abelian, every subgroup is normal. Hence $A_n \trianglelefteq G$ holds. Second, observe that $\langle \bigcup_{k \neq j} A_k \rangle = \{(x_1, x_2, \dots) \in G : x_j = 0\}$, i.e., the subgroup consisting of all sequences whose j -th coordinate is 0. Since $A_j = \{\mathbf{0}, \mathbf{e}_j\}$, where $\mathbf{e}_j$ has 1 in the j -th coordinate, we have

$$A_j \cap \left\langle \bigcup_{k \neq j} A_k \right\rangle = \{\mathbf{0}\}.$$

Thus the second condition holds. Third, the left-hand side G contains the element $(1, 1, 1, \dots) \in G$, i.e., the sequence with 1 in every coordinate. However, the subgroup

$$\left\langle \bigcup_{n=1}^{\infty} A_n \right\rangle$$

is generated by all $\mathbf{e}_n$, and consists precisely of all sequences with only finitely many coordinates equal to 1. Indeed, every element is a finite sum of the form

$$\mathbf{e}_{n_1} + \mathbf{e}_{n_2} + \cdots + \mathbf{e}_{n_m}.$$

Hence, the sequence $(1, 1, 1, \dots)$ is not contained in $\langle \bigcup_{n=1}^{\infty} A_n \rangle$. Hence $G \neq \langle \bigcup_{n=1}^{\infty} A_n \rangle$, so the third condition fails. Nevertheless, we trivially have $G \cong \prod_{n=1}^{\infty} A_n$, since G is exactly the direct product of the subgroups A_n (each $A_n \cong \mathbf{Z}_2$).

In this example, the key reason why the condition fails is that G contains elements for which the number of nonzero coordinates is not finite, which makes them impossible to be generated by elements of $\langle \bigcup_{n=1}^{\infty} A_n \rangle$. To resolve this issue, we give the following definition.

5.4.1 DEFINITION. Let K be a possibly infinite set and let $\{A_k : k \in K\}$ be a set of groups indexed by K . The *direct sum* denoted by $\bigoplus_{k \in K} A_k$ is the subgroup of $\prod_{k \in K} A_k$ consisting of all those elements for which there are only finitely many k with $a_k \neq 0$.

Under this definition, the finite direct sum and the direct product coincide. Therefore, the two propositions at the beginning of this section can both be stated with the direct product replaced by the direct sum. In general, the following proposition always holds:

5.4.3 PROPOSITION. *Let $\{A_k : k \in K\}$ be a set of normal subgroups of a group G . Then the following are equivalent:*

1. $G \cong \bigoplus_{k \in K} A_k$;
2. $G = \langle \bigcup_{k \in K} A_k \rangle$ and $A_j \cap \langle \bigcup_{k \neq j} A_k \rangle = \{1\}$;
3. every $g \in G$ has a unique expression of the form $g = \sum_{k \in K} a_k$ where $a_k \in A_k$ and $a_k \neq 0$ for only finitely many k .

Proof. (2) $\rightarrow$ (1) : if $\{a_k\}_K \in \bigoplus_{k \in K} A_k$, then $a_k = 0$ for all but a finite number of $k \in K$. Let K_0 be the finite set $\{k \in K : a_k \neq 0\}$. Then $\sum_{k \in K_0} a_k$ is a well-defined element of G since $A_j \cap A_i = \{0\}$ and they are normal subgroups (so that $a + b = b + a$ for any $a \in A_i, b \in A_j$). Consequently, the function $\varphi : \bigoplus_{k \in K} A_k \rightarrow G$ given by $\{a_k\}_K \mapsto \sum_{k \in K_0} a_k \in G$ is a homomorphism such that $\varphi \circ \iota_k(a_k) = a_k$ for $a_k \in A_k$ where $\iota_k : A_k \rightarrow \bigoplus_{k \in K} A_k$ given by $\iota_k(a) = \{a_k\}_K$ in which $a_j = 0, j \neq k$ and $a_k = a$ (clearly, a monomorphism). is the canonical injection.

Since G is generated by the subgroups A_k , every element a of G is a finite sum of elements from various A_k . Since element of A_k and A_j commute if $i \neq j$, a can be written as a sum $\sum_{k \in K_0} a_k$ where $a_k \in A_k$ and K_0 is some finite subset of K . Hence $\sum_{k \in K_0} \iota_k(a_k) \in \bigoplus_{k \in K} A_k$ and $\varphi(\sum_{k \in K_0} \iota_k(a_k)) = \sum_{k \in K_0} \varphi \circ \iota_k(a_k) = \sum_{k \in K_0} a_k = a$, therefore, φ is an epimorphism.

Suppose $\varphi(\{a_k\}) = \sum_{k \in K_0} a_k = 0 \in G$. WLOG, we may assume that $K_0 = \{1, 2, \dots, n\}$. Then $\sum_{k \in K_0} a_k = a_1 + a_2 + \dots + a_n = 0$ with $a_k \in A_k$. Hence $-a_1 = a_2 + a_3 + \dots + a_n \in A_1 \cap \langle \bigcup_{j \neq 1} A_j \rangle = \{0\}$ and therefore $a_1 = 0$. Repetition of this argument shows that $a_k = 0$ for all $k \in K_0$. Hence φ is a monomorphism.

(3) $\rightarrow$ (2) : Assume (3) holds: every $g \in G$ has a unique expression of the form $g = \sum_{k \in K} a_k$, $a_k \in A_k$, where only finitely many a_k are nonzero. First, we prove $G = \langle \bigcup_{k \in K} A_k \rangle$. By (3), every $g \in G$ can be written as a finite sum $\sum_{k \in K_0} a_k$, where $a_k \in A_k$ and K_0 is finite. Hence $g \in \langle \bigcup_{k \in K} A_k \rangle$, so $G \subseteq \langle \bigcup_{k \in K} A_k \rangle$. The reverse inclusion is immediate since each $A_k \subseteq G$. Thus $G = \langle \bigcup_{k \in K} A_k \rangle$.

Next, fix $j \in K$ and take any $x \in A_j \cap \langle \bigcup_{k \neq j} A_k \rangle$. Since $x \in A_j$, we have $x = a_j$ with $a_j \in A_j$. Also, since $x \in \langle \bigcup_{k \neq j} A_k \rangle$, we can write $x = \sum_{k \neq j} a_k$ with $a_k \in A_k$. Hence $a_j - \sum_{k \neq j} a_k = 0$, i.e., $a_j + \sum_{k \neq j} (-a_k) = 0$. By the uniqueness of expression for 0 in (3), all coefficients must be zero, so $a_j = 0$, and thus $x = 0$. Therefore

$$A_j \cap \langle \bigcup_{k \neq j} A_k \rangle = \{0\}.$$

(1) $\rightarrow$ (3) : Assume $G \cong \bigoplus_{k \in K} A_k$. Let $\varphi : \bigoplus_{k \in K} A_k \rightarrow G$ be a group isomorphism. Take any $g \in G$. Since φ is isomorphic, there exists a unique element $\{a_k\}_{k \in K} \in \bigoplus_{k \in K} A_k$ such that $\varphi(\{a_k\}) = g$. By definition of the direct sum, only finitely many a_k are nonzero. Thus

$$g = \varphi(\{a_k\}) = \varphi \left(\sum_{k \in K_0} \iota_k(a_k) \right) = \sum_{k \in K_0} \varphi \circ \iota_k(a_k) = \sum_{k \in K_0} a_k.$$

□

From the above proof process, when we say that G is the direct sum of certain subgroups, the homomorphism $\varphi : \bigoplus_{k \in K} A_k \rightarrow G$ is generally defined by

$$\{a_k\}_K \mapsto \sum_{k \in K_0} a_k \in G,$$

where K_0 is the finite set of indices for which $a_k \neq 0$. Under such a homomorphism, we can say that every element of G can always be decomposed as a sum of elements from these subgroups, i.e., we have decomposed the group G .

5.4.2 DEFINITION. A finite subset $X = \{x_1, \dots, x_n\}$ of nonzero elements of a group G is *independent* if, for all $m_i \in \mathbf{Z}$, $\sum_{i=1}^n m_i x_i = 0$ implies $m_i x_i = 0$ for each i . An infinite set X of nonzero elements in G is *independent* if every finite subset is independent.

5.4.4 LEMMA. Let G be abelian and $X \subset G$, then X is independent if and only if

$$\langle X \rangle = \bigoplus_{x \in X} \langle x \rangle.$$

Proof. Assume that X is independent. If $x_0 \in X$ and $y \in \langle x_0 \rangle \cap \langle X - \{x_0\} \rangle$, then $y = mx_0$ and $y = \sum m_i x_i$ where x_i are distinct elements of X not equal x_0 . Hence $-mx_0 + \sum m_i x_i = 0$, so that independence gives each term 0; in particular $0 = mx_0 = y$. Therefore, $\langle X \rangle = \bigoplus_{x \in X} \langle x \rangle$ by the proposition above.

Conversely, assume that $\langle X \rangle = \bigoplus_{x \in X} \langle x \rangle$. Take any finite subset $\{x_1, \dots, x_n\} \subset X$ and assume $m_1 x_1 + \dots + m_n x_n = 0$. Since every element of $\langle X \rangle$ has a unique expression as $\sum_{x \in X} a_x$ with $a_x \in \langle x \rangle$. The zero element has the expression $\sum_{x \in X} 0$, which forces $m_i x_i = 0$ for each i . Thus every finite subset is independent, so X is independent. $\square$

These proposition tells us that for a group to decompose into a direct product or direct sum of some subgroups, many conditions must be satisfied: $A_k \trianglelefteq G$, $A_j \cap \langle \bigcup_{k \neq j} A_k \rangle = \{0\}$, and $G = \langle \bigcup_{k \in K} A_k \rangle$. This is very restrictive, so there are very few groups that can be decomposed into a direct product or direct sum of some subgroups. In **Ab**, however, all subgroups are normal and the conditions in these proposition reduce to $A_j \cap \langle \bigcup_{k \neq j} A_k \rangle = \{0\}$, and $G = \langle \bigcup_{k \in K} A_k \rangle$. Moreover, in **Ab**, the direct sum and the coproduct are essentially the same concept, namely the following proposition (This fact is not actually used very often, but it does make one appreciate what an excellent property commutativity truly is.):

5.4.5 THEOREM. Let $\{A_k : k \in K\}$ be a set of abelian groups. Then $\bigoplus_{k \in K} A_k$ together with the canonical injections $\{\iota_k : A_k \rightarrow \bigoplus_{k \in K} A_k\}$ satisfies the universal property for the coproduct of $\{A_k : k \in K\}$.

Proof. $\square$

Free Abelian Groups

Let G be an abelian group written additively. A *basis* of an abelian group G is a subset A of G such that $G = \langle A \rangle$, and for distinct $x_1, \dots, x_k \in A$ and $m_i \in \mathbf{Z}$, $m_1 x_1 + m_2 x_2 + \dots + m_k x_k = 0 \rightarrow m_i = 0$ for every i .

This concept is analogous to the notion of a basis in a finite-dimensional vector space in linear algebra, but there are still differences. For instance, every finite-dimensional vector space necessarily has a basis, but for abelian groups this is not true; not every abelian group has a basis: Clearly, every finite cyclic group has no basis, because the expression of an element is not unique.

Then which abelian groups have a basis? We say that if it has a basis, then the expansion of every element is unique. If it is not unique, then there must exist a linear relation between at least two elements of the original basis. If we connect this with the concept of free groups and presentations, this means that when this set is taken as a generating set, its relation set is nonempty. Therefore, we may consider abelian groups with a basis to be “free”. So we attempt to define free abelian groups in the category **Ab** using the universal property as follows, and afterwards prove that our conjecture is correct, namely that every abelian group with a basis is a free abelian group.

5.4.3 DEFINITION. Let A be a nonempty set. In the concrete category $\mathbf{Ab}$, a *free abelian group* on the set A is the abelian group $F^{ab}(A)$ together with a function $\eta : A \rightarrow F^{ab}(A)$ satisfying the universal property: for all $G \in \mathbf{Ab}$ and function $f : A \rightarrow G$, there exists a unique group homomorphism $\varphi : F^{ab}(A) \rightarrow G$ such that the diagram commutes:

$$\begin{array}{ccc} F^{ab}(A) & \xrightarrow{\varphi} & G \\ \eta \uparrow & \nearrow f & \\ A & & \end{array}$$

The universal property of free abelian group guarantees that $F^{ab}(A)$ is unique up to isomorphism, if it exists; but we have not to prove it exists. We can certainly use a construction similar to what we did before, namely alphabets and words, to try to construct a free abelian group.

Recall that $A \cup A^{-1}$ is the alphabet, and every element of the free group $F(A)$ on A is of the form $a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n}$. Once the order of $a_1 \cdots a_n$ differs, they are regarded as two different words, i.e., distinct elements of the free group. For example, “dormitory” and “dirty room” are different elements. Intuitively, a free abelian group is obtained from a free group by adding the “commutativity property”; then we would regard “dormitory” and “dirty room” as the same element (indeed, some dormitories really are dirty!). In other words, after imposing the commutativity condition, we no longer care about the order of letters, but only about the number of occurrences of each letter. For instance, $a^2 b^3 a^2 c^3$ and $b^1 a^3 c^3 b^2 a^1$ are the same element, since in both the counts of a, b, c are: a appears 4 times, b appears 3 times, and c appears 3 times.

I believe the reader now sees what I am getting at: for a finite set $A = \{a_1, \dots, a_n\}$, each element of the free abelian group on A is uniquely determined by an integer tuple $(s_1, \dots, s_n)$. Intuitively, we may guess that

$$F^{ab}(A) \cong \underbrace{\mathbf{Z} \oplus \mathbf{Z} \oplus \cdots \oplus \mathbf{Z}}_{n \text{ times}}$$

.

5.4.6 PROPOSITION. $\bigoplus_{i=1}^n \mathbf{Z}$ is a free abelian group on $A = \{1, 2, \dots, n\}$.

Proof. Note that every element of $\bigoplus_{i=1}^n \mathbf{Z}$ can be written uniquely in the form $\sum_{i=1}^n m_i j(i)$ where $j : A \rightarrow \bigoplus_{i=1}^n \mathbf{Z}$ is defined by $j(i) := (0, \dots, 0, 1, 0, \dots, 0) \in \bigoplus_{i=1}^n \mathbf{Z}$:

$$\begin{aligned} (m_1, \dots, m_n) &= (m_1, 0, \dots, 0) + (0, m_2, 0, \dots, 0) + \cdots + (0, \dots, 0, m_n) \\ &= m_1(1, 0, \dots, 0) + m_2(0, 1, 0, \dots, 0) + \cdots + m_n(0, \dots, 0, 1) \\ &= m_i j(1) + \cdots + m_n j(n) \end{aligned}$$

and $(m_1, \dots, m_n) = (0, \dots, 0)$ if and only if all $m_i = 0$. Now, let $f : A \rightarrow G$ be any function from A to an abelian group G . Define the function $\varphi : \bigoplus_{i=1}^n \mathbf{Z} \rightarrow G$ by

$$\varphi \left(\sum_{i=1}^n m_i j(i) \right) := \sum_{i=1}^n m_i f(i),$$

indeed, we have no choice, this definition is forced by the needed commutativity of the diagram ($f(i) = \varphi(j(i))$) and by the homomorphism condition:

$$\begin{array}{ccc} \bigoplus_{i=1}^n \mathbf{Z} & \xrightarrow{\varphi} & G \\ j \uparrow & \nearrow f & \\ A & & \end{array}$$

Thus, φ is certainly uniquely determined and we just have to check that it is a homomorphism. This is where the commutativity of G enters:

$$\begin{aligned} \varphi\left(\sum_{i=1}^n m'_i j(i)\right) + \varphi\left(\sum_{i=1}^n m''_i j(i)\right) &= \sum_{i=1}^n m'_i f(i) + \sum_{i=1}^n m''_i f(i) = \sum_{i=1}^n (m'_i + m''_i) f(i) \\ &= \varphi\left(\sum_{i=1}^n (m'_i + m''_i) j(i)\right) = \varphi\left(\sum_{i=1}^n m'_i j(i) + \sum_{i=1}^n m''_i j(i)\right) \end{aligned}$$

□

What about a general set A ? That is, how do we consider the free abelian group on a set A that is not necessarily finite? Let us revisit the words in the free group on A : they are all of the form $a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n}$, where all letters in the word come from A and the word has finite length (i.e., the length of the word is finite). We regard the number of occurrences of any letter not appearing in the word as 0. Then how do we represent the number of occurrences of each letter? We consider a function $\alpha : A \rightarrow \mathbf{Z}$ that sends only finitely many elements of A to nonzero values. Thus every word corresponds to such a function (and conversely, such a function uniquely determines a word, of course, without regard to order). For example, “barry” corresponds to the function $\alpha : A \rightarrow \mathbf{Z}$ with $\alpha(b) = 1$, $\alpha(a) = 1$, $\alpha(r) = 2$, and $\alpha(y) = 1$.

Consider the Hom-set $\text{Hom}(A, \mathbf{Z})$ consists of all the functions from A into $\mathbf{Z}$ in the category **Set**. Let $\bigoplus_A \mathbf{Z}$ denote the set of all such functions, i.e., $\bigoplus_A \mathbf{Z} = \{\alpha \in \text{Hom}(A, \mathbf{Z}) : \alpha(a) \neq 0 \text{ for only finitely many } a \in A\}$. Intuitively, we have

$$F^{\text{ab}}(A) \cong \bigoplus_A \mathbf{Z}.$$

Here a minor detail needs to be handled, namely, to prove that $\bigoplus_A \mathbf{Z}$ is indeed an abelian group. We leave this to the reader as an exercise. Note that $\bigoplus_A \mathbf{Z}$ is the whole of $\text{Hom}(A, \mathbf{Z})$ if A is a finite set; and that $\bigoplus_A \mathbf{Z} \cong \bigoplus_{i=1}^n \mathbf{Z}$ if $A = \{1, 2, \dots, n\}$: indeed, $(m_1, \dots, m_n) \in \bigoplus_{i=1}^n \mathbf{Z}$ may be identified with a function $\{1, \dots, n\} \rightarrow \mathbf{Z}$ sending i to m_i .

5.4.7 PROPOSITION. $\bigoplus_A \mathbf{Z}$ is a free abelian group on the set A .

Proof. Note that every element of $\bigoplus_A \mathbf{Z}$ can be written uniquely as a *finite sum* of the form $\sum_{a \in A} m_a j(a)$, $m_a \neq 0$ for only finitely many a . In which, $j : A \rightarrow \bigoplus_A \mathbf{Z}$ is the function defined by sending $a \in A$ to the function $j_a : A \rightarrow \mathbf{Z}$ defined by

$$j_a(x) := \begin{cases} 1 & x = a \\ 0 & x \neq a \end{cases}$$

Suppose that $\alpha(a_i) = m_i$ for some $a_i \in A, m_i \in \mathbf{Z}, 1 \leq i \leq n$. Then $\alpha = \sum_{i=1}^n m_i j(a_i)$. The rest of the proof is entirely the same as the above case, so we shall not elaborate further here. □

5.4.2 EXAMPLE. Recall the definition of Grothendieck group associated with a commutative monoid $\langle M, + \rangle$. We have shown that $K(M) = M \times M / \sim$ satisfies the universal property of Grothendieck group with M : for any abelian group $A \in \mathbf{Ab}$ and the monoid homomorphism $f : M \rightarrow A$, there exists a unique homomorphism $\tilde{\varphi} : K(M) \rightarrow A$ such that the diagram commutes

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & K(M) \\ & \searrow f & \downarrow \tilde{\varphi} \\ & & A \end{array}$$

In fact, let $F^{\text{ab}}(M)$ be the free abelian group generated by M . We denote the generator of $F^{\text{ab}}(M)$ corresponding to an element $x \in M$ by (x) . And let B be the subgroup

generated by all the elements of the form $(x + y) - (x) - (y)$ where $x, y \in M$. Consider the quotient $F^{ab}(M)/B$, and let

$$\gamma : M \rightarrow F^{ab}(M)/B$$

be the function obtained by composing the injection of M into $F^{ab}(M)$ given by $x \mapsto (x)$ and the canonical projection $\pi : F^{ab}(M) \rightarrow F^{ab}(M)/B$. It is clear that γ is a homomorphism and satisfies the universal property, so that $K(M) \cong F^{ab}(M)/B$. (Looking again from this perspective at the problem of how to construct a group from a monoid in the most “economical” way becomes more appropriate.)

5.4.8 THEOREM. *If the abelian group G is generated by a subset A , then there is a surjective homomorphism from $F^{ab}(A) \rightarrow G$. In other words, every $G \in \mathbf{Ab}$ is the homomorphism image of a free abelian group.*

Proof. Consider the inclusion function $\iota : A \rightarrow G$. By the universal property of free abelian group, there is a group homomorphism $\varphi : F^{ab}(A) \rightarrow G$ such that $\varphi \circ \eta$ is the inclusion $\iota : A \rightarrow G$, then $\text{im } \varphi = G$ since $\text{im } \varphi$ contains every generator $a = \varphi(\eta(a))$ of G . $\square$

5.4.9 THEOREM. *Let G be an abelian group, then G has a nonempty basis if and only if G is a free abelian group on some set. In particular, if A is a basis of the abelian group G , then $G \cong F^{ab}(A)$.*

Proof. Assume G has a basis A , and let $\iota : A \rightarrow G$ be the inclusion function. Given a function $f : A \rightarrow G'$ in which G' is abelian. If $x \in G$, then $x = m_1x_1 + \cdots + m_kx_k$ since A generates G . If $x = n_1x_1 + \cdots + n_kx_k$, then $\sum_{i=1}^k (m_i - n_i)x_i = 0$, so that $n_i = m_i$ for every i since A is a basis. Now, consider the function $\varphi : G \rightarrow G'$ defined by

$$\varphi(x) = \varphi \left(\sum_{i=1}^k m_i x_i \right) = m_1 f(x_1) + \cdots + m_k f(x_k)$$

is well-defined and $\varphi \circ \iota = f$. Since G is abelian, φ is easily seen to be a homomorphism. Since A generates G , any homomorphism $G \rightarrow G'$ is determined by its restriction on A . Thus, if $\psi : G \rightarrow G'$ is a homomorphism such that $\psi \circ \iota = f$, then for any $x \in A$, $\psi(x) = \psi(\iota(x)) = f(x) = \varphi(x)$, whence $\psi = \varphi$ and φ is unique.

Conversely, if G is a free abelian group, then $G \cong \bigoplus_B \mathbf{Z}$ for some set B . Take the set $A = \{j_b \in \bigoplus_B \mathbf{Z} : j_b(x) = 1 \text{ if } x = b, j_b(x) = 0 \text{ if } x \neq b\}$. Then A is a basis of G . $\square$

5.4.3 EXAMPLE. If G_1, G_2 are free abelian, so is $G_1 \oplus G_2$.

Proof. Suppose that G_1 and G_2 are free abelian groups with bases X_1 and X_2 respectively. Let $\tilde{X}_1 = \{(x, 0) : x \in X_1\}$ and $\tilde{X}_2 = \{(0, x) : x \in X_2\}$. We claim that $Y = \tilde{X}_1 \cup \tilde{X}_2$ is a basis for $G_1 \oplus G_2$: let $(g_1, g_2) \in G_1 \oplus G_2$, then $g_1 = n_1x_1 + \cdots + n_kx_k$ and $g_2 = m_1x'_1 + \cdots + m_sx'_s$. Thus, $(g_1, g_2) = n_1(x_1, 0) + \cdots + n_k(x_k, 0) + m_1(0, x'_1) + \cdots + m_s(0, x'_s)$ for the unique choices of n_i and m_j . It implies that Y is a basis for $G_1 \oplus G_2$, so that it is free abelian. $\square$

5.4.10 THEOREM. *Any two bases of a free abelian group have the same cardinality.*

Proof. Suppose that $X = \{x_1, \dots, x_n\}$ is a basis of G with $|X| = n$. Then, by the definition of free abelian group, $F \cong \bigoplus_{i=1}^n \mathbf{Z}$. For any abelian group G' , define $2G := \{2g : g \in G\}$. We claim that $2G'$ is a subgroup of G' : indeed, $0 = 2 \cdot 0 \in 2G'$. If $2g, 2h \in 2G'$, then $2g + 2h = 2(g + h) \in 2G'$; also $-2g = 2(-g) \in 2G'$.

Under the isomorphism $G \cong \bigoplus_{i=1}^n \mathbf{Z}$, the subgroup $2G$ corresponds to $2G \cong \bigoplus_{i=1}^n 2\mathbf{Z}$ ($\varphi(2G) = 2(\bigoplus_{i=1}^n \mathbf{Z}) = \bigoplus_{i=1}^n 2\mathbf{Z}$). Therefore, by the third isomorphism theorem

$$G/2G \cong (\mathbf{Z}/2\mathbf{Z}) \oplus \cdots \oplus (\mathbf{Z}/2\mathbf{Z}) \cong \underbrace{\mathbf{Z}_2 \oplus \cdots \oplus \mathbf{Z}_2}_{n \text{ summands}}.$$

Thus, $|G/2G| = 2^n$. Now, let Y be any other basis of G , and suppose $|Y| = m$. Applying the same argument to the basis Y , we obtain that $|G/2G| = 2^m$. Hence $2^m = 2^n$, which implies $m = n$.

Assume G has an infinite basis X . Clearly $|X| \leq |G|$. We show that $|X| = |G|$. It remains to show $|F| \leq |X|$. Let $S = \bigcup_{n \geq 1} X^n$, where $X^n = X \times \cdots \times X$ (n factors). Since X is infinite, $|X^n| = |X|$ for each n , hence

$$|S| = \left| \bigcup_{n \geq 1} X^n \right| \leq \aleph_0 \cdot |X| = |X|.$$

The reverse inequality $|X| \leq |S|$ is obvious (take $n = 1$), so $|S| = |X|$. For each $s = (x_1, \dots, x_n) \in S$, define $G_s = \langle x_1, \dots, x_n \rangle$. If $y_1, \dots, y_t$ are the distinct elements among $x_1, \dots, x_n$, then since X is a basis,

$$G_s \cong y_1 \mathbf{Z} \oplus \cdots \oplus y_t \mathbf{Z} \cong \bigoplus_{i=1}^t \mathbf{Z},$$

so $|G_s| = \aleph_0$ for every $s \in S$. Every element of G is a finite linear combination of elements of X , hence belongs to some G_s . Thus $G = \bigcup_{s \in S} G_s$. Therefore,

$$|G| = \left| \bigcup_{s \in S} G_s \right| \leq \sum_{s \in S} |G_s| = |S| \cdot \aleph_0 = |X| \cdot \aleph_0 = |X|.$$

Combining with $|X| \leq |G|$, we get $|G| = |X|$ by Schröder-Bernstein. Now if Y is another infinite basis of G , applying the same result gives $|Y| = |G| = |X|$. Hence all infinite bases have the same cardinality. $\square$

5.4.4 DEFINITION. *Rank* of a free abelian group is defined to be the cardinality of the basis of the free abelian group. In particular, a free abelian group is finitely generated if and only if its rank is a finite number n .

5.4.11 THEOREM. *Let G be a free abelian group of a finite rank n and let H be a subgroup of G . Then H is a free abelian group and the rank of H is less than ($\leq$) the rank of G .*

5.4.12 LEMMA (Projective Property). *Let $f : H \rightarrow K$ be an epimorphism of abelian groups. If G is free abelian and if $g : G \rightarrow K$ is a homomorphism then there exists a homomorphism $\varphi : G \rightarrow H$ making the diagram commutes:*

$$\begin{array}{ccc} H & \xrightarrow{f} & K \\ \varphi \uparrow & \nearrow g & \\ G & & \end{array}$$

Proof. Let X be a basis of G . Since f is surjective, for each $x \in X$, there exists $h_x \in H$ such that $f(h_x) = g(x)$. Define a function $\alpha : X \rightarrow H$ defined by $\alpha(x) = h_x$. By the universal property of free abelian group, there exists a homomorphism $\varphi : G \rightarrow H$ such that the $\varphi(x) = \alpha(x) = h_x$ for all $x \in X$. It follows that $f \circ \varphi = g$ for they agree on a generating set of G . $\square$

5.4.13 COROLLARY. *Let G be an abelian group. If $H \leq G$ and G/H is free abelian, then H is a direct summand of G , that is, $G = H \oplus K$, where $K \leq G$ and $K \cong G/H$.*

Proof. Let $F = G/H$ and let $f : G \rightarrow F$ be the canonical projection. Consider the identity function $\text{id}_F : F \rightarrow F$. Since F has the projective property, there is a homomorphism $\varphi : F \rightarrow G$ such that $\pi \circ \varphi = \text{id}_F$. One may show that $G = \ker \pi \oplus \text{im} \varphi$: for any $x \in G$, let $\bar{g} = \pi(g) \in F = G/H$. Then $\pi(g - \varphi(\bar{g})) = \pi(g) - \pi \circ \varphi(\bar{g}) = \bar{g} - \bar{g} = 0$, hence $g - \varphi(\bar{g}) \in \ker \pi = H$ and $g = (g - \varphi(\bar{g})) + \varphi(\bar{g}) \in H + K$; let $x \in H \cap K$, then $x = \varphi(gH)$

for some $gH \in G/H$ and $\pi(x) = 0$, so $0 = \pi(x) = \pi(\varphi(gH)) = gH$ and $x = \varphi(0) = 0$, hence $H \cap K = \{0\}$ and $G = H \oplus K$.

$$\begin{array}{ccc} & G/H & \\ \varphi \swarrow & \downarrow \text{id}_{G/H} & \\ G & \xrightarrow{\pi} & G/H \end{array}$$

□

5.4.14 COROLLARY. *If $f : H \rightarrow K$ is an epimorphism of abelian groups and K is a free abelian group, then $H \cong K \oplus \ker f$.*

Proof. By the lemma applied to $G = K$ and $g = 1_K$, there exists a homomorphism $\varphi : K \rightarrow H$ such that $f \circ \varphi = 1_K$. Define $\Phi : K \oplus \ker f \rightarrow H$ by $\Phi(k, a) = \varphi(k) + a$. For any $k_1, k_2 \in K$ and $a_1, a_2 \in \ker f$, we have $\Phi((k_1, a_1) + (k_2, a_2)) = \varphi(k_1 + k_2) + a_1 + a_2 = \varphi(k_1) + \varphi(k_2) + a_1 + a_2 = \Phi(k_1, a_1) + \Phi(k_2, a_2)$, so Φ is a homomorphism.

If $\Phi(k, a) = 0$, then $\varphi(k) + a = 0$; applying f gives $k = f(\varphi(k)) = 0$, hence $a = 0$, so Φ is injective. For any $h \in H$, set $k = f(h)$ and $a = h - \varphi(k)$; then $f(a) = f(h) - f(\varphi(k)) = k - k = 0$, so $a \in \ker f$ and $h = \varphi(k) + a = \Phi(k, a)$, proving Φ is surjective. Thus Φ is an isomorphism, and since K and $\ker f$ are abelian, $H \cong K \oplus \ker f$.

$$\begin{array}{ccc} H & \xrightarrow{f} & K \\ \uparrow \varphi & \nearrow 1 & \\ K & & \end{array}$$

□

The proof of the above theorem: Since G is a free abelian group of rank n , $G \cong \bigoplus_{i=1}^n \mathbf{Z}$ and we may assume that $G = \bigoplus_{i=1}^n \mathbf{Z}$. It suffices to show that if $H \leq \bigoplus_{i=1}^n \mathbf{Z}$, then H is a free abelian group and $\text{rank } H \leq n$.

We apply induction on n . If $n = 1$, then $H = k\mathbf{Z}$ for some $k \geq 0$, so that $H = \{0\}$ or $H \cong \mathbf{Z}$. Assume that for some n every subgroup of $\bigoplus_{i=1}^n \mathbf{Z}$ is a free abelian group of rank $\leq n$, and consider the subgroup $H \leq \bigoplus_{i=1}^{n+1} \mathbf{Z}$. Define the function

$$f : \bigoplus_{i=1}^{n+1} \mathbf{Z} \rightarrow \mathbf{Z}, \quad f(m_1, \dots, m_{n+1}) = m_{n+1}$$

it is clear that f is a homomorphism and $\ker f = \{(m_1, \dots, m_n, 0) : m_i \in \mathbf{Z}\}$ and $\ker f \cong \bigoplus_{i=1}^n \mathbf{Z}$. Consider the epimorphism $f|_H : H \rightarrow \text{im}(f|_H)$. Since $\text{im}(f|_H) \subset \mathbf{Z}$, and $\text{im}(f|_H)$ is a free abelian group and so by the lemma we have

$$H \cong \text{im}(f|_H) \oplus \ker(f|_H),$$

we also have $\ker(f|_H) = \ker f \cap H$, it follows that $\ker(f|_H)$ is a subgroup of $\ker f$, and since $\ker f$ is a free abelian group of rank n , $\ker(f|_H)$ is a free abelian group of rank $\leq n$ by our hypothesis. Therefore,

$$H \cong \text{im}(f|_H) \oplus \ker(f|_H)$$

where $\text{im}(f|_H)$ is a free abelian group of rank ≤ 1 and $\ker(f|_H)$ is a free abelian group of rank $\leq n$. By the example 1, H is a free abelian group of rank $\leq n + 1$. □

In fact, this theorem also holds in the general case, where one considers cardinalities. However, in these notes we will never encounter the non-finitely generated case. Let G be a free abelian group and let H be a subgroup of G , then H is a free abelian group and $\text{rank}(H) \leq \text{rank}(G)$. (For reference, see pages 315-317 of the textbook: Rotman J. J. An introduction to the theory of groups[M]. Springer Science & Business Media, 2012.)

5.4.1 REMARK. This statement is false for the free group generated finitely. One may consider the multiplicative group G generated by the real matrices

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

If H is the set of all matrices in G whose diagonal entries are 1, then H is a subgroup that is not finitely generated.

Torsion Subgroups

Below we provide a new perspective on abelian groups, namely the torsion subgroup. It reveals that every finitely generated abelian group can always be decomposed as the direct sum of its torsion subgroup and a torsion-free part.

5.4.5 DEFINITION. Let G be a group, the *torsion subgroup* of G is

$$tG = \{x \in G : x \text{ has finite order}\}.$$

We say that G is *torsion* if $tG = G$ (so that every finite group is torsion), while G is *torsion-free* if $tG = \{0\}$. It is clear that tG is a subgroup when G is abelian.

5.4.15 PROPOSITION. Let G, H be abelian groups.

1. G/tG is torsion-free.
2. If $G \cong H$, then $tG \cong tH$ and $G/tG \cong H/tH$.

Proof. (1): Let $x + tG \neq 0 \in G/tG$, that is $x \notin tG$ so that x has infinite order. If $x + tG$ has finite order, then there is some $n > 0$ such that $0 + tG = n(x + tG) = nx + tG$; that is, $nx \in tG$. Thus, there is $m > 0$ with $0 = m(nx) = (mn)x$, contradiction x having the infinite order.

(2): If $\varphi : G \rightarrow H$ is a homomorphism and $x \in tG$, then $nx = 0$ for some $n > 0$ and $n\varphi(x) = \varphi(nx) = 0$; thus $\varphi(x) \in tH$ and $\varphi(tG) \subset tH$. In particular, if φ is an isomorphism, then the reverse inclusion $tH \subset \varphi(tG)$ holds as well; if $h \in tH$, then $h = \varphi(g)$ for some $g \in tG$ since isomorphism preserve orders of elements, and so $h = \varphi(g) \in \varphi(tG)$. Therefore $\varphi(tG) = tH$. $\square$

5.4.6 DEFINITION. Let p be a prime. An abelian group is *p-primary* if, for each $a \in G$, there is $k \geq 1$ with $p^k a = 0$. If we do want to specify the prime p , we say that G is *primary*. If G is an abelian group, then its *p-primary component* is

$$G_p = \{a \in G : p^k a = 0 \text{ for some } k \geq 1\}.$$

5.4.16 THEOREM (Primary Decomposition). Let G, H be torsion abelian groups.

1. G is the direct sum of its *p-primary components*: $G \cong \bigoplus_p G_p$;
2. G and H are isomorphic if and only if $G_p \cong H_p$ for every p .

Proof. (1): Let $x \in G$ has order $d > 1$, and let the prime factorization of d be $d = p_1^{f_1} \cdots p_t^{f_t}$. Define $r_i = d/p_i^{f_i}$, so that $p_i^{f_i} r_i = d$. It follows that $r_i x \in G_{p_i}$ for each i . Since $\gcd(r_1, \dots, r_t) = 1$, there are integers $s_1, \dots, s_t$ with $1 = \sum_i s_i r_i$. Therefore

$$x = \sum_{i=1}^t s_i r_i x \in G_{p_1} + \cdots + G_{p_t}.$$

Then we show that $G_{p_i} \cap \langle \bigcup_{j \neq i} G_{p_j} \rangle = \{0\}$. Let $x \in G_{p_i} \cap \langle \bigcup_{j \neq i} G_{p_j} \rangle$, then $p_i^l x = 0$ for some l and $ux = 0$ for some $u = \prod_{j \neq i} p_j^{g_j}$ (since $x = \sum_{j \neq i} y_j$ and $p_j^{g_j} y_j = 0$). But p_i^l and u are relatively prime, so there exist integers s and t such that $1 = sp_i^l + tu$. Therefore

$$x = (sp_i^l + tu)x = sp_i^l x + tux = 0.$$

(2): if $\varphi : G \rightarrow H$ is a homomorphism, then $\varphi(G_p) \subset H_p$ for every prime p : if $p^l x = 0$, then $0 = \varphi(p^l x) = p^l \varphi(x)$. If φ is also an isomorphism, then $\varphi^{-1} : H \rightarrow G$ is an isomorphism so that $\varphi^{-1}(H_p) \subset G_p$ for all p . It follows that each restriction $\varphi|_{G_p} : G_p \rightarrow H_p$ is an isomorphism. Conversely, given isomorphisms $\psi_p : G_p \rightarrow H_p$ for all p , there is an isomorphism $\Phi : \bigoplus_p G_p \rightarrow \bigoplus_p H_p$ given by $\sum_p a_p \mapsto \sum_p \psi_p(a_p)$. $\square$

5.4.17 LEMMA. *Every finitely generated torsion-free abelian group G is free abelian.*

Proof. We prove the statement by induction on n , where $G = \langle x_1, x_2, \dots, x_n \rangle$. If $n = 1$ and $G \neq 0$, then G is cyclic and $G \cong \mathbf{Z}$ because it is torsion-free.

Define $H = \{g \in G : mg \in \langle x_n \rangle \text{ for some integer } m\}$. It is clear that H is a subgroup of G . We claim that H is torsion-free: if $x \in G$ and $k(x + H) = 0$, then $kx \in H$ hence $m(kx) \in H$, and so $x \in H$. Since G/H is a torsion-free group that can be generated by fewer than n elements, it is free abelian by our hypothesis. By corollary 1 of the lemma 5.1.12, $G = F \oplus H$ where $F \cong G/H$, and so it suffices to show that $H \cong \mathbf{Z}$.

We claim that H is finitely generated: consider that homomorphism $\varphi : G \rightarrow H$ given by $(f + h) \mapsto h$, then $\ker \varphi = F$ and $H \cong G/\ker \varphi = G/F$. Since G is finitely generated, so is $G/F \cong H$.

If $g \in H$ and $g \neq 0$, then $mg = kx_n$ for some nonzero integer m and k . We claim that the function $\varphi : H \rightarrow \mathbf{Q}$ given by $g \mapsto k/m$ is a well-defined monomorphism: indeed, if $m_1 g = k_1 x_n$ and $m_2 g = k_2 x_n$, then

$$m_1 m_2 g = m_2 k_1 x_n, \quad m_1 m_2 g = m_1 k_2 x_n,$$

so $m_2 k_1 x_n = m_1 k_2 x_n$, that is, $(m_2 k_1 - m_1 k_2)x_n = 0$. Since G is torsion-free and $x_n \neq 0$, $m_2 k_1 - m_1 k_2 = 0$, that is, $k_1/m_1 = k_2/m_2$. Let $g, h \in H$ and $m_1 g = k_1 x_n$, $m_2 h = k_2 x_n$, then

$$(m_1 m_2)(g + h) = m_2(m_1 g) + m_1(m_2 h) = m_2 k_1 x_n + m_1 k_2 x_n = (m_2 k_1 + m_1 k_2)x_n;$$

hence $\varphi(g + h) = \frac{m_2 k_1 + m_1 k_2}{m_1 m_2} = k_1/m_1 + k_2/m_2 = \varphi(g) + \varphi(h)$. Assume $\varphi(g) = 0$ and $g \neq 0$, $mg = kx_n$, then $\varphi(g) = k/m = 0 \rightarrow k = 0$, hence $mg = 0$. Since G is torsion-free, $g = 0$, which is a contradiction, so that $g = 0$.

Now, H is isomorphic to a finitely generated subgroup of $\mathbf{Q}$ since φ is monomorphism, say $H = \langle a_1/b_1, \dots, a_t/b_t \rangle$. If $b = b_1 b_2 \dots b_t$, then the function $\psi : H \rightarrow \mathbf{Z}$ given by $h \mapsto bh$ is an injection since H is torsion-free. Therefore, H is isomorphic to a nonzero subgroup of $\mathbf{Z}$, and hence it is infinite cyclic. $\square$

In the finitely generated case, torsion-free abelian groups are equivalent to free abelian groups; this is a direct consequence of the lemma,

However, in the infinitely generated case, this equivalence no longer holds. A torsion-free abelian group need not be free. The reason is that in some torsion-free groups, certain elements cannot be expressed as finite linear combinations of basis elements. Such groups are not only non-free, but their structure is much more “complicated.” A typical and intuitive counterexample is the direct product $\prod_{i=1}^{\infty} \mathbf{Z}$ of copies of the integers, i.e., the group of all integer sequences. It is clearly torsion-free, since if a nonzero integer multiple of some integer sequence is the zero sequence, then the original sequence must itself be the zero sequence. However, it is not a free abelian group: consider the element $(1, 1, 1, \dots)$. In a free abelian group, every element can be expressed as an integer linear combination of only finitely many basis elements, and hence can have nonzero entries in only finitely many coordinates. But $(1, 1, 1, \dots)$ has entry 1 in every coordinate, so it cannot be generated by finitely many basis elements. Therefore, $\prod_{i=1}^{\infty} \mathbf{Z}$ is not free.

This also illustrates, from the opposite direction, that infinitely generated torsion-free abelian groups are far richer and more complex than free abelian groups.

Finitely Generated Abelian Groups

5.4.18 THEOREM (Simultaneous Bases). *Let H be a subgroup of finite index in a free abelian group F of finite rank n . Then there exist bases $\{y_1, \dots, y_n\}$ of F and $\{h_1, \dots, h_n\}$ of H such that $h_i \in \langle y_i \rangle$ for all i .*

Proof. We prove the statement by induction on n . If $n = 1$, then $F = \langle x_1 \rangle \cong \mathbf{Z}$, and $H = \langle kx_1 \rangle$ for some integer k . Assume that the statement is true for $n-1$. If $\{x_1, \dots, x_n\}$ is a basis of F , then each element $h \in H$ has coordinates. Choose an ordered basis and element h so that, among all such choices, the first coordinate of h is positive and minimal such. If $h = k_1x_1 + \dots + k_nx_n$, then we claim that $k_1 \mid k_i, i \geq 2$: assume $k_i = q_ik_1 + r_i$ where $0 \leq r_i < k_1$. Then

$$h = k_1(x_1 + q_2x_2 + \dots + q_nx_n) + r_2x_2 + \dots + r_nx_n.$$

Define $y_1 = x_1 + q_2x_2 + \dots + q_nx_n$, and $\{y_1, x_2, \dots, x_n\}$ is an ordered basis of F . Now $h = k_1y_1 + r_2x_2 + \dots + r_nx_n$. If $r_i \neq 0$ for some i , then the first coordinate of h relative to the ordered basis $\{x_i, y_1, \dots, x_n\}$ violates the minimality of our initial choice. Therefore $r_i = 0, i \geq 2$.

If $h' = m_1y_1 + m_2x_2 + \dots + m_nx_n$ is any element of H , we claim that $k_1 \mid m_1$. Assume $m_1 = qk_1 + r, 0 \leq r < k_1$, then $h' - qh \in H$ has first coordinate $r < k_1$, a contradiction. It follows that the function $\pi : H \rightarrow H$ given by $h' \mapsto m_1y_1$, is a retraction with image $\langle h \rangle$ (i.e., $\pi|_{\langle h \rangle} = \text{id}$). We show that $H = \langle h \rangle \oplus \ker \pi$: for any $h' \in H$, $\pi(h' - \pi(h')) = \pi(h') - \pi(h') = 0$, so that $h' - \pi(h') \in \ker \pi$, hence $H = \langle h \rangle + \ker \pi$; if $x \in \langle h \rangle \cap \ker \pi$, then $x = \pi(x) = 0$. $H = \langle h \rangle \oplus \ker \pi = \langle h \rangle \oplus (H \cap \langle x_2, \dots, x_n \rangle)$. Let $F' = \langle x_2, \dots, x_n \rangle$ and $H' = H \cap \langle x_2, \dots, x_n \rangle$. $\square$

5.4.19 COROLLARY (Basis Theorem). *Every finitely generated abelian group is a direct sum of some cyclic groups.*

Proof. Clearly, any finitely generated abelian group is finitely generated ($G = \langle G \rangle$). Let G be a finitely generated abelian group and $|G| = n$. Define an epimorphism $\varphi : \bigoplus_{i=1}^n \mathbf{Z} \rightarrow G$ by $e_i \mapsto g_i$. Then $G \cong \bigoplus_{i=1}^n \mathbf{Z} / \ker \varphi = F/R$. By the theorem above, there are bases $\{y_1, \dots, y_n\}$ and $\{h_1, \dots, h_n\}$ of F and R respectively, with $h_i = k_iy_i$ for each i . Then

$$F/R \cong \bigoplus_{i=1}^n \langle y_i \rangle / \bigoplus_{i=1}^n \langle k_iy_i \rangle \cong \bigoplus_{i=1}^n \langle y_i \rangle / \langle k_iy_i \rangle \cong \bigoplus_{i=1}^n \mathbf{Z}_{k_i}.$$

$\square$

From the Basis theorem, we have $G \cong \bigoplus_{i=1}^n \mathbf{Z}_{k_i}$ for some positive integers $k_1, \dots, k_n$. For each i , let $k_i = p_{i,1}^{f_{i,1}} p_{i,2}^{f_{i,2}} \dots p_{i,t_i}^{f_{i,t_i}}$ be the prime factorization of k_i . By the Chinese Remainder theorem, for each i ,

$$\mathbf{Z}_{k_i} \cong \mathbf{Z}_{p_{i,1}^{f_{i,1}}} \oplus \mathbf{Z}_{p_{i,2}^{f_{i,2}}} \oplus \dots \oplus \mathbf{Z}_{p_{i,t_i}^{f_{i,t_i}}}.$$

Substituting this into the direct sum gives $G \cong \bigoplus_{i=1}^n \bigoplus_{j=1}^{t_i} \mathbf{Z}_{p_{i,j}^{f_{i,j}}}$. Each $\mathbf{Z}_{p_{i,j}^{f_{i,j}}}$ is a cyclic p -group (a primary cyclic group), so G is a direct sum of primary cyclic groups. As for the proof of the Chinese Remainder Theorem, we shall not provide it here; the reader may refer to Wikipedia.

5.4.20 THEOREM (Fundamental Theorem). *Every finitely generated abelian group is a direct sum of primary and infinite cyclic groups.*

Proof. Clearly, G/tG is a finitely generated torsion-free group, hence G/tG is free abelian. By the corollary 1 of lemma 5.1.12, $G \cong tG \oplus F$ where $F \cong G/tG$. It is clear that tG is finitely generated. We claim that tG is finite: let $tG = \langle y_1, \dots, y_k \rangle$ in which the order of y_i is $n_i < \infty$, hence $|tG| \leq n_1 n_2 \dots n_k$. The basis theorem for finitely generated abelian groups says that tG is a direct sum of primary cyclic groups. $\square$

This theorem also tells us that any finitely generated abelian group can be split into two distinct parts: a torsion-free part and a torsion part.

5.5 Automorphism Groups

We have shown that the set of all the automorphisms of G under the operation of composition, $\text{Aut}_{\text{Grp}}(G)$ is a group. Indeed, it is a subgroup of the symmetric group S_G .

5.5.1 DEFINITION. An automorphism γ_g of G is *inner* if it is conjugation by some element g of G , i.e., the $\gamma_g(x) = gxg^{-1}$ for some $g \in G$; otherwise, it is *outer*. Denote the set of all inner automorphisms of G by $\text{Inn}(G)$.

Consider the function $G \rightarrow \text{Aut}(G)$ given by $g \mapsto \gamma_g$, note that $(gh)x(gh)^{-1} = g(hxh^{-1})g^{-1}$, i.e., $\gamma_{gh}(x) = (\gamma_g \circ \gamma_h)(x)$, so the function $G \rightarrow \text{Aut}(G)$ is a homomorphism. Its image is $\text{Inn}(G)$ and its kernel is the center of G ,

$$Z(G) = \{g \in G : gx = xg \text{ for all } x \in G\},$$

and so we obtain an isomorphism $G/Z(G) \cong \text{Inn}(G)$. In fact, $\text{Inn}(G)$ is a normal subgroup of $\text{Aut}(G)$: for $g \in G$ and $\alpha \in \text{Aut}(G)$, we have $\alpha \circ \gamma_g \circ \alpha^{-1} = \gamma_{\alpha(g)} \in \text{Inn}(G)$. The quotient group $\text{Aut}(G)/\text{Inn}(G)$ is called the *outer automorphism group* of G , denoted by $\text{Out}(G)$.

5.5.1 EXAMPLE. Let $G = \mathbf{F}_p^n$. The automorphisms of G as a commutative group are just the automorphisms of G as a vector space over $\mathbf{F}_p$; thus $\text{Aut}(G) = \text{GL}_n(\mathbf{F}_p)$. Since G is commutative, all nontrivial automorphisms of G are outer. In particular, we see that $\text{Aut}(\mathbf{Z}_2 \times \mathbf{Z}_2) = \text{GL}_2(\mathbf{F}_2)$.

5.5.2 EXAMPLE. $\text{Aut}(\mathbf{Z}) \cong \mathbf{Z}_2$. Let $G = \langle x \rangle$ be infinite cyclic. If $\varphi \in \text{Aut}(G)$, then $\varphi(x)$ must be a generator of G . Since the only generators of G are x and x^{-1} , there are only two automorphisms of G , and so $\text{Aut}(\mathbf{Z}) \cong \text{Aut}(G) \cong \mathbf{Z}_2$. Thus, an infinite group can have a finite automorphism group.

5.5.3 EXAMPLE. $\text{Aut}(G) = 1$ if and only if $|G| \leq 2$.

Proof. It is clear that $|G| \leq 2$ implies $\text{Aut}(G) = 1$. Conversely, assume that $\text{Aut}(G) = 1$. If $g \in G$, then $\gamma_g = 1$ if and only if $g \in Z(G)$; it follows that G is abelian. The function $g \mapsto g^{-1}$ is now an automorphism of G , so that every element of G has exponent 2. Then G is a vector space over $\mathbf{Z}_2$. If $|G| > 2$, then $\dim G \geq 2$ ($|G| = 2^{\dim G}$) and there exists a nonsingular linear transformation $\varphi : G \rightarrow G$ other than 1. $\square$

5.5.1 THEOREM (N/C Lemma). If $H \leq G$, then $C_G(H)N_G(H)$ and $N_G(H)/C_G(H)$ can be imbedded in $\text{Aut}(H)$, that is $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$.

Proof. Let $g \in G$ and let γ_g denote conjugation by g . Define $\theta : N_G(H) \rightarrow \text{Aut}(H)$ by $g \mapsto \gamma_g|_H$ (note that $\gamma_g|_H \in \text{Aut}(H)$ because $g \in N_G(H)$); θ is easily seen to be a homomorphism. The following statements are equivalent: $g \in \ker \theta$; $\gamma_g|_H$ is the identity on H ; $ghg^{-1} = h$ for all $h \in H$; $g \in C_G(H)$. By the first isomorphism theorem, $C_G(H) \trianglelefteq N_G(H)$ and $N_G(H)/C_G(H) \cong \text{im } \theta \leq \text{Aut}(H)$. $\square$

This theorem shows that any information we have about the automorphism group of a subgroup H of a group G translates into information about $N_G(H)/C_G(H)$. For example, if $H \cong \mathbf{Z}_2$, then $\text{Aut}(H) = 1$, thus, $N_G(H) = C_G(H)$; if in addition H is normal subgroup of G , then $C_G(H) = N_G(H) = G$, then $H \leq Z(G)$.

5.5.2 DEFINITION. A group G is *complete* if the homomorphism $G \rightarrow \text{Aut}(G)$ given by $g \mapsto \gamma_g$ is an isomorphism.

Thus, a group G is complete if and only if $Z(G) = 1$ (monomorphism) and every automorphism of G is inner (epimorphism). It follows that $G = G/Z(G) \cong \text{Inn}(G) = \text{Aut}(G)$.

5.5.2 THEOREM. S_n is complete unless $n = 2$ or $n = 6$.

5.5.3 LEMMA. Suppose that G is a group and S is a set of generators of G . If φ and ψ are two automorphisms of G that agree on S , then $\varphi = \psi$.

Proof. Let H be the largest subset of G on which φ and ψ agree. We show that H is a subgroup of G . Clearly $1 \in H$; suppose $g, h \in H$. We have $\varphi(gh) = \varphi(g)\varphi(h) = \psi(g)\psi(h) = \psi(gh)$. Thus, $gh \in H$; suppose $g \in H$, we have $\varphi(g^{-1}) = \varphi(g)^{-1} = \psi(g)^{-1} = \psi(g^{-1})$. Thus, $g^{-1} \in H$ and so H is indeed a subgroup of G . Since H contains S , $H = G$ and so $\varphi = \psi$. $\square$

5.5.4 LEMMA. An automorphism α of S_n preserves transpositions ($\varphi(\tau)$ is a transposition whenever τ is) if and only if α is inner.

Proof. If α is inner, i.e., they are conjugate, so that they have the same cycle type. The transpositions $(i \ i+1), 1 \leq i \leq n-1$ generates S_n . Suppose that $(a, b) = \alpha(i, i+1)$ and $(c, d) = \alpha(i+1, i+2)$. It is clear that $(i \ i+1)$ and $(i+1 \ i+2)$ do not commute. But α is an automorphism, it follows that $\alpha(i \ i+1) = (a \ b)$ and $\alpha(i+1 \ i+2) = (c \ d)$ do not commute, so that $\{a, b, c, d\}$ have three different elements. Possibly rearranging we may assume that $b = c$. Thus we may assume that there are $a_1, \dots, a_n$ such that $(a_i \ a_{i+1}) = \alpha(i \ i+1)$. Let $\tau(i) = a_i$. Then τ is a permutation of the first n natural numbers and α and α_τ agree on the generators $(i \ i+1)$ ($\alpha_\tau(x) = \tau x \tau^{-1}$). The lemma above implies that $\alpha = \alpha_\tau$ so that α is inner. $\square$

5.5.5 LEMMA. If $n \neq 2$ or $n \neq 6$, then S_n is not complete. ($S_2 \cong \mathbf{Z}_2$ is not complete since its center is itself. And this lemma does not say that S_6 is not complete.)

Proof. First, we shall show that the center of S_n is trivial ($n \neq 2$): assume $n \geq 3$ and $\sigma \neq 1 \in S_n$. Pick i such that $j = \sigma(i) \neq i$. Pick $j \notin \{i, j\}$ and let $\tau = (j \ k)$. Then $\tau\sigma\tau^{-1}$ sends i to k . Thus $\tau\sigma\tau^{-1} \neq \sigma$ so that σ does not belong to the center.

Let T_k denote the conjugacy class in S_n consists of all products of k disjoint transpositions. It is clear that a permutation in S_n is an involution (of order 2) if and only if it lies in some T_k . It follows that if $\theta \in \text{Aut}(S_n)$, then if $x \in T_1$, $\theta(x) \in T_i$ for some i . Moreover, $\theta(T_1)$ is a conjugacy class, which implies $\theta(T_1) = T_k$ for some k . We shall show that if $n \neq 6$, then $|T_k| \neq |T_1|$ for $k \neq 1$, then $\theta(T_1) = T_1$ and lemma above implies θ is inner.

Clearly $|T_1| = C_n^2 = n(n-1)/2$. Note that there are $C_n^2 C_{n-2}^2 \cdots C_{n-2k+2}^2$ k -tuples of disjoint transpositions. Since disjoint transpositions commute and there are $k!$ orderings obtained from any k -tuple, we have

$$|T_k| = [n(n-1)(n-2) \cdots (n-2k+1)]/(k!2^k).$$

The question whether $|T_1| = |T_k|$ leads to the question whether there is some $k > 1$ such that $(n-2)(n-3) \cdots (n-2k+1) = k!2^{k-1}$. Since the right side of this equality is positive, we must have $n \geq 2k$. Therefore, for fixed n ,

$$(n-2)(n-3) \cdots (n-2k+1) \geq (2k-2)(2k-3) \cdots (2k-2k+1) = (2k-2)!.$$

An easy induction shows that if $k \geq 4$, then $(2k-2)! > k!2^{k-1}$, and so $k = 2$ or $k = 3$. When $k = 2$, the right side is 4, and it easy to see that equality never holds; we may assume, therefore, that $k = 3$. Since $n \geq 2k$, we must have $n \geq 6$. If $n > 6$, then the $(n-2)(n-3) \cdots (n-2k+1) \geq 5 \times 4 \times 3 \times 2 = 120$ while $k!2^{k-1} = 3!2^2 = 24$. We have shown that if $n \neq 6$, then $|T_1| \neq |T_k|$ for all $k > 1$, as desired. $\square$

5.5.6 COROLLARY. *If S_6 contains an outer automorphism and θ is such an automorphism of S_6 (we have not shown that S_6 is not complete), and if $\tau \in S_6$ is a transposition, then $\theta(\tau)$ is a product of three disjoint transposition.*

Proof. Consider the equality $(n-2)(n-3)\cdots(n-2k+1) = k!2^{k-1}$. If $n = 6$, then the equality does not hold if $k \neq 3$; when $k = 3$, both sides equal 24. $\square$

In summary, the inner and outer automorphisms (if they exist) of $\text{Aut}(S_6)$ have a distinctive feature: inner automorphisms map transpositions to transpositions, while an outer automorphism, if it exists, maps a transposition to a product of three transpositions. Therefore, when no outer automorphism exists, by definition the outer automorphism group is trivial, of order 1. If an outer automorphism does exist, let one of them be denoted by φ . Then clearly this outer automorphism corresponds to a nontrivial coset in the outer automorphism group, and all outer automorphisms lie in this same coset (indeed, if ψ is also an outer automorphism, then $\psi^{-1}\varphi \in \text{Inn}(S_6)$). Hence we always have $|\text{Out}(S_6)| \leq 2$. This leads to the following conclusion.

5.5.7 LEMMA. $|\text{Out}(S_6)| = |\text{Aut}(S_6)/\text{Inn}(S_6)| \leq 2$. *Further the order is 2 if and only if there is an automorphism φ of S_6 which sends a transposition to a product of three disjoint transpositions.*

It is actually suprisingly involved to write down an automorphism φ which sends a transposition to a product of three disjoint transpositions. The problem is that there are too many choices. This lemma implies that if there exists an outer automorphism φ , then every other such has the form $\varphi\psi$ for some inner automorphism ψ . It follows that S_6 has exactly 720 outer automorphisms !

Now we consider all conjugacy classes in S_6 . We know that two elements in S_n are conjugate if and only if they have the same cycle type. Thus we divide them into the following conjugacy classes (types):

Type	Order	Size
[1, 1, 1, 1, 1, 1]	1	1
[2, 1, 1, 1, 1]	2	15
[2, 2, 1, 1]	2	45
[2, 2, 2]	2	15
[3, 1, 1, 1]	3	40
[3, 3]	3	40
[4, 1, 1]	4	90
[4, 2]	4	90
[5, 1]	5	144
[6]	6	120
[3, 2, 1]	6	120

Table 5.1: Conjugacy classes of S_6

Note that all of these conjugacy classes come in pairs (C_1, C_2) , where the order of the elements of C_1 and C_2 are the same and the size of C_1 and C_2 is the same, with 3 exceptions: [1, 1, 1, 1, 1, 1], [2, 2, 1, 1] and [5, 1]. [2, 1, 1, 1, 1] is paired with [2, 2, 2]; [3, 1, 1, 1] is paired with [3, 3]; [4, 1, 1] is paired with [4, 2]; [6] is paired with [3, 2, 1]. This suggests that 5-cycles play a special role.

5.5.8 THEOREM. $\text{Out}(S_6) \cong \mathbf{Z}_2$.

5.5.9 THEOREM. *If G is a nonabelian simple group, then $\text{Aut}(G)$ is complete.*

Proof. Let $I = \text{Inn}(G) \trianglelefteq \text{Aut}(G) = A$. Now $Z(G) = 1$ since G is simple and nonabelian; and so $I \cong G/Z(G) = G$. Now $Z(A) \leq C_A(I) = \{\psi \in A : \psi\gamma_g = \gamma_g\psi \text{ for all } g \in G\}$. We claim that $C_A(I) = 1$, it will then follow that $Z(A) = 1$. If $\psi\gamma_g = \gamma_g\psi$ for all $g \in G$, then $\gamma_g = \psi\gamma_g\psi^{-1} = \gamma_{\psi(g)}$. Therefore, $\psi(g)g^{-1} \in Z(G) = 1$ for all $g \in G$ and so $\psi = 1$.

(Here is a rather important conclusion: for any centreless group, its automorphism group is also centreless.)

It remains to show that every $\sigma \in \text{Aut}(A)$ is inner. Now $\sigma(I) \trianglelefteq A$ since $I \trianglelefteq A$; and so $I \cap \sigma(I) \trianglelefteq \sigma(I)$. But $\sigma(I) \cong I \cong G$ is simple, so that either $I \cap \sigma(I) = 1$ or $I \cap \sigma(I) = \sigma(I)$. Since both I and $\sigma(I)$ are normal, $[I, \sigma(I)] \leq I \cap \sigma(I)$. In the first case we have $[I, \sigma(I)] = 1$; that is $\sigma(I) \leq C_A(I) = 1$ and this contradicts $\sigma(I) \cong I$. Hence $I \cap \sigma(I) = \sigma(I)$, and so $\sigma(I) \leq I$. This inclusion holds for every $\sigma \in \text{Aut}(A)$; in particular, $\sigma^{-1}(I) \leq I$, and so $\sigma(I) = I$ for every $\sigma \in \text{Aut}(A)$. If $g \in G$, then $\gamma_g \in I$; there is thus $\alpha(g) \in G$ with

$$\sigma(\gamma_g) = \gamma_{\alpha(g)}.$$

It is clear that $\alpha : G \rightarrow G$ is a bijection. We now show that α is an automorphism of G ; that is, $\alpha \in A$. If $g, h \in G$, then $\sigma(\gamma_g \gamma_h) = \sigma(\gamma_{gh}) = \gamma_{\alpha(gh)}$. On the other hand, $\sigma(\gamma_g \gamma_h) = \sigma(\gamma_g) \sigma(\gamma_h) = \gamma_{\alpha(g)} \gamma_{\alpha(h)} = \gamma_{\alpha(g)\alpha(h)}$; hence $\alpha(gh) = \alpha(g)\alpha(h)$.

We claim that $\sigma = \Gamma_\alpha$, the conjugation by α . To this end, define $\tau = \sigma \circ \Gamma_\alpha^{-1}$. Observe that for all $h \in G$, we have

$$\begin{aligned} \tau(\gamma_h) &= \sigma \Gamma_\alpha^{-1}(\gamma_h) = \sigma(\alpha^{-1} \gamma_h \alpha) \\ &= \sigma(\gamma_{\alpha^{-1}(h)}) \\ &= \gamma_{\alpha \alpha^{-1}(h)} = \gamma_h. \end{aligned}$$

Thus, τ fixes every element in I . If $\beta \in A$, then for every $g \in G$, $\beta \gamma_g \beta^{-1} \in I$ implies $\beta \gamma_g \beta^{-1} = \tau(\beta \gamma_g \beta^{-1}) = \tau(\beta) \gamma_g \tau(\beta)^{-1}$. Hence $\tau(\beta) \beta^{-1} \in C_A(I) = 1$, and $\tau(\beta) = \beta$. Therefore, $\tau = 1$, $\sigma = \Gamma_\alpha$ and A is complete. $\square$

It follows, for every nonabelian simple group G , that $\text{Aut}(G) \cong \text{Aut}(\text{Aut}(G))$. There is a interesting problem, *The Automorphism Tower Problem*: If G is a centreless group ($Z(G) = 1$), then there is a natural imbedding of G into its automorphism group $\text{Aut}(G)$: $G \cong \text{Inn}(G) \trianglelefteq \text{Aut}(G)$. Moreover $\text{Aut}(G)$ is also centreless, and it can be imbedded into its automorphism group $\text{Aut}(\text{Aut}(G))$. This process may thus be iterated to give the *automorphism tower* of G :

$$G = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \cdots \trianglelefteq G_\alpha \trianglelefteq G_{\alpha+1} \trianglelefteq \cdots$$

such that for each ordinal α $G_{\alpha+1} = \text{Aut}(G_\alpha)$; and if α is a limit ordinal, then $G_\alpha = \bigcup_{\beta < \alpha} G_\beta$. The automorphism tower is said to terminate if there exists an ordinal α such that $G_{\alpha+1} = G_\alpha$. This occurs if and only if there exists an ordinal α such that $\text{Aut}(G_\alpha) = \text{Inn}(G_\alpha)$. In this case, the *height* $\tau(G)$ of automorphism tower is defined to be the least ordinal α such that $G_{\alpha+1} = G_\alpha$.

In 1939, Wielandt proved that if G is a finite centreless group, then its automorphism tower terminates after finitely many steps. However, there exist natural examples of infinite centreless groups whose automorphism towers do not terminate in finitely many steps. For example, the infinite dihedral group D_∞ . In 1984, S. Thomas proved that the automorphism tower of an arbitrary centreless group G terminates after at most $(2^{|G|})^+$ steps. (Here is a preprint on this problem: <https://sites.math.rutgers.edu/~stthomas/papers.html>)

This result tells us that every group can be embedded into a complete group. In fact, we already knew a stronger result before this, namely Cayley's theorem: every finite group can be embedded into some symmetric group S_n , and if $n \neq 2, 6$, then S_n is complete.

5.5.10 THEOREM. *If $N \trianglelefteq G$ and N is complete, then N is a direct factor of G ; that is, there is a normal subgroup Q of G with $G \cong N \times Q$.*

Proof. Define $Q = C_G(N) = \{g \in G : gn = ng \text{ for all } n \in N\}$. Now $N \cap Q \leq Z(N) = 1$, and so $N \cap Q = 1$. To see that $G = NQ$, take $g \in G$. Now $\gamma_g(N) = N$ because $N \trianglelefteq G$; and so $\gamma_g|_N \in \text{Aut}(N) = \text{Inn}(N)$. There is thus $n \in N$ with $gxg^{-1} = nxn^{-1}$ for all $x \in N$. Hence $n^{-1}g \in \bigcap_{x \in N} C_G(x) = C_G(N) = Q$, and so $g = n(n^{-1}g) \in NQ$. Finally, $Q \trianglelefteq G$, for $gQg^{-1} = n(n^{-1}g)Q(n^{-1}g)^{-1}n^{-1} = nQn^{-1}$, and $nQn^{-1} = Q$ since every element of Q commutes with n . Therefore, $G \cong N \times Q$. $\square$

The converse of this theorem is true, and we introduce a construction in order to prove it. Recall the proof of Cayley's theorem: let N acts on itself by left multiplication $\rho^l : N \times N \rightarrow N$ by $\rho^l(n, x) = nx$, the corresponding homomorphism $\sigma^l : N \rightarrow S_N$ is injective and $N \cong \text{im}\sigma^l$. Let $N^l := \text{im}\sigma^l = \{\sigma_n^l : N \rightarrow N, \sigma_n^l(x) = \rho^l(n, x) = nx, x \in N\}$ which is a subgroup of S_N . Similarly, let N acts on itself by right multiplicative $\rho^r(n, x) = xn^{-1}$, and let $N^r = \{\sigma_n^r : N \rightarrow N, \sigma_n^r(x) = \rho^r(n, x) = xn^{-1}, x \in N\}$ then N^r is also a subgroup of S_N isomorphic to N .

5.5.3 DEFINITION. The *holomorph* of a group N , denoted by $\text{Hol}(N)$ is the subgroup of S_N generated by N^l and $\text{Aut}(N)$; i.e., $\text{Hol}(N) := \langle N^l, \text{Aut}(N) \rangle$.

Note that, for all $n \in N$, $\sigma_n^r = \sigma_{n^{-1}}^l \gamma_n$ ($\sigma_{n^{-1}}^l(\gamma_n(x)) = \sigma_{n^{-1}}^l(nxn^{-1}) = n^{-1}n xn^{-1} = xn^{-1}$), so that $N^r \leq \text{Hol}(N)$; indeed, it is easy to see that $\text{Hol}(N) = \langle N^r, \text{Aut}(N) \rangle$.

5.5.11 LEMMA. *Let N be a group.*

1. $N^l \trianglelefteq \text{Hol}(N)$, $N^l \text{Aut}(N) = \text{Hol}(N)$ and $N^l \cap \text{Aut}(N) = 1$.
2. $\text{Hol}(N)/N^l \cong \text{Aut}(N)$;
3. $C_{\text{Hol}(N)}(N^l) = N^r$.

Proof. (1). It is easy to see that $\varphi\sigma_a^l\varphi^{-1} = \sigma_{\varphi(a)}^l$, and that it lies in N^l for every $a \in N$ and $\varphi \in \text{Aut}(N)$; since $\text{Hol}(N) = \langle N^l, \text{Aut}(N) \rangle$, it follows that $N^l \trianglelefteq \text{Hol}(N)$ and that $\text{Hol}(N) = N^l \text{Aut}(N)$. If $a \in N$, then $\sigma_a^l(1) = a$; therefore, $\sigma_a^l \in \text{Aut}(N)$ if and only if $a = 1$; that is $N^l \cap \text{Aut}(N) = 1$.

(2). $\text{Hol}(N)/N^l = N^l \text{Aut}(N)/N^l \cong \text{Aut}(N)/N^l \cap \text{Aut}(N) \cong \text{Aut}(N)$.

(3). If $a, b, x \in N$, then $\sigma_a^l \sigma_b^r(x) = a(xb^{-1})$ and $\sigma_b^r \sigma_a^l(x) = (ax)b^{-1}$, so that associativity gives $N^r \leq C_{\text{Hol}(N)}(N^l)$. For the reverse inclusion, assume that $\eta \in \text{Hol}(N)$ satisfies $\eta\sigma_a^l = \sigma_a^l\eta$ for all $a \in N$. Now $\eta = \sigma_b^l\varphi$ for some $b \in N, \varphi \in \text{Aut}(N)$. If $x \in N$, then $\eta\sigma_a^l(x) = \sigma_b^l\varphi\sigma_a^l(x) = b\varphi(a)\varphi(x)$ and $\sigma_a^l\eta(x) = \sigma_a^l\sigma_b^l\varphi(x) = ab\varphi(x)$. Hence, $b\varphi(a) = ab$ for all $a \in N$; that is $\varphi = \gamma_{b^{-1}}$. It follows that $\eta(x) = \sigma_b^l\varphi(x) = b(b^{-1}xb) = xb$, and so $\eta \in N^r$, as desired. $\square$

5.5.12 THEOREM. *If a group N is a direct factor whenever it is isomorphic to a normal subgroup of a group, then N is complete.*

Proof. We identify N with the subgroup $N^l \leq \text{Hol}(N)$. Since N^l is normal in $\text{Hol}(N)$, the hypothesis gives a subgroup B with $\text{Hol}(N) = N^l \times B$. Now $B \leq C_{\text{Hol}(N)}(N^l) = N^r$ since every element of B commutes with each element of N^l . It follows that if $\varphi \in \text{Aut}(N) \leq \text{Hol}(N)$, then $\varphi = \sigma_a^l \sigma_b^r$ for some $a, b \in N$. Hence $\varphi(x) = axb^{-1}$ for all $x \in N$. But now $axyb^{-1} = \varphi(x)\varphi(y) = axb^{-1}ayb^{-1}$, so that $1 = b^{-1}a$; therefore, $\varphi = \gamma_a \in \text{Inn}(N)$.

Next, we shall show that $Z(N) = 1$. Since $\text{Hol}(N) = N^l \times B$ and $B \leq N^r \leq \text{Hol}(N)$, one may show that $N^r = B \times (N^r \cap N^l)$. If $\varphi \in N^l \cap N^r$, then $\varphi = \sigma_a^l = \sigma_b^r$ for $a, b \in N$. For all $c \in N$, $\sigma_a^l(c) = \sigma_b^r(c)$ gives $ac = cb^{-1}$; if $c = 1$, then $a = b^{-1}$ from which it follows that $z \in Z(N)$. Therefore, $N^r \cap N^l = Z(N)$. Therefore, $N^r \cap N^l = Z(N)$ and $N \cong B \times Z(N)$. If $1 \neq \varphi \in \text{Aut}(Z(N))$, then it is easy to see that $\tilde{\varphi} : B \times Z(N) \rightarrow B \times Z(N)$ defined by $(b, z) \mapsto (b, \varphi(z))$ is an automorphism of N .

$\tilde{\varphi}$ must be outer, for conjugation by $(\beta, \zeta) \in B \times Z(N) \cong N$ sends (b, z) into $(\beta, \zeta)(b, z)(\beta^{-1}, \zeta^{-1}) = (\beta b \beta^{-1}, z)$. But we have shown that N has no outer automorphisms, so that $\text{Aut}(Z(N)) = 1$; and $|Z(N)| \leq 2$. If $Z(N) \cong \mathbf{Z}_2$, then it is isomorphic to a normal subgroup H of $\mathbf{Z}_4$ which is not a direct factor. But N is isomorphic to the normal subgroup $B \times H$ of $B \times \mathbf{Z}_4$ which is not a direct factor, contradicting the hypothesis. Therefore $Z(N) = 1$ and N is complete. $\square$

5.6 Extensions of Groups

Now we return to the general case of groups and discuss their direct products. We already know that among the equivalent conditions for G to be the direct product of its subgroups, there is one condition that is not automatically satisfied: these subgroups must be normal. Now we consider two subgroups N and H , requiring only that one of them, say N , be normal, and that they satisfy $G = NH$ and $N \cap H = \{1\}$. Let us examine what kind of relationship exists between G , H , and N in this situation.

5.6.1 DEFINITION. Consider a sequence of group homomorphisms

$$\cdots \xrightarrow{f_0} G_1 \xrightarrow{f_1} G_2 \xrightarrow{f_2} \cdots \xrightarrow{f_i} G_{i+1} \cdots,$$

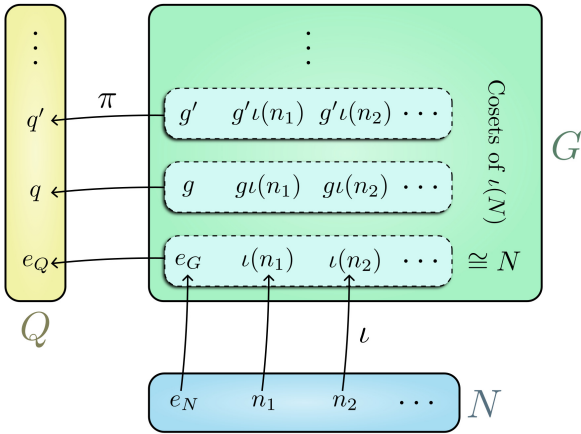
If for every i we have $\text{im}(f_i) = \ker(f_{i+1})$, then the sequence is called *exact*. We often abbreviate $\{1\}$ as 1, or, in additive notation, as 0. For example, for any homomorphism $\varphi : G \rightarrow G'$, the sequence $G \rightarrow G' \rightarrow 1$ is exact if and only if φ is surjective, and the sequence $1 \rightarrow G \rightarrow G'$ is exact if and only if φ is injective. Moreover, we always have the exact sequence

$$1 \longrightarrow \ker \varphi \xhookrightarrow{\iota} G \xrightarrow{\varphi} \text{im} \varphi \longrightarrow 1$$

5.6.2 DEFINITION. Let N, Q be groups. A group G is an *extension* of Q by N if there is an exact sequence of groups

$$1 \longrightarrow N \xhookrightarrow{\iota} G \xrightarrow{\pi} Q \longrightarrow 1$$

There is an alternative usage of the term *extension*, which calls the (middle) group G (not the exact sequence) an *extension* if it contains a normal subgroup N_1 with $N_1 \cong N$ and $G/N_1 \cong Q$. As do most people, we will use the term in both senses.



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However, this diagram does not reveal more information, such as why there are different group extensions of Q by N . Let us briefly explain: this is due to the different multiplicative relations between the fibres corresponding to elements of Q ; figuratively speaking, the ways in which the fibres are twisted differ. For example, consider elements $q, q' \in Q$ and their product qq' . Under the surjection π , these correspond to three distinct fibres. It is easy to see that the product of any element in the fibre over q and any element in the fibre over q' must lie in the fibre over qq' . Which particular element it is determines the structure of the extended group. (In this process, each fibre is a copy of N .)

5.6.1 EXAMPLE. Both $\mathbf{Z}_6$ and S_3 are extensions of $\mathbf{Z}_2$ by $\mathbf{Z}_3$. However, $\mathbf{Z}_6$ is an extension of $\mathbf{Z}_3$ by $\mathbf{Z}_2$, but S_3 is not such an extension since S_3 has no normal subgroup

of order 2:

$$\begin{aligned} 0 &\longrightarrow \mathbf{Z}_3 \xrightarrow{\iota} \mathbf{Z}_6 \xrightarrow{\varphi} \mathbf{Z}_2 \longrightarrow 0 \\ 0 &\longrightarrow \mathbf{Z}_3 \xrightarrow{\iota'} S_3 \xrightarrow{\varphi'} \mathbf{Z}_2 \longrightarrow 0 \end{aligned}$$

where $\iota : \mathbf{Z}_3 \rightarrow \mathbf{Z}_6$ is given by $\iota(0) = 0, \iota(1) = 2, \iota(2) = 4$ and $\varphi : \mathbf{Z}_6 \rightarrow \mathbf{Z}_2$ is given by $\varphi(a) = a \bmod 2$; ι' is given by $\iota'(1) = (1\ 2\ 3), \iota'(2) = (1\ 3\ 2)$ and φ' is given by $\varphi'(\sigma) = \text{sgn}(\sigma)$.

5.6.1 REMARK. Consider an extension $1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1$ where N is a normal subgroup of G , and $Q \cong G/N$. Such an extension gives rise to a homomorphism

$$\theta' : G \rightarrow \text{Aut}(N), \quad \theta'_g(n) = gng^{-1}$$

Now take any element $g \in G$, and let $q \in Q$ be its image in the quotient $Q \cong G/N$. If we replace g by $g' = gn_0$ for some $n_0 \in N$, then both elements have the same image q in Q . However, their conjugation actions differ by an inner automorphism of N :

$$\theta'(gn_0) = \theta'(g)\theta'(n_0);$$

and $\theta'(n_0)$ is precisely the inner automorphism given by n_0 , i.e., $\theta'(n_0) \in \text{Inn}(N)$. Therefore, the specific value of $\theta'(g)$ in $\text{Aut}(N)$ depends on the choice of the coset representative of g module N . But when we project $\theta'(g)$ to the quotient $\text{Out}(N) = \text{Aut}(N)/\text{Inn}(N)$, this projection depends only on the image q of g in Q , and is independent of the representative chosen. Hence, the extension induces a well-defined homomorphism on Q :

$$\theta : Q \rightarrow \text{Out}(N), \quad \theta(q) = \theta'(g)\text{Inn}(N),$$

where g is any preimage of q in G . This homomorphism θ depends only on the isomorphism class of the extension. Consider the commutative diagram:

$$\begin{array}{ccccccc} 1 & \longrightarrow & N & \xrightarrow{\iota_1} & G_1 & \xrightarrow{\pi_1} & Q \longrightarrow 1 \\ & & \parallel & & \downarrow \varphi & & \parallel \\ 1 & \longrightarrow & N & \xrightarrow{\iota_2} & G_2 & \xrightarrow{\pi_2} & Q \longrightarrow 1 \end{array}$$

Let $q \in Q$ and $g_1 \in G$ is a representative of q in G_1 . Since the diagram is commutative, $\pi_2(\varphi(g_1)) = \pi_1(g_1) = q$ and $\varphi(g_1)$ is a representative of q in G_2 . Now, for every $n \in N$,

$$\theta'_{1_{g_1}}(n) = \varphi(\theta'_{1_{g_1}}(n)) = \varphi(g_1 n g_1^{-1}) = \varphi(g_1) \varphi(n) \varphi(g_1)^{-1} = \varphi(g_1) n \varphi(g_1)^{-1} = \theta'_{2_{\varphi(g_1)}}(n);$$

so that $\theta'_{1_{g_1}} = \theta'_{2_{\varphi(g_1)}}$. Therefore, $\theta_1(q) = \theta'_1 \text{Inn}(N) = \theta'_2(\varphi(g_1)) \text{Inn}(N) = \theta_2(q)$ and $\theta_1 = \theta_2$. Thus, for a fixed θ , we denote by

$$\text{Ext}^1(Q, N)_\theta$$

the set of isomorphic classes of extensions that give rise to this prescribed θ . When both Q and N are abelian, the set $\text{Ext}^1(Q, N)_\theta$ admits a natural structure of an abelian group.

From the above example and our previous discussion, we see that given two groups, an extension of one by the other always exists (their direct product is one such extension). However, these extensions are not necessarily isomorphic. The *extension problem* thus amounts to classifying all extensions of two given groups.

We can better understand the Jordan-Hölder theorem in the light of this problem. Let a group G have a composition series

$$G = Q_0 \supsetneq Q_1 \supsetneq Q_2 \supsetneq \cdots \supsetneq Q_{n-1} \supsetneq Q_n = 1$$

and corresponding factor $JH(G) = \{G_0/G_1 = N_1, G_1/G_2 = N_2, \dots, G_{n-1}/G_n = N_n\}$. Since $Q_n = 1$, we have $Q_{n-1} = N_n$, but something more interesting happens at the next stage: $G_{n-2}/G_{n-1} = N_{n-1}$, so that Q_{n-2} is an extension of Q_{n-1} by N_{n-1} :

$$1 \longrightarrow Q_{n-1} \hookrightarrow Q_{n-2} \twoheadrightarrow N_{n-1} = G_{n-2}/G_{n-1} \longrightarrow 1$$

If we could solve the extension problem, then we would recapture Q_{n-2} from Q_{n-1} and N_{n-1} ; that is, from N_n and N_{n-1} . Once we have Q_{n-2} , we can attach Q_{n-3} in a similar manner, for $Q_{n-3}/Q_{n-2} = N_{n-2}$. Thus, a solution of the extension problem would allow us to recapture Q_{n-3} from N_n, N_{n-1}, N_{n-2} . Climbing up the composition series to $Q_0 = G$, we would recapture G from $N_n, \dots, N_1$. The group G is thus a “product” of the N_i , and the Jordan-Hölder theorem says that the simple groups N_i in this “factorization” of G are uniquely determined by G . We could thus survey all finite groups if we knew all finite simple groups and if we could solve the extension problem.

5.6.3 DEFINITION. It

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} Q \longrightarrow 1$$

is called a *split extension* (or a *splitting extension*) if there exists a group homomorphism $s : Q \rightarrow G$ such that $\pi \circ s = \text{id}_Q$. Such a homomorphism s is called a *splitting homomorphism* for the extension.

5.6.2 REMARK. Equivalently, G is a split extension of Q by N if and only if there exists a subgroup $Q' \leq G$ such that the epimorphism $\pi : G \rightarrow Q$ induces an isomorphism $Q' \rightarrow Q$: if $s : Q \rightarrow G$ splits the extension, let $Q' := \{s(q) : q \in Q\}$, then Q' is a subgroup of G . Now, consider $\pi|_{Q'} : Q' \rightarrow Q$, one may show that this is an isomorphism. Conversely, if $Q' \leq G$ and $\pi|_{Q'} : Q' \rightarrow Q$ is isomorphism, then it has the inverse morphism $s : Q \rightarrow Q'$ such that $\pi \circ s = \text{id}_Q$.

5.6.1 PROPOSITION. Let N be a normal subgroup of a group G . If $1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} Q \rightarrow 1$ is a split extension, where $s : Q \rightarrow G$ is the splitting homomorphism, then $\iota(N) \cap s(Q) = 1$ and $\iota(N)s(Q) = G$.

Proof. If $g \in \iota(N) \cap s(Q)$, then $g = \iota(a) = s(x)$ for some $a \in N$ and $x \in Q$. Now, $g = s(x)$ implies $\pi(g) = \pi s(x) = x$, while $g = \iota(a)$ implies $\pi(a) = \pi \iota(a) = 1$ by the definition of exact sequence. Therefore $x = 1$ and $g = s(x) = 1$.

If $g \in G$, then $\pi(g) = \pi s \pi(g)$, and so that $g(s\pi(g))^{-1} \in \ker \pi = \text{im } \iota$; hence, there is $a \in N$ with $g(s\pi(g))^{-1} = \iota(a)$, and so $g = \iota(a)s(\pi(g)) \in \iota(N)s(Q)$. $\square$

Now we return to a special case, discussing a group G and two of its subgroups N and Q , where N is normal while Q is not necessarily normal, satisfying $G = NQ$ and $N \cap Q = \{1\}$. We ask what special structure G has in this situation. We have the following proposition:

5.6.2 PROPOSITION. Let N be a normal subgroup of a group G , and let Q be a subgroup of G such that $G = NQ$ and $N \cap Q = 1$. Then G is a split extension of Q by N .

Proof. Consider the sequence

$$1 \longrightarrow N \hookrightarrow G \xrightarrow{\pi} Q \longrightarrow 1;$$

in which $N \hookrightarrow G$ is the inclusion. We define $\pi : G \rightarrow Q$ as follows. Since $G = NQ$ and $N \cap Q = \{1\}$, every element $g \in G$ can be uniquely expressed as a product $g = nq$ with $n \in N$ and $q \in Q$. We then set $\pi(g) = \pi(nq) = q$. It is clear that π is an epimorphism and $\ker \pi = N$, hence the sequence is exact. Moreover, it is obviously that $\pi|_Q : Q \rightarrow Q$ is an isomorphism. $\square$

If in the above proposition we also require that Q be normal, then by the well-known result, the extension G must be the direct product of N and Q , i.e., the extension is trivial. Since Q is now an arbitrary subgroup, we may preliminarily give such an extension G a similar name, calling it the *semidirect product* of N and Q , where the prefix “semi” naturally arises from the fact that Q is not necessarily normal.

Now, recall that as soon as N is normal, then every subgroup Q of G acts on N by conjugation: in fact conjugation determines a homomorphism

$$\sigma : Q \rightarrow \text{Aut}_{\text{Grp}}(N), \quad q \mapsto \sigma_q$$

where $\sigma_q : N \rightarrow N$ acts by $\sigma_q(n) := qnq^{-1} \in N$. If Q is normal and since $N \cap Q = 1$, then N and Q commute with each other, so that $\sigma_q(n) = qnq^{-1} = n$, that is $\sigma_q = \text{id}_N$ for every q , hence σ is trivial. Conversely, if σ is trivial, then $qnq^{-1} = n \in N$ for every $q \in Q$ and $n \in N$. We show that Q is normal in G : for any $g = nq \in G$ and $q' \in Q$:

$$gq'g^{-1} = nqq'q^{-1}n^{-1} := nq''n^{-1};$$

since $q''nq''^{-1} = n$, $q''n = nq''$, hence $nq''n^{-1} = q''$, as desired. Hence $Q \trianglelefteq G$. Therefore, we saw that the conjugation σ is trivial if and only if Q is normal (if $G = NQ$, $N \cap Q = 1$, $N \triangleleft G$), that is, G is the product of N and Q if and only if the conjugation σ is trivial.

In summary, if Q is not a normal subgroup, then the conjugation is nontrivial, and G will not be their direct product. Therefore, we shall next discuss what relationship G has with N and Q when this action is nontrivial.

Semidirect Products

When N and Q are subgroups of a group G , the set-product NQ might not be a subgroup. For example, in S_3 if $N = \langle (1\ 2) \rangle$ and $Q = \langle (1\ 3) \rangle$, then $NQ = \{(1), (1\ 2), (1\ 3), (1\ 3\ 2)\}$ has size 4 and is not a subgroup of S_3 . However, if N or Q is normal in G , then NQ is a subgroup. Taking $N \trianglelefteq G$, for example:

$$(n_1q_1)(n_2q_2) = (n_1q_1n_2q_1^{-1})(q_1q_2) \in NQ, \quad (n_1q_1)^{-1} = q_1^{-1}n_1^{-1} = (q_1^{-1}n_1^{-1}q_1)q_1^{-1} \in NQ$$

Notice that in the first formula there is a multiplication that draws our attention, namely $q_1n_2q_1^{-1}$. We write it as $\sigma_{q_1}(n_2)$. Then we have $(n_1q_1)(n_2q_2) = (n_1\sigma_{q_1}(n_2))(q_1q_2)$.

At this point, if Q is also a normal subgroup, and $G = NQ$ with $N \cap Q = 1$, then by what we have said before, G is precisely the direct product $N \times Q$, then we would have $\sigma_{q_1}(n_2) = n_2$ and $(n_1q_1)(n_2q_2) = (n_1n_2)(q_1q_2)$. Now, under the isomorphism

$$\varphi : N \times Q \rightarrow G, \quad (n, q) \mapsto nq,$$

this formula in the direct product $N \times Q$ becomes the standard multiplication of group $N \times Q$: $(n_1, q_1)(n_2, q_2) = (n_1n_2, q_1q_2) = (n_1\sigma_{q_1}(n_2), q_1q_2)$.

Now, we consider the general case: if $\sigma : Q \rightarrow \text{Aut}(N)$, $q \mapsto \sigma_q$ is an arbitrary homomorphism. Define an operation $\bullet_\sigma$ on the set-product $N \times Q$ as follows: for $n_1, n_2 \in N$, $q_1, q_2 \in Q$, let

$$(n_1, q_1) \bullet_\sigma (n_2, q_2) := (n_1\sigma_{q_1}(n_2), q_1q_2).$$

5.6.3 LEMMA. $\langle N \times Q, \bullet_\sigma, (1_N, 1_Q) \rangle$ is a group.

Proof. First, the action $\sigma : Q \rightarrow \text{Aut}(N)$ is a homomorphism, so that $\sigma_{q_1} \circ \sigma_{q_2} = \sigma(q_1q_2) = \sigma_{q_1q_2}$ and $\sigma_1 = \text{id}_N$. We show that the structure satisfies the group axioms:

- $(1, 1)$ is the identity: for any $n \in N, q \in Q$, $(n, q)(1, 1) = (n\sigma_q(1), q1) = (n, q)$; and $(1, 1)(n, q) = (1\sigma_1(n), 1q) = (n, q)$;
- Let (n, q) be any element in $N \rtimes Q$. We want to find a pair (n', q') that makes $(n, q)(n', q') = (1, 1)$, i.e., $(n\sigma_q(n'), qq') = (1, 1)$. For the second coordinates being equal, $q' = q^{-1}$. Then in the first coordinates: $n\sigma_q(n') = 1$ implies $\sigma_q(n') = n^{-1}$. Since σ_q is an isomorphism and $\sigma_{q^{-1}} = \sigma_q^{-1}$, $n' = \sigma_{q^{-1}}(n^{-1}) = (\sigma_{q^{-1}}(n))^{-1}$. We have solved for both n' and q' : if (n, q) has an inverse, it must be $(\sigma_{q^{-1}}(n^{-1}), q^{-1}) = ((\sigma_{q^{-1}}(n))^{-1}, q^{-1})$. One may check that this is precisely the inverse of (n, q) in $N \rtimes Q$.
- Associativity is easy to check.

□

5.6.4 DEFINITION. Let N, Q be groups, and let $\sigma : Q \rightarrow \text{Aut}(N)$ be a homomorphism. The group $\langle N \times Q, \bullet_\sigma, (1_N, 1_Q) \rangle$ is called a *semidirect product* of N and Q , and is denoted by $N \rtimes_\sigma Q$.

5.6.4 PROPOSITION. Let N, Q be groups, and let $\sigma : Q \rightarrow \text{Aut}(N)$ be a homomorphism. Let $G = N \rtimes_\sigma Q$ be the corresponding semidirect product of N and Q . Then $N \cong \{(n, 1) : n \in N\}$ by $n \mapsto (n, 1)$, $Q \cong \{(1, q) : q \in Q\}$ by $q \mapsto (1, q)$; and the copy of N in $N \rtimes_\sigma Q$ is a normal subgroup.

Proof. In $N \rtimes_\sigma Q$, the functions $n \mapsto (n, 1)$ and $q \mapsto (1, q)$ are obviously injective from N and Q into $N \rtimes_\sigma Q$. These multiply together like elements of N and Q do:

$$\begin{aligned} (n, 1)(n', 1) &= (n\sigma_1(n'), 1) = (nn', 1); \\ (1, q)(1, q') &= (1\sigma_q(1), qq') = (1, qq'). \end{aligned}$$

Therefore, these two functions are injective homomorphisms; the first statement is true. Let $N \times 1, 1 \times Q$ be the copies of N and Q in $N \rtimes_\sigma Q$. Consider the function $N \rtimes_\sigma Q \rightarrow Q$ defined by $(n, q) \mapsto q$, it is clear that it is a surjective homomorphism with kernel N , therefore $N \times 1$ is normal. □

5.6.3 REMARK. The homomorphism σ may be realized by conjugation in $N \rtimes_\sigma Q$, that is, for $(n, 1) \in N \times 1$ and $(1, q) \in 1 \times Q$, we have $(1, q)(n, 1)(1, q)^{-1} = (\sigma_q(n), q)(1, q^{-1}) = (\sigma_q(n)\sigma_q(1), 1) = (\sigma_q(n), 1)$. In particular, $(1, q)$ commutes with every $(n, 1)$ if and only if $q \in \ker \sigma$; and every $(1, q)$ and every $(n, 1)$ commute if and only if $\sigma : Q \rightarrow \text{Aut}(N)$ is trivial, that is $\sigma_q = \text{id}$ for every $q \in Q$:

$$(n, 1)(1, q) = (1, q)(n, 1) \leftrightarrow (n, q) = (\sigma_q(n), q) \leftrightarrow (n, q) = (\sigma_q(n), q).$$

Therefore $(1, q)$ commutes with all $(n, 1)$ if and only if $\sigma_q(n) = n$ for all $n \in N$, which means $q \in \ker \sigma$. That all $(1, q)$ and $(n, 1)$ commute means $\ker \sigma = Q$ which is another way of saying the action σ is trivial. , thus, $N \rtimes_\sigma Q$ is non-abelian whenever σ is not trivial. Even if N and Q are both abelian, if σ is nontrivial, then the semidirect product is non-abelian !

In fact, from the proof process, we have not only shown that the copy of N is a normal subgroup, but also that $NQ = G$ and $N \cap Q = \{1\}$ (for the copy), since for a single pair (n, q) in $N \rtimes_\sigma Q$: $(n, 1)(1, q) = (n\sigma_1(1), q) = (n, q)$, hence $N \rtimes_\sigma Q = NQ$ and it is clear that $N \cap Q = \{(1, 1)\}$. Therefore, by the last proposition of the previous section, the semidirect product $N \rtimes_\sigma Q$ of N and Q is actually a split extension of N by Q :

$$1 \longrightarrow N \longrightarrow N \rtimes_\sigma Q \xrightarrow{\pi} Q \longrightarrow 1;$$

$\xleftarrow{s}$

where $\pi : N \rtimes_\sigma Q \rightarrow Q$ is given by $\pi(n, q) = q$ and the splitting homomorphism $s : Q \rightarrow N \rtimes_\sigma Q$ is given by $s(q) = (1, q)$. Conversely:

5.6.5 PROPOSITION. Let N, Q be subgroups of a group G with N normal in G . Assume that $N \cap Q = 1$ and $G = NQ$, then there is a split extension of N by Q and the splitting homomorphism s . Let $\sigma : Q \rightarrow \text{Aut}(N)$ be defined by conjugation: for $q \in Q, n \in N$, $\sigma_q(n) = qnq^{-1}$. Then

$$G \cong N \rtimes_\sigma Q.$$

Proof. Define a function $\varphi : N \rtimes_\sigma Q \rightarrow G$ by $\varphi(n, q) = nq$. It is clear that φ is a bijection: for any $g \in G$, we may write g as nq for some $n \in N, q \in N$, hence φ is

surjective; since $N \cap Q = 1$, one may check that φ is injective. We need to verify that φ is a homomorphism, and indeed for any $n_1, n_2 \in N, q_1, q_2 \in Q$:

$$\begin{aligned} \varphi((n_1, q_1)(n_2, q_2)) &= \varphi(n_1 \sigma_{q_1}(n_2), q_1 q_2) \\ &= \varphi(n_1(q_1 n_2 q_1^{-1}), q_1 q_2) \\ &= n_1 q_1 n_2 (q_1^{-1} q_1) q_2 = (n_1 q_1)(n_2 q_2) \\ &= \varphi(n_1, q_1) \varphi(n_2, q_2). \end{aligned}$$

$$\begin{array}{ccccccc} 1 & \longrightarrow & N & \longrightarrow & N \rtimes_{\sigma} Q & \longrightarrow & Q \longrightarrow 1 \\ & & \parallel & & \downarrow \varphi & & \parallel \\ 1 & \longrightarrow & N & \longrightarrow & G & \longrightarrow & Q \longrightarrow 1 \end{array}$$

□

5.6.2 EXAMPLE. $\text{Aut}(S_6)$ is a semidirect product of S_6 by $\mathbf{Z}_2$. We shall show that there is an outer automorphism φ of S_6 which has order 2, then $\text{Inn}(S_6) \trianglelefteq \text{Aut}(S_6)$, $\text{Inn}(S_6) \langle \varphi \rangle = \text{Aut}(S_6)$ and $\text{Inn}(S_6) \cap \langle \varphi \rangle = 1$. The following φ satisfies the condition, though the computation is somewhat involved: define $\varphi \in \text{Aut}(S_6)$ by

$$\begin{aligned} (1\ 2) &\mapsto (1\ 5)(2\ 3)(4\ 6) \\ (1\ 3) &\mapsto (1\ 4)(2\ 6)(4\ 5) \\ (1\ 4) &\mapsto (1\ 3)(2\ 5)(5\ 6) \\ (1\ 5) &\mapsto (1\ 2)(3\ 6)(4\ 5) \\ (1\ 6) &\mapsto (1\ 6)(2\ 5)(3\ 4) \end{aligned}$$

5.6.3 EXAMPLE. The automorphism group of C_3 is isomorphic to the cyclic group C_2 : if $C_3 = \{1, y, y^2\}$, then the two automorphisms of C_3 are

$$\text{id} : \begin{cases} e \mapsto e, \\ y \mapsto y, \\ y^2 \mapsto y^2, \end{cases} \quad \theta : \begin{cases} e \mapsto e, \\ y \mapsto y^2, \\ y^2 \mapsto y. \end{cases}$$

Therefore, there are two homomorphisms $C_2 \rightarrow \text{Aut}(C_3)$: the trivial homomorphism, and the isomorphism sending the identity to id and the noidentity element to θ . The semidirect product corresponding to the trivial homomorphism is the direct product $C_3 \times C_2 \cong C_6$; the other semidirect product $C_3 \rtimes C_2$ is isomorphic to S_3 : this follows from the proposition above; let $N = \langle (1\ 2\ 3) \rangle$ and $Q = \langle (1\ 2) \rangle$, then $S_3 = NQ$ and $N \cap Q = \{1\}$.

5.6.4 EXAMPLE (Symmetric Groups). S_n is a semidirect product of A_n and C_2 . It is clear that $A_n \trianglelefteq S_n$ since A_n is the kernel of the sign homomorphism; let $\tau = (1\ 2)$, then $\langle \tau \rangle \cong C_2$. One may show that $A_n \cap \langle \tau \rangle = 1$, $S_n = A_n \langle \tau \rangle$. Hence $S_n \cong A_n \rtimes_{\sigma} C_2$ where the homomorphism σ is defined by conjugation.

5.6.5 EXAMPLE (Dihedral Groups). D_{2n} is a semidirect product of C_n and C_2 . Let $n \in \mathbf{N}$. Take the cyclic group $N := C_n$ and $Q := C_2$, with operation written multiplicative. Let τ be the nontrivial element of Q , that is $Q = \{1, \tau\}$, and define $\sigma : Q \rightarrow \text{Aut}(N)$ by

$$\sigma_1 = \text{id}, \quad \sigma_{\tau}(x) = x^{-1}, \quad x \in N.$$

The resulting semidirect product $N \rtimes_{\sigma} Q$ is the “dihedral group” D_{2n} : any element in $N \rtimes_{\sigma} Q$ is of the form (r^i, s^j) , $0 \leq i < n, j \in \{0, 1\}$ and the multiplication is

$$(r^i, s^j)(r^k, s^l) = (r^i \sigma_{s^j}(r^k), s^{j+l}) = (r^{i+(-1)^j k}, s^{j+l}),$$

let $r = (r, 1)$, $s = (1, s)$, then one may show that $r^n = 1$, $s^2 = 1$, $sr s^{-1} = r^{-1}$. Hence $D_{2n} \cong N \rtimes_{\sigma} Q \cong C_n \rtimes_{\sigma} C_2$.

6

Ring Theory

6.1 Rng and Ring

Rings

6.1.1 REMARK. Here is the motivation behind the definition of a ring: If G, H are abelian groups, then $\text{Hom}_{\text{Ab}}(G, H)$ is also an abelian group. In particular, if G is an abelian group, then so is the set of endomorphisms $\text{End}_{\text{Ab}}(G) = \text{Hom}_{\text{Ab}}(G, G)$. Since Ab is a category, the morphisms from an abelian group to itself may be composed with each other. Thus, two operations coexist in $\text{End}_{\text{Ab}}(G)$: addition and composition. In which addition and composition are “compatible”: for any $f, g, h \in \text{End}_{\text{Ab}}(G)$, it is easy to check that $f \circ (g + h) = f \circ g + f \circ h$ and $(f + g) \circ h = f \circ h + g \circ h$. By extracting the essential features of $\text{End}_{\text{Ab}}(G)$, we give the definition of a ring:

6.1.1 DEFINITION. A *ring* $\langle R, +, \cdot, 0, 1 \rangle$ is a set with two binary operations $R \times R \rightarrow R$: addition $(a, b) \mapsto a + b$ and multiplication $(a, b) \mapsto ab$ for which

- $\langle R, +, 0 \rangle$ is an abelian group.
- $\langle R, \cdot, 1 \rangle$ is a monoid.
- $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$ for every $a, b, c \in R$.

Some textbooks include a multiplicative identity in the definition of a ring, while others do not require one. The former convention originates from the *Bourbaki* school, whereas the latter follows the definition used earlier by the algebraist *Emmy Noether*. This may seem like a minor issue, but it gives rise to several consequences: for instance, with the notion of ideals to be introduced later, if we adopt the first definition, then an ideal is not a subring; if we adopt the second, then an ideal is indeed a subring. These notes follow the first definition.

As for the term “Rng” in the title, it refers to the absence of a multiplicative identity. In fact, many textbooks define a ring as not necessarily containing a unit element 1. However, for our purposes, the hom-set Hom must always have an identity element, so we introduce the less formal term “Rng” to distinguish our convention from that of other texts. Under this convention, the definition of a Rng is obtained simply by replacing the second condition in Definition 6.1.1 with the requirement that $\langle R, \cdot \rangle$ be a semigroup.

Let $\langle R, +, \cdot \rangle$ be a ring (rng), define the operation $\odot$ by $a \odot b := ba, a, b \in R$, then $\langle R^{\text{op}}, +, \odot \rangle$ is a ring (rng), and if $R^{\text{op}} = R$, that is $ab = ba$, we call R is a commutative ring (rng). The ring (rng) R is *commutative* if the multiplication is commutative.

In these notes, we mostly discuss rings (with identity), and only occasionally consider rngs. For convenience of writing, I will not repeatedly mention rngs hereafter, except in a few cases where extra emphasis is needed.

6.1.1 PROPOSITION. *Let R be a ring. Then*

1. $0a = a0 = 0$ for all $a \in R$;
2. $(-a)b = a(-b) = -(ab)$ for all $a, b \in R$;
3. $(-a)(-b) = ab$ for all $a, b \in R$;
4. $(na)b = a(nb) = n(ab)$ for all $n \in \mathbf{Z}, a, b \in R$;
5. $(\sum_{i=1}^n a_i) \left(\sum_{j=1}^m b_j \right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j$ for all $a_i, b_i \in R$.

Proof. Omitted. □

6.1.2 PROPOSITION (Binomial). *Let R be a ring, $n \in \mathbf{Z}$ and $a, b, a_1, \dots, a_s \in R$:*

1. *If $ab = ba$, then $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$;*
2. *If $a_i a_j = a_j a_i$ for all i, j , then*

$$(a_1 + \dots + a_s)^n = \sum_{i_1 + \dots + i_s = n} \frac{n!}{i_1! \dots i_s!} a_1^{i_1} a_2^{i_2} \dots a_s^{i_s}$$

Proof. Omitted. □

6.1.1 EXAMPLE. There is a ring that contains just one element, $0 = 1$; it is called the *zero ring*, denoted $R = \mathbf{0}$. In fact, $\mathbf{0}$ is the only ring where $1 = 0$, we shall often exclude this ring by imposing the condition “ $1 \neq 0$ ”:

Proof. First note that $0a = 0$: $0 = 0a - 0a = (0 - 0)a = 0a$. Assume that $1 = 0$ in R , and let a be any element. Then $a = 1a = 0a = 0$. The only element of R is $0 = 1$. □

6.1.2 EXAMPLE. The ring of integers, $\mathbf{Z}$ under the usual operations of addition and multiplication is a commutative ring. Similarly, the rational numbers, $\mathbf{Q}$, the real numbers, $\mathbf{R}$, and the complex numbers, $\mathbf{C}$ are commutative rings.

6.1.3 EXAMPLE. The quotient group $\mathbf{Z}/n\mathbf{Z}$ is a commutative ring under the operations of addition and multiplication of residue classes; that is, $[a]_n + [b]_n := [a + b]_n$ and $[a]_n \cdot [b]_n := [ab]_n$

6.1.4 EXAMPLE (Boolean Rings). Here is an example of a commutative ring arising from set theory. If A and B are subsets of a set X , then their *symmetric difference* is

$$A + B := (A \cup B) - (A \cap B).$$

$\mathcal{P}(X)$ is the set of all subsets of X . Define the addition on $\mathcal{P}(X)$ to be symmetric difference, and define multiplication on $\mathcal{P}(X)$ to be intersection. It is easy to show that $\mathcal{P}(X)$ is a commutative ring: the empty set $\emptyset$ is the zero element. We call $\mathcal{P}(X)$ a *Boolean Ring*.

6.1.5 EXAMPLE (The Real Hamilton Quaternions). Let $\mathbf{R}$ be the ring of real numbers, Let $\mathbf{H}$ be the additive abelian group $\mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}$ and write the elements of $\mathbf{H}$ as formal sums $(a_0, a_1, a_2, a_3) = a_0 1 + a_1 i + a_2 j + a_3 k$. Then $a_0 1 + a_1 i + a_2 j + a_3 k = b_0 1 + b_1 i + b_2 j + b_3 k$ if and only if $a_i = b_i$ for every i . We adopt the conventions that $a_0 1 \in \mathbf{H}$ is identified with $a_0 \in \mathbf{R}$ and that terms with zero coefficients may be omitted (for example, $4 + 2j := 4 \cdot 1 + 0i + 2j + 0k$ and $i := 0 + 1i + 0j + 0k$). Then addition in $\mathbf{H}$ is given by

$$(a_0 + a_1 i + a_2 j + a_3 k) + (b_0 + b_1 i + b_2 j + b_3 k) = (a_0 + b_0) + (a_1 + b_1)i + (a_2 + b_2)j + (a_3 + b_3)k.$$

and multiplication is defined by expanding it using the distributive law and simplifying using the relations $i^2 = j^2 = k^2 = -1$, $ij = -ji = k$, $jk = -kj = i$, $ki = -ik = j$. Under this multiplication, $\mathbf{H}$ is a noncommutative ring.

6.1.6 EXAMPLE. Let R be a ring and S the set of non-negative integers. R^S is the set of all functions from S into R . Now, we will define the addition and multiplication on R^S . We may denote every function of R^S by an infinite sequence $f = (f_0, f_1, f_2, \dots)$ of elements f_i in R . If $g = (g_0, g_1, g_2, \dots)$, $g_i \in R$, $f + g$ is the infinite sequence given by

$$f + g := (f_0 + g_0, f_1 + g_1, f_2 + g_2, \dots);$$

we define the multiplication in R^S by associating with each pair of f and g in R^S : $(fg)_n = \sum_{i=0}^n f_i g_{n-i}$, $n = 0, 1, 2, \dots$. Thus fg is the infinite sequence given by

$$fg := (f_0 g_0, f_0 g_1 + f_1 g_0, f_0 g_2 + f_1 g_1 + f_2 g_0, \dots).$$

For all $f, g, h \in R^S$ and each n ,

$$[(fg)h]_n = \sum_{i=0}^n (fg)_i h_{n-i} = \sum_{i=0}^n \left(\sum_{j=0}^i f_j g_{i-j} \right) h_{n-i} = \sum_{i=0}^n \sum_{j=0}^i f_j g_{i-j} h_{n-i}.$$

On the other hand,

$$[f(gh)]_n = \sum_{j=0}^n f_j (gh)_{n-j} = \sum_{j=0}^n f_j \left(\sum_{k=0}^{n-j} g_k h_{n-j-k} \right) = \sum_{j=0}^n \sum_{k=0}^{n-j} f_j g_k h_{n-j-k}.$$

Let $i = j + k$. Both double sums range over all $0 \leq j \leq n$, $0 \leq k \leq n - j$, so they are equal. Hence $(fg)h = f(gh)$. One may show that R^S is a ring under the addition and multiplication, and that the function $\mathbf{1} = (1, 0, 0, \dots)$ serves as an identity for R^S .

The function $(0, 1, 0, \dots, 0, \dots)$ plays a distinguished role in what follows and we shall consistently denote it by x , and we call x an *indeterminate*. Throughout this chapter, x will never be used to denote an element of a ring. The product of x with itself n times will be denoted by x^n and we shall put $x^0 = 1$. Then

$$x^2 = (0, 0, 1, 0, \dots), \quad x^3 = (0, 0, 0, 1, 0, \dots)$$

and in general for each integer $k \geq 0$, $(x^k)_k = 1$ and $(x^k)_n = 0$ for all non-negative integers $n \neq k$. For $a \in R$ and $k \geq 0$, we define ax^k to be the element of R^S given by

$$(ax^k)_n = \begin{cases} a, & n = k \\ 0, & n \neq k \end{cases}$$

that is, $ax^k = (0, \dots, 0, a, 0, \dots)$ with a in the k -th position. (A more convenient and correct approach is to denote $(a, 0, 0, \dots, 0)$ simply by a . Then, from the fact that $(0, \dots, 0, a, 0, \dots) = (a, 0, 0, \dots) x^k$, we can write $(0, \dots, 0, a, 0, \dots)$ as ax^k .) With this notation, the element $f = (f_0, f_1, f_2, \dots)$ is frequently written

$$f = \sum_{n=0}^{\infty} f_n x^n.$$

This notation may not very “legal”, but it is very convenient for dealing with the operations. When used, it must be remembered that it is purely formal. There are no “infinite sums” in *Algebra* and the power series notation is not intended to suggest anything about convergence. Let $R[[x]]$ denote the set of all formal power series over R . Clearly $R[[x]] := R^S$ is a ring. We now define the polynomial ring $R[x]$ over the ring R :

6.1.3 COROLLARY. *Formal power series $\sum f_n x^n$ and $\sum g_n x^n$ in R^S are equal if and only if $f_n = g_n$ for all n .*

Proof. This is merely a restatement of the definition of equality of sequences, rephrased in the usual notation for formal power series. $\square$

6.1.2 DEFINITION. Let R be a ring, and R^S is the ring of all functions from R into S . Let $R[x]$ be the set of all functions that have only finitely many nonzero terms. An element of $R[x]$ is called a *polynomial* with coefficients in a ring R .

It is clear that $R[x] = \{f \in R^S : \exists N \geq 0, \text{ s.t. } f_n = 0, n > N\}$. Moreover, $f \in R[x]$ if and only if there is an integer $n \geq 0$ such that $f_n \neq 0$ and such that $f_k = 0$ for all integers $k > n$; this integer (when it exists) is obviously unique and is called the *degree of f* . We denote the degree of a polynomial f by $\deg f$. If f is a non-zero polynomial of degree n it follows that

$$f = f_0x^0 + f_1x + f_2x^2 + \cdots + f_nx^n, \quad f_n \neq 0.$$

This expression is obviously unique. After a simple calculation, we can express the addition and multiplication of polynomials in $R[x]$, when written in the above form, as follows:

$$\begin{aligned} \sum_{i=0}^n f_i x^i + \sum_{i=0}^n g_i x^i &= \sum_{i=0}^n (f_i + g_i) x^i \\ \left(\sum_{i=0}^n f_i x^i \right) \left(\sum_{j=0}^m g_j x^j \right) &= \sum_{k=0}^{m+n} c_k x^k, \quad c_k = \sum_{i+j=k} f_i g_j. \end{aligned}$$

It is clear that $R[x]$ with these operations is a ring. We shall call polynomials of the form cx^0 *constants*; they form a “copy” of R in $R[x]$; and frequently write c for cx^0 .

Since $R[x]$ is itself a ring, we may apply the same construction again to obtain the polynomial ring in a new y with coefficients in $R[x]$: $R[x][y]$. Using the polynomial notation, we write $F = F_0 + F_1y + F_2y^2 + \cdots + F_ny^n$, where $y = (0, 1_{R[x]}, 0, \dots)$ and $F_j = f_{0j} + f_{1j}x + f_{2j}x^2 + \cdots + f_{m_jj}x^{m_j}$. Then

$$F = \sum_{j=0}^n F_j y^j = \sum_{j=0}^n \left(\sum_{i=0}^{m_j} f_{ij} x^i \right) y^j = \sum_{i=0}^m \sum_{j=0}^n f_{ij} x^i y^j,$$

where $m = \max\{m_0, \dots, m_n\}$ and we set $f_{ij} = 0$ for all $i > m_j$ so that all coefficients f_{ij} lie in the same finite range $0 \leq i \leq m, 0 \leq j \leq n$.

Similarly, we may define $R[x_1][x_2] \cdots [x_n]$. For convenience of notation and to distinguish polynomials in different numbers of x_i , we shall use $f(x)$ to denote a polynomial $f \in R[x]$, and $f(x, y)$ to denote a polynomial $f \in R[x][y]$. However, when we say that $f(x)$ is a polynomial, we mean a formal polynomial, not a polynomial function. With this convention, no ambiguity arises.

6.1.7 EXAMPLE (Monoid Ring). Given a monoid $\langle M, \cdot \rangle$ and a ring R . We can construct monoid rings $R[M]$ in the same way we construct polynomial rings. More precisely, as follows: let $R[M]$ be the set of all functions $f : M \rightarrow R$ of *finite support* ($f(g)$ is nonzero for only finitely many elements g). Define the addition and multiplication as follows:

$$\begin{aligned} f + g : m &\mapsto f(m) + g(m); \\ fg : m &\mapsto \sum_{uv=m} f(u)g(v). \end{aligned}$$

The functions such as $f : M \rightarrow R$ are sometimes written as the formal linear combination of elements of M with “coefficients” in R :

$$f := \sum_{m \in M} f(m) \cdot m.$$

If $f \in R[M]$, then it means that $f(m) \neq 0$ for only finitely many m . When using this

notation, the two operations above can be written as follows:

$$\begin{aligned} \left(\sum_{m \in M} f(m) \cdot m \right) + \left(\sum_{m \in M} g(m) \cdot m \right) &= \sum_{m \in M} (f(m) + g(m)) \cdot m \\ \left(\sum_{m \in M} f(m) \cdot m \right) \cdot \left(\sum_{m \in M} g(m) \cdot m \right) &= \sum_{m \in M} \sum_{uv=m} (f(u)g(v)) \cdot m. \end{aligned}$$

This notation has no practical meaning other than reflecting its value at m ; it is merely intended to match the infinite sum notation $\sum_{n=0}^{\infty} f_n x^n$ that we defined earlier. (Since M need not be countable, a function $f : M \rightarrow R$ cannot in general be written as a sequence. Consequently, we cannot introduce a concrete element analogous to $x = (0, 1, 0, \dots)$ to mark positions; instead, we must use abstract symbols $m \in M$ as positional indicators. Conversely, the notation x^n in the polynomial ring $R[x]$ may also be understood as an abstract positional marker, which becomes concrete only when $M = \mathbf{N} = S$ admits a natural ordering and countability.)

Let R be a ring, and consider the monoid (group) $\prod_{i=1}^n \mathbf{N}$ under the additive notation. The elements of $R[\prod_{i=1}^n \mathbf{N}]$ are the functions $f : \prod_{i=1}^n \mathbf{N} \rightarrow R$ such that $f(\mathbf{u}) \neq 0$ for at most a finite number of elements $\mathbf{u}$ of $\prod_{i=1}^n \mathbf{N}$. And the addition and multiplication are defined by

$$(f + g)(\mathbf{u}) = f(\mathbf{u}) + g(\mathbf{u}), \quad (fg)(\mathbf{u}) = \sum_{\mathbf{v} + \mathbf{w} = \mathbf{u}} f(\mathbf{v})g(\mathbf{w}).$$

Let $\epsilon_i = (0, \dots, 0, 1, 0, \dots, 0) \in \prod_{i=1}^n \mathbf{N}$ where 1 is the i -th coordinate of ϵ_i . If $k \in \mathbf{N}$, let $k\epsilon_i = (0, \dots, 0, k, 0, \dots, 0)$. Then every element of $\prod_{i=1}^n \mathbf{N}$ may be written in the form $k_1\epsilon_1 + k_2\epsilon_2 + \dots + k_n\epsilon_n$.

6.1.4 PROPOSITION. *Let $x_i \in R[\prod_{i=1}^n \mathbf{N}]$ be defined by $x_i(\epsilon_i) = 1$ and $x_i(\mathbf{u}) = 0$ for $\mathbf{u} \neq \epsilon_i$. Then for every $f \in R[\prod_{i=1}^n \mathbf{N}]$, there exist unique $f_{k_1 k_2 \dots k_n} \in R$ and nonzero for at most a finite number of $(k_1, \dots, k_n) \in \prod_{i=1}^n \mathbf{N}$ such that*

$$f = \sum_{(k_1, \dots, k_n) \in \prod_{i=1}^n \mathbf{N}} f_{k_1 k_2 \dots k_n} x_1^{k_1} \dots x_n^{k_n}.$$

(This sum is also a formal sum. If a function takes the value $a \in R$ only at $(0, 0, \dots, 0)$ and takes the value 0 everywhere else, then we simply denote it by a . Then the expression $ax_1^{k_1} \dots x_n^{k_n}$, $a \in R$, becomes meaningful, just as in our previous treatment.)

Proof. Let $f_{k_1 \dots k_n} := f(k_1, \dots, k_n) \in R$. Then consider the function $x_1^{k_1} x_2^{k_2} \dots x_n^{k_n}$, one may show that $(x_1^{k_1} \dots x_n^{k_n})(k_1, \dots, k_n) = (x_1^{k_1} \dots x_n^{k_n})(k_1\epsilon_1 + \dots + k_n\epsilon_n) = 1_R$ and $(x_1^{k_1} \dots x_n^{k_n})(\mathbf{u}) = 0$ for all $\mathbf{u} \neq k_1\epsilon_1 + \dots + k_n\epsilon_n$. Then $f = \sum f_{k_1 \dots k_n} x_1^{k_1} \dots x_n^{k_n}$. For the uniqueness, $\sum ax_1^{k_1} \dots x_n^{k_n} = 0$ implies $a = 0$. $\square$

We denote $R[\prod_{i=1}^n \mathbf{N}]$ by $R[x_1, \dots, x_n]$. Intuitively, $R[x_1, \dots, x_n]$ seems to be “the same as” $R[x_1][x_2] \dots [x_n]$. In mathematical language, they are isomorphic in some sense, and the reader will naturally guess that this means ring isomorphism. To this end, we introduce the concept of ring homomorphism and then address this problem.

Homomorphisms

For two rings $(R, +_R, \cdot_R)$ and $(S, +_S, \cdot_S)$, a *ring homomorphism* $\varphi : R \rightarrow S$ is first of all a group homomorphism from the group $(R, +_R)$ to $(S, +_S)$; but a mapping considered purely at the group level may completely ignore the multiplication structure of R and S . In other words, it must be “compatible” with the multiplication. The natural way is: the

function φ must preserve both operations and identity:

$$\begin{aligned}\varphi(a + b) &= \varphi(a) + \varphi(b), \\ \varphi(ab) &= \varphi(a)\varphi(b) \text{ for any } a, b \in R, \\ \varphi(1_R) &= 1_S\end{aligned}$$

We shall not dwell further here on the motivation, since the thought process is almost identical to that for introducing group homomorphisms. Moreover, we can also use commutative diagrams to show that, under this definition of ring homomorphisms, one may define the category **Ring** of rings, whose hom-set $\text{Hom}(R, S)$ is precisely the set of all ring homomorphisms from R to S . (Here we strictly need the restriction of a Grothendieck universe $\mathbf{U}$, just as in the earlier definition of **Mon**. However, apart from being more rigorous, this does not seem to have much other significance.)

6.1.2 REMARK. The zero ring $\mathbf{0}$ is clearly final (there exists exactly one! morphism $R \rightarrow \mathbf{0}$ for every ring R) in **Ring**. However, note that it is not initial: because of the requirement that ring homomorphisms send 1 to 1, the only rings to which zero ring map homomorphically are the zero rings (ring).

The category **Ring** does have initial objects: the ring of integers $\mathbf{Z}$ under the usual operations is initial in **Ring**. Indeed, for every ring R we can define a group homomorphism $\varphi : \mathbf{Z} \rightarrow R$ by

$$\varphi(n) = n \cdot 1_R, \quad n \in \mathbf{Z}.$$

But φ is in fact a ring homomorphism, since $\varphi(1) = 1_R$, and $\varphi(mn) = (mn)1_R = m(n1_R) = (m1_R) \cdot (n1_R) = \varphi(m)\varphi(n)$, where “=” holds by the distributivity axiom. This ring homomorphism is unique, since it is determined by the requirement that $\varphi(1) = 1_R$ and by the fact that φ preserves addition. Thus, for every ring R there exists a unique ring homomorphism $\mathbf{Z} \rightarrow R$ showing that $\mathbf{Z}$ is initial in **Ring**.

6.1.5 PROPOSITION. Let R, S be two rings. A ring homomorphism $\varphi : R \rightarrow S$ is an isomorphism if and only if φ is a bijection in the functional sense.

Proof. One implication is immediate. For the other implication, assume that $\varphi : R \rightarrow S$ is a bijective ring homomorphism. As a bijection, φ has an inverse in **Set**: $\varphi^{-1} : S \rightarrow R$; we simply need to check that this is a ring homomorphism. Let $s_1, s_2 \in S$, and let $r_1 = \varphi^{-1}(s_1), r_2 = \varphi^{-1}(s_2)$ be the corresponding elements of R . Then $\varphi^{-1}(s_1 + s_2) = \varphi^{-1}(\varphi(r_1) + \varphi(r_2)) = \varphi^{-1}(\varphi(r_1 + r_2)) = r_1 + r_2 = \varphi^{-1}(s_1) + \varphi^{-1}(s_2)$; $\varphi^{-1}(s_1 s_2) = \cdots = \varphi^{-1}(s_1)\varphi^{-1}(s_2)$; $\cdots$ $\square$

6.1.8 EXAMPLE. $\text{End}_{\text{Ab}}(\mathbf{Z}) \cong \mathbf{Z}$.

Proof. Consider the function $\varphi : \text{End}_{\text{Ab}}(\mathbf{Z}) \rightarrow \mathbf{Z}$ defined by $\varphi(\alpha) = \alpha(1)$ for all group homomorphisms $\alpha : \mathbf{Z} \rightarrow \mathbf{Z}$. Then φ is a group homomorphism: the addition in $\text{End}_{\text{Ab}}(\mathbf{Z})$ is defined so that $\forall n \in \mathbf{Z}, (\alpha + \beta)(n) = \alpha(n) + \beta(n)$; in particular

$$\varphi(\alpha + \beta) = (\alpha + \beta)(1) = \alpha(1) + \beta(1) = \varphi(\alpha) + \varphi(\beta).$$

Next, we show that φ is a ring homomorphism, it suffices to show that $\varphi(\alpha \circ \beta) = \varphi(\alpha)\varphi(\beta)$ and $\varphi(\text{id}_{\mathbf{Z}}) = 1$. For $\alpha, \beta \in \text{End}_{\text{Ab}}(\mathbf{Z})$, denote $\alpha(1) = a \in \mathbf{Z}$, then $\alpha(n) = n\alpha(1) = na = a n$ for all $n \in \mathbf{Z}$; in particular, $\alpha(\beta(1)) = a\beta(1) = \alpha(1)\beta(1)$. Therefore

$$\varphi(\alpha \circ \beta) = (\alpha \circ \beta)(1) = \alpha(\beta(1)) = \alpha(1)\beta(1) = \varphi(\alpha)\varphi(\beta).$$

Also $\varphi(\text{id}_{\mathbf{Z}}) = \text{id}_{\mathbf{Z}}(1) = 1$. Finally, φ has an inverse: for $a \in \mathbf{Z}$, let $\psi(a)$ be the group homomorphism $\alpha : \mathbf{Z} \rightarrow \mathbf{Z} \in \text{End}_{\text{Ab}}(\mathbf{Z})$ defined by

$$\alpha_a(n) := an, \quad \forall n \in \mathbf{Z}.$$

One may show that ψ is a ring homomorphism and inverse to φ . Therefore φ is a ring homomorphism, verifying the statement. $\square$

I believe that when reading through our construction of the polynomial ring, the reader will inevitably be reminded of the same method we used to solve the problem of free abelian groups. Specifically, we previously defined $\bigoplus_A \mathbf{Z}$ as the set of all finitely supported functions from a set A to the integers $\mathbf{Z}$. When A is taken to be the natural numbers $\mathbf{N}$, this becomes $\mathbf{Z}[x]$. Hence $\mathbf{Z}[x]$ naturally satisfies the universal property of free abelian groups: namely, when regarded as groups, for any abelian group G , there exists a unique group homomorphism $\varphi : \mathbf{Z} \rightarrow G$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathbf{Z}[x] & \xrightarrow{\varphi} & G \\ j \uparrow & \nearrow f & \\ \mathbf{N} & & \end{array}$$

where $j : \mathbf{N} \rightarrow \mathbf{Z}[x]$ is given by $j(n) = x^n$. Then $\varphi(x^n) = f(n)$. If we now discuss the case of rings, adding the multiplicative structure of rings, i.e., taking G to be a ring (commutative), then if φ is a ring homomorphism, we must have

$$\varphi(x^n) = \varphi(x)^n = f(1)^n, \quad n = 0, 1, \dots$$

In other words, once we require φ to be a ring homomorphism, we actually only need to know the value of x in S , namely $f(x)$, in order to determine $f(0), f(1), f(2), \dots$. Therefore, defining the values of f on the whole of $\mathbf{N}$ is completely redundant, and may even cause the induced group homomorphism to fail to be a ring homomorphism:

$$\varphi(x^{m+n}) = f(m+n) \neq f(m)f(n) = \varphi(x^m)\varphi(x^n).$$

Since the only truly meaningful information in a function $f : \mathbf{N} \rightarrow S$ (under the condition that $f(m+n) = f(m)f(n)$ holds) is the value $f(1)$, why not be more direct and consider only set-theoretic functions from a singleton set to S and from a singleton set to $\mathbf{Z}[x]$, and then discuss the universal property in that setting? (Moreover, this approach truly allows us to take “arbitrary” f without worrying about whether the equality $f(m+n) = f(m)f(n)$ holds.) That is, we have the following property:

6.1.6 PROPOSITION. *Let $A = \{a\}$. Define $j : A \rightarrow \mathbf{Z}[x]$ sending a to x . For any commutative ring S and set-function $i : A \rightarrow S$, there exists a unique ring homomorphism such that the diagram commutes:*

$$\begin{array}{ccc} \mathbf{Z}[x] & \xrightarrow{\varphi} & S \\ j \uparrow & \nearrow i & \\ A & & \end{array}$$

Proof. Assume $i(a) = r \in S$. Since $\mathbf{Z}$ is initial in the category **Ring**, there exists a unique ring homomorphism $\psi : \mathbf{Z} \rightarrow S$. If the diagram commutes, then $\varphi(x) = \varphi(j(a)) = i(a) = r$. Define $\varphi : \mathbf{Z}[x] \rightarrow S$, $\varphi(z) = \psi(z)$, $z \in \mathbf{Z}$ (These equality holds automatically) and

$$\varphi\left(\sum_{i=0}^n f_i x^i\right) = \sum_{i=0}^n \psi(f_i) s^i,$$

φ is well-defined because the uniqueness of coefficients. Let us prove that φ is a ring homomorphism. First $\varphi(\mathbf{1}) = \psi(1) = 1$ because ψ is a ring homomorphism. Second, if $g = g_0 + g_1 x + \dots + g_m x^m$, then

$$\begin{aligned} \varphi(f + g) &= \varphi\left(\sum (f_i + g_i) x^i\right) = \sum \psi(f_i + g_i) s^i \\ &= \sum (\psi(f_i) + \psi(g_i)) s^i = \sum \psi(f_i) s^i + \sum \psi(g_i) s^i \\ &= \varphi(f) + \varphi(g). \end{aligned}$$

Third, let $fg = \sum_k c_k x^k$ where $c_k = \sum_{i+j=k} f_i g_j$. Then

$$\begin{aligned}\varphi(fg) &= \varphi\left(\sum_k c_k x^k\right) = \sum \psi(c_k) s^k \\ &= \sum \psi\left(\sum f_i g_j\right) s^k = \sum \left(\sum \psi(f_i) \psi(g_j)\right) s^k\end{aligned}$$

On the other hand

$$\varphi(f)\varphi(g) = \left(\sum \psi(f_i) s^i\right) \left(\sum \psi(g_j) s^j\right) = \sum_k \left(\sum_{i+j=k} \psi(f_i) \psi(g_j)\right) s^k.$$

Uniqueness of φ is obvious: if $\theta : \mathbf{Z}[x] \rightarrow S$ is a ring homomorphism such that the diagram commutes, then $\theta(f_0 + f_1 x + \cdots + f_n x^n) = \psi(f_0) + \psi(f_1) s + \cdots + \psi(f_n) s^n$. $\square$

6.1.7 COROLLARY. *For any commutative ring S and homomorphism $\psi : \mathbf{Z} \rightarrow S$ (in fact, this homomorphism is unique.), let s be any element of S , then there exists a unique homomorphism $\varphi : \mathbf{Z}[x] \rightarrow S$ such that $\varphi(x) = s$ and the diagram commutes (in fact, this diagram commutes automatically, independently of φ .) :*

$$\begin{array}{ccc}\mathbf{Z}[x] & \xrightarrow{\varphi} & S \\ \uparrow \iota & \nearrow \psi & \\ \mathbf{Z} & & \end{array}$$

6.1.8 THEOREM. *Let R be a commutative ring. For any commutative ring R and homomorphism $\psi : R \rightarrow S$, let $s_1, \dots, s_n$ be any elements of S , then there exists a unique homomorphism $\varphi : R[x_1] \cdots [x_n] \rightarrow S$ such that $\varphi(x_i) = s_i$ for $i = 1, 2, \dots, n$ and the diagram commutes:*

$$\begin{array}{ccc}R[x_1] \cdots [x_n] & \xrightarrow{\varphi} & S \\ \uparrow \iota & \nearrow \psi & \\ R & & \end{array}$$

(Define a category $\mathbf{C}_R$ whose objects are all the $n+2$ -tuples $(\psi, S, s_1, \dots, s_n)$ where S is a commutative ring and $s_i \in S$ and $\psi : R \rightarrow S$ is a ring homomorphism; a morphism from $(\psi_1, S, s_1, \dots, s_n)$ to $(\psi_2, K, k_1, \dots, k_n)$ is a homomorphism $\zeta : S \rightarrow K$ such that $\zeta\psi_1 = \psi_2$ and $\zeta(s_i) = k_i, i \leq n$. This theorem shows that $(\iota, R[x_1] \cdots [x_n], s_1, \dots, s_n)$ is initial in $\mathbf{C}_R$.)

Proof. The proof is by induction on n . If $n = 1$, the proof is almost identical to that of the above proposition, so we shall not elaborate further here. We have completed the proof of the base step. For the inductive step, define $A = R[x_1] \cdots [x_n]$; the inductive hypothesis gives a homomorphism $\phi : A \rightarrow S$ with $\phi(x_i) = s_i, i \leq n$ and $\phi(r) = \psi(r)$ for all $r \in R$. The base step gives a homomorphism $\Phi : A[x_{n+1}] \rightarrow S$ with $\Phi(x_{n+1}) = s_{n+1}$ and $\Phi(a) = \phi(a)$ for all $a \in A$. The result follows because $R[x_1] \cdots [x_{n+1}], \Phi(x_i) = \phi(x_i) = s_i, i \leq n, \Phi(x_{n+1}) = s_{n+1}$ and $\Phi(r) = \phi(r) = \varphi(r), r \in R$:

$$\begin{array}{ccc}R[x_1] \cdots [x_n][x_{n+1}] & \xrightarrow{\Phi} & S \\ \uparrow \tilde{\iota} & \nearrow \phi & \\ R[x_1] \cdots [x_n] & \nearrow \psi & \\ \uparrow \iota & & \\ R & & \end{array}$$

$\square$

6.1.9 EXAMPLE. In particular, consider the homomorphism $\text{id}_R : R \rightarrow R$, the universal property induces a unique homomorphism $\varphi : R[x_1] \cdots [x_n] \rightarrow R$ for any $r_1, \dots, r_n \in R$ and $\varphi(x_i) = r_i$. Given a polynomial $f = \sum f_{\mathbf{k}} \mathbf{x}^{\mathbf{k}} \in R[x_1] \cdots [x_n]$, it is clear that $\varphi(f) = \sum f_{\mathbf{k}} \mathbf{r}^{\mathbf{k}} \in R$. This homomorphism is called the *evaluation at $\mathbf{r}$* .

6.1.9 THEOREM. *Let R be a commutative ring. Then*

$$R[x_1, \dots, x_n] \cong R[x_1] \cdots [x_n].$$

Proof. The first method is naturally to construct an isomorphism directly, while the second method is to use the universal property to solve the problem: Let $\psi : R \rightarrow S$ be a homomorphism and let $s_1, \dots, s_n \in S$, there exists a homomorphism $\varphi_1 : R[x_1] \cdots [x_n] \rightarrow S$ such that $\varphi_1(r) = \psi(r), r \in R$ and $\varphi_1(x_i) = s_i, i \leq n$ by the theorem above. Let $f = \sum_{(k_1, \dots, k_n)} f_{k_1 \dots k_n} x_1^{k_1} \cdots x_n^{k_n} \in R[x_1, \dots, x_n]$. Since f has finite support, the above sum is finite. Define

$$\varphi(f) := \sum_{(k_1, \dots, k_n)} \psi(f_{k_1 \dots k_n}) s_1^{k_1} \cdots s_n^{k_n}.$$

This is well-defined by the uniqueness of the representation of f in $R[x_1, \dots, x_n]$. Clearly, $\varphi(r) = \psi(r)$ for every $r \in R$, and $\varphi(x_i) = s_i$ for every $i = 1, \dots, n$. One may show that φ is indeed a homomorphism and the uniqueness of φ . $\square$

Under this theorem, we no longer care about what the x_i specifically represent. Therefore, somewhat informally, we may say that they are the same, and moreover: Since $R[x_1, x_2] = R[x_1][x_2]$, $R[x_1, x_2][x_3] = R[x_1][x_2][x_3] = R[x_1, x_2, x_3]$. It is clear that

$$R[x_1, \dots, x_n] = R[x_1, \dots, x_k][x_{k+1}, \dots, x_n] = R[x_{k+1}, \dots, x_n][x_1, \dots, x_k].$$

Integral Domain and Field

The groups $\mathbf{Z}/n\mathbf{Z}$ endowed with the multiplication defined as $[a] \cdot [b] := [ab]$ is a ring. In the special rings $\mathbf{Z}, \mathbf{Q}, \mathbf{R}$, the cancellation of multiplication holds for the nonzero elements (Of course the cancellation of addition is automatic since rings are groups), but the cancellation of multiplication fails in general since cannot “cancel 0” by the proposition above. Worse, in the ring $\mathbf{Z}/6\mathbf{Z}$, for example, $[2]_6 \cdot [4]_6 = [8]_6 = [2]_6 = [2]_6 \cdot [1]_6$ even though $[4]_6 \neq [1]_6$. Therefore, for many familiar rings, they do not truly possess the “cancellation of multiplication”.

6.1.3 DEFINITION. An element a in a ring R is a *left zero divisor* (resp. *right zero divisor*) if there exist elements $b \neq 0$ in R such that $ab = 0$ (resp. $ba = 0$). A *zero divisor* is an element which is both a left and a right zero divisor.

The element 0 is a zero-divisor in all nonzero rings R ; the zero ring $\mathbf{0}$ is the only ring without zero-divisors. Logically speaking, the cancellation law for multiplication does hold in the zero ring, since the condition $b \neq 0$ is false in the zero ring.

6.1.10 PROPOSITION. *In a ring R , $a \in R$ is not a left (resp. right) zero divisor if and only if left (resp. right) multiplication by a is an injective function $R \rightarrow R$. In other words, a is not a left (resp. right) zero divisor if multiplicative left-cancellation (resp. right-cancellation) by the element a holds in R .*

Proof. Assume that a is not the left zero divisor and $ab = ac$ for $b, c \in R$. Then by distributivity, $a(b - c) = ab - ac = 0$, which implies that $b - c = 0$ since a is not a left zero divisor, that is $b = c$. This proves that left multiplication is injective in this case. Conversely, if a is a left zero divisor, then there exists an element $b \neq 0$ such that $ab = 0 = a \cdot 0$, which implies the left multiplication is not injective! $\square$

Although cancellation for multiplication does not hold completely in all rings (meaning for arbitrary elements), there are nevertheless certain rings in which the cancellation law holds almost entirely: for example, $\mathbf{Z}$, $\mathbf{Q}$, and $\mathbf{R}$ have only the trivial zero divisor 0, unlike $\mathbf{Z}/6\mathbf{Z}$, where besides the zero element, $[2]$ and $[3]$ are also zero divisors. Such rings deserve a special name:

6.1.4 DEFINITION. An *integral domain* is a commutative ring R ($1 \neq 0$) such that

$$ab = 0 \rightarrow a = 0 \text{ or } b = 0, \quad \forall a, b \in R.$$

A zero divisor that is nonzero is called a *nontrivial zero divisor*. So that, integral domains are those commutative rings ($1 \neq 0$) with no nontrivial zero divisors. By proposition above, multiplicative cancellation by nonzero elements holds in integral domains. The rings $\mathbf{Z}$, $\mathbf{Q}$, $\mathbf{R}$ are all integral domains. As we have seen, some $\mathbf{Z}/n\mathbf{Z}$ are not integral domains.

Some people may have a mistaken understanding of cancellation for multiplication (the cancellation law for addition in groups was introduced in Section 3.1 when we discussed groups). For example, from $ab = ac$ one concludes $b = c$. The proof we give is: $ab - ac = a(b - c) = 0$ implies $b - c = 0$, hence $b = c$. But if one understands it differently, one might multiply both sides of $ab = ac$ on the left by a^{-1} , and then naturally obtain $b = c$ (this is exactly the step we used to derive the cancellation law for addition in groups). The problem is that we have not said that a^{-1} is the inverse of a ; in fact, we have not even said that a has an inverse, since in a ring R , the structure $\langle R, \cdot \rangle$ is only a monoid, not a group. For this reason, we may also introduce the notion of a *unit*:

6.1.5 DEFINITION. An element u of a ring R is a *left unit* (resp. *right unit*) if there exist some $v \in R$ such that $uv = 1$ (resp. $vu = 1$). A *unit* is an element which is both a left and a right unit.

6.1.11 PROPOSITION. In a ring R :

1. u is a left (resp. right) unit if and only if left (resp. right) multiplication by u is a surjective function from R onto R ;
2. if u is a left (resp. right) unit, then right (resp. left) multiplication by u is an injective function from R into R ; that is, u is not a right (resp. left) zero divisor;

Proof. (1). Denote by $\rho_u : R \rightarrow R$ left multiplication by u , so that $\rho_u(r) = ur$. If u is a left unit, then there exists $v \in R$ such that $uv = 1$, then

$$\rho_u \circ \rho_v(r) = \rho(vr) = u(vr) = r.$$

That is, ρ_v is a right inverse to ρ_u , and therefore ρ_u is surjective. Conversely, if ρ_u is surjective, then there exists a v such that $1 = \rho_u(v) = uv$, so that u is a left unit.

(2). Denote by $\lambda_u : R \rightarrow R$ right multiplication by u , so that $\lambda_u(r) = ru$. Assume u is a left unit, and let v be such that $uv = 1$, then

$$\lambda_v \circ \lambda_u(r) = \lambda_v(ru) = (ru)v = r.$$

That is, λ_v is a left inverse to λ_u , and therefore λ_u is injective, which implies that u is not a right zero divisor by proposition 6.2.1. $\square$

Every ring has a unit, including the zero ring (Of course, this remark is somewhat superfluous.). However, even in familiar rings, the units are often quite scarce. For example, in $\mathbf{Z}$, the only units are 1 and -1 , whereas in $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{C}$, every nonzero element is a unit. For such special rings, we may also give them special names.

6.1.6 DEFINITION. A *division ring* is a ring in which every nonzero element is a unit. A *field* is a nonzero commutative division ring.

Since every unit is not a zero divisor by the proposition above, a field is an integral domain. However, the converse does not hold; for example, as is well known, $\mathbf{Z}$ is an integral domain, but its only units are ± 1 , so it is not a field. Fortunately, in special cases, this does hold.

6.1.12 THEOREM. *If R is a finite integral domain, then R is a field.*

Proof. Let a be a nonzero element of R . By the cancellation law, the function $r \mapsto ar$ is an injective function. Since R is finite, this function is also surjective. In particular, there is some $b \in R$ such that $ab = 1$, i.e., a is a unit in R . Since a was an arbitrary element, R is a field. $\square$

There is another wonderful theorem below, known as *Wedderburn's little theorem*, but with the material covered so far in this note book, it is still difficult to complete its proof. A widely known proof is due to *Ernst Witt*; one may refer to: https://pub.math.leidenuniv.nl/~finkelnergh//seminarium/eindige_delingsring_is_lichaam.pdf. If I have the energy to improve these notes later, I will add the relevant content in an appropriate place in a subsequent chapter.

6.1.13 THEOREM. *If R is a finite division ring, then R is a field.*

6.1.10 EXAMPLE ($\mathbf{Z}/n\mathbf{Z}$). We denote the group of units of the ring $\mathbf{Z}/n\mathbf{Z}$ by $(\mathbf{Z}/n\mathbf{Z})^\times$. A class $[m]_n$ is a unit if and only if right multiplication by $[m]_n$ is surjective, if and only if the function $[a]_n \mapsto [a]_n[m]_n = [am]_n = a \cdot [m]_n$ is surjective, if and only if $[m]_n$ generates $\langle \mathbf{Z}/n\mathbf{Z}, + \rangle$. Now, we show that $[m]_n$ generates $\mathbf{Z}/n\mathbf{Z}$ if and only if $\gcd(m, n) = 1$:

Proof. If $[m]_n$ generates $\langle \mathbf{Z}/n\mathbf{Z} \rangle$, then $[1]_n = [km]_n$ for some $k \in \mathbf{Z}$, that is, $km - 1 = qn$ for some integer q . By the Bézout's theorem, $\gcd(m, n) = 1$. Conversely, if $\gcd(m, n) = 1$, then there exist $x, y \in \mathbf{Z}$ such that $mx + ny = 1$, and $mx \equiv 1 \pmod{n}$, that is $[1]_n = [mx]_n = x \cdot [m]_n$. Since $[1]_n$ may generate the group $\langle \mathbf{Z}/n\mathbf{Z}, + \rangle$, so is $[m]_n$. $\square$

In particular, those n for which all the nonzero elements of $\mathbf{Z}/n\mathbf{Z}$ are units (that is, for which $\mathbf{Z}/n\mathbf{Z}$ is a field) are precisely those $n \in \mathbf{Z}$ for which $\gcd(m, n) = 1$ for all m that are not multiples of n ; this is the case if and only if n is a prime.

6.1.11 EXAMPLE. If R is a ring of all functions from the closed interval $[0, 1]$ to $\mathbf{R}$ under the functional addition and multiplication, then the units of R are the functions that are not zero at any point. If f is not a unit and not zero then f is a zero divisor because if we define

$$g(x) = \begin{cases} 0, & f(x) \neq 0 \\ 1, & f(x) = 0 \end{cases}$$

then $g \neq \mathbf{0}$ but $fg(x) = f(x)g(x) = 0$ for all x . This shows that every element of R is either a unit or a zero divisor.

6.1.12 EXAMPLE. Let D be a rational number that is not a perfect square in $\mathbf{Q}$ and define $\mathbf{Q}(\sqrt{D}) = \{a + b\sqrt{D} : a, b \in \mathbf{Q}\}$ as a subset of $\mathbf{C}$. This set is clearly a ring under the usual addition and multiplication. It is easy to show that the assumption that D is not a square implies that every element of $\mathbf{Q}(\sqrt{D})$ may be written uniquely in the form $a + b\sqrt{D}$. This assumption also implies that if a and b are not both 0, then $a^2 - Db^2 \neq 0$, and since $(a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2$ it follows that if $a + b\sqrt{D} \neq 0$, then $\frac{a - b\sqrt{D}}{a^2 - Db^2}$ is the inverse of $a + b\sqrt{D}$ in $\mathbf{Q}(\sqrt{D})$. This shows that every nonzero element in this commutative ring is a unit, i.e., $\mathbf{Q}(\sqrt{D})$ is a field.

Let R be a commutative ring. Then it is clear that $R[x_1, \dots, x_n]$ is a commutative ring. Then, if R is an integral domain, is $R[x_1, \dots, x_n]$ also an integral domain? What about the case when R is a field?

6.1.14 PROPOSITION. *Let R be an integral domain, and let $f, g \neq \mathbf{0} \in R[x]$. Then*

1. $fg \neq \mathbf{0}$ and $\deg fg = \deg f + \deg g$;
2. $R[x]$ is an integral domain.

Proof. (1). Let $f = (f_0, \dots, f_n, 0, \dots)$ and $g = (g_0, \dots, g_m, 0, \dots)$ in which $f_n, g_m \neq 0$. For any $k \geq 1$, $(fg)_k = \sum_{i=0}^k f_i g_{k-i}$. Clearly, each term in $\sum_{i=0}^{m+n+\ell} f_i g_{m+n+\ell-i}$ is 0. Since R is an integral domain, $f_n, g_m \neq 0$ implies $f_n g_m \neq 0$. (2). This follows from (1) because the product of two nonzero polynomials is now nonzero. $\square$

Moreover, if f is a unit in $R[x]$; that is, there exists $g \neq \mathbf{0}$ such that $fg = \mathbf{1} = (1, 0, 0, \dots)$. Then $\deg f + \deg g = \deg \mathbf{1} = 0$, so both $\deg f$ and $\deg g$ are 0, that is, $f = (f_0, 0, \dots), g = (g_0, 0, \dots)$ in which $f_0, g_0 \neq 0 \in R$. Since R is an integral domain, $f_0, g_0 \neq 0$ are units. Therefore, under the homomorphism $R \rightarrow R[x], r \mapsto (r, 0, \dots)$, one may say that “the units of $R[x]$ are the units of R ”.

As for the second question, if R is a field, then $R[x]$ is clearly not a field. In fact, for any nonzero ring R , $R[x]$ is certainly not a field, because in any case x is not a unit. Otherwise, suppose there exists $f = (f_0, f_1, \dots)$ such that $xf(x) = 1$. Then a simple calculation yields $1 = 0 \in R$, which implies that R is the zero ring, a contradiction.

6.2 Quotient Rings

Subrings and Ideals

6.2.1 DEFINITION. A *subring* of a ring R is a set S of R that preserves the structure. Equivalently, the inclusion function $S \hookrightarrow R$ is a ring homomorphism.

It is immediately checked that $S \subset R$ is a subring if it is a subgroup of $\langle R, + \rangle$, it is a submonoid of $\langle R, \cdot \rangle$. As a nonexample, the zero ring $\mathbf{0}$ is not a subring of a nonzero ring R because it does not contain 1_R .

6.2.1 PROPOSITION. Let $\varphi : R \rightarrow S$ be a ring homomorphism. If A is a subring of R , then $\varphi(A) = \{\varphi(a) : a \in A\}$ is a subring of S . If B is a subring of S , then $\varphi^{-1}(B) = \{b \in R : \varphi(b) \in B\}$ is a subring of R .

Proof. Omitted. $\square$

Recall how we used that proposition to obtain two special subgroups. In the categorical setting of groups, considering R, S (in additive notation) and a homomorphism $\varphi : R \rightarrow S$, the trivial subgroup $\{0\}$ is always a subgroup of S , and the kernel $\ker \varphi$ defined accordingly is always a subgroup of R . We would like to use the same proposition to obtain an analogous notion of kernel in **Ring** to that in **Grp**. However, in the ring setting, $\{0\}$ is not a subring of S (since subrings are required to contain the multiplicative identity), so there is no notion of a trivial subring analogous to the trivial subgroup $\{0\}$. Moreover, the kernel $\ker \varphi$ obtained in the group homomorphism sense is indeed not a subring of R . Even if we try to force the definition by using the universal property of kernels in **Grp** to define kernels of ring homomorphisms in **Ring**, we find ourselves unable to proceed, because there is no zero morphism!

This prompts us to re-examine the concept of kernel. In the next chapter, we shall begin the second part of category theory, with the main goal of introducing a more general notion of kernel, namely the **equaliser**. Before that, let us set aside this question and focus on learning ring theory.

6.2.2 DEFINITION. Let $\varphi : R \rightarrow S$ be a ring homomorphism. We define

- $\ker \varphi = \{r \in R : \varphi(r) = 0_S\}$ the *kernel* of φ ;
- $\operatorname{im} \varphi = \{\varphi(r) : r \in R\}$ the *image* of φ .

6.2.2 PROPOSITION. *Let $\varphi : R \rightarrow S$ be a ring homomorphism, then the following are equivalent:*

1. φ is a monomorphism;
2. $\ker \varphi = \{0\}$;
3. φ is an injection in function sense.

Proof. The proofs of (2) $\rightarrow$ (3) and (3) $\rightarrow$ (1) are quite trivial, so we omit them directly. (1) $\rightarrow$ (2): assume that $\varphi : R \rightarrow S$ is a monomorphism and $r \in \ker \varphi$. Applying the universal property of polynomial ring $\mathbf{Z}[x]$, we obtain unique ring homomorphism $\text{ev}_r : \mathbf{Z}[x] \rightarrow R$ such that $\text{ev}_r(x) = r$ and $\text{ev}_0 : \mathbf{Z}[x] \rightarrow R$ such that $\text{ev}_0(x) = 0$. Consider the parallel ring homomorphisms:

$$\begin{array}{ccccc} \mathbf{Z}[x] & \xrightarrow{\text{ev}_r} & R & \xrightarrow{\varphi} & S \\ & \searrow \text{ev}_0 & \uparrow \psi & \nearrow & \\ \mathbf{Z} & \xrightarrow{\iota} & \mathbf{Z} & & \end{array}$$

Since $\varphi(r) = 0 = \varphi(0)$, the two compositions $\varphi \circ \text{ev}_r, \varphi \circ \text{ev}_0$ agree by the universal property of $\mathbf{Z}[x]$ (they agree on $\mathbf{Z}$ and they agree on x). Hence $\text{ev}_r = \text{ev}_0$ since φ is a monomorphism. Therefore,

$$r = \text{ev}_r(x) = \text{ev}_0(x) = 0.$$

□

This proposition may induce the reader to believe that “a ring homomorphism is an epimorphism if and only if it is a surjective function”. A surjective homomorphism is certainly an epimorphism in **Ring**, since it is already an epimorphism in **Set**; but unlike in **Set**, **Grp**, epimorphisms need not be surjective in **Ring**. Indeed, consider the inclusion homomorphism of rings

$$\iota : \mathbf{Z} \hookrightarrow \mathbf{Q},$$

ι is not surjective, and hence it is not an epimorphism in **Set** or **Grp**; but it is an epimorphism of rings ! Indeed, for any ring R and homomorphisms α_1, α_2 . Assume that $\alpha_1 \circ \iota = \alpha_2 \circ \iota$, i.e., α_1, α_2 are parallel ring homomorphisms

$$\mathbf{Z} \xrightarrow{\iota} \mathbf{Q} \xrightarrow[\alpha_2]{\alpha_1} R$$

α_1, α_2 agree on $\mathbf{Z}$ since $\mathbf{Z}$ is initial in **Ring**, say $\alpha = \alpha_1|_{\mathbf{Z}} = \alpha_2|_{\mathbf{Z}}$. They must agree on $\mathbf{Q}$: because for any $p, q \in \mathbf{Z}, q \neq 0$,

$$\alpha_i \left(\frac{p}{q} \right) = \alpha_i(pq^{-1}) = \alpha_i(p)\alpha_i(q)^{-1} = \alpha(p)\alpha(q)^{-1}, \quad i = 1, 2,$$

Thus, in **Ring**, a homomorphism may well be both monomorphism and an epimorphism without being an isomorphism !

6.2.3 DEFINITION. Let R be a ring. A subgroup I of $\langle R, + \rangle$ is a *left ideal* (resp. *right ideal*) of R if $rI \subset I$ (resp. $Ir \subset I$) for all $r \in R$; that is (a two-side ideal is a subgroup I which is both a left and right ideal),

$$\forall r \in R, a \in I, \quad ra \in I \text{ (resp. } ar \in I).$$

If the ring R is commutative, then we no longer need to distinguish between left and right ideals; they are uniformly called ideals.

The ring R itself and $\{0\}$ are always two-side ideals in a ring R . An (left, right, two-side) ideal $I \neq R$ is *proper* if $I \neq R$. Two-sided ideals are close to being subrings: they are subgroups, and they are closed with respect to multiplication. But the only two-sided ideal of a ring R containing 1_R is R itself: this is an immediate consequence of the “absorption properties” stated in the definition. Thus, two-sided ideals are in general not subrings; they are subrings.

6.2.1 EXAMPLE. Let $\varphi : R \rightarrow S$ be a homomorphism, then $\ker \varphi$ is a two-sided ideal of R ; and $\operatorname{im} \varphi$ need not be an (left, right) ideal of S . Moreover, if R and S are not zero rings, then $\ker \varphi$ is proper.

Proof. $\ker \varphi$ is an additive subgroup of R ; moreover, if $a \in \ker \varphi$ and $r \in R$, then $\varphi(ar) = \varphi(a)\varphi(r) = 0 \cdot \varphi(r) = 0$. Similarly $\varphi(ra) = 0$. Hence $\ker \varphi$ is an ideal. If S is not the zero ring, i.e., $1 \neq 0$, then $\ker \varphi$ is a proper ideal in R (the identity $1 \notin \ker \varphi$). Consider the homomorphism $\iota : \mathbf{Z} \rightarrow \mathbf{Q}$, $\operatorname{im} \iota = \mathbf{Z}$ however $1/2 \cdot 1 = 1/2 \notin \mathbf{Z}$. $\square$

6.2.2 EXAMPLE.

1. If a left (or right) ideal I of a ring R contains 1 , then $I = R$. Indeed, if I contains a unit u , then $I = R$, since then I contains $u^{-1}u = 1$.
2. If R is a field, then the only ideals I in R are $\{0\}$ and R itself: if $I \neq \{0\}$, it contains some nonzero element, and every nonzero element in a field is a unit. Conversely, assume that R is a nonzero commutative ring whose only ideals are R itself and $\{0\}$. If $a \in R$ and $a \neq 0$, then $Ra := \{ra : r \in R\}$ is a nonzero ideal, and so $\{ra : r \in R\} = R$; hence $1 = ra$ for some $r \in R$; that is a is a unit, and so R is a field.

Let $a \in R$ be any element of a ring R . Then the subset $I = Ra$ of R is a left ideal of R . Indeed, for all $r \in R$, we have $rI = rRa = \{rr'a : r' \in R\} \subset Ra$. Similarly, $aR := \{ar : r \in R\}$ is a right ideal. In the commutative case, these two subsets coincide and are denoted $(a) := Ra = aR$.

In general, let RA denote the set of all finite sums of elements of the form ra with $r \in R$ and $a \in A$, i.e., $RA = \{r_1a_1 + \cdots + r_na_n : r_i \in R, a_i \in A, n \in \mathbf{N}\}$. Similarly, $AR := \{a_1r_1 + \cdots + a_nr_n : a_i \in A, r_i \in R, n \in \mathbf{N}\}$. It is clear that RA is a left ideal and AR is a right ideal. In the commutative case, these two subsets coincide and are denoted $(A) := RA = AR$.

6.2.4 DEFINITION. Let R be a ring and A be a subset of R . The *left ideal* (resp. *right ideal*) generated by the subset A of R is the left ideal RA (resp. AR).

6.2.3 PROPOSITION. Let R be a ring, and let $\{I_\alpha\}_{\alpha \in \Lambda}$ be a set of left ideals (resp. right ideals) of R . Then $\bigcap_{\alpha \in \Lambda} I_\alpha$ is also a left ideal (resp. right ideal).

Proof. First, we show that $I = \bigcap_{\alpha \in \Lambda} I_\alpha$ is a subgroup of R . Since $0 \in I_\alpha$ for all $\alpha \in \Lambda$, $0 \in I$. Take $a, b \in I$, then $a, b \in I_\alpha$ for all α , hence $a - b \in I_\alpha$ for all $\alpha \in \Lambda$; that is $a - b \in \bigcap_{\alpha \in \Lambda} I_\alpha = I$. Now we show that I has the left absorption property: omitted. $\square$

6.2.4 PROPOSITION. $RA = \bigcap_{A \subset I, I \text{ a left ideal}} I$, $AR = \bigcap_{A \subset I, I \text{ a right ideal}} I$.

Proof. Omitted. $\square$

Just as in our discussion of cyclic subgroups, we are interested in (left, right, two-sided) ideals generated by a single element or by finitely many elements. An ideal generated by a single element is called the principal (left, right, two-sided) ideal generated by a . In the commutative case, it is simply called a *principal ideal*.

6.2.3 EXAMPLE. We have shown that all the subgroups of $\mathbf{Z}$ are of the form $n\mathbf{Z}$ for a nonnegative integer n . One may show that all the subgroups are in fact ideals of the ring $\langle \mathbf{Z}, +, \cdot \rangle$. Moreover, it is clear that every ideal $n\mathbf{Z} = (n) = (-n)$, so that every ideal

of $\mathbf{Z}$ is principal. For positive integers n, m , $n\mathbf{Z} \subset m\mathbf{Z}$ if and only if m divides n in $\mathbf{Z}$ ($m \mid n \rightarrow n = mq, nk = m(qk) \in m\mathbf{Z}$; conversely, $n \in m\mathbf{Z} \rightarrow n = mq$). Furthermore, the ideal (n, m) generated by two nonzero integers n and m must be principal and hence

$$(m, n) = (d)$$

for some positive integer. This integer is manifestly the gcd of m and n because $m \in (d)$ and $n \in (d)$, then $d \mid m$ and $d \mid n$; $d \in (m, n)$ implies $d = um + vn$, hence $c \mid m, c \mid n$ implies $c \mid d$. In particular, n and m are relatively prime if and only if $(m, n) = (1)$.

6.2.4 EXAMPLE. We show that the ideal $(2, x)$ generated by 2 and x in $\mathbf{Z}[x]$ is not a principal ideal. Observe that $(2, x) = \{2p(x) + xq(x) : p(x), q(x) \in \mathbf{Z}[x]\}$ and so this ideal consists precisely of the polynomials with integer whose constant term is even; in particular, this is a proper ideal.

Assume that $(2, x) = (a(x))$ for some $a(x) \in \mathbf{Z}[x]$. Since $2 \in (a(x))$, there must be some $p(x)$ such that $2 = p(x)a(x)$. Then $0 = \deg 2 = \deg p(x)a(x) = \deg p(x) + \deg a(x)$ since $\mathbf{Z}$ is an integral domain and $2 \neq 0$, hence $p(x)$ and $a(x)$ must be constant polynomials; and $p(x), a(x) \in \{\pm 1, \pm 2\}$. If $a(x) = \pm 1$, then $(a(x)) = \mathbf{Z}[x]$, contradicting the fact $(a(x))$ is proper. The only possibility is $a(x) = \pm 2$. But now, $x \in (a(x)) = (2) = (-2)$ and so $x = 2q(x)$ for some $q(x) \in \mathbf{Z}[x]$, which is impossible. This contradiction proves that $(2, x)$ is not principal.

Quotient Rings

Let I be a subgroup of the abelian group $\langle R, + \rangle$ of a ring R . Subgroups of abelian groups are automatically normal, so we have a quotient group R/I , whose elements are the cosets of I : $r + I$. Further, we have a surjective group homomorphism

$$\pi : R \rightarrow R/I, \quad r \mapsto r + I.$$

Just as in the following universal property of quotient groups, the above π satisfies that universal property. Now when we consider defining π in the category **Ring**, we naturally require π to be a ring homomorphism. In this case, assuming π is a ring homomorphism, we are forced to include multiplications such as:

$$(a + I) \cdot (b + I) = \varphi(a) \cdot \varphi(b) = \varphi(ab) = ab + I.$$

6.2.1 REMARK. Let H be a normal subgroup of a group G . Then for every group homomorphism $\varphi : G \rightarrow G'$ such that $H \subset \ker \varphi$, there exists a unique group homomorphism $\tilde{\varphi} : G/H \rightarrow G'$ such that the diagram commutes:

$$\begin{array}{ccc} G/H & \xrightarrow{\tilde{\varphi}} & G' \\ \uparrow \pi & \nearrow \varphi & \\ G & & \end{array}$$

Therefore, we are forced to define the multiplication in R/I as follows:

$$(a + I)(b + I) := ab + I.$$

If this multiplication is well-defined, then it naturally becomes a ring, with associativity inherited from R itself and multiplicative identity $1 + I$; moreover, π then becomes a ring homomorphism. Unfortunately, this definition is not well-defined in general, as shown by the following example.

6.2.5 EXAMPLE. Take $\mathbf{Z}$ as a subgroup of $\mathbf{Q}$, then $0 + \mathbf{Z} = 1 + \mathbf{Z} = \mathbf{Z} \in \mathbf{Q}/\mathbf{Z}$, and $1/2 + \mathbf{Z}$ is another coset. However

$$(0 + \mathbf{Z}) \cdot (1/2 + \mathbf{Z}) = 0 \cdot 1/2 + \mathbf{Z} = \mathbf{Z} \neq 1/2 + \mathbf{Z} = 1 \cdot 1/2 + \mathbf{Z} = (1 + \mathbf{Z}) \cdot (1/2 + \mathbf{Z}).$$

Now, let us add the condition that this multiplication is also well-defined; that is, $(a_1 + I)(b_1 + I) = (a_2 + I)(b_2 + I)$ if $a_1 + I = a_2 + I, b_1 + I = b_2 + I$, and see what happens in that case: if $a \in I$, then $a + I = I$, hence for any $r \in R$,

$$I = (0 + I)(r + I) = (a + I)(r + I) = ar + I,$$

that is, $ar \in I$ for any $r \in R, a \in I$, which implies I is a right ideal. Similarly, I is a left ideal. Hence I is a two-sided ideal. Conversely, if I is a two-sided ideal of R , we show that the multiplication is well-defined: if $a_1 + I = a_2 + I, b_1 + I = b_2 + I$, then $a_2 - a_1 \in I, b_2 - b_1 \in I$, and

$$a_2b_2 - a_1b_1 = a_2b_2 - a_2b_1 + a_2b_1 - a_1b_1 = a_2(b_2 - b_1) + (a_2 - a_1)b_1 \in I.$$

This says precisely that $a_1b_1 + I = a_2b_2 + I$, proving that the operation is well-defined. Summarizing, we have verified that R/I is a ring, in such a way that the canonical projection $\pi : R \rightarrow R/I$ is a ring homomorphism, if and only if I is a two-sided ideal of R .

6.2.5 PROPOSITION. *Let I be a two-sided ideal of a ring R . Then for any ring homomorphism $\varphi : R \rightarrow S$ such that $I \subset \ker \varphi$, there exists a unique ring homomorphism $\tilde{\varphi} : R/I \rightarrow S$ such that the diagram commutes:*

$$\begin{array}{ccc} R/I & \xrightarrow{\tilde{\varphi}} & S \\ \uparrow \pi & \nearrow \varphi & \\ R & & \end{array}$$

Proof. Omitted. □

By a simple comparison, the description of this proposition is exactly the same as the description we use to define quotient objects in concrete categories. Although it may be hard to accept, we still call R/I a *quotient ring* of R .

6.2.6 THEOREM. *If $\varphi : R \rightarrow S$ is a ring homomorphism, then*

$$R/\ker \varphi \cong \text{im} \varphi;$$

in fact, there is a unique isomorphism $\tilde{\varphi} : R/\ker \varphi \rightarrow \text{im} \varphi$ such that $\varphi = \iota \circ \tilde{\varphi} \circ \pi$ where $\iota : \text{im} \varphi \rightarrow S$ is the inclusion homomorphism and $\pi : R \rightarrow R/\ker \varphi$ is the canonical projection:

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ \downarrow \pi & & \uparrow \iota \\ R/\ker \varphi & \xrightarrow{\tilde{\varphi}} & \text{im} \varphi \end{array}$$

Proof. By the first isomorphism theorem, there is an isomorphism of abelian groups $\tilde{\varphi} : R/\ker \varphi \rightarrow \text{im} \varphi$ unique such that $\varphi = \iota \circ \tilde{\varphi} \circ \pi$; equivalently, $\tilde{\varphi}(a + \ker \varphi) = \varphi(a)$ for all $a \in R$. One may show that $\tilde{\varphi}$ is indeed a ring homomorphism. □

6.2.6 EXAMPLE. Let R be a commutative ring. We show that $R[x]/(x) \cong R$. Consider the evaluation at 0 $\varphi_0 : R[x] \rightarrow R$, then $\ker \varphi_0 = \{\sum f_i x^i \in R[x] : \sum f_i 0^i = 0\} = \{\sum f_i x^i \in R[x] : f_0 = 0\} = (x)$. By the first isomorphism theorem, $R[x]/(x) \cong R$.

When we studied quotient groups earlier, we had a very useful theorem, the correspondence theorem, which reveals a bijective correspondence between subgroups or normal subgroups of a group and those of its quotient group. Similarly, ideals of a ring have the same conclusion. Listing it separately might not be interesting, so we make a generalization to show that any surjective homomorphism also has an analogous conclusion.

Observe that for any ring homomorphism $\varphi : R \rightarrow S$, there is a corresponding preimage map between ideals (one may show that $\varphi^{-1}(I_S)$ is a two-sided ideal of R):

$$\mathcal{S} = \{\text{two-sided ideals of } R_2\} \rightarrow \mathcal{R} = \{\text{two-sided ideals of } R_1\}, \\ I_S \mapsto \varphi^{-1}(I_S).$$

If we take φ to be the inclusion map of a subring $R_1 \hookrightarrow R_2$, then this becomes $I_2 \mapsto I_2 \cap R_1$. Now consider the other extreme, namely the case of a quotient homomorphism or a surjective homomorphism

6.2.7 PROPOSITION. *Let $\varphi : R \rightarrow S$ be a surjective homomorphism, and let $\mathcal{R}_\varphi$ be the set of all two-sided ideals of R which contains $\ker \varphi$; and let $\mathcal{S}$ be the set of all two-sided ideals of S . Then the function $\tilde{\varphi}$ is a bijection*

$$\tilde{\varphi} : \mathcal{R}_\varphi \rightarrow \mathcal{S}, \quad I_R \mapsto \varphi(I_R),$$

Proof. Let $I_R \in \mathcal{R}_\varphi$, i.e., $I_R \subset R$ and $I_R \supset \ker \varphi$. We claim $\varphi(I_R)$ is a two-sided ideal of S . Since I_R is an additive subgroup of R and φ is a group homomorphism, $\varphi(I_R)$ is an additive subgroup of S . For any $s \in S$ and $y \in \varphi(I_R)$, choose $r \in R$ with $\varphi(r) = s$ (by surjectivity) and $x \in I_R$ with $\varphi(x) = y$. Then

$$sy = \varphi(r)\varphi(x) = \varphi(rx) \in \varphi(I_R), \\ ys = \varphi(x)\varphi(r) = \varphi(xr) \in \varphi(I_R),$$

Thus $\varphi(I_R)$ is a two-sided ideal of S , so $\tilde{\varphi}$ is well-defined. Let $I_S \in \mathcal{S}$, i.e., $I_S \subset S$. Consider $I_R := \varphi^{-1}(I_S)$. By the standard preimage property, I_R is a two-sided ideal of R . Also,

$$\ker \varphi = \varphi^{-1}(0) \subset \varphi^{-1}(I_S) = I_R,$$

since $0 \in I_S$. Hence $I_R \in \mathcal{R}_\varphi$. Therefore $\tilde{\varphi}$ is surjective. Let $I_1, I_2 \in \mathcal{R}_\varphi$ and suppose $\tilde{\varphi}(I_1) = \tilde{\varphi}(I_2)$, i.e., $\varphi(I_1) = \varphi(I_2)$. We prove $I_1 = I_2$.

Take any $x \in I_1$. Then $\varphi(x) \in \varphi(I_1) = \varphi(I_2)$, so there exists $y \in I_2$ with $\varphi(x) = \varphi(y)$. Thus $\varphi(x - y) = 0$, so $x - y \in \ker \varphi$. Since $\ker \varphi \subset I_2$, we get $x - y \in I_2$, and therefore $x = y + (x - y) \in I_2$. Hence $I_1 \subset I_2$. Similarly, $I_2 \subset I_1$. Thus $I_1 = I_2$, so $\tilde{\varphi}$ is injective. $\square$

Moreover, consider the surjective composition homomorphism $\psi : R \xrightarrow{\varphi} S \xrightarrow{\pi} S/I_S$, $\ker \psi = \{r \in R : \psi(r) = 0_{S/I_S}\} = \{r \in R : \varphi(r) \in I_S\} = \varphi^{-1}(I_S)$. By the first isomorphism theorem, $R/\varphi^{-1}(I_S) \cong S/I_S$. We show that $\mathbf{Z}[x]/(\mathbf{2}, x) \cong \mathbf{Z}/2\mathbf{Z}$: consider the composition of evaluation at 0 and the canonical projection $\mathbf{Z}[x] \rightarrow \mathbf{Z} \rightarrow \mathbf{Z}/(2)$, then

$$\frac{\mathbf{Z}[x]}{(\mathbf{2}, x)} = \frac{\mathbf{Z}[x]}{\varphi_0^{-1}((2))} \cong \mathbf{Z}/(2) = \mathbf{Z}/2\mathbf{Z}.$$

6.2.8 THEOREM. *Let I be an ideal of a ring, and let J be an ideal of R containing I . Then J/I is an ideal of R/I , and*

$$(R/I)/(J/I) \cong R/J.$$

Proof. Consider the canonical projection $\pi : R \rightarrow R/J$. Then $\ker \pi = J$. Since $I \subset J = \ker \pi$, the universal property of quotient ring induce a unique ring homomorphism $\varphi : R/I \rightarrow R/J$ and $\varphi(r + I) = r + J$. It is clear that φ is surjective. Since $\ker \varphi = \{r + I : \varphi(r + I) = J\} = \{r + I : r \in J\} = J/I$, the first isomorphism implies $(R/I)/(J/I) \cong R/J$. $\square$

6.2.7 EXAMPLE. Let R be a commutative ring, and let $a, b \in R$. Consider the principal ideal (a) of R . Denote by $\bar{b}$ the class of b in $R/(a)$, i.e., $\bar{b} = b + (a)$. Then $(\bar{b}) = (a, b)/(a) \subset R/(a)$: indeed, $(\bar{b}) = \{rb + (a) : r \in R\}$ and $(a, b)/(a) = \{ra + sb + (a) : r, s \in R\} = \{sb + (a) : s \in R\}$. By the theorem above,

$$(R/(a))/(\bar{b}) = \frac{R/(a)}{(a, b)/(a)} \cong R/(a, b).$$

6.2.9 PROPOSITION (Division Algorithm). *If k is a field and $f, g \in k[x]$ with $f \neq 0$, then there exist unique polynomials $q, r \in k[x]$ such that*

$$g = qf + r,$$

such that either $r = 0$ or $\deg r < \deg f$ (We also assign to the zero polynomial the so-called degree $-\infty$, so that $r = 0$ naturally satisfies $\deg r < \deg f$).

Proof. We prove the existence of such q and r . If $g = 0$, define $q = 0$ and $r = 0$; if f is a nonzero constant s_0 , then it is a unit, and we can get $q = s_0^{-1}g$ and $r = 0$. Thus, we may assume that $\deg g$ is defined and that $\deg f > 0$. Let

$$f = s_n x^n + \cdots + s_0, \quad g = t_m x^m + \cdots + t_0.$$

The last normalizing condition: we may assume that $\deg g \geq \deg f$; that is $m \geq n$; otherwise, we may set $q = 0$ and $r = g$.

We prove that q and r exist by induction on $m = \deg g \geq 0$. For the base step $m = 0$, we have $g = t_0$; set $q = 0$ and $r = g$. Note that $\deg r = \deg g = 0 < \deg f$ for f is not constant. For the inductive step, define

$$f = g - t_m s_n^{-1} x^{m-n} f.$$

Notice that $\deg g < \deg g$. Now $g = t_m s_n^{-1} x^{m-n} f + h$. If $h = 0$, we are done. If $h \neq 0$, then $\deg h < \deg g$, and the inductive hypothesis gives q' and r with $h = q'f + r$, where $\deg r < \deg f$. In the latter case,

$$g = (q' + t_m s_n^{-1} x^{m-n})f + r.$$

To prove the uniqueness of q and r , assume that $g = q'f + r'$ whose $\deg r' < \deg f$. Then $(q - q')f = r' - r$. If $r' \neq r$, then each side has a degree. Since $k[x]$ is an integral domain, $\deg((q - q')f) = \deg(q - q') + \deg f \geq \deg f$, while $\deg(r' - r) \leq \max\{\deg r', \deg r\} < \deg f$, a contradiction. Hence $r' = r$ and $(q - q')f = 0$. As $f \neq 0$, it follows that $q - q' = 0$ and $q = q'$. $\square$

The hypothesis that k is a field is much too strong; the existence and uniqueness of quotient and remainder holds in $R[x]$ for any commutative ring R as long as the leading “coefficient” of f is a unit in R (in this case, f is not a zero divisor of $R[x]$).

6.2.10 PROPOSITION. *Let R be a commutative ring, let f be a polynomial whose leading coefficient is a unit (in particular, f is a monic polynomial), and let g be any polynomial in $R[x]$. There exist unique polynomials q and r in $R[x]$ such that*

$$g(x) = f(x)q(x) + r(x), \quad \deg r < \deg f.$$

Recall the process of our discussion on the universal property of $\mathbf{Z}[x]$. At the very beginning, we noticed the fact that $\mathbf{Z}[x]$ is precisely the direct sum $\bigoplus_{i=1}^{\infty} \mathbf{Z}$ that we discussed earlier in the chapter on free abelian groups. Its elements are functions of finite support from $\mathbf{N}$ to $\mathbf{Z}$. For one such function, say $(r_0, r_1, \dots, r_{d-1}, 0, \dots)$, we actually only care about the first d values of the sequence, that is, we consider the element $(r_0, \dots, r_{d-1})$ in the direct sum $\bigoplus_{i=1}^d \mathbf{Z}$. In fact, we can have the following function:

$$\iota_d : \bigoplus_{i=1}^d \mathbf{Z} \rightarrow \mathbf{Z}[x], \quad \iota_d((r_0, \dots, r_{d-1})) = r_0 + r_1 x + \cdots + r_{d-1} x^{d-1}.$$

It is clear that ι_d is an injective homomorphism of abelian groups, hence an isomorphism onto its image, and this consists precisely of the polynomials of degree $< d$. More generally, the above $\mathbf{Z}$ can be replaced by any ring R in the discussion. Under the analogous homomorphism $\iota_d : \bigoplus_{i=1}^d R \rightarrow R[x]$, we may simply use $\bigoplus_{i=1}^d R$ to denote the set of all polynomials with coefficients in R of degree less than d .

6.2.11 PROPOSITION. *Let R be a commutative ring, and let $f \in R[x]$ be a monic polynomial of degree d . Then the function*

$$\varphi_d : R[x] \rightarrow \bigoplus_{i=1}^d R$$

defined by sending $g(x) \in R[x]$ to the remainder of the division of $g(x)$ by $f(x)$ induces an isomorphism of abelian groups

$$R[x]/(f(x)) \cong \bigoplus_{i=1}^d R.$$

Proof. The function φ is well-defined by the lemma above, and it is surjective since it has a right inverse ι_d defined above. We show that φ is a group homomorphism. Indeed, if

$$g_1(x) = f(x)q_1(x) + r_1(x) \text{ and } g_2(x) = f(x)q_2(x) + r_2(x)$$

with $\deg r_1 < d, \deg r_2 < d$, then $g_1(x) + g_2(x) = f(x)(q_1(x) + q_2(x)) + (r_1(x) + r_2(x))$ and $\deg r_1 + r_2 < d$; this implies $\varphi(g_1(x) + g_2(x)) = r_1(x) + r_2(x) = \varphi(g_1(x)) + \varphi(g_2(x))$. By the first isomorphism theorem for groups, then, φ_d induces an isomorphism

$$R[x]/\ker \varphi \cong \bigoplus_{i=1}^d R.$$

On the other hand, $\varphi(g(x)) = \mathbf{0}$ if and only if $g(x) = f(x)q(x)$ for some $q(x) \in R[x]$, that is, if and only if $g(x)$ is in the principal ideal $(f(x))$. This shows that $\ker \varphi = (f(x))$. $\square$

6.2.8 EXAMPLE. Assume $f(x)$ is monic of degree 1: $f(x) = x - a$ for some $a \in R$. Then the remainder of $g(x)$ after division by $f(x)$ is simply the “evaluation” $g(a)$: indeed,

$$g(x) = (x - a)q(x) + \mathbf{r}.$$

for some $\mathbf{r} = r \in R$ (the remainder must have degree < 1 ; hence it is a constant); evaluating at a gives that

$$g(a) = (a - a)q(a) + r = 0 \cdot q(a) + r = r.$$

6.2.2 REMARK. Since $(f(x))$ is an ideal, the abelian group $R[x]/(f(x))$ is a ring, and one may use the isomorphism to define a new ring structure onto the abelian group $\bigoplus_{i=1}^d R$. Different $f(x)$ may lead to different multiplicative structures. That is, $\bigoplus_{i=1}^d R$ can be equipped with different multiplications to obtain ring structures. Specifically, let us consider the cases $d = 1$ and $d = 2$. If $d = 1$, then all the structures are isomorphic. If $d = 2$, and let $f(x) = x^2 + 1$. This proposition implies

$$R \oplus R \cong R[x]/(x^2 + 1);$$

what multiplication does this isomorphism induce on $R \oplus R$? Take two elements $(a_0, a_2), (b_0, b_1)$ of $R \oplus R$. Then $(a_0, a_1) = \varphi(a_0 + a_1x), (b_0, b_1) = \varphi(b_0 + b_1x)$. Now,

$$\begin{aligned} (a_0 + a_1x)(b_0 + b_1x) &= a_0b_0 + (a_0b_1 + a_1b_0)x + a_1b_1x^2 \\ &= (x^2 + 1)a_1b_1 + ((a_0b_0 - a_1b_1) + (a_0b_1 + a_1b_0)x) \end{aligned}$$

which shows that $\varphi((a_0 + a_1x)(b_0 + b_1x)) = (a_0b_0 - a_1b_1, a_0b_1 + a_1b_0)$. Therefore, the multiplication induced on $R \oplus R$ by this procedure is defined by

$$(a_0, a_1) \cdot (b_0, b_1) = (a_0b_0 - a_1b_1, a_0b_1 + a_1b_0).$$

This recipe may seem somewhat arbitrary, but note that upon taking $R = \mathbf{R}$, the ring of real numbers, and identifying pairs $(x, y) \in \mathbf{R} \oplus \mathbf{R}$ with complex numbers $x + iy$, the multiplication we obtained on $\mathbf{R} \oplus \mathbf{R}$ matches precisely the usual multiplication in $\mathbf{C}$. Therefore,

$$\mathbf{R}[x]/(x^2 + 1) \cong \mathbf{C}.$$

6.3 Localizations

A Prime and Maximal Ideals

6.3.1 DEFINITION. Let R be a commutative ring and let $I \neq R$ be a proper ideal.

1. I is a *prime ideal* if $xy \in I$ implies $x \in I$ or $y \in I$;
2. I is a *maximal ideal* if $J \subset I$ for all ideals J of R .

Denote by $\text{Spec } R$ and $\text{MaxSpec } R$ the sets consisting of prime ideals and maximal ideals, respectively, which are called the prime spectrum and the maximal ideal spectrum.

A general ring need not have maximal ideals: the zero ring $\mathbf{0}$ has no maximal ideals, hence any result involving maximal ideals forces a ring to be nonzero. But in fact, this is the only ring that has no maximal ideals. The next proposition shows that any nonzero ring always possess maximal ideals.

6.3.1 PROPOSITION. $\text{MaxSpec } R \neq \emptyset$ unless $R = \mathbf{0}$.

Proof. It suffices to show that every proper ideal I of $R \neq 0$ is contained in a maximal ideal. Let $\mathcal{S}$ be the set of all proper ideals of R which contain I . Then $\mathcal{S}$ is nonempty and is partially ordered by inclusion. If $\mathcal{C}$ is a chain in $\mathcal{S}$, define J to be the union of all ideals in $\mathcal{C}$:

$$J = \bigcup_{A \in \mathcal{C}} A.$$

We first show that J is an ideal. Clearly J is nonempty because $\mathcal{C}$ is nonempty ($0 \in J$). If $a, b \in J$, then there are ideals $A, B \in \mathcal{C}$ such that $a \in A$ and $b \in B$. By definition of a chain either $A \subset B$ or $B \subset A$. In either case $a - b \in J$. Moreover, since each $A \in \mathcal{C}$ is closed under multiplication by elements of R , so is J . This proves J is an ideal.

If J is not a proper ideal, then $1 \in J$. In this case, by definition of J we must have $1 \in A$ for some $A \in \mathcal{C}$. This is a contradiction because each A is a proper ideal. This proves that each chain has an upper bound (i.e., J) in $\mathcal{S}$. By Zorn's lemma, $\mathcal{S}$ has maximal element which is therefore a maximal proper ideal containing I . $\square$

6.3.1 REMARK. It is worth noting that this proposition holds only for rings as we have defined them in **Ring**. If we consider **Rng** (rings without identity), it need not hold. A typical example is the ring $\langle \mathbf{Q}, +, \cdot \rangle$, where $+$ is the usual addition and $\cdot$ is defined by $ab = 0$ for all $a, b \in \mathbf{Q}$. This is a typical **Rng**. Clearly, every subgroup is an ideal, so there is no so-called maximal ideal.

Moreover, the prime spectrum $\text{Spec } R$ of every nonzero ring R is nonempty, because *every maximal ideal is a prime ideal*. This is the first primary goal of this section, and the following propositions will establish this fact.

6.3.2 LEMMA. For any ring homomorphism $\varphi : R \rightarrow S$, the function $\mathcal{S} \rightarrow \mathcal{R}$ maps the prime ideal of S to the prime ideal of R ; that is, φ induces a well-defined function

$$\varphi^\# : \text{Spec } S \rightarrow \text{Spec } R, \quad I_S \mapsto \varphi^{-1}(I_S).$$

Proof. Assume I_S is a prime ideal of S . If $xy \in \varphi^{-1}(I_S)$, then $\varphi(x)\varphi(y) \in I_S$, hence $x \in \varphi^{-1}(I_S)$ or $y \in \varphi^{-1}(I_S)$, which means $\varphi^{-1}(I_S)$ is a prime ideal of R . $\square$

Now, consider the canonical projection $\pi : R \rightarrow R/I$ which is a surjective homomorphism. Then the induced bijection $\tilde{\varphi} : \mathcal{R}_\varphi \rightarrow \mathcal{S}$ says that if $J \subset R$ is an ideal of R which contains I , then $\varphi(J) = J/I$ is an ideal of S . Clearly, if J is maximal in R if and only if J/I is maximal in R/I . But what if J is a prime ideal, or J/I is a prime ideal? This also holds:

the lemma above states that if $J/I \in \text{Spec} R/I$, then $\pi^{-1}(J/I) = J \in \text{Spec} R$. Assume that $J \in \text{Spec} R$ and $J \supset I$, then $(x+I)(y+I) \in J/I$ if and only if $xy \in J$, which implies $J/I \in \text{Spec} R/I$.

6.3.3 PROPOSITION. *Let R be a commutative ring and let I be a proper ideal of R . Then an ideal $J \supset I$ is prime (maximal) in R if and only if J/I is prime (maximal) in R/I . In particular,*

1. R/I is an integral domain if and only if I is prime in R ;
2. R/I is a field if and only if I is maximal in R .

Proof. □

6.3.4 COROLLARY. *Every maximal ideal is a prime ideal.*

6.3.1 EXAMPLE.

1. The principal ideals generated by primes in $\mathbf{Z}$ are both prime and maximal ideals.
2. The ideal $(2, x)$ is a maximal ideal in $\mathbf{Z}[x]$ since its quotient ring “is” a field $\mathbf{Z}/2\mathbf{Z}$. The ideal (x) is not a maximal ideal since $(2) \subset (2, x) \subset \mathbf{Z}[x]$. It can also be understood in this way: $\mathbf{Z}[x]/(x) \cong \mathbf{Z}$, and $\mathbf{Z}$ is not a field, which implies (x) is not maximal in $\mathbf{Z}[x]$. But (x) is a prime ideal, because $\mathbf{Z}$ is an integral domain.

At this point, however, $\text{Spec} R$ is only a set. To turn it into a geometric object, we would like to regard the prime ideals as points of a space and define a suitable topology on this set.

The basic idea is to think of the elements of the ring R as functions on some space. This point of view is familiar when R is a ring such as $\mathbf{C}[x, y]$: every polynomial can indeed be regarded as a function (polynomial function) on the plane $\mathbf{C}^2$. If a function f vanishes at a point, then we write

$$f(a, b) = 0.$$

To say that $f(a, b) = 0$ is equivalent to saying that the original polynomial $f \in \mathbf{C}[x, y]$ (rather than the polynomial function we regard it as) belongs to the ideal $(x - a, y - b)$. In other words, we will say that the polynomial f vanishes at the ideal $(x - a, y - b)$ to indicate that the corresponding polynomial function f vanishes at the point $(a, b) \in \mathbf{C}^2$. On the other hand, this ideal is prime, since clearly $(x - a, y - b)$ is maximal, and $\mathbf{C}[x, y]/(x - a, y - b) \cong \mathbf{C}$ is a field, so it is prime.

Of course, the reader might wonder whether every prime ideal of $\mathbf{C}[x, y]$ corresponds uniquely to a point $(a, b) \in \mathbf{C}^2$. The answer is no; for example, the ideal $(y - x^2)$ is prime, but $f \in (y - x^2)$ means that the polynomial function f vanishes on the entire parabola $y = x^2$, which does not correspond to a single point, but rather to a unique curve. If we abstractly regard this curve as a point, then it would not be incorrect to say that a prime ideal corresponds to a point. In this case, we should say that f vanishes at the prime ideal $(y - x^2)$ to indicate that the polynomial function f vanishes on $y = x^2$.

In summary, we know that a polynomial function f vanishing on some subset of $\mathbf{C}^2$ (such as a point, a curve, or the plane) can actually be described by saying that f belongs to the corresponding prime ideal. Let us simply say that the polynomial f vanishes at the “point” $\mathfrak{p}$ (a prime ideal). (Since the author has not yet delved into commutative algebra and algebraic geometry, this understanding is personal.) Thus, generalizing to an arbitrary commutative ring R , we can certainly regard an element $f \in R$ as a function f , and an element of $\text{Spec} R$ as a point, and say that f vanishes at the $\mathfrak{p}$ if $f \in \mathfrak{p}$.

In the classical plane $\mathbf{C}^2$, we usually regard the “zero set” as a closed set. For example: The zero set of a single polynomial f , $Z(f) = \{(a, b) \in \mathbf{C}^2 : f(a, b) = 0\}$, is a closed set (a curve or some points); the common zero set of several polynomials $f_1, \dots, f_n$, $Z(f_1, \dots, f_n) = \{(a, b) \in \mathbf{C}^2 : f_1(a, b) = \dots = f_n(a, b) = 0\}$, is also a closed set (an

algebraic variety). After we replace “points” with “prime ideals,” the definition of the zero set becomes:

“The function f vanishes at the prime ideal $\mathfrak{p}$ ” $\longleftrightarrow f \in \mathfrak{p}$.

Therefore, for any ideal $\mathfrak{a} \subset \mathbf{C}[x, y]$ (which represents a set of functions), their *common vanishing locus* should be:

$$V(\mathfrak{a}) := \{\mathfrak{p} \in \text{Spec } \mathbf{C}[x, y] : \mathfrak{a} \subset \mathfrak{p}\}.$$

That is, the set of *all prime ideals containing* $\mathfrak{a}$. Thus, for an arbitrary commutative ring R , we declare the subsets $V(\mathfrak{a}) = \{\mathfrak{p} \in \text{Spec } R : \mathfrak{a} \subset \mathfrak{p}\}$, $\mathfrak{a} \subset R$, to be the closed sets of a topology on $\text{Spec } R$. This is called the *Zariski topology*.

6.3.5 LEMMA. *Let R be a commutative ring.*

1. *If $\mathfrak{a}$ and $\mathfrak{b}$ are two ideals of R , then $V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$.*
2. *If $\{\mathfrak{a}_i\}$ is any set of ideals of R , then $V(\sum \mathfrak{a}_i) = \bigcap V(\mathfrak{a}_i)$.*
3. *If $\mathfrak{a}$ and $\mathfrak{b}$ are two ideals of R , then $V(\mathfrak{a}) \subset V(\mathfrak{b})$ if and only if $\sqrt{\mathfrak{a}} \supset \sqrt{\mathfrak{b}}$.*

Proof.

(1). Clearly, if $\mathfrak{p} \supset \mathfrak{a}$ or $\mathfrak{p} \supset \mathfrak{b}$, then $\mathfrak{p} \supset \mathfrak{a}\mathfrak{b}$. Conversely, if $\mathfrak{p} \supset \mathfrak{a}\mathfrak{b}$ and if $\mathfrak{p} \not\supset \mathfrak{b}$ for examine, then there is a $b \in \mathfrak{b}$ such that $b \notin \mathfrak{p}$. Now for any $a \in \mathfrak{a}$, $ab \in \mathfrak{p}$, so we must have $a \in \mathfrak{a}$ since $\mathfrak{p}$ is a prime ideal. Thus $\mathfrak{p} \supset \mathfrak{a}$.

(2). One may show that the sum $\sum \mathfrak{a}_i = \{a_{i_1} + \cdots + a_{i_n} : a_k \in \mathfrak{a}_{i_k}, n \in \mathbf{N}\}$ is the smallest ideal of R containing all $\mathfrak{a}_i$, i.e., the intersection of all the ideals that containing $\bigcup \mathfrak{a}_i$. If $\mathfrak{p} \in \bigcap V(\mathfrak{a}_i)$, then $\mathfrak{a}_i \subset \mathfrak{p}$ for each i , so that $\sum \mathfrak{a}_i \subset \mathfrak{p}$, i.e., $\mathfrak{p} \in V(\sum \mathfrak{a}_i)$. Conversely, if $\mathfrak{p} \in V(\sum \mathfrak{a}_i)$, then $\sum \mathfrak{a}_i \subset \mathfrak{p}$. Hence $\mathfrak{a}_i \subset \mathfrak{p}$, which implies $\mathfrak{p} \in \bigcap V(\mathfrak{a}_i)$.

(3). One may show that the radical $\sqrt{\mathfrak{a}} = \{r \in R : r^n \in \mathfrak{a} \text{ for some } n \geq 1\}$ is the intersection of all prime ideals containing $\mathfrak{a}$: if $x \in \sqrt{\mathfrak{a}}$, then there exists $n \geq 1$ such that $x^n \in \mathfrak{a}$. If $\mathfrak{p}$ is a prime ideal containing $\mathfrak{a}$, then $x^n \in \mathfrak{a} \subset \mathfrak{p}$. Since $\mathfrak{p}$ is prime, $x \in \mathfrak{p}$, which implies $x \in \bigcap_{\mathfrak{a} \subset \mathfrak{p}} \mathfrak{p}$. Conversely, assume there exist $x \in \bigcap_{\mathfrak{a} \subset \mathfrak{p}} \mathfrak{p}$ but $x \notin \sqrt{\mathfrak{a}}$. Let $\mathcal{S} = \{\mathfrak{b} \subset R : \mathfrak{a} \subset \mathfrak{b} \neq R, x^n \notin \mathfrak{b}, \forall n \geq 1\}$. $\mathcal{S}$ is nonempty since $\mathfrak{a} \in \mathcal{S}$. Let $\mathcal{C}$ be a chain of $\mathcal{S}$, and let

$$\mathfrak{j} = \bigcup_{\mathfrak{b} \in \mathcal{C}} \mathfrak{b},$$

if $x^k \in \mathfrak{j}$ for some k , then $x^k \in \mathfrak{b}$ which is a contradiction, so that every chain of $\mathcal{S}$ have upper bound $\mathfrak{j}$. By Zorn's lemma, $\mathcal{S}$ has a maximal element, $\mathfrak{p}$. Next, we show that $\mathfrak{p}$ is a prime ideal: assume there exist $a, b \notin \mathfrak{p}$ but $ab \in \mathfrak{p}$. Since $\mathfrak{p}$ is the maximal element in $\mathcal{S}$, the ideal $\mathfrak{p} + (a) \notin \mathcal{S}$, then $x^n \in \mathfrak{p} + (a)$ for some n . Similarly, $x^m \in \mathfrak{p} + (b)$ for some m . Let $x^n = p_1 + r_1a$, $x^m = p_2 + r_2b$, then $x^{n+m} = (p_1 + r_1a)(p_2 + r_2b)$. Since $ab \in \mathfrak{p}$, it is clear that $x^{n+m} \in \mathfrak{p}$, contradicting the fact that $\mathfrak{p} \in \mathcal{S}$. This shows that $\mathfrak{p}$ is indeed a prime ideal not containing x , which is a contradiction. So $x \in \bigcap_{\mathfrak{a} \subset \mathfrak{p}} \mathfrak{p}$ implies $x \in \sqrt{\mathfrak{a}}$. $\square$

Now we define a topology on $\text{Spec } R$ by taking the subsets of the form $V(\mathfrak{a})$ to be the closed subsets. Note that $V(R) = \emptyset$; $V((0)) = \text{Spec } R$; and the lemma shows that finite unions and arbitrary intersections of sets of the form $V(\mathfrak{a})$ are again of that form. Hence they do form the set of closed sets for a topology on $\text{Spec } R$.

B Localization

Let R be a ring which is not a field. The algebraic idea of “localization” is then to make more (or even all) nonzero elements invertible by introducing fractions, in the same way as one passes from the integers $\mathbf{Z}$ to the rational numbers $\mathbf{Q}$.

Let us have a more precise look at this particular example: in order to construct the rational numbers from the integers we start with $R = \mathbf{Z}$ and let $S = \mathbf{Z} \setminus \{0\}$ be the subset of the elements of R that we would like to become invertible. On the subset $R \times S$ we then consider the equivalence relations

$$(r, s) \sim (r', s') \leftrightarrow rs' - r's = 0$$

and denote the equivalence class of a pair (r, s) by r/s . The set of “fractions” is then obviously $\mathbf{Q}$, and we can define addition and multiplication on it in the expected way by

$$r/s + r'/s' := \frac{rs' + r's}{ss'} \text{ and } r/s \cdot r'/s' := \frac{rr'}{ss'}.$$

Let X be topological space and let $R = C(X)$ be the commutative ring of all continuous functions from X into $\mathbf{R}$. In the same way in (1), we can ask if it makes sense to consider fractions of such continuous functions, i.e.,

$$\frac{f}{g} : x \mapsto \frac{f(x)}{g(x)}.$$

Of course, for global functions on all of X this does not work generally, since g will in general have zeros somewhere, so that the value of the rational function is “ill-defined” at these points. But if we consider functions on subsets of X we can allow such fractions. The most important example from the point of view of localization is the following: for a fixed point $a \in X$ let $S = \{f \in C(X) : f(a) \neq 0\}$ be the set of all continuous functions that do not vanish at a . Then the fractions f/g for $f \in R$ and $g \in S$ can be thought of as a continuous functions that are well-defined at a .

In fact, the best way to interpret the fractions f/g with $f \in R$ and $g \in S$ is as “functions on an arbitrarily small neighborhood of a ”, assuming that we have a suitable topology on X : although the only point at which these functions are guaranteed to be well-defined is a , for every such function there is a neighborhood on which it is defined as well, and to a certain extent the continuous functions f and g can be recovered from the values of the function on such a neighborhood. This is because continuity ensures that if $g(a) \neq 0$, then there exists an open neighborhood U of a such that g is non-zero on all of U ; hence $\frac{f}{g}$ is a well-defined continuous function on U . In this sense we will refer to functions of the form f/g with $f \in R$ and $g \in S$ as “local functions” at a . In fact, this geometric interpretation is the origin of the name “localization” for the process of constructing fractions described in this chapter.

6.3.2 DEFINITION. Let R be a commutative ring, and let S be a subset of R . S is called a *multiplicative subset* of R if S is a monoid with respect to the multiplication of R .

6.3.2 EXAMPLE.

1. For every nonzero $r \in R$, the set $S = \{1, r, r^2, \dots\}$ is a multiplicative subset of R .
2. $R^\times$, i.e., the set of all nonzero divisors of R is a multiplicative subset of R .
3. If $\mathfrak{p}$ is a prime ideal of R , then $S = R \setminus \mathfrak{p}$ is a multiplicative subset of R . Moreover generally, if $\{\mathfrak{p}_i\}$ is a set of prime ideals of R , we have the multiplicative subset $S = R \setminus \bigcup \mathfrak{p}_i$.

Below, we construct the localization $S^{-1}R$ with respect to the multiplicative subset S as follows. Let S be a multiplicative subset of a commutative ring R , we define a relation on the set $R \times S$:

$$(r, s) \sim (r', s') \leftrightarrow \exists t \in S, t(rs' - r's) = 0$$

One may show that $\sim$ is an equivalence relation. (As for why we add a t here rather than simply requiring $rs' - r's = 0$ as we stated at the beginning, it is precisely because

S may contain zero divisors, which causes the proof of transitivity to fail. Specifically, if we only require $(rs' - r's) = 0$, then $(r, s) \sim (r', s')$ and $(r', s') \sim (r'', s'')$ imply that

$$s'(rs'' - r''s) = s''(rs' - r's) + s(r's'' - r''s') = 0$$

, this does not imply $rs'' - r''s = 0$ if S contains zero divisors.) The set of all equivalence classes of $R \times S$ under $\sim$ will be denoted by $S^{-1}R$. The equivalence class of $(r, s) \in R \times S$ will be denoted r/s . If we accept this notation in the operations within $\mathbf{Q}$, it is quite natural to define the following operation:

$$\begin{aligned} r/s + r'/s' &:= rs'r's/ss' \\ r/s \cdot r'/s' &:= rr'/ss' \end{aligned}$$

6.3.6 PROPOSITION. $\langle S^{-1}R, +, \cdot \rangle$ is a commutative ring.

where the zero element is $0 = 0/s$ and the identity element is $1 = s/s$, where s can be chosen arbitrarily. Thus we obtain $r/s = 0$ if and only if $\exists t \in S, tr = 0 \in R$, which means $S^{-1}R$ is a zero ring if and only if $\exists s \in S, sR = 0$ if and only if $0 \in S$. Therefore, some people are accustomed to adding the prerequisite that a multiplicative subset does not contain 0. In most cases, we are also only interested in $S^{-1}R$ being a nonzero ring. To make a distinction, we add $0 \notin S$ to indicate that $S^{-1}R$ is a nonzero ring.

On the other hand, $r \mapsto r/1$ gives a ring homomorphism $\iota : R \rightarrow S^{-1}R$. Note that the image of $s \in S$ lies in $S^{-1}R$, whose inverse is simply $1/s$. The localization should be considered together with the homomorphism $R \rightarrow S^{-1}R$. From a categorical point of view, the localization construction $R \rightarrow S^{-1}R$ is the “most economical” way to make the elements of S invertible, and it is uniquely characterized by the following property:

6.3.7 PROPOSITION. *Let R be a commutative ring and S be a multiplicative subset of R . Then the ring $S^{-1}R$ with the homomorphism $\iota : R \rightarrow S^{-1}R$ satisfies the universal property: for any commutative ring A and homomorphism $\varphi : R \rightarrow A$ that $\varphi(S) \subset A^\times$, then there exists a unique homomorphism $\tilde{\varphi} : S^{-1}R \rightarrow A$ such that the diagram commutes:*

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & A \\ \downarrow \iota & \nearrow \tilde{\varphi} & \\ S^{-1}R & & \end{array}$$

Proof. Since $\varphi(S) \subset A^\times$, every form $\varphi(s)$ has the inverse $\varphi(s)^{-1}$ in the ring A . Define $\tilde{\varphi} : S^{-1}R \rightarrow A$ by $\tilde{\varphi}(r/s) = \varphi(r)\varphi(s)^{-1}$. This function is well-defined since if $r/s = r'/s'$, then $t(rs' - r's) = 0$ for some $t \in S$. Then $\varphi(t)(\varphi(rs') - \varphi(r's)) = 0 \in A$, so $\varphi(rs') - \varphi(r's) = 0$ since $\varphi(t)$ is a unit in A , and therefore $\varphi(r)\varphi(s)^{-1} = \varphi(r')\varphi(s')^{-1}$. It is immediate that $\tilde{\varphi}$ is a ring homomorphism and the diagram commutes.

For the uniqueness, $\tilde{\varphi}$ is unique since every element of $S^{-1}R$ can be written as a product $(r/1)(d/1)^{-1}$. The value of $\tilde{\varphi}$ on each element of the form $x/1$ is uniquely determined by φ , namely $\tilde{\varphi}(x/1) = \tilde{\varphi}(\iota(x)) = \varphi(x)$. Since $\tilde{\varphi}$ is a ring homomorphism, its value on u^{-1} for any unit u is uniquely determined by $\tilde{\varphi}(u)$. Thus, $\tilde{\varphi}$ is uniquely determined on every element of $S^{-1}R$. $\square$

Localization with respect to a multiplicative subset makes all elements of S invertible. Moreover, just like the sign-preserving property of continuous functions in calculus, there are still some elements in the localization of S that are not originally in S but are also invertible. Specifically,

6.3.8 PROPOSITION. *Let R be a commutative ring and let S be a multiplicative subset of R ($0 \notin S$). r/s is invertible if and only if $\exists r' \in R$ such that $rr' \in S$.*

Proof. If $rr' \in S$, then $r/s \cdot r's/rr' = rr's/srr' = 1$. Conversely, if there exists r'/s' such that $r/s \cdot r'/s' = 1$, then $rr'/ss' = t/t, t \in S$ i.e., $trr' = tss'$, hence $r(tr') \in S$. $\square$

6.3.9 PROPOSITION. *Let R be a commutative ring and S be a multiplicative subset of R . $\iota : R \rightarrow S^{-1}R$ is injective if and only if S contains no zero divisors of R .*

Proof. □

In particular, let S be the set of all elements which are not zero divisors in $R \setminus \{0\}$, we denote the corresponding localization by $R \rightarrow \text{Frac}(R)$, and the ring $\text{Frac}(R)$ is called the *total ring of fractions*.

If R is an integral domain, then in this case, $\text{Frac}(R)$ is nothing but the localization with respect to $S = R \setminus \{0\}$. In this situation, $\text{Frac}(R)$ is naturally a field, since every nonzero element is invertible, as follows from the above proposition. In this case, we call $\text{Frac}(R)$ the *fraction field*.

Next, we study the effect of localization on ideals. Given a multiplicative subset S with $0 \notin S$. For any ideal $I \subset R$, define

$$S^{-1}I := \{r/s : r \in I, s \in S\} \subset S^{-1}R.$$

From the ring structure of $S^{-1}R$, it is easy to verify that $S^{-1}I$ is in fact the ideal generated by the image of I under $\iota : R \rightarrow S^{-1}R$, this induces a function $\iota_{\text{ext}} : \text{Id}(R) \rightarrow \text{Id}(S^{-1}R)$, $\iota_{\text{ext}}(I) = S^{-1}I$.

On the other hand, ι induces a function $\iota_{\text{cot}} : \text{Id}(S^{-1}R) \rightarrow \text{Id}(R)$, $J \mapsto I = \iota^{-1}(J) := \{r \in R : r/1 \in J\}$, where J is an ideal of $S^{-1}R$, and the right-hand side is precisely the preimage of J under $\iota : R \rightarrow S^{-1}R$.

Now consider

$$\text{Id}(R) = \{I : \text{ideals of } R\} \xrightleftharpoons[\iota_{\text{cot}}]{\iota_{\text{ext}}} \text{Id}(S^{-1}R) = \{J : \text{ideals of } S^{-1}R\}$$

We claim that the composition $\iota_{\text{ext}} \circ \iota_{\text{cot}}$ in the above diagram is the identity: clearly $\iota_{\text{cot}}(J) = I$ implies $S^{-1}I \subset J$; conversely, for any $r/s \in J$, we have $r/1 = r/s \cdot s/1 \in J$, hence $r \in I$, and so $J \subset S^{-1}I$. Therefore, $\iota_{\text{ext}}(I) = S^{-1}I = J$, that is, $\iota_{\text{ext}} \circ \iota_{\text{cot}}(J) = J$. Therefore, every ideal in $S^{-1}R$ is of the form $S^{-1}I$ for some ideal I of R .

Then for any $J \in \text{Id}(S^{-1}R)$, taking $I = \iota_{\text{cot}}(J) = \iota^{-1}(J)$, then $\iota_{\text{ext}}(I) = S^{-1}I = J$ which implies ι_{ext} is surjective; moreover, if $\iota_{\text{cot}}(J_1) = \iota_{\text{cot}}(J_2) = I$, then $S^{-1}I = J_1 = J_2$, which implies ι_{cot} is injective. However, if we do not restrict the types of ideals considered, the composition $\iota_{\text{cot}} \circ \iota_{\text{ext}}$ in the above diagram need not be the identity, as can be seen from the following proof.

6.3.10 PROPOSITION. *Let S be a commutative subset of R and $0 \notin S$.*

1. *For any ideal I , $S^{-1}I = S^{-1}R$ if and only if $I \cap S \neq \emptyset$.*
2. *The function ι_{ext} induces a bijection $\{I \in \text{Spec}R : I \cap S = \emptyset\} \rightarrow \text{Spec}S^{-1}R$.*

Proof. (1). If $S^{-1}I = S^{-1}R$, then $1 = r/s$ for some $r \in I, s \in S$, that is, $tr = ts$ for some $t \in S$. Since S is multiplicative, $ts \in S$; and $ts \in I$ since I is an ideal. So $ts \in I \cap S$. Conversely, if $I \cap S \neq \emptyset$, and $s_0 \in I \cap S$. For any $r/s \in S^{-1}R$, $r/s = r/s \cdot s_0/s_0$. Clearly, $a/s \in S^{-1}I$ and $S^{-1}R \subset S^{-1}I$.

(2). Assume $I \in \text{Spec}R$ and $I \cap S = \emptyset$. Then $S^{-1}I$ is proper in $S^{-1}R$. Moreover, $r_1/s_1 \cdot r_2/s_2 \in S^{-1}I$ if and only if $r_1r_2 \in I$ if and only if $r_1 \in I$ or $r_2 \in I$, which means $S^{-1}I$ is a prime ideal of $S^{-1}R$. Now, consider the homomorphism $\iota : R \rightarrow S^{-1}R$, the preimage of $S^{-1}I \in \text{Spec}S^{-1}R$ is

$$\{r \in R : \exists a \in I, s \in S, r/1 = a/s\} = \{r \in R : \exists s \in S, rs \in I\}.$$

The preimage I is indeed a prime ideal by passing the induced function $\iota^\# : \text{Spec}S^{-1}R \rightarrow \text{Spec}R$ given by $J \rightarrow \iota_{\text{cot}}(J) = \iota^{-1}(J)$. Since $I \cap S = \emptyset$, the preimage $\{r \in R : \exists s \in S, rs \in I\}$ is indeed I . Then $\iota_{\text{cot}}(\iota_{\text{ext}}(I)) = I$. We have already proved that for every prime ideal $J \subset S^{-1}R$, $\iota_{\text{ext}}(\iota_{\text{cot}}(J)) = S^{-1}(\iota^{-1}(J)) = J$. Therefore, when we restrict both ends of ι_{ext} to the sets consisting of prime ideals, it becomes a bijection. □

6.3.3 EXAMPLE. Let $\mathfrak{p}$ be a prime ideal of R and take the multiplicative subset $S = R \setminus \mathfrak{p}$. Consider the localization $S^{-1}R$. The set $\{\mathfrak{a} \in \text{Spec} R : \mathfrak{a} \cap (R \setminus \mathfrak{p}) = \emptyset\} = \{\mathfrak{a} \in \text{Spec} R : \mathfrak{a} \subset \mathfrak{p}\}$. The ideal $S^{-1}\mathfrak{p}$ is clearly prime. We show that it is the only maximal ideal of $S^{-1}R$: If M is a maximal ideal of $S^{-1}R$, then M is prime, whence $M = S^{-1}\mathfrak{q}$ for some prime ideal $\mathfrak{q}$ of R with $\mathfrak{q} \subset \mathfrak{p}$. But $\mathfrak{q} \subset \mathfrak{p}$ implies $S^{-1}\mathfrak{q} \subset S^{-1}\mathfrak{p}$ and $S^{-1}\mathfrak{p} \neq S^{-1}R$ (because $\mathfrak{p} \cap S = \emptyset$), we must have $S^{-1}\mathfrak{q} = S^{-1}\mathfrak{p}$. Therefore for $S^{-1}\mathfrak{p}$ is the unique maximal ideal in $S^{-1}R$.

6.4 Factorizations

A UFDs

Let R be an integral domain, a nonzero element x is said to *divide* y or be a divisor of y if there exists an element $a \in R$ such that $y = ax$, and written $x \mid y$; that is,

$$x \mid y \iff \exists a \in R, y = ax.$$

Clearly, $\exists a \in R, y = ax$ if and only if $(y) \subset (x)$, hence $x \mid y \iff (y) \subset (x)$. If $x \mid y$ and $y \mid x$, then $(y) = (x)$ which implies $y = ux$ for some unit $u \in R^\times$. Moreover, let $u \in R^\times$, then $x \mid y$ if and only if $u \cdot x \mid y$. Therefore, when we consider the question of “divisibility”, the monoid of units $R^\times$ can be “ignored”. Specifically, we can consider the following quotient monoid:

$$\mathcal{P} := (R \setminus \{0\})/R^\times.$$

Unless otherwise specified, let $\dot{x}$ denote the image of $x \in R \setminus \{0\}$ under the canonical projection $\pi : R \setminus \{0\} \rightarrow \mathcal{P}$. Now, define a binary relation $\leq$ on $\mathcal{P}$ by

$$\dot{y} \leq \dot{x} \iff x \mid y,$$

We prove that $\leq$ is a partial ordering on $\mathcal{P}$. First, the relation is well-defined, if $\dot{x} = \dot{x}'$ and $\dot{y} = \dot{y}'$, then $x' = ux, y' = vy$ for some units $u, v \in R^\times$. Now, if $x \mid y$, say $y = ax$, then $y' = vy = vax = (vau^{-1})x'$, so $x' \mid y'$. Clearly, the reflexivity is true. For the antisymmetry, suppose $\dot{x} \leq \dot{y}, \dot{y} \leq \dot{x}$. By definition, $x = uy$ for some unit $u \in R^\times$, hence $\dot{x} = \dot{y}$. For the transitivity, suppose $\dot{x} \leq \dot{y}, \dot{y} \leq \dot{z}$. Then $x = ay, y = bz$ for some units $a, b \in R^\times$, and $x = (ab)z$, thus $\dot{x} \leq \dot{z}$. Therefore, $\langle \mathcal{P}, \leq \rangle$ is partial.

Assume $\{\dot{x}, \dot{y}\} \subset \mathcal{P}$, and if $\{\dot{x}, \dot{y}\}$ has a supremum $\dot{d}$, that is, $\dot{d} \mid \dot{x}, \dot{d} \mid \dot{y}$, and for any $\dot{z}$ satisfying this property we have $\dot{d} \leq \dot{z}$, i.e. $z \mid d$. We call the preimage d the greatest common divisor of x, y . If $\dot{d} = \dot{1}$, we say x and y are coprime.

6.4.1 DEFINITION. Let R be an integral domain. A nonzero element $r \in R$ is *irreducible* in R if $r \notin R^\times$ and $d \mid r$ implies $(d) = (r)$ or $d \in R^\times$; that is $d \mid r$ implies $\dot{d} = \dot{r}$ or $\dot{d} = \dot{1}$. If every element $\dot{r}$ of $\mathcal{P}$ may be written as

$$\dot{r} = \prod_{i=1}^n \dot{p}_i, n \geq 1,$$

where $\dot{p}_i \in \mathcal{P}$ are irreducible and the set $\{\dot{p}_1, \dots, \dot{p}_n\}$ (disregarding order but counting multiplicities) is unique, then R is called a *unique factorization domain* (UFD). the $\dot{p}_1, \dots, \dot{p}_n$ (or their preimages $p_1, \dots, p_n \in R$) are called the irreducible factors of $\dot{r}$ (or of its preimage $r \in R$). By convention, $n = 0 \iff \dot{r} = \dot{1}$.

6.4.2 DEFINITION. Let R be an integral domain. A nonzero element $p \in R$ is prime in R if $p \notin R^\times$ and $p \mid ab \iff (p \mid a) \vee (p \mid b)$; that is $\dot{a}\dot{b} \leq \dot{p} \iff \dot{a} \leq \dot{p} \vee \dot{b} \leq \dot{p}$.

In the integral domain R , p is prime if and only if (p) is prime, indeed: assume p is prime, then (p) is proper since $p \notin R^\times$, and if $ab \in (p)$, then $ab = rp$ for some $r \in R$,

that is $p \mid ab$, which implies $p \mid a$ or $p \mid b$, i.e., $a \in (p)$ or $b \in (p)$. Conversely, if (p) is prime, then $p \notin R^\times$ otherwise (p) is proper. If $p \mid ab, a, b \in R$, i.e., $ab \in (p)$, then $a \in (p)$ or $b \in (p)$, that is, $p \mid a$ or $p \mid b$.

6.4.1 PROPOSITION. *Every prime element is always irreducible.*

Proof. Suppose p is prime and $p = ab$, then $p \mid a$ or $p \mid b$, say $p \mid a$, i.e., $a = pr$ for some $r \in R$. This implies $p = ab = prb$ so $rb = 1$ (since R is an integral domain) and $b \in R^\times$. This shows that p is irreducible. $\square$

6.4.2 PROPOSITION. *In a UFD., every irreducible element is prime.*

Proof. Let p be an irreducible in R and assume $p \mid ab$ for some $a, b \in R$ or $ab = pc$ for some $c \in R$. Writing a and b as a product of irreducibles, $\bar{a} = \prod_{i=1}^n \bar{p}_i, \bar{b} = \prod_{j=1}^m \bar{q}_j$. The equality $ab = pc$ shows that the irreducible factors on the right must be irreducible factors of ab , that is, $\bar{p} = \bar{p}_i$ or $\bar{p} = \bar{q}_j$ for some i, j . In any case, this implies that $p \mid a$ or $p \mid b$, completing the proof. $\square$

Not every irreducible element in an integral domain is prime. However, if every irreducible element in an integral domain is prime, then this integral domain has a special property: if a nonzero element has an irreducible decomposition, then that decomposition is unique. That is, the property that all irreducible elements are prime characterizes the uniqueness of irreducible decompositions. The proof is as follows: suppose that

$$\bar{r} = \bar{p}_1 \cdots \bar{p}_m = \bar{q}_1 \cdots \bar{q}_n,$$

where all p_i and q_j are irreducible. Since p_1 is prime and $p_1 \mid q_1 \cdots q_n$, we have $p_1 \mid q_i$ for some i . Since q_i is irreducible, $(p_1) = (q_i)$. After reordering the factors, we may assume that $(p_1) = (q_1)$. Hence $\bar{p}_1 = \bar{q}_1$ and we can cancel them on both sides (since $\mathcal{P}$ is a monoid). Repeating the argument, we obtain $m = n$ and, after a permutation of the indices,

$$\bar{p}_i = \bar{q}_i, \quad i = 1, \dots, m$$

If we can further characterize the existence of irreducible decompositions, then we can use these two properties to equivalently define a UFD. How can we guarantee the existence of irreducible decompositions? To this end, let us again look backwards at what special property the existence of irreducible decompositions in a UFD entails.

6.4.3 DEFINITION. Let $\langle P, \leq \rangle$ be a partially ordered set. We say that P satisfies the *ascending chain condition* (abbreviated as **ACC**) if every ascending chain

$$a_1 \leq a_2 \leq a_3 \leq \cdots$$

of elements in P eventually stabilizes; that is, there exists $N \geq 1$ such that $a_N = a_{N+1} = a_{N+2} = \cdots$. If we are given an integral domain R , and let $\mathcal{P}_0$ be the set of all principal ideals of R , with the relation $\subset$, then clearly $\langle \mathcal{P}_0, \subset \rangle$ is a partially ordered set. We say that $\mathcal{P}_0$ satisfies **ACC**, and denote this by saying that R satisfies **ACCP**.

6.4.3 PROPOSITION. *Every UFD satisfies **ACCP**.*

Proof. Suppose, to the contrary, that R does not satisfy **ACCP**. Then there exists a strictly ascending chain of principal ideals

$$(a_1) \subsetneq (a_2) \subsetneq (a_3) \subsetneq \cdots$$

Thus, for every $i \geq 1$, we have $a_i \in (a_{i+1})$, so there exists $b_i \in R$ such that $a_i = a_{i+1}b_i$. Since the inclusion $(a_i) \subsetneq (a_{i+1})$ is strict, b_i cannot be a unit. Hence

$$a_1 = a_2b_1 = a_3b_2b_1 = \cdots = a_{n+1}b_n \cdots b_2b_1$$

for every $n \geq 1$. Since R is a UFD, a_1 has a finite factorization into irreducible elements. On the other hand, each b_i is a nonunit, and therefore has an irreducible factorization.

Consequently, the equality $a_1 = a_{n+1}b_n \cdots b_1$ gives a factorization of a_1 containing at least n irreducible factors coming from $b_1, \dots, b_n$. Since n is arbitrary, a_1 would have irreducible factorizations of arbitrarily large length, contradicting the uniqueness of factorization in a UFD. $\square$

Not every integral domain satisfies the **ACCP**. However, if an integral domain R satisfies **ACCP**, then every nonzero nonunit element of R has an irreducible factorization. That is, the property that R satisfies **ACCP** characterizes the existence of irreducible decompositions. The proof is as follows: let $r \in R$ be a nonzero nonunit. Suppose, to the contrary, that r cannot be written as a product of irreducible elements. So we may write $r = a_1b_1$ where $a_1 \notin R^\times$. a_1 also cannot be written as a product of irreducibles, Hence we may write $a_1 = a_2b_2$ where $a_2 \notin R^\times$. Continuing in this way, we obtain

$$r = a_1b_1, \quad a_1 = a_2b_2, \quad a_2 = a_3b_3, \quad \dots,$$

where every $b_i \notin R^\times$. This gives a strictly ascending chain of principal ideals

$$(a_1) \subsetneq (a_2) \subsetneq (a_3) \subsetneq \cdots,$$

contradicting the fact that R satisfies **ACCP**. Indeed, since $r = a_1b_1$, we have $(r) \subseteq (a_1)$. If $(r) = (a_1)$, then $r = ua_1$ for some unit $u \in R^\times$. Comparing this with $r = a_1b_1$, we obtain $a_1b_1 = ua_1$. Since R is an integral domain and $a_1 \neq 0$, cancellation gives $b_1 = u$, contradicting the fact that b_1 is a nonunit. Thus $(r) \subsetneq (a_1)$. The same argument gives

$$(a_i) \subsetneq (a_{i+1})$$

6.4.4 THEOREM. *Let R be an integral domain, then R is a UFD if and only if*

1. (Uniqueness) every irreducible element of R is prime;
2. (Existence) R satisfies **ACCP**.

6.4.5 PROPOSITION. *Let R be a UFD. Suppose S is a multiplicative subset of R with $0 \notin S$. Then the localization $S^{-1}R$ is also a UFD.*

Proof. Consider the localization $\text{Frac}(R) := (R \setminus \{0\})^{-1}R = \{r/s : r \in R, s \in R \setminus \{0\}\}$. Clearly, for any multiplicative subset S and the corresponding localization $S^{-1}R$, we may define a natural “inclusion”: $S^{-1}R \rightarrow \text{Frac}(R)$ by $r/s \mapsto r/s$. Hence $S^{-1}R$ is an integral domain.

Now classify the irreducible elements p of R into two types. (i) p divides some element of S ; by proposition, this is equivalent to p ($p/1$) being invertible in $S^{-1}R$. (ii) p divides no element of S ; in this case p remains irreducible in $S^{-1}R$, as shown below. First, its image is not invertible. Second, suppose $p = \frac{r}{s} \frac{r'}{s'}$; since p is prime, $ss'p = rr'$ implies $p \mid r$ or $p \mid r'$; assume $p \mid r$, then from $1 = \frac{r/p}{s} \cdot \frac{r'}{s'}$ we see that $\frac{r'}{s'}$ is invertible in $S^{-1}R$.

Next, we show that $S^{-1}R$ is a UFD, assume a nonzero element $r/s \in S^{-1}R, r \in R, s \in S$. In R , write the irreducible decomposition $\overset{\circ}{r} = \prod_{i=1}^m \overset{\circ}{p}_i \cdot \prod_{j=1}^n \overset{\circ}{q}_j$ where p_i, q_j are irreducible elements of types (i) and (ii), respectively. Then

$$r/s = \prod_{i=1}^m \overset{\circ}{p}_i/1 \cdot \prod_{j=1}^n \overset{\circ}{q}_j/1 \cdot 1/s = \prod_{j=1}^n \overset{\circ}{q}_j/1,$$

since each $p_i/1$ and $1/s$ is invertible in $S^{-1}R$. Thus every nonzero element of $S^{-1}R$ can be written as a finite product of irreducible elements. It remains to prove the uniqueness of the factorization: since (q_i) is prime (q_i is prime) and $(q_i) \cap S = \emptyset$, $S^{-1}(q_i)$ is prime in $S^{-1}R$, which implies $q_i/1$ is prime, hence the decomposition is unique. $\square$

6.4.6 LEMMA. *Let R be a PID. Then R satisfies **ACCP**.*

Proof. For any ascending chain of (principal) ideals $\mathfrak{a}_1 \subset \mathfrak{a}_2 \subset \cdots$, define $\mathfrak{a} = \bigcup_{i=1}^{\infty} \mathfrak{a}_i$, clearly $\mathfrak{a}$ is an ideal and it may be written as $\mathfrak{a} = (a)$ for some $a \in R$. Because $a \in \mathfrak{a} = \bigcup \mathfrak{a}_i$, there exists some $N \geq 1$ such that $a \in \mathfrak{a}_N$. Hence $(a) \subset \mathfrak{a}_N$. Therefore $\mathfrak{a}_N = (a)$. Now for any $k \geq N$, since $\mathfrak{a}_N \subset \mathfrak{a}_k \subset \mathfrak{a} = \mathfrak{a}_N$, we get $\mathfrak{a}_k = \mathfrak{a}_N$. $\square$

6.4.7 THEOREM. *Every PID is UFD.*

Proof. By the equivalent characterization of UFDs, we only need to prove that in a PID every irreducible element is prime, and that it satisfies **ACCP**; the latter has already been proved in the previous lemma. Here we only prove the first assertion. Let p be an irreducible element, we claim that (p) is maximal. Assume there exists an ideal (d) such that

$$(p) \subset (d) \subset R.$$

Then $d \mid p$. Since d is irreducible, it follows that $(d) = (p)$ or $(d) = R$. Hence (p) is maximal. Every maximal ideal is prime, it follows that (p) is prime; and p is prime. $\square$

6.4.8 PROPOSITION. *Let R be PID, and let (a) be a nonzero ideal in R . Then (a) is prime if and only if it is maximal.*

Proof. Let $I = (a)$ be a prime ideal in R , with $a \neq 0$, and assume that $(a) \subset (b)$ for an ideal of (b) . We have that $a = bc$ for some $c \in R$. But then $b \in (a)$ or $c \in (a)$ since (a) is prime. If $b \in (a)$, then $(b) \subset (a)$ and $(a) = (b)$ follows. If $c \in (a)$, then $c = da$ for some $d \in R$. But then $a = bc = bda$, from which $bd = 1$ since cancellation by the nonzero a holds in R . This implies that b is a unit, and hence $(b) = R$. $\square$

6.4.4 DEFINITION. Let R be an integral domain. A function $N : R \setminus \{0\} \rightarrow \mathbf{N}$ is called a *norm* on R . A *Euclidean norm* on an integral domain R is a norm satisfying the following property: for all $a \in R$ and all nonzero $b \in R$ there exist $q, r \in R$ such that

$$a = qb + r$$

with either $r = 0$ or $N(r) < N(b)$. The element q is called the quotient and the element r the remainder of the division. R is called a *Euclidean domain* (ED) if there is a Euclidean norm N on R .

6.4.9 THEOREM. *Every ED is PID.*

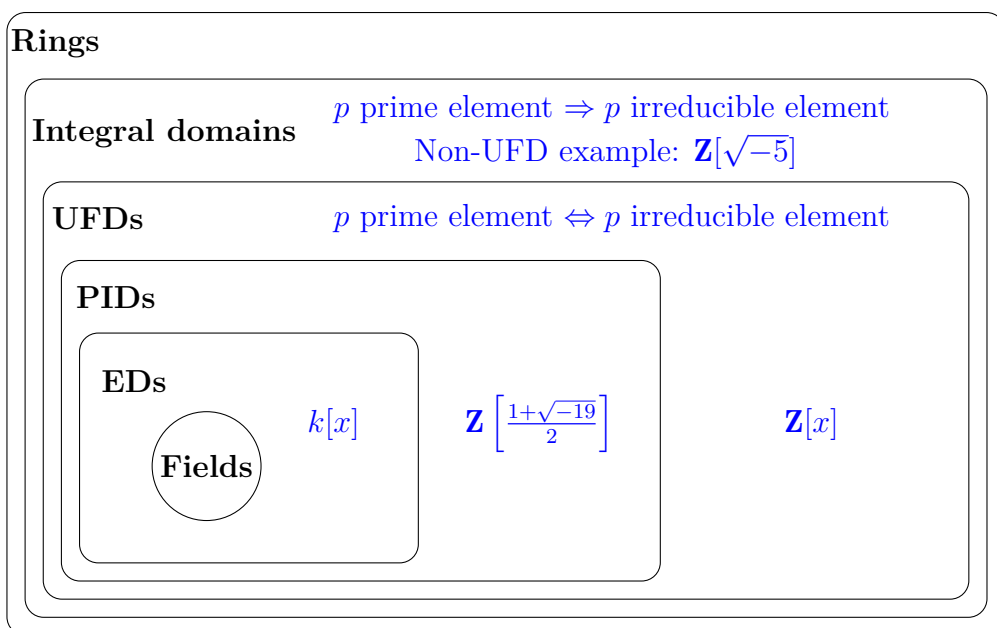
Proof. Let I be an ideal of a Euclidean domain R ; we have to prove that I is principal. If $I = \{0\}$, there is nothing to show, assume $I \neq \{0\}$. Let $b \in I$ be an element with the smallest norm (This must exist, because $\mathbf{N}$ is a well-ordered set). Then we show that $I = (b)$; therefore I is principal, as needed. Clearly $(b) \subset I$, we only need to show that $I \subset (b)$.

For this, let $a \in I$ and apply division with remainder: we have $a = qb + r$ for some $q, r \in R$ with $r = 0$ or $N(r) < N(b)$. But $r = a - qb \in I$; by the minimality of $N(b)$, we cannot have $N(r) < N(b)$. Therefore $r = 0$, showing that $a = qb \in (b)$, as needed. $\square$

6.4.1 EXAMPLE. If k is a field, then $k[x]$ is a ED with Euclidean norm given by $N(f) = \deg f$, hence also a PID. Indeed, if k is a field, the Division Algorithm of polynomials holds in $k[x]$: if $f \neq 0$, then for any $g \in k[x]$, there exist unique $q, r \in k[x]$ such that

$$g = qf + r, \quad r = 0 \text{ or } N(r) = \deg r < \deg f = N(f).$$

The converse to this is also true. Assume $R[x]$ is a PID. Since R is a subring of $R[x]$ under the homomorphism $\iota : R \rightarrow R[x], r \mapsto \mathbf{r} = (r, 0, \dots)$ and $R[x]$ is an integral domain, R must be an integral domain. The ideal (x) is a nonzero prime ideal in $R[x]$ since $R[x]/(x) \cong R$ is an integral domain. Since $R[x]$ is a PID, the prime ideal (x) is maximal, hence the quotient R is a field.



In summary, we have proved that every Euclidean domain is a PID, and every PID is a UFD. The converses are not necessarily true in either case, and examples for these are not so easy to construct. The most typical examples of a PID that is not a Euclidean domain come from certain quadratic integer rings; the most typical example of a UFD that is not a PID is the polynomial ring $\mathbf{Z}[x]$. Our next goal is to introduce these two examples. For the second example, we will give a more detailed treatment and devote a separate subsection to it.

6.4.2 EXAMPLE (Quadratic Integer Rings). Let D be a squarefree integer. It is immediate from the addition and multiplication that the subset $\mathbf{Z}[\sqrt{D}] = \{a + b\sqrt{D} : a, b \in \mathbf{Z}\}$ forms a subring of the quadratic field $\mathbf{Q}(\sqrt{D})$ defined earlier. If $D \equiv 1 \pmod{4}$, then the slightly larger subset

$$\mathbf{Z}\left[\frac{1+\sqrt{D}}{2}\right] = \left\{a + b\frac{1+\sqrt{D}}{2} : a, b \in \mathbf{Z}\right\}$$

is also a subring: closure under addition is immediate and $(a + b\frac{1+\sqrt{D}}{2})(c + d\frac{1+\sqrt{D}}{2}) = (ac + bd\frac{D-1}{4}) + (ad + bc + bd)\frac{1+\sqrt{D}}{2}$ together with the congruence on D shows closure under multiplication. Define

$$\mathcal{O} = \mathcal{O}_{\mathbf{Q}(\sqrt{D})} = \mathbf{Z}[\omega] = \{a + b\omega : a, b \in \mathbf{Z}\}, \quad \omega = \begin{cases} \sqrt{D}, & D \equiv 2, 3 \pmod{4} \\ \frac{1+\sqrt{D}}{2}, & D \equiv 1 \pmod{4} \end{cases}$$

called the *ring of integers* in the quadratic field $\mathbf{Q}(\sqrt{D})$. The terminology comes from the fact that the elements of the subring $\mathcal{O}$ of the field $\mathbf{Q}(\sqrt{D})$ have many properties analogous to those of the subring of integers $\mathbf{Z}$ in the field $\mathbf{Q}$.

Define a function $N : \mathbf{Q}(\sqrt{D}) \setminus \{0\} \rightarrow \mathbf{Q}$ by $N(a + b\sqrt{D}) = (a + \sqrt{D})(a - \sqrt{D}) = a^2 - Db^2 \in \mathbf{Q}$, which, as previously mentioned, is nonzero if $a + b\sqrt{D} \neq 0$. It is easy to check that $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in \mathbf{Q}(\sqrt{D})$. On the subring $\mathcal{O}$, it is also easy to see that the function is given by

$$N(a + b\omega) = (a + b\omega)(a + b\bar{\omega}) = \begin{cases} a^2 - Db^2, & D \equiv 2, 3 \pmod{4} \\ a^2 + ab + \frac{1-D}{4}b^2, & D \equiv 1 \pmod{4}. \end{cases}$$

We may use this function to characterize the units in $\mathcal{O}$. If $\alpha \in \mathcal{O}$ has $N(\alpha) = \pm 1$, the previous formula shows that $(a + b\omega)^{-1} = \pm(a + b\bar{\omega})$, which is again an element of $\mathcal{O}$ and so α is a unit in $\mathcal{O}$. Conversely, suppose α is a unit in $\mathcal{O}$, say $\alpha\beta = 1$ for some $\beta \in \mathcal{O}$. Then $N(\alpha)N(\beta) = N(\alpha\beta) = 1$. Since both $N(\alpha)$ and $N(\beta)$ are integers, each must be ± 1 . Hence

the element α is a unit if and only if $N(\alpha) = \pm 1$.

6.4.3 EXAMPLE. $\mathbf{Z}[\sqrt{-5}]$ is an example of an integer domain that is not a UFD. Now, $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$. We show that $2, 3, 1 + \sqrt{-5}$ and $1 - \sqrt{-5}$ are irreducible. Assume $2 = ab, a, b \in \mathbf{Z}[\sqrt{-5}], N(2) = N(a)N(b)$,

6.4.4 EXAMPLE. In general $\mathcal{O}$ is not Euclidean with respect to these functions. In the special case when $D = -1$ we obtain the ring $\mathbf{Z}[i]$ of *Gaussian integers*, which are the complex numbers $a + bi \in \mathbf{C}$ with a and b both integers. These numbers were originally introduced by Gauss in 1800. The Gaussian integers $\mathbf{Z}[i]$ is a ED with respect to the function $N(a + bi) = a^2 + b^2$:

Proof. Let $\alpha = a + bi, \beta = c + di$ be two elements of $\mathbf{Z}[i]$ with $\beta \neq 0$. Then in the field $\mathbf{Q}(i)$ we have $\alpha/\beta = r + si$ where $r = (ac + bd)/(c^2 + d^2)$ and $s = (bc - ad)/(c^2 + d^2)$ are rational numbers. Let p be an integer closest to the rational number r and let q be integer closest to the rational number s , so that both $|r - p|$ and $|s - q|$ are at most $1/2$. The Division Algorithm follows immediately once we show

$$\alpha = (p + qi)\beta + \gamma, \quad \text{for some } \gamma \in \mathbf{Z}[i] \text{ with } N(\gamma) \leq 1/2N(\beta).$$

Let $\theta = (r - p) + (s - q)i$ and set $\gamma = \beta\theta$. Then $\gamma = \alpha - (p + qi)\beta$, so that $\gamma \in \mathbf{Z}[i]$ is a Gaussian integer and $\alpha = (p + qi)\beta + \gamma$. Since $N(\theta) = (r - p)^2 + (s - q)^2$ is at most $1/4 + 1/4 = 1/2$, hence $N(\gamma) = N(\theta)N(\beta) \leq 1/2N(\beta)$. $\square$

The last result in this subsection will be used to prove that not every PID is a Euclidean Domain and relates the PID with another weakening of the Euclidean condition.

6.4.5 DEFINITION. Let R be an integral domain. A norm $N : R \setminus \{0\} \rightarrow \mathbf{N}$ is called a *Dedekind-Hasse norm* on R if it satisfies the following property: for all nonzero $a, b \in R$, either $a \in (b)$ or there exist $s, t \in R$ such that

$$N(sa - tb) < N(b).$$

Note that R is Euclidean with respect to a norm N if it is always possible to satisfy the Dedekind-Hasse condition with $s = 1$, so this is indeed a weakening of the Euclidean condition. Next, we prove that the Dedekind-Hasse norm can be used to equivalently characterize PIDs. From this perspective, the proposition that every ED is necessarily a PID becomes quite evident.

6.4.10 PROPOSITION. *An integral domain R is a PID if and only if R has a Dedekind-Hasse norm.*

Proof. Let I be any nonzero ideal of R and let b be a nonzero element of I with the smallest norm. We show that $I = (b)$. Suppose that a is any nonzero element in I , so that the ideal (a, b) is contained in I . Then the Dedekind-Hasse condition on N implies that $a \in (b)$, so $I = (b)$.

Conversely, if R is a PID, then R is a UFD. Define the norm N by setting $N(u) = 1$ if u is a unit, and $N(a) = 2^n$ if $a = p_1 \cdots p_n$ where p_i are irreducibles in R (well defined since the number of irreducible factors of a is unique). Clearly, $N(ab) = N(a)N(b)$. To show that N is Dedekind-Hasse, suppose that a, b are nonzero elements of R . Then the ideal $(a, b) = (r)$ for some r since R is a PID. If $a \notin (b)$, then also $r \notin (b)$. Since $b = xr$ for some $x \in R$, it follows that x is not a unit in R and so $N(b) = N(x)N(r) > N(r)$. Hence (a, b) contains a nonzero element with norm strictly smaller than the norm of b . $\square$

For $D = 1, 2, 3, 7, 11, 19, 43, 67, 163$, $\mathcal{O}_{\mathbf{Q}(\sqrt{-D})}$ is a PID.

For $D = 19, 43, 67, 163$, $\mathbf{Z}\left[\frac{1+\sqrt{-D}}{2}\right]$ is a PID, but not a ED.

6.4.5 EXAMPLE. We show that the quadratic integer ring $R = \mathbf{Z}[(1 + \sqrt{-19})/2]$ is a PID but not a ED. We show that the norm $N(a + b(1 + \sqrt{-19})/2) = a^2 + ab + 5b^2$ defined on R is a Dedekind-Hasse norm.

Proof. Suppose α, β are nonzero elements of R and $\alpha \notin (\beta)$ ($\alpha/\beta \notin R$). We must show that there are elements $s, t \in R$ with $N(s\alpha - t\beta) < N(\beta)$. If we consider this inequality and the original function N over the field $\mathbf{Q}(\sqrt{-19})$, then once we prove that

$$N\left(\frac{\alpha}{\beta}s - t\right) < 1$$

we can use its multiplicativity to obtain that the inequality holds in $\mathbf{Q}(\sqrt{-19})$, and then restrict to R to obtain the desired conclusion. Write $\frac{\alpha}{\beta} = \frac{a+b\sqrt{-19}}{c} \in \mathbf{Q}(\sqrt{-19})$ with integers a, b, c having no common divisor and with $c > 1$ (since β is assumed not to divide α). Since a, b, c have no common divisor there are integers x, y, z with $ax + by + cz = 1$. Write $ay - 19bx = cq + r$ for some q, r with $|r| \leq c/2$ and let $s = y + x\sqrt{-19}$ and $t = q - z\sqrt{-19}$. Then a quick computation shows that

$$N\left(\frac{\alpha}{\beta}s - t\right) = \frac{(ay - 19bx - cq)^2 + 19(ax + by + cz)^2}{c^2} \leq \frac{1}{4} + \frac{19}{c^2}$$

and so the inequality $N(\frac{\alpha}{\beta}s - t) < 1$ is satisfied with this s and t if $c \geq 5$. The target inequality also holds for $c = 2, 3, 4$; for the detailed process, see page 282 of *Dummit and Foote's Abstract Algebra*. $\square$

B Gauss's Lemma

The core of this subsection is to prove the following theorem: if R is a UFD, then $R[x]$ is also a UFD. On this basis, $\mathbf{Z}[x]$ becomes a typical UFD, and since $(2, x)$ in $\mathbf{Z}[x]$ is not a principal ideal, it is not a PID; thus this provides a typical example of a UFD that is not a PID. Proving this theorem requires several steps, which are divided into two major parts: the first part is to prove Gauss's lemma, and the second part is to prove the theorem.

6.4.11 LEMMA. *Let I be an ideal of a ring R ; and define $I[x] := \{a_0 + a_1x + \cdots + a_dx^d \in R[x] : a_i \in I\}$. Clearly, $I[x]$ is an ideal of $R[x]$. In fact, it is the ideal generated by the "subset" I of $R[x]$. Then*

$$\frac{R[x]}{I[x]} \cong \frac{R}{I}[x].$$

Proof. Consider the function $\varphi : R[x] \rightarrow (R/I)[x]$ giving by reducing each of the coefficients of a polynomial modulo I . The definition of addition and multiplication in these two rings shows that φ is a ring homomorphism. The kernel is precisely the set of polynomials each of whose coefficients is an element of I , which is to say that

$$\ker \varphi = I[x]$$

$\square$

6.4.12 COROLLARY. *If I is a prime ideal of R , then $I[x]$ is prime in $R[x]$.*

Proof. Since I is a prime ideal, R/I is an integral domain. Hence $(R/I)[x]$ is an integral domain. This shows if I is a prime ideal of R , then $I[x]$ is a prime ideal of $R[x]$. $\square$

6.4.6 EXAMPLE. Let $R = \mathbf{Z}$ and consider the ideal $n\mathbf{Z}$ of $\mathbf{Z}$. Then the isomorphism above can be written

$$\frac{\mathbf{Z}[x]}{n\mathbf{Z}[x]} \cong (\mathbf{Z}/n\mathbf{Z})[x]$$

and the natural projection function of $\mathbf{Z}[x]$ into $\mathbf{Z}/n\mathbf{Z}$ by reducing the coefficients modulo n is a ring homomorphism. If n is composite, then clearly the quotient ring is not an integral domain. If, however, n is a prime p , then $\mathbf{Z}/p\mathbf{Z}$ is a field and so $(\mathbf{Z}/p\mathbf{Z})[x]$ is an integral domain.

6.4.6 DEFINITION. Let R be a commutative ring, and let $f = a_0 + a_1x + \cdots + a_dx^d \in R[x]$ be a polynomial. We define

- f is *very primitive* if for all prime ideals $\mathfrak{p}$ of R , $f \notin \mathfrak{p}[x]$.
- f is *primitive* if for all principal prime ideals $\mathfrak{p}$ of R , $f \notin \mathfrak{p}[x]$.

Now, consider the quotient $R/\mathfrak{p}$, it is an integral domain, one may consider the field of fractions $\kappa(\mathfrak{p}) = \text{Frac}(R/\mathfrak{p})$. Define a polynomial $\bar{f}_{\mathfrak{p}}$ in $\kappa(\mathfrak{p})[x]$

$$\bar{f}_{\mathfrak{p}} := \bar{a}_0 + \bar{a}_1x + \cdots + \bar{a}_dx^d \in \kappa(\mathfrak{p})[x]$$

6.4.13 LEMMA. Let R be a commutative ring. And let $f, g \in R[x]$. Then

$$fg \text{ is (very) primitive} \iff \text{both } f \text{ and } g \text{ are (very) primitive.}$$

Proof.

$$\begin{aligned} fg \text{ primitive} &\iff \forall \mathfrak{p} \text{ prime and principal } fg \notin \mathfrak{p}[x] \\ &\iff \forall \mathfrak{p} \text{ prime and principal } f \notin \mathfrak{p}[x] \text{ and } g \notin \mathfrak{p}[x] \\ &\iff f \text{ is primitive and } g \text{ is primitive} \end{aligned}$$

□

6.4.14 LEMMA. Let R be a commutative ring and $f = a_0 + a_1x + \cdots + a_dx^d \in R[x]$.

- f is *very primitive* if and only if $(a_0, \dots, a_d) = (1)$.
- If R is a UFD, then f is *primitive* if and only if $\gcd(a_0, \dots, a_d) = 1$.

Proof. If $(a_0, \dots, a_d) = (1)$, then no prime ideal can contain all coefficients a_i and it follows that f is very primitive (if $\exists \mathfrak{p}$ such that $f \in \mathfrak{p}[x]$, then $a_i \in \mathfrak{p}$). Conversely, if f is very primitive, then the coefficients of f are not all contained in any one prime ideal, and in particular they are not all contained in any maximal ideal. Thus, $(a_0, \dots, a_d) = (1)$ in this case, since every proper ideal is contained in a maximal ideal.

$f \notin \mathfrak{p}[x]$ for any principal prime $\mathfrak{p}$ if and only if no $\mathfrak{p}$ can contain all coefficients a_i . In a UFD, $\gcd(a_0, \dots, a_d) \neq 1$ if and only if $\exists q \in R$ irreducible such that $(a_0, \dots, a_d) \subset (q)$: assume $\gcd(a_0, \dots, a_d) = g \notin R^\times$, then $\dot{g} = \dot{q}_1 \cdots \dot{q}_n$ for unique irreducible elements $q_1, \dots, q_n$. Then $\dot{a}_i = \dot{q}_1 \cdots \dot{q}_n \dot{b}_i$ for each i and $a_i \in (q_1)$. Hence $(a_0, \dots, a_d) \subset (q_1)$. Conversely, if $\exists q \in R$ irreducible such that $(a_0, \dots, a_d) \subset (q)$, then $q \mid a_i$ for each i and $\gcd(a_0, \dots, a_d) \neq 1$. Now, in UFD, every irreducible element is prime, which implies (q) is prime. That is, $\gcd a_0, \dots, a_d \neq 1$ if and only if $\exists$ principal prime (q) such that $(a_0, \dots, a_d) \subset (q)$ if and only if f is not primitive. □

6.4.7 DEFINITION. Let R be a UFD, the *content* of a nonzero polynomial $f \in R[x]$ denoted cont_f is the gcd of its coefficients.

6.4.15 LEMMA. Let R be a UFD. and let $f \in R[x]$. Then $(f) = (\text{cont}_f)(\underline{f})$ where $\underline{f}$ is primitive. Furthermore, if $(f) = (c)(g)$ with $c \in R$ and g primitive, then $(c) = (\text{cont}_f)$.

Proof. Let $f = a_0 + a_1x + \cdots + a_dx^d$ and $\text{cont}_f = \gcd(a_0, \dots, a_d) = c$. Then $c \mid a_i$ for each i and $a_i = c \cdot b_i$. Clearly $\gcd(b_0, \dots, b_d) = 1$ which implies $\underline{f} = b_0 + \cdots + b_dx^d$ is primitive. Since for any principal ideals (a) and (b) , $(ab) = (a)(b)$, $f = \text{cont}_f \cdot \underline{f}$ implies $(f) = (\text{cont}_f)(\underline{f})$.

Assume $(f) = (c)(g)$ with $c \in R$ and g primitive. This means $f = ucg$ for some unit $u \in R^\times$ (Since R is a UFD (integral domain), the units of $R[x]$ are the units of R). Clearly $\text{cont}_f = \text{cont}_{ucg} = uc$. Hence $(\text{cont}_f) = (c)$, as needed. □

6.4.16 THEOREM (Gauss's Lemma). Let R be UFD, and let $f, g \in R[x]$. Then

$$(\text{cont}_{fg}) = (\text{cont}_f)(\text{cont}_g).$$

Proof. $(fg) = (f)(g) = ((\text{cont}_f)(\underline{f}))((\text{cont}_g)(\underline{g})) = (\text{cont}_f)(\text{cont}_g)(\underline{f}\underline{g})$ with $\text{cont}_f, \text{cont}_g$ primitive. Then $\underline{f}\underline{g}$ is primitive. By the lemma above, it follows that $(\text{cont}_f)(\text{cont}_g)$ is the content of fg , as needed. $\square$

6.4.17 COROLLARY. *Let R be a UFD. If $(f) \subset (g)$, then $(\text{cont}_f) \subset (\text{cont}_g)$.*

Proof. $f = gh$ for some $h \in R[x]$. Hence $(\text{cont}_f) = (\text{cont}_g)(\text{cont}_h) \subset (\text{cont}_g)$. $\square$

Now let us return to the main line: proving that $R[x]$ is a UFD if R is also a UFD. In order to prove that $R[x]$ is UFD whenever R is UFD, we have to verify that $R[x]$ satisfies **ACCP** and that every irreducible element in $R[x]$ is prime. The general idea is to reduce these questions to matters in $F[x]$ where $F = \text{Frac}(R)$ is the field of fractions of R : as we know, $F[x]$ is a UFD (ED). The following lemma captures the most crucial ingredient of the intersection between $R[x]$ and $F[x]$:

6.4.18 LEMMA. *Let R be a UFD, and let $F = \text{Frac}(R)$ be its field of fractions. For nonzero $f, g \in R[x]$, denote by $(f), (g)$ the principal ideal of $R[x]$, and denote by $(f)_F, (g)_F$ the principal ideals of $F[x]$. Assume*

- $(\text{cont}_g) \subset (\text{cont}_f)$;
- $(g)_F \subset (f)_F$.

Then $(g) \subset (f)$.

Proof. Since $(g)_F \subset (f)_F$, we have $g = fh$ for some $h \in F[x]$. Write $h = \frac{a}{b}\underline{h}$ where $a, b \in R$ and $\underline{h} \in R[x]$ is primitive. We then have $bg = af\underline{h}$ in $R[x]$ ($F[x]$). By Gauss's lemma and since $\underline{h}$ is primitive,

$$(a\text{cont}_f) = (b\text{cont}_g);$$

and since $(\text{cont}_g) \subset (\text{cont}_f)$ by hypothesis, we obtain $(a\text{cont}_f) \subset (b\text{cont}_f)$. Since R is an integral domain and $(\text{cont}_f) \neq (0)$ this implies $a = bc$ for some $c \in R$. But then $h = \frac{a}{b}\underline{h} = c\underline{h} \in R[x]$, and $g = fh \in (f)$; that is $(g) \subset (f)$, as needed. $\square$

6.4.19 PROPOSITION. *Let R be a UFD and let $F = \text{Frac}(R)$ be its field of fractions. Let $f \in R[x]$ be a nonconstant, irreducible polynomial. Then f is irreducible as an element of $F[x]$.*

Proof. First note that f is primitive; otherwise we could factor out its content and f would not be irreducible. Next, assume that $f = gh$ with $g, h \in F[x]$. Let $c, d \in F$ such that

$$g = c\underline{g}, \quad h = d\underline{h}$$

and $\underline{g}, \underline{h}$ are primitive polynomials in $R[x]$. Then $\underline{g}\underline{h}$ is also primitive; thus $(\text{cont}_{\underline{g}\underline{h}}) = (1) = (\text{cont}_f)$; further, $(f)_F = (gh)_F$ since $cd \neq 0$ is a unit in F . By lemma above, $(f) = (\underline{g}\underline{h})$ in $R[x]$, that is, $f = u\underline{g}\underline{h}$ with $u \in R[x]$ a unit. Since f is irreducible in $R[x]$, this implies that either $\underline{g}$ or $\underline{h}$ is a unit in $R[x]$. But then g or h were units in $F[x]$, as needed. $\square$

6.4.20 COROLLARY. *Let R be a UFD and $F = \text{Frac}(R)$ be its field of fractions. Let $f \in R[x]$ be a nonconstant polynomial. Then f is irreducible in $R[x]$ if and only if it is irreducible in $F[x]$ and primitive.*

6.4.21 THEOREM. *If R is a UFD, so is $R[x]$.*

Proof. (1).**ACCP**: Let $(f_1) \subset (f_2) \subset (f_3) \subset \dots$ be an ascending chain of principal ideals of $R[x]$. This induces an ascending chain of principal ideals

$$(\text{cont}_{f_1}) \subset (\text{cont}_{f_2}) \subset (\text{cont}_{f_3}) \subset \dots$$

in the UFD R which implies $(\text{cont}_{f_i}) = (\text{cont}_{f_{i+1}})$ for some $i \geq 0$. On the other hand, in the ED $F[x]$, we have

$$(f_1)_F \subset (f_2)_F \subset (f_3)_F \subset \cdots$$

as a sequence of ideals in $F[x]$; since $F[x]$ is a UFD (ED), this sequence stabilizes. Therefore $(f_i)_F = (f_{i+1})_F$ for some $i \geq 0$. By the lemma above, $(f_i) = (f_{i+1})$ for some $i \geq 0$; that is, the given chain of principal ideals stabilizes, as needed.

Next, we consider an irreducible element f of $R[x]$. We show that (f) is a prime ideal. If f is irreducible and constant, then f is prime in R since R is UFD, and it follows that f is prime in $R[x]$. Thus we may assume that f is nonconstant and irreducible in $R[x]$.

By the proposition above, f is irreducible as an element of $F[x]$; since $F[x]$ is a UFD, $(f)_F$ is prime in $F[x]$. Consider the composition

$$\rho : R[x] \hookrightarrow F[x] \twoheadrightarrow \frac{F[x]}{(f)_F}.$$

We claim that $\ker \rho = (f)$. Indeed, $(f) \subset \ker \rho$ is clear; for the other inclusion, note that $\rho(g) = 0$ implies g is divisible by f in $F[x]$: that is $(g)_F \subset (f)_F$; and we have $(\text{cont}_g) \subset (\text{cont}_f)$ since f is primitive. Then $(g) \subset (f)$ in $R[x]$, i.e., g is divisible by f in $R[x]$, as needed. Since $\ker \rho = (f)$,

$$\frac{R[x]}{(f)} \cong \text{im } \rho \subset \frac{F[x]}{(f)_F}$$

Since the ring on the right is an integral domain (because $(f)_F$ is prime in $F[x]$), so is the ring on the left. This shows that (f) is prime in $R[x]$. $\square$

7

Category Theory II

7.1 Functors

7.2 Limits

A Equaliser and Kernel